GRADE 8 COMMON CORE MATH

Answer Key

Main • Backward Branches • Forward 1 (Grade 9) • Forward 2 (Grade 10)

65 question sets • 1625 questions

Page numbers match the Question Collection.

Every page shows its question ID (for example S40-F2-Q3).

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SET 1 • 8.NS.A.1 • THE NUMBER SYSTEM

Classify numbers as rational or irrational

MAIN	Grade 8 main question	**8.NS.A.1**
BACKWARD 1	Write a fraction as a terminating or repeating decimal	**7.NS.A.2.d**
BACKWARD 2	Write rational numbers as quotients of integers	**7.NS.A.2.b**
FORWARD 1	Decide whether sums and products are rational or irrational	**HSN-RN.B.3**
FORWARD 2	Extend the number system to complex numbers using i	**HSN-CN.A.1**

SET 1 • 8.NS.A.1

MAIN

Grade 8 main question • Grade 8 • The Number System • 8.NS.A.1

Which number is irrational?

A. $\sqrt{49}$

B. $\frac{-7}{3}$

C. 0.1212... (the digits 12 repeat)

D. $\sqrt{15}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q1/5

8

QUESTION ID

S1-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. √15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q1/5

Is 0.375 rational or irrational? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q2/5

9

QUESTION ID

S1-M-Q2

TYPE

Short answer

CORRECT ANSWER

Rational. It equals 3/8, a quotient of integers (its decimal ends).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q2/5

Is √50 rational or irrational? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q3/5

10

QUESTION ID

S1-M-Q3

TYPE

Short answer

CORRECT ANSWER

Irrational. 50 is not a perfect square, so √50 cannot be written as a quotient of integers; its decimal neither ends nor repeats.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q3/5

True or false?

Every number whose decimal expansion repeats is rational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q4/5

11

QUESTION ID

S1-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q4/5

True or false?

π is rational because it is often written as 3.14.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q5/5

12

QUESTION ID

S1-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 3.14 is only an approximation; π is irrational.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q5/5

SET 1 • 8.NS.A.1

BACKWARD 1

Write a fraction as a terminating or repeating decimal • Grade 7 • The Number System • 7.NS.A.2.d

Write $\frac{5}{8}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q1/5

14

QUESTION ID

S1-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0.625

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q1/5

Write $\frac{2}{3}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q2/5

15

QUESTION ID

S1-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0.666... (the 6 repeats)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q2/5

Which fraction has a repeating decimal?

A. $\frac{3}{5}$

B. $\frac{1}{6}$

C. $\frac{1}{4}$

D. $\frac{7}{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q3/5

16

QUESTION ID

S1-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q3/5

True or false?

$\frac{4}{11}$ = 0.3636... (the digits 36 repeat)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q4/5

17

QUESTION ID

S1-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\frac{1}{3} = 0.3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q5/5

18

QUESTION ID

S1-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 1/3 = 0.333..., a repeating decimal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q5/5

SET 1 • 8.NS.A.1

BACKWARD 2

Write rational numbers as quotients of integers • Grade 7 • The Number System • 7.NS.A.2.b

Write $-2\frac{1}{4}$ as a quotient of two integers.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q1/5

20

QUESTION ID

S1-B2-Q1

TYPE

Short answer

CORRECT ANSWER

−9/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q1/5

Write 0.7 as a quotient of two integers.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q2/5

21

QUESTION ID

S1-B2-Q2

TYPE

Short answer

CORRECT ANSWER

7/10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q2/5

Which number is equal to $\frac{-5}{6}$?

A. $\frac{5}{6}$

B. $\frac{-5}{-6}$

C. $\frac{6}{-5}$

D. $\frac{5}{-6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q3/5

22

QUESTION ID

S1-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 5/(−6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q3/5

True or false?

-12 can be written as $\frac{-12}{1}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q4/5

23

QUESTION ID

S1-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q4/5

True or false?

$\frac{-3}{0}$ is a rational number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q5/5

24

QUESTION ID

S1-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Division by 0 is undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q5/5

SET 1 • 8.NS.A.1

FORWARD 1

Decide whether sums and products are rational or irrational • High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3

Explain why 4 + √3 is irrational. (Suppose 4 + √3 = r for some rational number r, and see what that would say about √3.)

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q1/5

26

QUESTION ID

S1-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

4 + √3 is irrational.

GRADING NOTE REQUIRED WORK: An argument like this: if 4 + √3 = r with r rational, then √3 = r − 4, a difference of two rational numbers, which is rational. But √3 is irrational, so 4 + √3 cannot be rational.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F1-Q1**

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q1/5

Is √8 × √2 rational or irrational? Give its value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q2/5

27

QUESTION ID

S1-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Rational; √16 = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q2/5

Which product is irrational?

A. $\sqrt{3} \times \sqrt{12}$

B. $3\sqrt{5}$

C. $0 \times \sqrt{7}$

D. $\frac{2}{3} \times 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q3/5

28

QUESTION ID

S1-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $3\sqrt{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q3/5

True or false?

The sum of two rational numbers is always rational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q4/5

29

QUESTION ID

S1-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q4/5

True or false?

The product of two irrational numbers is always irrational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q5/5

30

QUESTION ID

S1-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{2} \times \sqrt{2} = 2$, which is rational.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F1-Q5**

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q5/5

SET 1 • 8.NS.A.1

FORWARD 2

Extend the number system to complex numbers using i • High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1

The number i has $i^2 = -1$.

Write $\sqrt{-16}$ in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q1

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q1/5

32

QUESTION ID

S1-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F2-Q1**

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q1/5

The number i has $i^2 = -1$.

What is i^4?

$i^4 =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q2

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q2/5

33

QUESTION ID

S1-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F2-Q2**

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q2/5

The number i has $i^2 = -1$.

Which number is a solution of $x^2 = -9$?

A. 3

B. -3

C. 9i

D. 3i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q3

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q3/5

34

QUESTION ID

S1-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q3

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q3/5

True or false?

Every real number a is also a complex number, a + 0i.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q4

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q4/5

35

QUESTION ID

S1-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q4

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q4/5

True or false?

$\sqrt{-25}$ is a real number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q5

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q5/5

36

QUESTION ID

S1-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{-25} = 5i$, which is not real.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q5

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q5/5

SET 2 • 8.NS.A.1 • THE NUMBER SYSTEM

Convert a repeating decimal into a fraction

MAIN	Grade 8 main question	**8.NS.A.1**
BACKWARD 1	Use long division to write a fraction as a repeating decimal	**7.NS.A.2.d**
BACKWARD 2	Multiply decimals by powers of 10	**5.NBT.A.2**
FORWARD 1	Solve linear equations with letter coefficients that come from repe.	**HSA-REI.B.3**
FORWARD 2	Write a geometric sequence from a repeating decimal	**HSF-BF.A.2**

SET 2 • 8.NS.A.1

MAIN

Grade 8 main question • Grade 8 • The Number System • 8.NS.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Write 0.444... (the 4 repeats) as a fraction. Show your work.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q1/5

39

QUESTION ID

S2-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

4/9

GRADING NOTE REQUIRED WORK: Any valid method. Example: let x = 0.444...; then 10x = 4.444...; subtracting gives 9x = 4, so x = 4/9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q1/5

Write 0.2727... (the digits 27 repeat) as a fraction in simplest form.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q2/5

40

QUESTION ID

S2-M-Q2

TYPE

Short answer

CORRECT ANSWER

3/11

GRADING NOTE 27/99 is equal but is not in simplest form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q2/5

Which fraction is equal to 0.1666... (the 6 repeats)?

A. $\frac{1}{16}$

B. $\frac{1}{6}$

C. $\frac{16}{100}$

D. $\frac{6}{10}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q3/5

41

QUESTION ID

S2-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q3/5

True or false?

2.333... (the 3 repeats) is equal to $\frac{7}{3}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q4/5

42

QUESTION ID

S2-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q4/5

True or false?

0.0909... (the digits 09 repeat) is equal to $\frac{9}{100}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q5/5

43

QUESTION ID

S2-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is equal to 9/99 = 1/11.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q5/5

SET 2 • 8.NS.A.1

BACKWARD 1

Use long division to write a fraction as a repeating decimal • Grade 7 • The Number System • 7.NS.A.2.d

Write $\frac{7}{9}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q1/5

45

QUESTION ID

S2-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0.777... (the 7 repeats)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q1/5

Write $\frac{5}{12}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q2/5

46

QUESTION ID

S2-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0.41666... (the 6 repeats)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q2/5

Which decimal is equal to $\frac{3}{11}$?

A. 0.27 (the decimal ends)

B. 0.3

C. 0.2727... (the digits 27 repeat)

D. 0.311

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q3/5

47

QUESTION ID

S2-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 0.2727... (the digits 27 repeat)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q3/5

True or false?

$\frac{1}{9} = 0.111...$ (the 1 repeats)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q4/5

48

QUESTION ID

S2-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q4/5

True or false?

$\frac{5}{6} = 0.56$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q5/5

49

QUESTION ID

S2-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 5/6 = 0.8333...

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q5/5

SET 2 • 8.NS.A.1

BACKWARD 2

Multiply decimals by powers of 10 • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Multiply.

0.454 × 100

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q1/5

51

QUESTION ID

S2-B2-Q1

TYPE

Short answer

CORRECT ANSWER

45.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q1/5

Multiply.

3.27 × 10

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q2/5

52

QUESTION ID

S2-B2-Q2

TYPE

Short answer

CORRECT ANSWER

32.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q2/5

Multiply.

0.0081 × 1,000

A. 81

B. 8.1

C. 0.81

D. 0.081

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q3/5

53

QUESTION ID

S2-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 8.1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q3/5

True or false?

$2.5 \times 100 = 250$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q4/5

54

QUESTION ID

S2-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q4/5

True or false?

$0.36 \times 10 = 36$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q5/5

55

QUESTION ID

S2-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q5/5

SET 2 • 8.NS.A.1

FORWARD 1

Solve linear equations with letter coefficients that come from repeating decimals • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Let x = 0.ddd..., where the digit d (from 1 to 9) repeats. Then 10x = d + x.

Solve for x in terms of d.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q1/5

57

QUESTION ID

S2-F1-Q1

TYPE

Short answer

CORRECT ANSWER

d/9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q1/5

Let x = 0.ababab..., where the two digits ab repeat, and let n be the two-digit number ab. Then 100x = n + x.

Solve for x in terms of n.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q2/5

58

QUESTION ID

S2-F1-Q2

TYPE

Short answer

CORRECT ANSWER

n/99

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q2/5

Solve for x in terms of k. Assume k ≠ 1.

kx = x + 5

A. x = 5 ÷ (k − 1)

B. x = 5 − k

C. x = 5 ÷ k

D. x = 5k − 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q3/5

59

QUESTION ID

S2-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. x = 5 ÷ (k − 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q3/5

True or false?

If $1{,}000x = 123 + x$, then $x = \frac{123}{999} = \frac{41}{333}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q4/5

60

QUESTION ID

S2-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q4/5

True or false?

For every number m, the equation mx = 2x + 3 has exactly one solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q5/5

61

QUESTION ID

S2-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE When m = 2, it becomes 0 = 3, which has no solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q5/5

SET 2 • 8.NS.A.1

FORWARD 2

Write a geometric sequence from a repeating decimal • High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2

0.333... = 0.3 + 0.03 + 0.003 + ... The terms 0.3, 0.03, 0.003, ... form a geometric sequence.

What is the common ratio?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q1/5

63

QUESTION ID

S2-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.1

GRADING NOTE 1/10 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q1/5

Write an explicit formula for the nth term of the sequence 0.3, 0.03, 0.003, ...

a(n) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q2/5

64

QUESTION ID

S2-F2-Q2

TYPE

Short answer

CORRECT ANSWER

0.3(0.1)^(n − 1)

GRADING NOTE Equivalent forms such as 3(0.1)^n are correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q2**

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q2/5

Which recursive formula gives the sequence 0.27, 0.0027, 0.000027, ...?

A. $a(1) = 0.27;\ a(n) = 100 \cdot a(n-1)$

B. $a(1) = 0.27;\ a(n) = 0.01 \cdot a(n-1)$

C. $a(1) = 0.27;\ a(n) = a(n-1) - 0.01$

D. $a(1) = 0.0027;\ a(n) = 0.01 \cdot a(n-1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q3/5

65

QUESTION ID

S2-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $a(1) = 0.27;\ a(n) = 0.01 \cdot a(n-1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q3/5

True or false?

The sequence 0.5, 0.05, 0.005, ... is geometric with a common ratio of $\frac{1}{10}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q4/5

66

QUESTION ID

S2-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q4/5

True or false?

The sequence 0.4, 0.04, 0.004, ... is arithmetic.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q5/5

67

QUESTION ID

S2-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The terms have a common ratio, not a common difference.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q5/5

SET 3 • 8.NS.A.2 • THE NUMBER SYSTEM

Estimate the value of an irrational number

MAIN	Grade 8 main question	**8.NS.A.2**
BACKWARD 1	Find squares of whole numbers and decimals	**6.EE.A.1**
BACKWARD 2	Round decimals to the nearest tenth	**5.NBT.A.4**
FORWARD 1	Solve quadratic equations by taking square roots, exactly and ap…	**HSA-REI.B.4.b**
FORWARD 2	Find arc lengths and sector areas exactly in terms of π and appro…	**HSG-C.B.5**

SET 3 • 8.NS.A.2

MAIN

Grade 8 main question • Grade 8 • The Number System • 8.NS.A.2

Between which two whole numbers is √40?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q1/5

70

QUESTION ID

S3-M-Q1

TYPE

Short answer

CORRECT ANSWER

6 and 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q1/5

Estimate √20 to the nearest tenth. Explain how you know.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q2/5

71

QUESTION ID

S3-M-Q2

TYPE

Short answer

CORRECT ANSWER

4.5. $4.4^2 = 19.36$ and $4.5^2 = 20.25$, and 20 is closer to 20.25.

GRADING NOTE Grade both: 4.5, and a reason that compares squares of tenths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q2/5

Which is the best estimate of √75 to the nearest tenth?

A. 8.7

B. 8.6

C. 37.5

D. 9.1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q3/5

72

QUESTION ID

S3-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 8.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q3/5

True or false?

√2 is between 1.4 and 1.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q4/5

73

QUESTION ID

S3-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q4/5

True or false?

√90 is between 8 and 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q5/5

74

QUESTION ID

S3-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $9^2 = 81$ and $10^2 = 100$, so √90 is between 9 and 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q5/5

SET 3 • 8.NS.A.2

BACKWARD 1

Find squares of whole numbers and decimals • Grade 6 • Expressions & Equations • 6.EE.A.1

Find the value.

7^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q1/5

76

QUESTION ID

S3-B1-Q1

TYPE

Short answer

CORRECT ANSWER

49

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q1/5

Find the value.

1.5^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q2/5

77

QUESTION ID

S3-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q2/5

Find the value.

12^2

A. 24

B. 121

C. 122

D. 144

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q3/5

78

QUESTION ID

S3-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 144

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q3/5

True or false?

$8^2 = 64$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q4/5

79

QUESTION ID

S3-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q4/5

True or false?

$3^2 = 6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q5/5

80

QUESTION ID

S3-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q5/5

SET 3 • 8.NS.A.2

BACKWARD 2

Round decimals to the nearest tenth • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4

Round 4.472 to the nearest tenth.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q1/5

82

QUESTION ID

S3-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q1/5

Round 8.660 to the nearest tenth.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q2/5

83

QUESTION ID

S3-B2-Q2

TYPE

Short answer

CORRECT ANSWER

8.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q2/5

Round 3.162 to the nearest tenth.

A. 3.1

B. 3.0

C. 3.2

D. 3.16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q3/5

84

QUESTION ID

S3-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q3/5

True or false?

6.325 rounded to the nearest tenth is 6.3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q4/5

85

QUESTION ID

S3-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q4/5

True or false?

2.449 rounded to the nearest tenth is 2.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q5/5

86

QUESTION ID

S3-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q5/5

SET 3 • 8.NS.A.2

FORWARD 1

Solve quadratic equations by taking square roots, exactly and approximately • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Solve $x^2 = 30$. Give the exact solutions, then round each to the nearest tenth.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q1/5

88

QUESTION ID

S3-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$x = \sqrt{30}$ and $x = -\sqrt{30}$; about 5.5 and −5.5

GRADING NOTE Both solutions are required, exact and rounded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F1-Q1**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q1/5

Solve $x^2 - 7 = 0$. Give the exact solutions.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q2/5

89

QUESTION ID

S3-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$x = \sqrt{7}$ and $x = -\sqrt{7}$

GRADING NOTE Both solutions are required; $x = \pm\sqrt{7}$ is accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F1-Q2**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q2/5

Solve $(x - 1)^2 = 5$.

A. $x = 1 + \sqrt{5}$ or $x = 1 - \sqrt{5}$

B. $x = 1 + \sqrt{5}$ only

C. $x = 6$ or $x = -4$

D. $x = \sqrt{6}$ or $x = -\sqrt{6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q3/5

90

QUESTION ID

S3-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $x = 1 + \sqrt{5}$ or $x = 1 - \sqrt{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F1-Q3**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q3/5

True or false?

The solutions of $x^2 = 12$ are $x = 2\sqrt{3}$ and $x = -2\sqrt{3}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q4/5

91

QUESTION ID

S3-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q4/5

True or false?

$x^2 = -9$ has the real solutions 3 and −3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q5/5

92

QUESTION ID

S3-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE No real number squared is negative, so there is no real solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q5/5

SET 3 • 8.NS.A.2

FORWARD 2

Find arc lengths and sector areas exactly in terms of π and approximately • High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5

⚑ No high-school standard directly continues this skill at this step. This branch uses the nearest related high-school standard (placed at Grade 10 in this bank).

A circle has a radius of 6 cm.

Find the length of a 90° arc exactly in terms of π, then to the nearest tenth.

Arc length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q1

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q1/5 • NEAREST RELATED

94

QUESTION ID

S3-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3π cm, about 9.4 cm

GRADING NOTE Both the exact and the rounded value are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q1

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q1/5 • NEAREST RELATED

A circle has a radius of 4 inches.

Find the area of a 45° sector exactly in terms of π, then to the nearest tenth.

Sector area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q2

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q2/5 • NEAREST RELATED

95

QUESTION ID

S3-F2-Q2

TYPE

Short answer

CORRECT ANSWER

2π square inches, about 6.3 square inches

GRADING NOTE Both the exact and the rounded value are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F2-Q2**

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q2/5 • NEAREST RELATED

A circle has a radius of 3 m.

To the nearest tenth, how long is a 60° arc?

A. 6.3 m

B. 9.4 m

C. 3.1 m

D. 18.8 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q3

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q3/5 • NEAREST RELATED

96

QUESTION ID

S3-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3.1 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q3

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

The arc of a semicircle with a radius of 1 is π units long, about 3.14 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q4/5 • NEAREST RELATED

97

QUESTION ID

S3-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

A 180° sector of a circle with a radius of 2 has an area of 4π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q5

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q5/5 • NEAREST RELATED

98

QUESTION ID

S3-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its area is half of 4π, which is 2π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q5

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q5/5 • NEAREST RELATED

SET 4 • 8.NS.A.2 • THE NUMBER SYSTEM

Locate and compare irrational numbers on a number line

MAIN	Grade 8 main question	**8.NS.A.2**
BACKWARD 1	Read inequalities as positions on a number line	**6.NS.C.7.a**
BACKWARD 2	Position decimals on a number line	**6.NS.C.6.c**
FORWARD 1	Report irrational values to a level of accuracy that fits the situati…	**HSN-Q.A.3**
FORWARD 2	Use coordinates to find exact lengths and perimeters with radicals	**HSG-GPE.B.7**

SET 4 • 8.NS.A.2

MAIN

Grade 8 main question • Grade 8 • The Number System • 8.NS.A.2

Plot √5 on the number line. Use an estimate to the nearest tenth.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q1/5

101

QUESTION ID

S4-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

A point at about 2.2: between 2 and 2.5, closer to 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q1/5

Which list is in order from least to greatest?

A. 2, √5, √3, 1.8

B. √3, 1.8, 2, √5

C. √3, 2, 1.8, √5

D. 1.8, √3, 2, √5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q2/5

102

QUESTION ID

S4-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. √3, 1.8, 2, √5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q2/5

Write <, >, or =.

π ___ √10

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q3/5

103

QUESTION ID

S4-M-Q3

TYPE

Short answer

CORRECT ANSWER

<

GRADING NOTE π is about 3.14 and √10 is about 3.16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q3/5

Which number is the best name for point P: √2, √3, or √6?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q4/5

104

QUESTION ID

S4-M-Q4

TYPE

Short answer

CORRECT ANSWER

√3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q4/5

True or false?

√30 is to the left of 5.5 on a number line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q5/5

105

QUESTION ID

S4-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE √30 is about 5.48.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q5/5

SET 4 • 8.NS.A.2

BACKWARD 1

Read inequalities as positions on a number line • Grade 6 • The Number System • 6.NS.C.7.a

2.4 is to the right of 2.25 on a number line.

Write an inequality that shows this.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q1

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q1/5

107

QUESTION ID

S4-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2.4 > 2.25

GRADING NOTE 2.25 < 2.4 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q1

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q1/5

Which number is farthest to the left on a number line?

A. 0.5

B. −1.9

C. −2.05

D. −2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q2

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q2/5

108

QUESTION ID

S4-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. −2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q2

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q2/5

What does 1.7 < 1.75 tell you about the positions of the two numbers on a number line?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q3

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q3/5

109

QUESTION ID

S4-B1-Q3

TYPE

Short answer

CORRECT ANSWER

1.7 is to the left of 1.75.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q3

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q3/5

True or false?

$-1.4 > -1.41$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q4/5

110

QUESTION ID

S4-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q4/5

True or false?

3.09 > 3.1

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q5

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q5/5

111

QUESTION ID

S4-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q5

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q5/5

SET 4 • 8.NS.A.2

BACKWARD 2

Position decimals on a number line • Grade 6 • The Number System • 6.NS.C.6.c

Plot 1.4 and 2.75 on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q1/5

113

QUESTION ID

S4-B2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

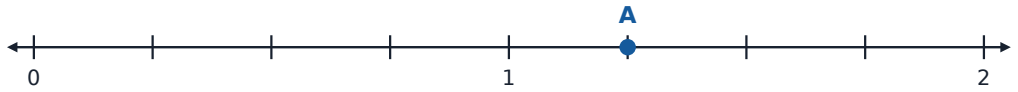
Points at 1.4 (a little past the 1.25 mark) and 2.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q1/5

What decimal is at point A?



Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q2/5

114

QUESTION ID

S4-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q2/5

Which decimal is between 1.7 and 1.8?

A. 1.81

B. 1.73

C. 1.69

D. 1.08

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q3/5

115

QUESTION ID

S4-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1.73

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q3/5

True or false?

2.5 is halfway between 2 and 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q4/5

116

QUESTION ID

S4-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q4/5

True or false?

0.9 is to the right of 1.1 on a number line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q5/5

117

QUESTION ID

S4-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q5/5

SET 4 • 8.NS.A.2

FORWARD 1

Report irrational values to a level of accuracy that fits the situation • High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3

A square garden has an area of 50 square meters, so its side is √50 m. A tape measure reads to the nearest centimeter.

Report the side length to that accuracy, in meters.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q1/5

119

QUESTION ID

S4-F1-Q1

TYPE

Short answer

CORRECT ANSWER

7.07 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q1/5

A coin has a diameter of 4 cm. Its circumference is 4π cm.

Report the circumference to the nearest tenth of a centimeter.

Circumference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q2/5

120

QUESTION ID

S4-F1-Q2

TYPE

Short answer

CORRECT ANSWER

12.6 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q2/5

Which value of √2 is given to the nearest thousandth?

A. 1.41

B. 1.414

C. 1.4142

D. 1.42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q3/5

121

QUESTION ID

S4-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1.414

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q3/5

True or false?

Reporting √3 as 1.7 is accurate to the nearest tenth.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q4/5

122

QUESTION ID

S4-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q4/5

True or false?

Writing more digits of √5 always makes a measured length more accurate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q5/5

123

QUESTION ID

S4-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A reported length cannot be more accurate than the measuring tool.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q5/5

SET 4 • 8.NS.A.2

FORWARD 2

Use coordinates to find exact lengths and perimeters with radicals • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

Find the exact perimeter of the triangle with vertices (0, 0), (2, 0), and (0, 2).

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q1/5

125

QUESTION ID

S4-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4 + 2√2 units

GRADING NOTE The exact form is required; a decimal approximation alone is not accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F2-Q1**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q1/5

Find the exact length of the segment from (1, 1) to (4, 3).

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q2/5

126

QUESTION ID

S4-F2-Q2

TYPE

Short answer

CORRECT ANSWER

√13 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F2-Q2**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q2/5

A square has vertices (0, 0), (1, 2), (−1, 3), and (−2, 1).

What is its perimeter?

A. 2√5 units

B. 20 units

C. 4√5 units

D. 4√3 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q3/5

127

QUESTION ID

S4-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 4√5 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q3/5

True or false?

The distance from $(-1, -1)$ to $(2, 3)$ is 5 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q4/5

128

QUESTION ID

S4-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q4/5

True or false?

A rectangle with vertices (0, 0), (3, 0), (3, 1), and (0, 1) has a diagonal 4 units long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q5/5

129

QUESTION ID

S4-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The diagonal is √10 units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q5/5

SET 59 • 8.NS.A.2 • THE NUMBER SYSTEM

Estimate the values of expressions that contain irrational numbers

MAIN	Grade 8 main question	**8.NS.A.2**
BACKWARD 1	Evaluate expressions with decimal values	**6.EE.A.2.c**
BACKWARD 2	Add, subtract, and multiply multi-digit decimals	**6.NS.B.3**
FORWARD 1	Solve quadratic equations and estimate their irrational solutions	**HSA-REI.B.4.b**
FORWARD 2	Derive the quadratic formula by completing the square	**HSA-REI.B.4.a**

SET 59 • 8.NS.A.2

MAIN

Grade 8 main question • Grade 8 • The Number System • 8.NS.A.2

Estimate 2√5 + 1 to the nearest tenth. Show how you estimated √5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 59 • MAIN • Q1/5

132

QUESTION ID

S59-M-Q1

TYPE

Short answer

CORRECT ANSWER

About 5.5. √5 ≈ 2.24 ($2.2^2 = 4.84$ and $2.3^2 = 5.29$; $2.24^2 \approx 5.02$), so $2\sqrt{5} + 1 \approx 4.48 + 1 = 5.48 \approx 5.5$.

GRADING NOTE Grade both: 5.5, and an estimate of √5 between 2.2 and 2.3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 59 • MAIN • Q1/5

Estimate π^2 to the nearest whole number.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 59 • MAIN • Q2/5

133

QUESTION ID

S59-M-Q2

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE $\pi \approx 3.14$, and $3.14^2 \approx 9.86$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 59 • MAIN • Q2/5

Which is the best estimate of √2 + √3 to the nearest tenth?

A. 5.0

B. 3.1

C. 3.6

D. 2.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 59 • MAIN • Q3/5

134

QUESTION ID

S59-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 3.1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 59 • MAIN • Q3/5

Between which two whole numbers is $3\sqrt{7}$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 59 • MAIN • Q4/5

135

QUESTION ID

S59-M-Q4

TYPE

Short answer

CORRECT ANSWER

7 and 8. $3\sqrt{7} = \sqrt{63}$, and $7^2 = 49 < 63 < 64 = 8^2$.

GRADING NOTE Any correct reasoning is accepted, for example $\sqrt{7} \approx 2.65$ so $3\sqrt{7} \approx 7.9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 59 • MAIN • Q4/5

True or false?

$5 - \sqrt{2}$ is greater than 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 59 • MAIN • Q5/5

136

QUESTION ID

S59-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{2} \approx 1.41$, so $5 - \sqrt{2} \approx 3.59$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 59 • MAIN • Q5/5

SET 59 • 8.NS.A.2

BACKWARD 1

Evaluate expressions with decimal values • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Evaluate 2x + 1 when x = 2.24.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 59 • BACKWARD 1 • Q1/5

138

QUESTION ID

S59-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5.48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 59 • BACKWARD 1 • Q1/5

Evaluate x^2 when $x = 3.14$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 59 • BACKWARD 1 • Q2/5

139

QUESTION ID

S59-B1-Q2

TYPE

Short answer

CORRECT ANSWER

9.8596

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 59 • BACKWARD 1 • Q2/5

Evaluate 3x − 1 when x = 1.7.

A. 3.1

B. 4.1

C. 5.1

D. 2.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 59 • BACKWARD 1 • Q3/5

140

QUESTION ID

S59-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4.1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 59 • BACKWARD 1 • Q3/5

True or false?

When x = 1.4, x + 1.7 = 3.1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 59 • BACKWARD 1 • Q4/5

141

QUESTION ID

S59-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 59 • BACKWARD 1 • Q4/5

True or false?

When x = 2.6, 3x = 6.8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 59 • BACKWARD 1 • Q5/5

142

QUESTION ID

S59-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 3 × 2.6 = 7.8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 59 • BACKWARD 1 • Q5/5

SET 59 • 8.NS.A.2

BACKWARD 2

Add, subtract, and multiply multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Add.

1.414 + 1.732

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 59 • BACKWARD 2 • Q1/5

144

QUESTION ID

S59-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3.146

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 59 • BACKWARD 2 • Q1/5

Multiply.

2×2.236

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 59 • BACKWARD 2 • Q2/5

145

QUESTION ID

S59-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4.472

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 59 • BACKWARD 2 • Q2/5

Subtract.

$3.162 - 1$

A. 2.062

B. 4.162

C. 3.062

D. 2.162

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 59 • BACKWARD 2 • Q3/5

146

QUESTION ID

S59-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2.162

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 59 • BACKWARD 2 • Q3/5

True or false?

$9.86 + 0.14 = 10$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 59 • BACKWARD 2 • Q4/5

147

QUESTION ID

S59-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 59 • BACKWARD 2 • Q4/5

True or false?

$3 \times 2.65 = 7.65$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 59 • BACKWARD 2 • Q5/5

148

QUESTION ID

S59-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3 \times 2.65 = 7.95$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 59 • BACKWARD 2 • Q5/5

SET 59 • 8.NS.A.2

FORWARD 1

Solve quadratic equations and estimate their irrational solutions • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Solve $x^2 - 4x + 1 = 0$. Give the exact solutions, then estimate each to the nearest tenth.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 59 • FORWARD 1 • Q1/5

150

QUESTION ID

S59-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$x = 2 + \sqrt{3} \approx 3.7$ and $x = 2 - \sqrt{3} \approx 0.3$

GRADING NOTE Both solutions are required, exact and estimated.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S59-F1-Q1**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 59 • FORWARD 1 • Q1/5

Solve $(x + 1)^2 = 8$. Give the exact solutions, then estimate each to the nearest tenth.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 59 • FORWARD 1 • Q2/5

151

QUESTION ID

S59-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$x = -1 + 2\sqrt{2} \approx 1.8$ and $x = -1 - 2\sqrt{2} \approx -3.8$

GRADING NOTE Both solutions are required, exact and estimated. $-1 \pm \sqrt{8}$ is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 59 • FORWARD 1 • Q2/5

Which is the best estimate of the positive solution of $x^2 + 2x - 4 = 0$?

A. 1.2

B. −3.2

C. 2.2

D. 3.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 59 • FORWARD 1 • Q3/5

152

QUESTION ID

S59-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 1.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 59 • FORWARD 1 • Q3/5

True or false?

The solutions of $x^2 = 2x + 2$ are $1 + \sqrt{3}$ and $1 - \sqrt{3}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 59 • FORWARD 1 • Q4/5

153

QUESTION ID

S59-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 59 • FORWARD 1 • Q4/5

True or false?

The positive solution of $x^2 = 6$ is about 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 59 • FORWARD 1 • Q5/5

154

QUESTION ID

S59-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{6} \approx 2.4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 59 • FORWARD 1 • Q5/5

SET 59 • 8.NS.A.2

FORWARD 2

Derive the quadratic formula by completing the square • High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a

Complete the square to solve $x^2 + 6x + c = 0$ for x in terms of c. Assume $c \leq 9$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 59 • FORWARD 2 • Q1/5

156

QUESTION ID

S59-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

$x = -3 + \sqrt{(9 - c)}$

and

$x = -3 - \sqrt{(9 - c)}$

GRADING NOTE REQUIRED WORK: The completed square must be shown: $x^2 + 6x + 9 = 9 - c$, so $(x + 3)^2 = 9 - c$; then $x + 3 = \pm\sqrt{(9 - c)}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 59 • FORWARD 2 • Q1/5

Start with $ax^2 + bx + c = 0$ ($a \neq 0$). Divide by a, subtract c/a, and add $(b/(2a))^2$ to both sides.

What equation do you get?

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 59 • FORWARD 2 • Q2/5

157

QUESTION ID

S59-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$(x + b/(2a))^2 = (b^2 - 4ac)/(4a^2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S59-F2-Q2**

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 59 • FORWARD 2 • Q2/5

Which is the quadratic formula for $ax^2 + bx + c = 0$ ($a \neq 0$)?

A. $x = (b \pm \sqrt{(b^2 - 4ac)}) \div (2a)$

B. $x = (-b \pm \sqrt{(b^2 - 4ac)}) \div (2a)$

C. $x = (-b \pm \sqrt{(b^2 + 4ac)}) \div (2a)$

D. $x = -b \pm \sqrt{(b^2 - 4ac)} \div 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 59 • FORWARD 2 • Q3/5

158

QUESTION ID

S59-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $x = (-b \pm \sqrt{(b^2 - 4ac)}) \div (2a)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 59 • FORWARD 2 • Q3/5

True or false?

For $x^2 - 2x - 1 = 0$, the quadratic formula gives $x = 1 \pm \sqrt{2}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 59 • FORWARD 2 • Q4/5

159

QUESTION ID

S59-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 59 • FORWARD 2 • Q4/5

True or false?

Completing the square on $x^2 + 10x = 3$ gives $(x + 5)^2 = 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 59 • FORWARD 2 • Q5/5

160

QUESTION ID

S59-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It gives $(x + 5)^2 = 28$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 59 • FORWARD 2 • Q5/5

SET 5 • 8.EE.A.1 • EXPRESSIONS & EQUATIONS

Use the product and quotient properties of integer exponents

MAIN	Grade 8 main question	**8.EE.A.1**
BACKWARD 1	Evaluate whole-number exponents	**6.EE.A.1**
BACKWARD 2	Write powers of 10 with whole-number exponents	**5.NBT.A.2**
FORWARD 1	Multiply and divide monomials using exponent properties	**HSA-SSE.A.2**
FORWARD 2	Use exponent properties to rewrite and interpret exponential funct	**HSF-IF.C.8.b**

SET 5 • 8.EE.A.1

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.A.1

Write as a single power.

$3^4 \times 3^5$

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q1/5

163

QUESTION ID

S5-M-Q1

TYPE

Short answer

CORRECT ANSWER

3^9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q1/5

Write as a single power.

$7^{10} \div 7^{4}$

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q2/5

164

QUESTION ID

S5-M-Q2

TYPE

Short answer

CORRECT ANSWER

7^{6}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q2/5

Which expression is equal to 2^6?

A. $2^{12} \div 2^2$

B. 2×6

C. $2^2 \times 2^3$

D. $2^2 \times 2^4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q3/5

QUESTION ID

S5-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $2^2 \times 2^4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q3/5

True or false?

$5^3 \times 5^2 = 5^5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q4/5

166

QUESTION ID

S5-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q4/5

True or false?

$4^8 \div 4^2 = 4^4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q5/5

167

QUESTION ID

S5-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $4^8 \div 4^2 = 4^6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q5/5

SET 5 • 8.EE.A.1

BACKWARD 1

Evaluate whole-number exponents • Grade 6 • Expressions & Equations • 6.EE.A.1

Find the value.

2^5

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q1/5

169

QUESTION ID

S5-B1-Q1

TYPE

Short answer

CORRECT ANSWER

32

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the value.

10^4

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q2/5

170

QUESTION ID

S5-B1-Q2

TYPE

Short answer

CORRECT ANSWER

10,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q2/5

Find the value.

3^4

A. 81

B. 64

C. 12

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q3/5

171

QUESTION ID

S5-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 81

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q3/5

True or false?

$6^2 = 36$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q4/5

172

QUESTION ID

S5-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q4/5

True or false?

$2^3 = 6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q5/5

QUESTION ID

S5-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q5/5

SET 5 • 8.EE.A.1

BACKWARD 2

Write powers of 10 with whole-number exponents • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Write 10 × 10 × 10 × 10 as a power of 10.

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q1/5

175

QUESTION ID

S5-B2-Q1

TYPE

Short answer

CORRECT ANSWER

10^4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q1/5

Write 1,000,000 as a power of 10.

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q2/5

176

QUESTION ID

S5-B2-Q2

TYPE

Short answer

CORRECT ANSWER

10^6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q2/5

What is the value of 10^5?

A. 10,000

B. 50

C. 100,000

D. 1,000,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q3/5

177

QUESTION ID

S5-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 100,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q3/5

True or false?

$10^3 = 1{,}000$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q4/5

178

QUESTION ID

S5-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q4/5

True or false?

$10^2 \times 10 = 10^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q5/5

179

QUESTION ID

S5-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $10^2 \times 10 = 1{,}000 = 10^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q5/5

SET 5 • 8.EE.A.1

FORWARD 1

Multiply and divide monomials using exponent properties • High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Simplify.

$(3x^4)(5x^2)$

Simplified: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q1

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q1/5

181

QUESTION ID

S5-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$15x^6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q1

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q1/5

Simplify. Assume $x \neq 0$.

$$\frac{12x^7}{4x^3}$$

Simplified: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q2

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q2/5

182

QUESTION ID

S5-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$3x^4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q2

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q2/5

Simplify.

$x^5 \cdot x \cdot x^2$

A. x^7

B. $3x^8$

C. x^{10}

D. x^8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q3

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q3/5

183

QUESTION ID

S5-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x^8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q3

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q3/5

True or false?

$(2a^3)(4a^2) = 8a^5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q4

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q4/5

184

QUESTION ID

S5-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q4

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q4/5

True or false?

Assume $y \neq 0$. $\frac{y^9}{y^3} = y^3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q5

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q5/5

185

QUESTION ID

S5-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $y^9 \div y^3 = y^6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q5

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q5/5

SET 5 • 8.EE.A.1

FORWARD 2

Use exponent properties to rewrite and interpret exponential functions • High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b

f(t) = 2^(t + 3)

Use a property of exponents to write f(t) in the form a · 2^t.

f(t) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q1/5

187

QUESTION ID

S5-F2-Q1

TYPE

Short answer

CORRECT ANSWER

8 · 2^t

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q1/5

A population is P(t) = 200 · 2^t after t years.

By what factor does the population grow each year, and what is P(5)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q2/5

188

QUESTION ID

S5-F2-Q2

TYPE

Short answer

CORRECT ANSWER

It doubles (factor 2) each year;
P(5) = 6,400

GRADING NOTE Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q2/5

$g(t) = 5^{(t - 1)}$

Which expression is equal to g(t)?

A. $5^t - 5$

B. $5 \div 5^t$

C. $5^t \div 5$

D. $5^t - 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q3/5

189

QUESTION ID

S5-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $5^t \div 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q3/5

True or false?

If f(x) = 5 · 2^x, then f(x + 1) = 2 · f(x).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q4/5

190

QUESTION ID

S5-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q4/5

True or false?

If f(x) = 2^x, then f(3) · f(4) = f(12).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q5/5

191

QUESTION ID

S5-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE f(3) · f(4) = $2^3 \cdot 2^4 = 2^7$ = f(7).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q5/5

SET 6 • 8.EE.A.1 • EXPRESSIONS & EQUATIONS

Use the power-of-a-power property and zero and negative exponents

MAIN	Grade 8 main question	**8.EE.A.1**
BACKWARD 1	Find powers of fractions and decimals	**6.EE.A.1**
BACKWARD 2	Multiply signed numbers	**7.NS.A.2.a**
FORWARD 1	Explain why rational exponents are defined so the properties of ex…	**HSN-RN.A.1**
FORWARD 2	Rewrite exponential expressions with the properties of exponents	**HSA-SSE.B.3.c**

SET 6 • 8.EE.A.1

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.A.1

Write as a single power.

$(5^2)^3$

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q1/5

194

QUESTION ID

S6-M-Q1

TYPE

Short answer

CORRECT ANSWER

5^6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q1/5

Find the value.

4^{-2}

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q2/5

195

QUESTION ID

S6-M-Q2

TYPE

Short answer

CORRECT ANSWER

1/16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q2/5

Which expression is equal to 1?

A. 0^9

B. 9^0

C. $1^{-9} \times 9$

D. 9^{-1}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q3/5

196

QUESTION ID

S6-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 9^0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q3/5

True or false?

$3^{-2} \times 3^{5} = 27$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q4/5

197

QUESTION ID

S6-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q4/5

True or false?

$(2^3)^2 = 2^5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q5/5

198

QUESTION ID

S6-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(2^3)^2 = 2^6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q5/5

SET 6 • 8.EE.A.1

BACKWARD 1

Find powers of fractions and decimals • Grade 6 • Expressions & Equations • 6.EE.A.1

Find the value.

$(\frac{1}{2})^3$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q1/5

200

QUESTION ID

S6-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the value.

0.3^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q2/5

201

QUESTION ID

S6-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0.09

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q2/5

Find the value.

$(\frac{2}{3})^2$

A. $\frac{4}{3}$

B. $\frac{4}{9}$

C. $\frac{2}{9}$

D. $\frac{4}{6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q3/5

202

QUESTION ID

S6-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4/9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q3/5

True or false?

$0.1^3 = 0.001$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q4/5

QUESTION ID

S6-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q4/5

True or false?

$(\frac{3}{4})^2 = \frac{6}{8}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q5/5

204

QUESTION ID

S6-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(3/4)^2 = 9/16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q5/5

SET 6 • 8.EE.A.1

BACKWARD 2

Multiply signed numbers • Grade 7 • The Number System • 7.NS.A.2.a

Multiply.

$(-2)(-2)(-2)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q1/5

206

QUESTION ID

S6-B2-Q1

TYPE

Short answer

CORRECT ANSWER

−8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q1/5

Multiply.

$(-3)(-3)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q2/5

207

QUESTION ID

S6-B2-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q2/5

Multiply.

$(-1)(-1)(-1)(-1)$

A. -1

B. -4

C. 1

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q3/5

208

QUESTION ID

S6-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q3/5

True or false?

$(-5)(-5) = 25$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q4/5

209

QUESTION ID

S6-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q4/5

True or false?

$(-2)(-2)(-2)(-2) = -16$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q5/5

210

QUESTION ID

S6-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q5/5

SET 6 • 8.EE.A.1

FORWARD 1

Explain why rational exponents are defined so the properties of exponents still hold • High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1

If the power-of-a-power property is to stay true for rational exponents, then $(5^{1/2})^2 = 5^{(1/2) \cdot 2} = 5$.

Explain what number $5^{1/2}$ must be.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q1/5

212

QUESTION ID

S6-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$5^{1/2}$ must be a positive number whose square is 5, so $5^{1/2} = \sqrt{5}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q1/5

If the power-of-a-power property is to stay true, what must 8^(1/3) equal? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q2/5

213

QUESTION ID

S6-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2, because $(8^{1/3})^3 = 8^{(1/3)\cdot 3} = 8$, and 2 is the number whose cube is 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q2/5

Which number is equal to 9^(3/2)?

A. 6

B. 27

C. 13.5

D. 81

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q3/5

214

QUESTION ID

S6-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q3/5

True or false?

Because $(8^{(1/3)})^3 = 8$, $8^{(1/3)} = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q4/5

215

QUESTION ID

S6-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q4/5

True or false?

x^(1/2) means x ÷ 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q5/5

216

QUESTION ID

S6-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE x^(1/2) is the square root of x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F1-Q5**

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q5/5

SET 6 • 8.EE.A.1

FORWARD 2

Rewrite exponential expressions with the properties of exponents • High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c

Rewrite 1.21^t as a power of 1.1.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q1

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q1/5

218

QUESTION ID

S6-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1.1^(2t)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q1

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q1/5

A value grows by a factor of 1.04 each month, so after t years it has grown by (1.04^12)^t.

Write the yearly growth factor, rounded to the nearest hundredth.

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q2

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q2/5

219

QUESTION ID

S6-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1.60

GRADING NOTE 1.04^12 ≈ 1.601, about 60% growth per year.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q2

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q2/5

Which expression is equivalent to 8^x?

A. 2^x ÷ 3

B. 3^(2x)

C. 2^(3x)

D. 2^(x + 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q3

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q3/5

220

QUESTION ID

S6-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2^(3x)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q3

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q3/5

True or false?

$(0.5)^{(-t)} = 2^t$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q4

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q4/5

221

QUESTION ID

S6-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q4

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q4/5

True or false?

$4^x \times 4^x = 16^{(2x)}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q5

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q5/5

222

QUESTION ID

S6-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $4^x \times 4^x = 4^{(2x)} = 16^x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q5

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q5/5

SET 7 • 8.EE.A.2 • EXPRESSIONS & EQUATIONS

Solve equations of the form $x^2 = p$ and $x^3 = p$

MAIN	Grade 8 main question	**8.EE.A.2**
BACKWARD 1	Solve one-step multiplication equations	**6.EE.B.7**
BACKWARD 2	Find squares and cubes of whole numbers	**6.EE.A.1**
FORWARD 1	Solve quadratic equations by taking square roots	**HSA-REI.B.4.b**
FORWARD 2	Find a dimension from a volume with square and cube roots	**HSG-GMD.A.3**

SET 7 • 8.EE.A.2

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.A.2

Solve for x. Give all solutions.

$x^2 = 49$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q1/5

225

QUESTION ID

S7-M-Q1

TYPE

Short answer

CORRECT ANSWER

x = 7 and x = −7

GRADING NOTE Both solutions are required; x = ±7 is accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q1/5

Solve for x.

$x^3 = 64$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q2/5

226

QUESTION ID

S7-M-Q2

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q2/5

Solve for x.

$x^2 = \frac{9}{16}$

A. $x = \frac{3}{4}$ only

B. $x = \frac{3}{4}$ or $x = -\frac{3}{4}$

C. $x = \frac{9}{32}$

D. $x = \frac{81}{256}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q3/5

227

QUESTION ID

S7-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. x = 3/4 or x = −3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q3/5

Solve for x.

$x^3 = -27$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q4/5

228

QUESTION ID

S7-M-Q4

TYPE

Short answer

CORRECT ANSWER

−3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q4/5

True or false?

$x^2 = 10$ has the two solutions $x = \sqrt{10}$ and $x = -\sqrt{10}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q5/5

229

QUESTION ID

S7-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q5/5

SET 7 • 8.EE.A.2

BACKWARD 1

Solve one-step multiplication equations • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

4x = 64

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q1/5

231

QUESTION ID

S7-B1-Q1

TYPE

Short answer

CORRECT ANSWER

16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q1/5

Solve for x.

$\frac{x}{3} = 9$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q2/5

232

QUESTION ID

S7-B1-Q2

TYPE

Short answer

CORRECT ANSWER

27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q2/5

Solve for x.

9x = 45

A. x = 36

B. x = 5

C. x = 405

D. x = 54

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q3/5

233

QUESTION ID

S7-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. x = 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q3/5

True or false?

x = 12 is the solution of 3x = 36.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q4/5

234

QUESTION ID

S7-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q4/5

True or false?

x = 5 is the solution of 7x = 42.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q5/5

235

QUESTION ID

S7-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q5/5

SET 7 • 8.EE.A.2

BACKWARD 2

Find squares and cubes of whole numbers • Grade 6 • Expressions & Equations • 6.EE.A.1

Find the value.

9^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q1/5

237

QUESTION ID

S7-B2-Q1

TYPE

Short answer

CORRECT ANSWER

81

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q1/5

Find the value.

5^3

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q2/5

238

QUESTION ID

S7-B2-Q2

TYPE

Short answer

CORRECT ANSWER

125

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q2/5

Which whole number cubed is 8?

A. 3

B. 4

C. 8

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q3/5

239

QUESTION ID

S7-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q3/5

True or false?

$10^4 = 10{,}000$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q4/5

240

QUESTION ID

S7-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q4/5

True or false?

$4^2 = 8$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q5/5

241

QUESTION ID

S7-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q5/5

SET 7 • 8.EE.A.2

FORWARD 1

Solve quadratic equations by taking square roots • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Solve for x. Give all solutions.

$2x^2 = 50$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q1/5

243

QUESTION ID

S7-F1-Q1

TYPE

Short answer

CORRECT ANSWER

x = 5 and x = −5

GRADING NOTE Both solutions are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q1/5

Solve for x. Give all solutions.

$(x + 2)^2 = 36$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q2/5

244

QUESTION ID

S7-F1-Q2

TYPE

Short answer

CORRECT ANSWER

x = 4 and x = −8

GRADING NOTE Both solutions are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F1-Q2**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q2/5

Solve for x.

$x^2 - 9 = 0$

A. x = 9 or x = −9

B. x = 3 only

C. x = 3 or x = −3

D. x = 4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q3/5

245

QUESTION ID

S7-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 3 or x = −3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q3/5

True or false?

The solutions of $3x^2 = 27$ are 3 and −3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q4/5

246

QUESTION ID

S7-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q4/5

True or false?

The solutions of $(x - 1)^2 = 16$ are 5 and 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q5/5

247

QUESTION ID

S7-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solutions are 5 and −3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q5/5

SET 7 • 8.EE.A.2

FORWARD 2

Find a dimension from a volume with square and cube roots • High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

Every edge of a cube is multiplied by the same factor, and the volume is multiplied by 27.

By what factor was each edge multiplied? Explain.

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q1

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q1/5

249

QUESTION ID

S7-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3, because the volume is multiplied by the cube of the factor and $\sqrt[3]{27} = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q1

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q1/5

A sphere has a volume of 36π cubic centimeters.

($V = \frac{4}{3}\pi r^3$)

What is its radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q2

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q2/5

250

QUESTION ID

S7-F2-Q2

TYPE

Short answer

CORRECT ANSWER

3 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q2

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q2/5

A cylinder has a volume of 200π cubic units and a height of 8 units.

What is its radius?

A. 10 units

B. 25 units

C. 5 units

D. √200 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q3

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q3/5

251

QUESTION ID

S7-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 5 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q3

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q3/5

True or false?

A cone with a volume of 12π cubic units and a height of 4 units has a radius of 3 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q4

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q4/5

252

QUESTION ID

S7-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q4

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q4/5

True or false?

A cube with a volume of 1,000 cubic centimeters has edges 100 cm long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q5

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q5/5

253

QUESTION ID

S7-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Each edge is 10 cm.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q5

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q5/5

SET 8 • 8.EE.A.2 • EXPRESSIONS & EQUATIONS

Evaluate square roots and cube roots and recognize irrational roots

MAIN	Grade 8 main question	**8.EE.A.2**
BACKWARD 1	Find factor pairs and square numbers	**4.OA.B.4**
BACKWARD 2	Find the volume of a cube	**5.MD.C.5.b**
FORWARD 1	Simplify square roots by factoring out perfect squares	**HSN-RN.A.2**
FORWARD 2	Solve quadratic equations by completing the square	**HSA-REI.B.4.a**

SET 8 • 8.EE.A.2

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.A.2

Find the value.

√144

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q1/5

256

QUESTION ID

S8-M-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q1/5

Find the value.

$\sqrt[3]{125}$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q2/5

257

QUESTION ID

S8-M-Q2

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q2/5

Which square root is irrational?

A. $\sqrt{\frac{1}{4}}$

B. $\sqrt{0.09}$

C. $\sqrt{20}$

D. $\sqrt{36}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q3/5

258

QUESTION ID

S8-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. √20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q3/5

True or false?

$\sqrt[3]{(-8)} = -2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q4/5

259

QUESTION ID

S8-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q4/5

True or false?

√2 can be written as a fraction of two whole numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q5/5

260

QUESTION ID
S8-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE √2 is irrational.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q5/5

SET 8 • 8.EE.A.2

BACKWARD 1

Find factor pairs and square numbers • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List all the factor pairs of 16.

Factor pairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q1/5

262

QUESTION ID

S8-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1 × 16, 2 × 8, and 4 × 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q1/5

What whole number times itself is 49?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q2/5

263

QUESTION ID

S8-B1-Q2

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q2/5

Which number has a factor pair made of two equal factors?

A. 30

B. 24

C. 18

D. 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q3/5

264

QUESTION ID

S8-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q3/5

True or false?

81 has a factor pair made of two equal factors.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q4/5

265

QUESTION ID

S8-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q4/5

True or false?

12 has a factor pair made of two equal factors.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q5/5

266

QUESTION ID

S8-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q5/5

SET 8 • 8.EE.A.2

BACKWARD 2

Find the volume of a cube • Grade 5 • Measurement & Data • 5.MD.C.5.b

A cube has edges 3 units long.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q1/5

268

QUESTION ID

S8-B2-Q1

TYPE

Short answer

CORRECT ANSWER

27 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q1/5

A cube has edges 5 cm long.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q2/5

269

QUESTION ID

S8-B2-Q2

TYPE

Short answer

CORRECT ANSWER

125 cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q2/5

Which cube has a volume of 64 cubic units?

A. A cube with 32-unit edges

B. A cube with 4-unit edges

C. A cube with 16-unit edges

D. A cube with 8-unit edges

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q3/5

270

QUESTION ID

S8-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. A cube with 4-unit edges

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q3/5

True or false?

A cube with 2-unit edges has a volume of 8 cubic units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q4/5

271

QUESTION ID

S8-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q4/5

True or false?

A cube with 10-unit edges has a volume of 30 cubic units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q5/5

272

QUESTION ID

S8-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q5/5

SET 8 • 8.EE.A.2

FORWARD 1

Simplify square roots by factoring out perfect squares • High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2

Simplify.

√48

Simplified: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q1/5

274

QUESTION ID

S8-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4√3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F1-Q1**

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q1/5

Simplify.

√200

Simplified: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q2/5

275

QUESTION ID

S8-F1-Q2

TYPE

Short answer

CORRECT ANSWER

10√2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F1-Q2**

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q2/5

Which expression is equal to √18?

A. 2√3

B. 9√2

C. 3√2

D. 6√3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q3/5

276

QUESTION ID

S8-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3√2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q3/5

True or false?

$\sqrt{50} = 5\sqrt{2}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q4/5

277

QUESTION ID

S8-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q4/5

True or false?

$\sqrt{27} = 9\sqrt{3}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q5/5

278

QUESTION ID

S8-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{27} = 3\sqrt{3}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q5/5

SET 8 • 8.EE.A.2

FORWARD 2

Solve quadratic equations by completing the square • High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a

What number goes in the blank?

$x^2 + 6x + 9 = (x + ___)^2$

Blank: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q1/5

280

QUESTION ID

S8-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F2-Q1**

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q1/5

Solve by completing the square. Show the completed square, then give both exact solutions.

$x^2 + 4x = 1$

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q2/5

281

QUESTION ID

S8-F2-Q2

TYPE

Answer with required work

CORRECT ANSWER

$x = -2 + \sqrt{5}$ and $x = -2 - \sqrt{5}$

GRADING NOTE REQUIRED WORK: The completed square must be shown: $x^2 + 4x + 4 = 1 + 4$, so $(x + 2)^2 = 5$; then $x + 2 = \pm\sqrt{5}$. Both roots are required. Correct roots without the completed square are not full credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q2/5

Which number completes the square?

$x^2 - 10x + ___$

A. 5

B. −25

C. 100

D. 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q3/5

282

QUESTION ID

S8-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q3/5

True or false?

$x^2 + 8x + 16 = (x + 4)^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q4/5

283

QUESTION ID

S8-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q4/5

True or false?

Completing the square on $x^2 + 2x = 3$ gives $(x + 1)^2 = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q5/5

284

QUESTION ID

S8-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It gives $(x + 1)^2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q5/5

SET 9 • 8.EE.A.3 • EXPRESSIONS & EQUATIONS

Estimate very large and very small quantities as a single digit times a power of 10

MAIN	Grade 8 main question	**8.EE.A.3**
BACKWARD 1	Multiply and divide by powers of 10	**5.NBT.A.2**
BACKWARD 2	Round whole numbers to any place	**4.NBT.A.3**
FORWARD 1	Choose a level of accuracy for very large and very small quantities	**HSN-Q.A.3**
FORWARD 2	Use base-10 logarithms to solve exponential equations	**HSF-LE.A.4**

SET 9 • 8.EE.A.3

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.A.3

Estimate 412,000,000 as a single digit times a power of 10.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q1/5

287

QUESTION ID

S9-M-Q1

TYPE

Short answer

CORRECT ANSWER

4×10^8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q1/5

Estimate 0.00079 as a single digit times a power of 10.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q2/5

288

QUESTION ID

S9-M-Q2

TYPE

Short answer

CORRECT ANSWER

8×10^{-4}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q2/5

A country has about 3×10^8 people. A city has about 6×10^5 people.

About how many times as many people live in the country?

A. 2

B. 50

C. 500

D. 5,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q3/5

289

QUESTION ID

S9-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 500

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q3/5

A whale has a mass of about 2×10^{5} kg. A mouse has a mass of about 2×10^{-2} kg.

About how many times as large is the whale's mass?

Times as large: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q4/5

290

QUESTION ID

S9-M-Q4

TYPE

Short answer

CORRECT ANSWER

10,000,000

GRADING NOTE 10^{7} is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q4/5

True or false?

6×10^{-3} is greater than 9×10^{-4}.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q5/5

291

QUESTION ID

S9-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q5/5

SET 9 • 8.EE.A.3

BACKWARD 1

Multiply and divide by powers of 10 • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Multiply.

4×10^3

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q1/5

293

QUESTION ID

S9-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q1/5

Divide.

$7 \div 10^2$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q2/5

294

QUESTION ID

S9-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0.07

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q2/5

What is the value of 3×10^5?

A. 15

B. 3,000,000

C. 300,000

D. 30,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q3/5

295

QUESTION ID

S9-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 300,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q3/5

True or false?

$6 \div 10^3 = 0.006$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q4/5

296

QUESTION ID

S9-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q4/5

True or false?

$8 \times 10^2 = 8{,}000$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q5/5

297

QUESTION ID

S9-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q5/5

SET 9 • 8.EE.A.3

BACKWARD 2

Round whole numbers to any place • Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3

Round 412,000 to the nearest hundred thousand.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q1/5

299

QUESTION ID

S9-B2-Q1

TYPE

Short answer

CORRECT ANSWER

400,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q1/5

Round 6,750 to the nearest thousand.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q2/5

300

QUESTION ID

S9-B2-Q2

TYPE

Short answer

CORRECT ANSWER

7,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q2/5

Round 58,300 to the nearest ten thousand.

A. 50,000

B. 59,000

C. 60,000

D. 58,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q3/5

301

QUESTION ID

S9-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 60,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q3/5

True or false?

245,000 rounded to the nearest hundred thousand is 200,000.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q4/5

302

QUESTION ID

S9-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q4/5

True or false?

8,500 rounded to the nearest thousand is 8,000.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q5/5

303

QUESTION ID

S9-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q5/5

SET 9 • 8.EE.A.3

FORWARD 1

Choose a level of accuracy for very large and very small quantities • High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3

Earth's mean radius is about 6,371 km.

Write it to one significant digit as a single digit times a power of 10.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q1/5

305

QUESTION ID

S9-F1-Q1

TYPE

Short answer

CORRECT ANSWER

6×10^3 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q1/5

A scale reads to the nearest 0.1 kg. One bag of rice measures 2.3 kg, and a truck carries 4,000 such bags.

Report the total mass in scientific notation, with no more accuracy than the scale supports.

Total mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q2/5

306

QUESTION ID

S9-F1-Q2

TYPE

Short answer

CORRECT ANSWER

About 9.2 × 10³ kg

GRADING NOTE 9.2 × 10³ kg (two significant digits). Reports such as 9.200 × 10³ kg claim more accuracy than the scale gives.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q2/5

A runner's time over 100 m was measured to the nearest second as 12 s. A calculator gives $100 \div 12 = 8.333333$.

Which report of the speed fits the accuracy of the measurement?

A. About 8 m/s

B. 83 m/s

C. 8.33 m/s

D. 8.333333 m/s

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q3/5

QUESTION ID

S9-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. About 8 m/s

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q3/5

True or false?

Reporting 7,900,000,000 people as 8×10^9 is accurate to one significant digit.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q4/5

308

QUESTION ID

S9-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q4/5

True or false?

A distance known to be about 3×10^8 m is reported more accurately as 300,000,000.0 m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q5/5

309

QUESTION ID

S9-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Writing more digits does not make an estimate more accurate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q5/5

SET 9 • 8.EE.A.3

FORWARD 2

Use base-10 logarithms to solve exponential equations • High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

A bacteria count is 10^t thousand after t hours.

Use a logarithm to find when the count reaches 1,000,000 (1,000 thousand).

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q1

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q1/5

311

QUESTION ID

S9-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3 hours

($t = \log_{10}(1{,}000) = 3$)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q1

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q1/5

Solve 5 · 10^x = 2,000. Write x as a logarithm, then use a calculator to round it to the nearest tenth.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q2

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q2/5

312

QUESTION ID

S9-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$x = \log_{10}(400) \approx 2.6$

GRADING NOTE Both the logarithm and the rounded value are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F2-Q2**

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q2/5

Between which two whole numbers is $\log_{10}(4{,}500)$?

A. 2 and 3

B. 45 and 46

C. 4 and 5

D. 3 and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q3

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q3/5

313

QUESTION ID

S9-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3 and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F2-Q3**

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q3/5

True or false?

$\log_{10}(10^{-5}) = -5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q4

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q4/5

314

QUESTION ID

S9-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q4

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q4/5

True or false?

$\log_{10}(50)$ is between 5 and 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q5

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q5/5

315

QUESTION ID

S9-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $10^1 = 10$ and $10^2 = 100$, so $\log_{10}(50)$ is between 1 and 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q5

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q5/5

SET 10 • 8.EE.A.4 • EXPRESSIONS & EQUATIONS

Multiply and divide numbers written in scientific notation

MAIN	Grade 8 main question	**8.EE.A.4**
BACKWARD 1	Multiply decimals	**5.NBT.B.7**
BACKWARD 2	Multiply by powers of 10	**5.NBT.A.2**
FORWARD 1	Use units to guide multistep calculations in scientific notation	**HSN-Q.A.1**
FORWARD 2	Model exponential growth and decay with scientific notation	**HSF-LE.A.1.c**

SET 10 • 8.EE.A.4

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.A.4

Multiply. Write the answer in scientific notation.

$(4 \times 10^{6})(2.5 \times 10^{3})$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q1/5

318

QUESTION ID

S10-M-Q1

TYPE

Short answer

CORRECT ANSWER

1×10^{10}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q1/5

Divide. Write the answer in scientific notation.

$(6.3 \times 10^{8}) \div (9 \times 10^{2})$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q2/5

319

QUESTION ID

S10-M-Q2

TYPE

Short answer

CORRECT ANSWER

7×10^{5}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q2/5

Light travels about 3×10^5 km per second.

How far does it travel in 5×10^2 seconds?

A. 1.5×10^7 km

B. 6×10^2 km

C. 1.5×10^8 km

D. 8×10^7 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q3/5

320

QUESTION ID

S10-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1.5×10^8 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q3/5

True or false?

$(2 \times 10^{-3})(3 \times 10^{-4}) = 6 \times 10^{-7}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q4/5

321

QUESTION ID

S10-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q4/5

True or false?

$(8 \times 10^9) \div (4 \times 10^3) = 2 \times 10^3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q5/5

322

QUESTION ID

S10-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The quotient is 2×10^6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q5/5

SET 10 • 8.EE.A.4

BACKWARD 1

Multiply decimals • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Multiply.

2.5 × 4

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q1/5

324

QUESTION ID

S10-B1-Q1

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q1/5

Multiply.

1.2 × 3.5

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q2/5

325

QUESTION ID

S10-B1-Q2

TYPE

Short answer

CORRECT ANSWER

4.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q2/5

Multiply.

6×0.5

A. 6.5

B. 3

C. 0.3

D. 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q3/5

326

QUESTION ID

S10-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q3/5

True or false?

$2.4 \times 0.5 = 1.2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q4/5

327

QUESTION ID

S10-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q4/5

True or false?

$3.5 \times 2 = 6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q5/5

328

QUESTION ID

S10-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q5/5

SET 10 • 8.EE.A.4

BACKWARD 2

Multiply by powers of 10 • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Multiply.

2.5×10^4

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q1/5

330

QUESTION ID

S10-B2-Q1

TYPE

Short answer

CORRECT ANSWER

25,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q1/5

Multiply.

3.1×10^2

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q2/5

331

QUESTION ID

S10-B2-Q2

TYPE

Short answer

CORRECT ANSWER

310

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q2/5

Multiply.

0.6×10^3

A. 60

B. 0.0006

C. 6,000

D. 600

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q3/5

332

QUESTION ID

S10-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 600

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q3/5

True or false?

$7 \times 10^3 = 7{,}000$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q4/5

333

QUESTION ID

S10-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q4/5

True or false?

$4.2 \times 10 = 4.20$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q5/5

334

QUESTION ID

S10-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q5/5

SET 10 • 8.EE.A.4

FORWARD 1

Use units to guide multistep calculations in scientific notation • High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1

A drive transfers 2×10^8 bytes per second.

How many seconds does it take to transfer 6×10^{10} bytes?

Seconds: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q1/5

336

QUESTION ID

S10-F1-Q1

TYPE

Short answer

CORRECT ANSWER

300

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q1/5

Light travels 3×10^8 meters per second.

How many meters does it travel in 1 hour (3.6×10^3 seconds)? Use scientific notation.

Meters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q2/5

337

QUESTION ID

S10-F1-Q2

TYPE

Short answer

CORRECT ANSWER

1.08×10^{12} m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-F1-Q2**

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q2/5

The Sun is about 1.5×10^{11} m from Earth. Light travels 3×10^{8} m per second.

About how long does sunlight take to reach Earth?

A. 500 seconds

B. 4.5×10^{19} seconds

C. 5×10^{19} seconds

D. 50 seconds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q3/5

338

QUESTION ID

S10-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 500 seconds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q3/5

True or false?

A flow of 5×10^3 liters per minute is 3×10^5 liters per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q4/5

339

QUESTION ID

S10-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q4/5

True or false?

6×10^{-3} km is equal to 6×10^{-6} m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q5/5

340

QUESTION ID

S10-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 6×10^{-3} km = 6 m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q5/5

SET 10 • 8.EE.A.4

FORWARD 2

Model exponential growth and decay with scientific notation • High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c

A colony of 2×10^3 bacteria triples every hour.

How many bacteria are there after 4 hours? Use scientific notation.

Bacteria: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q1

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q1/5

342

QUESTION ID

S10-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1.62×10^5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-F2-Q1**

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q1/5

A quantity of 5×10^9 is cut in half each year.

What is it after 3 years? Use scientific notation.

Amount: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q2

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q2/5

343

QUESTION ID

S10-F2-Q2

TYPE

Short answer

CORRECT ANSWER

6.25×10^8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-F2-Q2**

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q2/5

A population of 4×10^6 grows by 10% each year.

What is the population after 2 years?

A. 4.84×10^6

B. 8×10^6

C. 4.4×10^6

D. 4.8×10^6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q3

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q3/5

344

QUESTION ID

S10-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 4.84×10^6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q3

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q3/5

True or false?

Doubling 3×10^5 ten times gives about 3.07×10^8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q4

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q4/5

345

QUESTION ID

S10-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q4

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q4/5

True or false?

A quantity that grows by a constant percent rate increases by the same amount each year.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q5

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q5/5

346

QUESTION ID

S10-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q5

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q5/5

SET 11 • 8.EE.A.4 • EXPRESSIONS & EQUATIONS

Add and subtract numbers written in scientific notation

MAIN	Grade 8 main question	**8.EE.A.4**
BACKWARD 1	Add decimals	**5.NBT.B.7**
BACKWARD 2	Write a number times a power of 10 in standard form	**5.NBT.A.2**
FORWARD 1	Add and subtract polynomials	**HSA-APR.A.1**
FORWARD 2	Add and subtract functions — nearest related (no direct Grade 10+)	**HSF-BF.A.1.b**

SET 11 • 8.EE.A.4

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.A.4

Add. Write the answer in scientific notation.

$(3.2 \times 10^5) + (4.5 \times 10^4)$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q1/5

349

QUESTION ID

S11-M-Q1

TYPE

Short answer

CORRECT ANSWER

3.65×10^5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q1/5

Subtract. Write the answer in scientific notation.

$(9 \times 10^{7}) - (6 \times 10^{6})$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q2/5

350

QUESTION ID

S11-M-Q2

TYPE

Short answer

CORRECT ANSWER

8.4×10^{7}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q2/5

Add. Write the answer in scientific notation.

$(1.5 \times 10^{-3}) + (2 \times 10^{-4})$

A. 1.7×10^{-4}

B. 3.5×10^{-3}

C. 1.7×10^{-3}

D. 3.5×10^{-7}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q3/5

351

QUESTION ID

S11-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1.7×10^{-3}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q3/5

True or false?

$(5 \times 10^8) + (5 \times 10^8) = 1 \times 10^9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q4/5

352

QUESTION ID

S11-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q4/5

True or false?

$(4 \times 10^6) - (3 \times 10^5) = 1 \times 10^1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q5/5

353

QUESTION ID

S11-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The difference is 3.7×10^6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q5/5

SET 11 • 8.EE.A.4

BACKWARD 1

Add decimals • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Add.

3.2 + 0.45

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q1/5

355

QUESTION ID

S11-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3.65

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q1/5

Add.

8.4 + 1.6

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q2/5

356

QUESTION ID

S11-B1-Q2

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q2/5

Add.

1.5 + 0.2

A. 0.17

B. 1.7

C. 1.52

D. 3.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q3/5

357

QUESTION ID

S11-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q3/5

True or false?

$2.75 + 0.25 = 3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q4/5

358

QUESTION ID

S11-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q4/5

True or false?

4.6 + 0.04 = 5

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q5/5

359

QUESTION ID

S11-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q5/5

SET 11 • 8.EE.A.4

BACKWARD 2

Write a number times a power of 10 in standard form • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Write 6×10^6 in standard form.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q1/5

361

QUESTION ID

S11-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6,000,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q1/5

Write 0.6×10^7 in standard form.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q2/5

362

QUESTION ID

S11-B2-Q2

TYPE

Short answer

CORRECT ANSWER

6,000,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q2/5

Which number is equal to 45×10^4?

A. 450,000

B. 4,500

C. 45,000

D. 4,500,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q3/5

363

QUESTION ID

S11-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 450,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q3/5

True or false?

$3.2 \times 10^2 = 320$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q4/5

364

QUESTION ID

S11-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q4/5

True or false?

$0.5 \times 10^3 = 50$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q5/5

365

QUESTION ID

S11-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q5/5

SET 11 • 8.EE.A.4

FORWARD 1

Add and subtract polynomials • High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Add.

$(3x^2 + 5x) + (4x^2 - 2x)$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q1

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q1/5

367

QUESTION ID

S11-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$7x^2 + 3x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q1

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q1/5

Subtract.

$(6a^3 - 2a + 1) - (2a^3 + 5a - 4)$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q2

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q2/5

368

QUESTION ID

S11-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$4a^3 - 7a + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q2

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q2/5

Add.

$(x^2 - 4) + (3x^2 + x)$

A. $4x^2 + x - 4$

B. $3x^4 + x - 4$

C. $4x^2 - 3$

D. $4x^2 + x + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q3

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q3/5

369

QUESTION ID

S11-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $4x^2 + x - 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q3

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q3/5

True or false?

$(2y^2 + 3) - (y^2 - 3) = y^2 + 6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q4

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q4/5

370

QUESTION ID

S11-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q4

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q4/5

True or false?

$(5n - 2) - (3n - 4) = 2n - 6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q5

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q5/5

371

QUESTION ID

S11-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The difference is 2n + 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q5

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q5/5

SET 11 • 8.EE.A.4

FORWARD 2

Add and subtract functions • High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b

⚑ No high-school standard directly continues this skill at this step. This branch uses the nearest related high-school standard (placed at Grade 10 in this bank).

Revenue is R(x) = 25x and cost is C(x) = 4,000 + 9x for x items.

Write the profit function P(x) = R(x) − C(x).

P(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q1/5 • NEAREST RELATED

373

QUESTION ID

S11-F2-Q1

TYPE

Short answer

CORRECT ANSWER

16x − 4,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q1/5 • NEAREST RELATED

The profit function is P(x) = 16x − 4,000.

Find P(1,000).

P(1,000) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q2/5 • NEAREST RELATED

374

QUESTION ID

S11-F2-Q2

TYPE

Short answer

CORRECT ANSWER

12,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q2/5 • NEAREST RELATED

$f(x) = 3x^2$ and $g(x) = x^2 - 5$.

Which expression is $(f + g)(x)$?

A. $2x^2 + 5$

B. $4x^2 - 5$

C. $3x^4 - 5$

D. $4x^2 + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q3/5 • NEAREST RELATED

375

QUESTION ID

S11-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $4x^2 - 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

If f(x) = 2x and g(x) = x + 1, then (f − g)(3) = 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q4/5 • NEAREST RELATED

376

QUESTION ID

S11-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

If $f(x) = x + 4$ and $g(x) = 2x$, then $(f + g)(x) = 2x^2 + 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q5/5 • NEAREST RELATED

377

QUESTION ID

S11-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(f + g)(x) = 3x + 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q5/5 • NEAREST RELATED

SET 12 • 8.EE.A.4 • EXPRESSIONS & EQUATIONS

Choose units for scientific notation and read calculator notation

MAIN	Grade 8 main question	**8.EE.A.4**
BACKWARD 1	Convert metric units	**5.MD.A.1**
BACKWARD 2	Divide by powers of 10	**5.NBT.A.2**
FORWARD 1	Convert units of very large and very small quantities	**HSN-Q.A.1**
FORWARD 2	Define the quantities and units a model needs	**HSN-Q.A.2**

SET 12 • 8.EE.A.4

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.A.4

A calculator shows 4.7E8.

Write the number in scientific notation and in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q1/5

380

QUESTION ID

S12-M-Q1

TYPE

Short answer

CORRECT ANSWER

4.7×10^8

$= 470{,}000{,}000$

GRADING NOTE Grade both forms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q1/5

A calculator shows 2.05E-6.

Write the number in standard form.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q2/5

381

QUESTION ID

S12-M-Q2

TYPE

Short answer

CORRECT ANSWER

0.00000205

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q2/5

The distance from Earth to the Moon is about 3.84×10^8 m.

Which unit is the most sensible for this distance?

A. Kilometers

B. Centimeters

C. Millimeters

D. Micrometers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q3/5

382

QUESTION ID

S12-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Kilometers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q3/5

A dust particle has a mass of 7.53×10^{-10} kg.

Write the mass in micrograms. (1 μg = 10^{-9} kg)

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q4/5

383

QUESTION ID

S12-M-Q4

TYPE

Short answer

CORRECT ANSWER

0.753 μg

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q4/5

True or false?

A calculator display of 3E-4 means −30,000.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q5/5

384

QUESTION ID

S12-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 3E-4 means $3 \times 10^{-4} = 0.0003$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q5/5

SET 12 • 8.EE.A.4

BACKWARD 1

Convert metric units • Grade 5 • Measurement & Data • 5.MD.A.1

How many grams are in 3.5 kilograms?

Grams: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q1

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q1/5

386

QUESTION ID

S12-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3,500

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q1

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q1/5

How many meters are in 4,200 millimeters?

Meters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q2

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q2/5

387

QUESTION ID

S12-B1-Q2

TYPE

Short answer

CORRECT ANSWER

4.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q2

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q2/5

How many centimeters are in 6 meters?

A. 60

B. 0.06

C. 6,000

D. 600

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q3

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q3/5

388

QUESTION ID

S12-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 600

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q3

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q3/5

True or false?

5 km = 5,000 m

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q4

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q4/5

389

QUESTION ID

S12-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q4

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q4/5

True or false?

250 cm = 25 m

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q5

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q5/5

390

QUESTION ID

S12-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q5

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q5/5

SET 12 • 8.EE.A.4

BACKWARD 2

Divide by powers of 10 • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Divide.

$3{,}840 \div 10^3$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q1/5

392

QUESTION ID

S12-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3.84

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q1/5

Divide.

$75 \div 10^2$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q2/5

393

QUESTION ID

S12-B2-Q2

TYPE

Short answer

CORRECT ANSWER

0.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q2/5

Divide.

$6 \div 10^4$

A. 0.006

B. 0.00006

C. 60,000

D. 0.0006

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q3/5

394

QUESTION ID

S12-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 0.0006

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q3/5

True or false?

520 ÷ 10 = 52

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q4/5

395

QUESTION ID

S12-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q4/5

True or false?

$9 \div 10^3 = 0.09$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q5/5

396

QUESTION ID

S12-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q5/5

SET 12 • 8.EE.A.4

FORWARD 1

Convert units of very large and very small quantities • High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1

Convert 3.6×10^6 mm to kilometers.

Kilometers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q1/5

398

QUESTION ID

S12-F1-Q1

TYPE

Short answer

CORRECT ANSWER

3.6 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q1/5

A server sends 4.5×10^9 bytes per minute.

Write this rate in gigabytes per second. (1 GB = 10^9 bytes)

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q2/5

399

QUESTION ID

S12-F1-Q2

TYPE

Short answer

CORRECT ANSWER

0.075 GB per second

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q2/5

A virus is 1.2×10^{-7} m wide.

What is its width in nanometers? (1 nm = 10^{-9} m)

A. 1,200 nm

B. 120 nm

C. 12 nm

D. 0.12 nm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q3/5

400

QUESTION ID

S12-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 120 nm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q3/5

True or false?

7×10^5 g = 700 kg

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q4/5

401

QUESTION ID

S12-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q4/5

True or false?

2.5×10^{-3} km = 2.5 cm

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q5/5

402

QUESTION ID

S12-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 2.5×10^{-3} km = 2.5 m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q5/5

SET 12 • 8.EE.A.4

FORWARD 2

Define the quantities and units a model needs • High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2

You want to estimate how much water a city of 2×10^6 people uses each day.

Which quantities do you need, and with what units?

Quantities: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q1

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q1/5

404

QUESTION ID

S12-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The number of people and the water used per person per day (liters per person per day).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q1

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q1/5

A model for the cost of running a fan is $C = p \cdot w \cdot h$, with C in dollars.

Define p, w, and h, with units, so that the units work out to dollars.

Definitions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q2

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q2/5

405

QUESTION ID

S12-F2-Q2

TYPE

Short answer

CORRECT ANSWER

p = the price of electricity (dollars per kilowatt-hour), w = the fan's power (kilowatts), h = the time it runs (hours).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q2

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q2/5

A model will predict how long it takes to fill a pool with a hose.

Which pair of quantities does it need?

A. The pool's volume (liters) and the hose's flow rate (liters per minute)

B. The hose's length (meters) and its color

C. The number of swimmers and the pool's depth (meters)

D. The pool's depth (meters) and the water's temperature (°C)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q3

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q3/5

406

QUESTION ID

S12-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. The pool's volume (liters) and the hose's flow rate (liters per minute)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q3

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q3/5

True or false?

In the model $C = p \cdot w \cdot h$, if p is in dollars per kilowatt-hour, w in kilowatts, and h in hours, then C is in dollars.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q4

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q4/5

QUESTION ID

S12-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q4

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q4/5

True or false?

A model of a city's daily water use needs only the city's population; the units of the other quantities do not matter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q5

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q5/5

408

QUESTION ID

S12-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It also needs the water used per person per day, with its units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q5

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q5/5

SET 13 • 8.EE.B.5 • EXPRESSIONS & EQUATIONS

Graph a proportional relationship and interpret the unit rate as the slope

MAIN	Grade 8 main question	**8.EE.B.5**
BACKWARD 1	Explain points on the graph of a proportional relationship	**7.RP.A.2.d**
BACKWARD 2	Find the constant of proportionality from a graph	**7.RP.A.2.b**
FORWARD 1	Calculate and interpret an average rate of change	**HSF-IF.B.6**
FORWARD 2	Compare properties of functions represented in different ways	**HSF-IF.C.9**

SET 13 • 8.EE.B.5

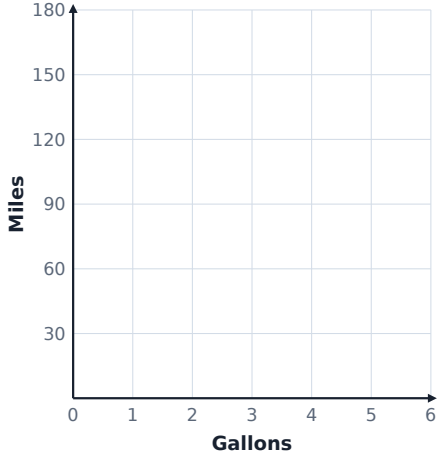
MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.B.5

A car uses gas at a constant rate: 3 gallons for every 90 miles.

Graph the relationship between gallons (x) and miles (y). What is the slope, and what does it mean?

Slope and meaning: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q1/5

411

QUESTION ID

S13-M-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A straight line through (0, 0) and points such as (1, 30) and (3, 90)

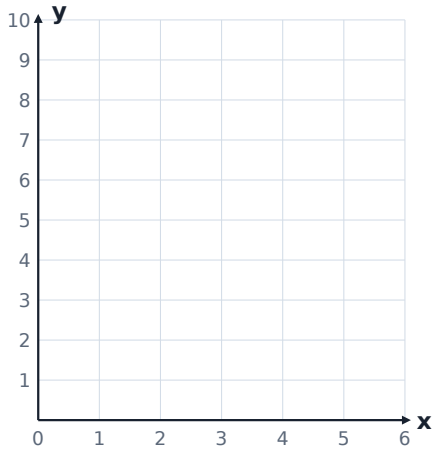
Written: 30; the car travels 30 miles per gallon.

GRADING NOTE Grade both: the line through the origin with slope 30, and the slope with its meaning.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q1/5

Graph y = 1.5x on the grid for x from 0 to 6.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q2/5

412

QUESTION ID

S13-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

A straight line through (0, 0), (2, 3), (4, 6), and (6, 9)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

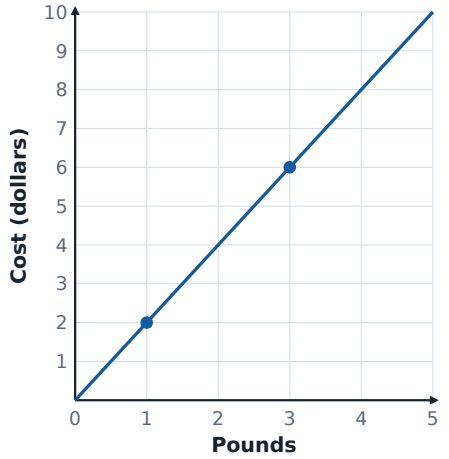
ID S13-M-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q2/5

The graph shows the cost of apples.

What is the slope, and what does it represent?

Slope and meaning: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q3/5

413

QUESTION ID

S13-M-Q3

TYPE

Short answer

CORRECT ANSWER

2; the apples cost $2 per pound.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q3/5

The graph of a proportional relationship passes through (4, 10).

What is the unit rate?

A. 14

B. 2.5

C. 0.4

D. 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q4/5

414

QUESTION ID

S13-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q4/5

True or false?

On graphs of distance (y) against time (x) that pass through the origin, a steeper line means a greater speed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q5/5

415

QUESTION ID

S13-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q5/5

SET 13 • 8.EE.B.5

BACKWARD 1

Explain points on the graph of a proportional relationship • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d

A proportional graph of hours (x) and pay in dollars (y) passes through (1, 15).

What does this point mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q1/5

417

QUESTION ID

S13-B1-Q1

TYPE

Short answer

CORRECT ANSWER

The pay is $15 for 1 hour (the unit rate).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q1/5

A proportional graph shows cups of flour (x) and cookies (y).

What does the point (0, 0) mean?

A. Each cup makes 0 cookies.

B. 0 cups of flour make 0 cookies.

C. The recipe starts with some cookies.

D. There is no graph.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q2/5

418

QUESTION ID

S13-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 0 cups of flour make 0 cookies.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q2/5

A proportional graph of minutes (x) and pages read (y) passes through (5, 20).

What does this point mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q3/5

419

QUESTION ID

S13-B1-Q3

TYPE

Short answer

CORRECT ANSWER

20 pages are read in 5 minutes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q3/5

True or false?

On the graph of a proportional relationship, the point (1, r) tells you the unit rate is r.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q4/5

420

QUESTION ID

S13-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q4/5

True or false?

On a graph of hours (x) and miles (y), the point (3, 12) means 3 miles in 12 hours.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q5/5

421

QUESTION ID

S13-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It means 12 miles in 3 hours.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q5/5

SET 13 • 8.EE.B.5

BACKWARD 2

Find the constant of proportionality from a graph • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b

A proportional graph passes through (0, 0) and (5, 20).

What is the constant of proportionality?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q1/5

423

QUESTION ID

S13-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q1/5

Which point shows k = 3 on the graph of a proportional relationship?

A. (3, 6)

B. (6, 2)

C. (3, 1)

D. (2, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q2/5

424

QUESTION ID

S13-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. (2, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q2/5

A proportional graph passes through (8, 2).

What is the constant of proportionality?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q3/5

425

QUESTION ID

S13-B2-Q3

TYPE

Short answer

CORRECT ANSWER

0.25

GRADING NOTE 1/4 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q3/5

True or false?

A line through (0, 0) and (10, 25) has a constant of proportionality of 2.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q4/5

426

QUESTION ID

S13-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q4/5

True or false?

A line through (0, 0) and (6, 3) has a constant of proportionality of 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q5/5

427

QUESTION ID

S13-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $k = 3 \div 6 = 0.5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q5/5

SET 13 • 8.EE.B.5

FORWARD 1

Calculate and interpret an average rate of change • High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6

$f(x) = 3x + 2$

What is the average rate of change of f from $x = 1$ to $x = 5$?

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q1/5

429

QUESTION ID

S13-F1-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q1/5

The table shows a trip.

What is the average rate of change from 2 hours to 5 hours?

Hours	0	2	5
Miles	0	110	260

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q2/5

430

QUESTION ID

S13-F1-Q2

TYPE

Short answer

CORRECT ANSWER

50 miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q2/5

$h(t) = t^2$

What is the average rate of change of h from t = 1 to t = 4?

A. 5

B. 3

C. 16

D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q3/5

431

QUESTION ID

S13-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q3/5

True or false?

For a linear function, the average rate of change is the same over every interval.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q4/5

QUESTION ID

S13-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q4/5

True or false?

For $f(x) = x^2$, the average rate of change from $x = 0$ to $x = 2$ is 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q5/5

433

QUESTION ID
S13-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It is $(4 - 0) \div 2 = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q5/5

SET 13 • 8.EE.B.5

FORWARD 2

Compare properties of functions represented in different ways • High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9

$f(x) = 2.5x$. The table shows $g(x) = x^2$.

Which function has the greater average rate of change from $x = 0$ to $x = 4$? Give both rates.

x	0	1	2	3	4
g(x)	0	1	4	9	16

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q1/5

435

QUESTION ID

S13-F2-Q1

TYPE

Short answer

CORRECT ANSWER

g: its average rate is 16 ÷ 4 = 4, and f's is 2.5.

GRADING NOTE Both rates and the comparison are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q1/5

f(x) = 40 + 5x and g(x) = 30(1.2)^x.

Which function has the greater initial value? Which is greater at x = 10? Round values to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q2/5

436

QUESTION ID

S13-F2-Q2

TYPE

Short answer

CORRECT ANSWER

f has the greater initial value (40 > 30); g is greater at x = 10 (about 185.8, while f(10) = 90).

GRADING NOTE Both comparisons are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q2/5

f(x) = 2^x and g(x) = 10x.

For which x-value is f(x) > g(x)?

A. x = 3

B. x = 4

C. x = 6

D. x = 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q3/5

437

QUESTION ID

S13-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q3/5

True or false?

A linear function with a slope of 4 eventually grows more slowly than f(x) = 2^x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q4/5

438

QUESTION ID

S13-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q4/5

True or false?

Two functions with the same y-intercept must be the same function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q5/5

439

QUESTION ID

S13-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q5/5

SET 14 • 8.EE.B.5 • EXPRESSIONS & EQUATIONS

Compare two proportional relationships represented in different ways

MAIN	Grade 8 main question	**8.EE.B.5**
BACKWARD 1	Write an equation for a proportional relationship	**7.RP.A.2.c**
BACKWARD 2	Compute unit rates with fractions	**7.RP.A.1**
FORWARD 1	Create linear and quadratic equations in two variables and graph t	**HSA-CED.A.2**
FORWARD 2	Compare linear and exponential growth	**HSF-LE.A.3**

SET 14 • 8.EE.B.5

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.B.5

Runner A's distance is d = 6.2t (miles, hours). Runner B's graph is a line through (0, 0) and (2, 13).

Which runner is faster?

A. Runner A

B. Runner B

C. They run at the same speed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q1/5

442

QUESTION ID

S14-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Runner B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q1/5

Plan A costs c = 0.12m dollars for m minutes. The table shows Plan B.

Which plan costs less per minute, and by how much?

Minutes	100	250	400
Cost ($)	15	37.5	60

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q2/5

443

QUESTION ID

S14-M-Q2

TYPE

Short answer

CORRECT ANSWER

Plan A, by $0.03 per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q2/5

Pump A fills 45 gallons in 3 minutes at a constant rate.
Pump B fills according to g = 14m.

Which pump is faster, and by how much?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q3/5

444

QUESTION ID

S14-M-Q3

TYPE

Short answer

CORRECT ANSWER

Pump A, by 1 gallon per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q3/5

True or false?

The graph of y = 0.8x is less steep than the graph of a proportional relationship through (5, 5).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q4/5

445

QUESTION ID

S14-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q4/5

True or false?

A proportional relationship through (3, 7) has a greater unit rate than y = 2.5x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q5/5

446

QUESTION ID

S14-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 7 ÷ 3 ≈ 2.33, which is less than 2.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q5/5

SET 14 • 8.EE.B.5

BACKWARD 1

Write an equation for a proportional relationship • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c

Pencils cost $0.45 each.

Write an equation for the cost c of n pencils.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q1/5

448

QUESTION ID

S14-B1-Q1

TYPE

Short answer

CORRECT ANSWER

c = 0.45n

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q1/5

The table shows a proportional relationship.

Write an equation for y in terms of x.

x	2	4	6
y	7	14	21

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q2/5

449

QUESTION ID

S14-B1-Q2

TYPE

Short answer

CORRECT ANSWER

y = 3.5x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q2/5

A car travels 55 miles per hour.

Which equation gives the distance d after t hours?

A. $d = \frac{t}{55}$

B. $d = 55t$

C. $t = 55d$

D. $d = t + 55$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q3/5

QUESTION ID

S14-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. d = 55t

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q3/5

True or false?

A recipe uses 6 cups of water for every cup of rice. The equation y = 6x gives the cups of water y for x cups of rice.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q4/5

451

QUESTION ID

S14-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q4/5

True or false?

A printer prints 30 pages in 2 minutes. The equation p = 30m gives the pages p printed in m minutes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q5/5

452

QUESTION ID

S14-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The equation is p = 15m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q5/5

SET 14 • 8.EE.B.5

BACKWARD 2

Compute unit rates with fractions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

Ty walks $\frac{3}{4}$ mile in $\frac{1}{4}$ hour.

How many miles per hour is this?

Miles per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q1/5

454

QUESTION ID

S14-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q1/5

$\frac{2}{5}$ pound of nuts costs $2.

What is the price per pound?

Price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q2/5

455

QUESTION ID

S14-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q2/5

A hose fills $\frac{1}{2}$ gallon in $\frac{1}{6}$ minute.

How many gallons per minute is this?

A. $\frac{1}{12}$

B. 6

C. $\frac{1}{3}$

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q3/5

456

QUESTION ID

S14-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q3/5

True or false?

Walking $\frac{1}{3}$ mile in $\frac{1}{6}$ hour is a speed of 2 miles per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q4/5

457

QUESTION ID

S14-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q4/5

True or false?

A drink uses $\frac{3}{4}$ cup of juice for every $\frac{1}{2}$ cup of water. That is $\frac{3}{8}$ cup of juice per cup of water.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q5/5

458

QUESTION ID

S14-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 3/2 cups of juice per cup of water.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q5/5

SET 14 • 8.EE.B.5

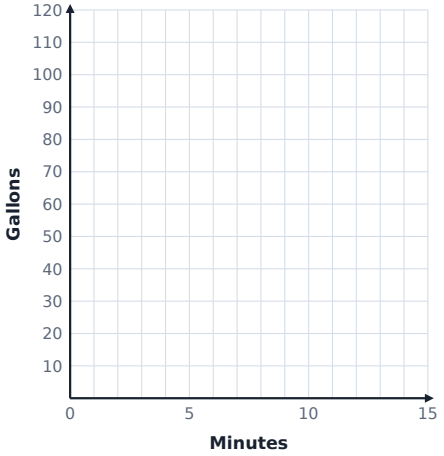
FORWARD 1

Create linear and quadratic equations in two variables and graph them with labels and scales • High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2

Tank A has 20 gallons and gains 5 gallons per minute. Tank B starts empty and gains 7 gallons per minute.

Write an equation for each tank (y gallons after x minutes). Graph both for $0 \le x \le 15$, label each line, and mark where they meet.

Equations: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q1

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q1/5

460

QUESTION ID

S14-F1-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: Line A from (0, 20) to (15, 95) and line B from (0, 0) to (15, 105), each labeled, meeting at (10, 70)

Written: Tank A: y = 5x + 20; Tank B: y = 7x

GRADING NOTE Grade both: the two equations, and the two labeled lines meeting at (10, 70).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q1

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q1/5

A rectangle has a perimeter of 40 m.

Write an equation for its area A in terms of its width w, and give the values of w that make sense.

Equation and domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q2

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q2/5

461

QUESTION ID

S14-F1-Q2

TYPE

Short answer

CORRECT ANSWER

A = w(20 − w), for 0 < w < 20

GRADING NOTE Equivalent forms such as $A = 20w - w^2$ are correct. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q2

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q2/5

A ball's height is $h = -5t^2 + 20t$ meters after t seconds, for $0 \le t \le 4$.

Which axes and scales fit a graph of this equation?

A. Horizontal: time from 0 to 4 seconds; vertical: height from −20 to 0 meters

B. Horizontal: time t from 0 to 4 seconds; vertical: height h from 0 to 20 meters

C. Horizontal: height from 0 to 4 meters; vertical: time from 0 to 20 seconds

D. Horizontal: time from 0 to 20 seconds; vertical: height from 0 to 4 meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q3

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q3/5

462

QUESTION ID

S14-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Horizontal: time t from 0 to 4 seconds; vertical: height h from 0 to 20 meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q3

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q3/5

True or false?

The graphs of y = 5x + 20 and y = 7x intersect at (10, 70).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q4

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q4/5

463

QUESTION ID

S14-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q4

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q4/5

True or false?

For a rectangle with a perimeter of 40 m, the area $A = w(20 - w)$ is greatest when $w = 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q5

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q5/5

464

QUESTION ID

S14-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is greatest when w = 10 (a 10 m by 10 m square).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q5

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q5/5

SET 14 • 8.EE.B.5

FORWARD 2

Compare linear and exponential growth • High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

f(x) = 100 + 50x and g(x) = 100(1.5)^x.

Which is greater at x = 5? Give both values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q1

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q1/5

466

QUESTION ID

S14-F2-Q1

TYPE

Short answer

CORRECT ANSWER

g: g(5) = 759.375, while f(5) = 350

GRADING NOTE Exact values are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q1

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q1/5

Which function eventually has the greater values:
y = 1,000x or y = 2^x?

A. It depends only on the y-intercepts

B. y = 1,000x

C. y = 2^x

D. Neither; they are always equal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q2

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q2/5

467

QUESTION ID

S14-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. y = 2^x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q2

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q2/5

Which function in the table is exponential? Explain.

x	0	1	2	3
f(x)	2	4	8	16
g(x)	2	4	6	8

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q3

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q3/5

468

QUESTION ID

S14-F2-Q3

TYPE

Short answer

CORRECT ANSWER

f, because it doubles each time x increases by 1 (g increases by 2 each time).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q3

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q3/5

True or false?

A quantity that increases exponentially eventually exceeds a quantity that increases linearly.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q4

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q4/5

469

QUESTION ID

S14-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q4

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q4/5

True or false?

At x = 2, 3^x is greater than 10x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q5

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q5/5

470

QUESTION ID

S14-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3^2 = 9$, which is less than 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q5

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q5/5

SET 15 • 8.EE.B.6 • EXPRESSIONS & EQUATIONS

Use similar triangles to explain why the slope is the same between any two points on a line

MAIN	Grade 8 main question	**8.EE.B.6**
BACKWARD 1	Use a scale factor to find a length in a scaled copy	**7.G.A.1**
BACKWARD 2	Write an equation from a table of values	**6.EE.C.9**
FORWARD 1	Use similarity transformations to decide whether triangles are sim	**HSG-SRT.A.2**
FORWARD 2	Use slopes of parallel and perpendicular lines	**HSG-GPE.B.5**

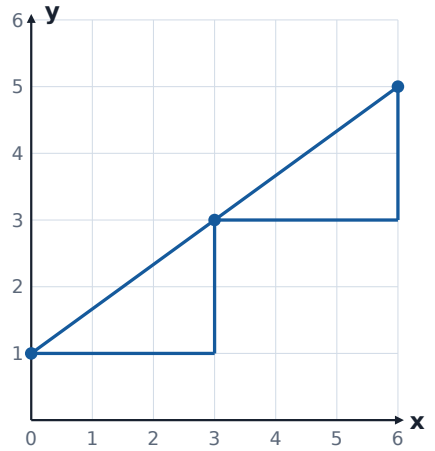
SET 15 • 8.EE.B.6

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.B.6

Two slope triangles are drawn under the line. Explain why they give the same slope.

Explanation: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q1/5

473

QUESTION ID

S15-M-Q1

TYPE

Short answer

CORRECT ANSWER

The triangles are similar (they have the same angles), so their rise-to-run ratios are equal: 2/3 = 2/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q1/5

A slope triangle on a line has a run of 4 and a rise of 6.
A larger slope triangle on the same line has a run of 10.

Use similar triangles to find its rise.

Rise: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q2/5

474

QUESTION ID

S15-M-Q2

TYPE

Short answer

CORRECT ANSWER

15 (6/4 = rise/10)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q2/5

Why are two slope triangles on the same non-vertical line similar?

A. Their hypotenuses are always equal.

B. Each has a right angle, and their angles where the line meets the horizontal legs are equal (corresponding angles).

C. Their legs are always equal.

D. They always have equal areas.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q3/5

475

QUESTION ID

S15-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Each has a right angle, and their angles where the line meets the horizontal legs are equal (corresponding angles).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q3/5

The points (2, 1), (5, 3), and (11, 7) lie on one line.

Find the slope using (2, 1) and (5, 3), and again using (5, 3) and (11, 7). Explain why the two results must agree.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q4/5

476

QUESTION ID

S15-M-Q4

TYPE

Short answer

CORRECT ANSWER

2/3 and 4/6 = 2/3. The two slope triangles are similar, so rise ÷ run is the same.

GRADING NOTE Grade both slopes and the similar-triangle reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q4/5

True or false?

Any two slope triangles drawn on the same non-vertical line are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q5/5

477

QUESTION ID

S15-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q5/5

SET 15 • 8.EE.B.6

BACKWARD 1

Use a scale factor to find a length in a scaled copy • Grade 7 • Geometry • 7.G.A.1

A right triangle has legs of 2 and 3. In a scaled copy, the leg of 2 becomes 8.

How long is the other leg in the copy?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q1/5

479

QUESTION ID

S15-B1-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q1/5

A right triangle has legs of 4 and 6. A scaled copy has a leg of 6 matching the leg of 4.

How long is the other leg in the copy?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q2/5

480

QUESTION ID

S15-B1-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q2/5

Which pair of right triangles are scaled copies of each other (matching the shorter legs)?

A. Legs 2 and 5; legs 4 and 7

B. Legs 2 and 5; legs 6 and 12

C. Legs 2 and 5; legs 3 and 6

D. Legs 2 and 5; legs 4 and 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q3/5

481

QUESTION ID

S15-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Legs 2 and 5; legs 4 and 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q3/5

True or false?

A scaled copy with a scale factor of 3 has every length 3 times as long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q4/5

482

QUESTION ID

S15-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q4/5

True or false?

Right triangles with legs 3 and 4 and legs 6 and 9 are scaled copies of each other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q5/5

483

QUESTION ID

S15-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q5/5

SET 15 • 8.EE.B.6

BACKWARD 2

Write an equation from a table of values • Grade 6 • Expressions & Equations • 6.EE.C.9

Write an equation for y in terms of x.

x	1	2	3
y	6	7	8

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q1/5

485

QUESTION ID

S15-B2-Q1

TYPE

Short answer

CORRECT ANSWER

y = x + 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q1/5

Write an equation for y in terms of x.

x	1	2	3
y	3	6	9

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q2/5

QUESTION ID

S15-B2-Q2

TYPE

Short answer

CORRECT ANSWER

y = 3x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q2/5

Which equation matches the table?

x	0	1	2
y	1	3	5

A. y = 2x

B. y = x + 1

C. y = 3x

D. y = 2x + 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q3/5

QUESTION ID

S15-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. y = 2x + 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q3/5

True or false?

For y = 2x + 4, y = 10 when x = 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q4/5

488

QUESTION ID

S15-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q4/5

True or false?

For y = x + 7, y = 14 when x = 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q5/5

489

QUESTION ID

S15-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q5/5

SET 15 • 8.EE.B.6

FORWARD 1

Use similarity transformations to decide whether triangles are similar • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Triangle ABC has sides 3, 4, and 5. Triangle DEF has sides 6, 8, and 10.

Are the triangles similar? If so, what is the scale factor from ABC to DEF?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q1/5

491

QUESTION ID

S15-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes; the scale factor is 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q1/5

On the line through (0, 1), (3, 3), and (9, 7), slope triangle T1 has vertices (0, 1), (3, 1), (3, 3). Slope triangle T2 has vertices (3, 3), (9, 3), (9, 7).

Describe a translation followed by a dilation that maps T1 onto T2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q2/5

492

QUESTION ID

S15-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Translate T1 3 units right and 2 units up, so (0, 1) → (3, 3), (3, 1) → (6, 3), (3, 3) → (6, 5). Then dilate about (3, 3) with a scale factor of 2: (6, 3) → (9, 3) and (6, 5) → (9, 7).

GRADING NOTE Any correct sequence that names the translation, the center, and the scale factor is accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q2/5

Which information is enough to show that two triangles are similar?

A. One pair of equal sides

B. Two pairs of equal angles

C. Equal perimeters

D. One pair of equal angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q3/5

493

QUESTION ID

S15-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Two pairs of equal angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q3/5

True or false?

If one triangle is a dilation of another, their corresponding sides are proportional.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q4/5

494

QUESTION ID

S15-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q4/5

True or false?

Any two right triangles are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q5/5

495

QUESTION ID

S15-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q5/5

SET 15 • 8.EE.B.6

FORWARD 2

Use slopes of parallel and perpendicular lines • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Line p has a slope of $\frac{3}{4}$.

What is the slope of a line perpendicular to p?

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q1/5

497

QUESTION ID

S15-F2-Q1

TYPE

Short answer

CORRECT ANSWER

−4/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-F2-Q1**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q1/5

Write an equation of the line through (0, 2) that is parallel to y = −3x + 7.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q2/5

498

QUESTION ID

S15-F2-Q2

TYPE

Short answer

CORRECT ANSWER

y = −3x + 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q2/5

Which line is perpendicular to $y = 2x - 5$?

A. $y = -2x - 5$

B. $y = 2x + 1$

C. $y = -\frac{1}{2}x + 1$

D. $y = \frac{1}{2}x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q3/5

499

QUESTION ID

S15-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $y = -(1/2)x + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q3/5

True or false?

Parallel lines that are not vertical have equal slopes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q4/5

500

QUESTION ID

S15-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q4/5

True or false?

Lines with slopes 3 and −3 are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q5/5

501

QUESTION ID

S15-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Perpendicular slopes multiply to −1; 3 × (−3) = −9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-F2-Q5**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q5/5

SET 60 • 8.EE.B.6 • EXPRESSIONS & EQUATIONS

Use similar triangles to derive y = mx for a line through the origin

MAIN	Grade 8 main question	**8.EE.B.6**
BACKWARD 1	Represent a proportional relationship with an equation	**7.RP.A.2.c**
BACKWARD 2	Find the constant of proportionality from a point or a table	**7.RP.A.2.b**
FORWARD 1	Show that a dilation centered at the origin maps a line through th…	**HSG-SRT.A.1**
FORWARD 2	Use rotated slope triangles to explain the slopes of parallel and p…	**HSG-GPE.B.5**

SET 60 • 8.EE.B.6

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.B.6

A line passes through the origin and (4, 6). For any other point (x, y) on the line (x ≠ 0), the triangle with vertices (0, 0), (x, 0), (x, y) is similar to the triangle with vertices (0, 0), (4, 0), (4, 6).

Use this to write the equation of the line.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 60 • MAIN • Q1/5

504

QUESTION ID

S60-M-Q1

TYPE

Short answer

CORRECT ANSWER

y = 1.5x (because y ÷ x = 6 ÷ 4 for x ≠ 0, and the origin also fits)

GRADING NOTE y = (3/2)x is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 60 • MAIN • Q1/5

A line passes through the origin with slope m. Use a slope triangle from (0, 0) to a point (x, y) on the line ($x \neq 0$) to explain why every point on the line satisfies $y = mx$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 60 • MAIN • Q2/5

505

QUESTION ID

S60-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

y = mx

GRADING NOTE REQUIRED WORK: The slope triangle with vertices (0, 0), (x, 0), (x, y) is similar to the triangle with run 1 and rise m, so $y \div x = m \div 1$ and $y = mx$. The point (0, 0) also satisfies $y = m \cdot 0$. Any equivalent explanation is accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 60 • MAIN • Q2/5

A line through the origin contains the point (5, 2).

Which equation does a similar-triangle argument give for the line?

A. $y = 2x + 5$

B. $y = x + 2$

C. $y = \frac{5}{2}x$

D. $y = \frac{2}{5}x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 60 • MAIN • Q3/5

506

QUESTION ID

S60-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. y = (2/5)x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 60 • MAIN • Q3/5

A line through the origin has a slope of −3. The point (2, y) is on the line.

Use a slope triangle to find y, then write the equation of the line.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 60 • MAIN • Q4/5

507

QUESTION ID

S60-M-Q4

TYPE

Short answer

CORRECT ANSWER

y = −6; the line is y = −3x

GRADING NOTE Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 60 • MAIN • Q4/5

True or false?

The point (6, 4) is on the line through the origin and (3, 2), because $4 \div 6 = 2 \div 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 60 • MAIN • Q5/5

508

QUESTION ID

S60-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 60 • MAIN • Q5/5

SET 60 • 8.EE.B.6

BACKWARD 1

Represent a proportional relationship with an equation • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c

Each notebook costs $3.

Write an equation for the cost c of n notebooks.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 60 • BACKWARD 1 • Q1/5

510

QUESTION ID

S60-B1-Q1

TYPE

Short answer

CORRECT ANSWER

c = 3n

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 60 • BACKWARD 1 • Q1/5

The table shows a proportional relationship.

Write an equation for y in terms of x.

x	2	5	8
y	5	12.5	20

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 60 • BACKWARD 1 • Q2/5

511

QUESTION ID

S60-B1-Q2

TYPE

Short answer

CORRECT ANSWER

y = 2.5x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 60 • BACKWARD 1 • Q2/5

A car goes 28 miles on each gallon of gas.

Which equation gives the miles m for g gallons?

A. m = 28g

B. g = 28m

C. m = 28 ÷ g

D. m = g + 28

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 60 • BACKWARD 1 • Q3/5

512

QUESTION ID

S60-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. m = 28g

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 60 • BACKWARD 1 • Q3/5

True or false?

The equation y = 7x describes a proportional relationship with a constant of proportionality of 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 60 • BACKWARD 1 • Q4/5

513

QUESTION ID

S60-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 60 • BACKWARD 1 • Q4/5

True or false?

The equation y = x + 4 describes a proportional relationship.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 60 • BACKWARD 1 • Q5/5

514

QUESTION ID

S60-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its graph does not pass through the origin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 60 • BACKWARD 1 • Q5/5

SET 60 • 8.EE.B.6

BACKWARD 2

Find the constant of proportionality from a point or a table • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b

A proportional relationship passes through (6, 9).

What is the constant of proportionality?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 60 • BACKWARD 2 • Q1/5

516

QUESTION ID

S60-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1.5

GRADING NOTE 3/2 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 60 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The table shows a proportional relationship.

What is the constant of proportionality?

x	4	10
y	2	5

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 60 • BACKWARD 2 • Q2/5

517

QUESTION ID

S60-B2-Q2

TYPE

Short answer

CORRECT ANSWER

0.5

GRADING NOTE 1/2 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 60 • BACKWARD 2 • Q2/5

What is the constant of proportionality in y = 12x?

A. 6

B. $\frac{1}{12}$

C. 12

D. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 60 • BACKWARD 2 • Q3/5

518

QUESTION ID

S60-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 60 • BACKWARD 2 • Q3/5

True or false?

The graph of a proportional relationship with constant k passes through (1, k).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 60 • BACKWARD 2 • Q4/5

519

QUESTION ID

S60-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 60 • BACKWARD 2 • Q4/5

True or false?

A line through (0, 0) and (4, 10) has a constant of proportionality of 0.4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 60 • BACKWARD 2 • Q5/5

520

QUESTION ID

S60-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $k = 10 \div 4 = 2.5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 60 • BACKWARD 2 • Q5/5

SET 60 • 8.EE.B.6

FORWARD 1

Show that a dilation centered at the origin maps a line through the origin onto itself • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1

A dilation with center (0, 0) and a scale factor of 3 is applied to the point (2, 5) on the line y = 2.5x.

What is the image, and is it on the same line? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 60 • FORWARD 1 • Q1/5

522

QUESTION ID

S60-F1-Q1

TYPE

Short answer

CORRECT ANSWER

(6, 15); yes, because 15 = 2.5 · 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 60 • FORWARD 1 • Q1/5

Explain why a dilation centered at the origin with scale factor k (k > 0) maps the line y = mx onto itself.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 60 • FORWARD 1 • Q2/5

523

QUESTION ID

S60-F1-Q2

TYPE

Answer with required work

CORRECT ANSWER

The image of the line y = mx is the line y = mx.

GRADING NOTE REQUIRED WORK: A point (x, mx) maps to (kx, kmx) = (X, mX) with X = kx, which is on y = mx; every point of the line is the image of a point of the line. Any equivalent explanation is accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F1-Q2**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 60 • FORWARD 1 • Q2/5

A dilation centered at (0, 0) with a scale factor of 2 is applied to the line y = 3x + 1.

What is the image?

A. y = 3x + 1

B. y = 3x + 2

C. y = 6x + 2

D. y = 6x + 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 60 • FORWARD 1 • Q3/5

524

QUESTION ID

S60-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. y = 3x + 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 60 • FORWARD 1 • Q3/5

True or false?

A dilation centered at the origin maps every line through the origin onto itself.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 60 • FORWARD 1 • Q4/5

525

QUESTION ID

S60-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 60 • FORWARD 1 • Q4/5

True or false?

A dilation always maps a line onto a line with a different slope.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 60 • FORWARD 1 • Q5/5

526

QUESTION ID

S60-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A dilation maps a line to a parallel line or to the same line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 60 • FORWARD 1 • Q5/5

SET 60 • 8.EE.B.6

FORWARD 2

Use rotated slope triangles to explain the slopes of parallel and perpendicular lines • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

A slope triangle for line p has a run of 3 and a rise of 2. Rotating it 90° counterclockwise gives a slope triangle for a line q perpendicular to p.

What are the run and rise of the rotated triangle, and what is the slope of q?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 2 • Q1/5

528

QUESTION ID

S60-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Run −2 and rise 3 (or run 2 and rise −3); the slope of q is −3/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 2 • Q1/5

Use slope triangles to explain why two parallel, non-vertical lines have equal slopes.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 2 • Q2/5

529

QUESTION ID

S60-F2-Q2

TYPE

Short answer

CORRECT ANSWER

A transversal makes equal corresponding angles with the two lines, so their slope triangles have the same angles and are similar; similar triangles have equal rise ÷ run.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 2 • Q2/5

Line p has a slope of $\frac{4}{5}$.

What is the slope of a line perpendicular to p?

A. $-\frac{5}{4}$

B. $\frac{5}{4}$

C. $-\frac{4}{5}$

D. $\frac{4}{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 2 • Q3/5

530

QUESTION ID

S60-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. −5/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 2 • Q3/5

True or false?

Lines with slopes 2 and $-\frac{1}{2}$ are perpendicular, because $2 \times (-\frac{1}{2}) = -1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 2 • Q4/5

531

QUESTION ID

S60-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 2 • Q4/5

True or false?

Two lines whose slopes multiply to 1 are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 2 • Q5/5

532

QUESTION ID

S60-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Perpendicular slopes multiply to −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 2 • Q5/5

SET 61 • 8.EE.B.6 • EXPRESSIONS & EQUATIONS

Use similar triangles to derive y = mx + b for a line that crosses the y-axis at b

MAIN	Grade 8 main question	**8.EE.B.6**
BACKWARD 1	Find horizontal and vertical distances between points	**6.NS.C.8**
BACKWARD 2	Solve equations of the form px + q = r	**7.EE.B.4.a**
FORWARD 1	Construct linear functions from a point and a slope or from two po…	**HSF-LE.A.2**
FORWARD 2	Describe vertical and horizontal translations of linear functions	**HSF-BF.B.3**

SET 61 • 8.EE.B.6

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.B.6

A line has a slope of 2 and crosses the y-axis at (0, 3).

Use a slope triangle from (0, 3) to a point (x, y) on the line ($x \neq 0$) to explain why $y = 2x + 3$.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 61 • MAIN • Q1/5

535

QUESTION ID

S61-M-Q1

TYPE

Short answer

CORRECT ANSWER

The slope triangle has run x and rise y − 3, so (y − 3) ÷ x = 2. Then y − 3 = 2x and y = 2x + 3. The point (0, 3) also fits.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 61 • MAIN • Q1/5

The line y = mx is translated b units up.

Explain why the image is the line y = mx + b.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 61 • MAIN • Q2/5

536

QUESTION ID

S61-M-Q2

TYPE

Short answer

CORRECT ANSWER

Each point (x, mx) moves to (x, mx + b), so every point of the image satisfies y = mx + b.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 61 • MAIN • Q2/5

A line crosses the y-axis at (0, −4) and has a slope of $\frac{1}{2}$.

For a point (x, y) on the line with x ≠ 0, which equation does the slope triangle from (0, −4) give?

A. $(y - 4) \div x = \frac{1}{2}$

B. $(y + 4) \div x = 2$

C. $(y + 4) \div x = \frac{1}{2}$

D. $y \div x = \frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 61 • MAIN • Q3/5

537

QUESTION ID

S61-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (y + 4) ÷ x = 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 61 • MAIN • Q3/5

A line crosses the y-axis at (0, −2) and passes through (5, 8).

Use a slope triangle to find the slope, then write the equation in the form y = mx + b.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 61 • MAIN • Q4/5

538

QUESTION ID

S61-M-Q4

TYPE

Short answer

CORRECT ANSWER

y = 2x − 2 (the slope triangle has run 5 and rise 10)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 61 • MAIN • Q4/5

True or false?

The line y = 3x + 4 is the image of y = 3x after a translation 4 units to the right.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 61 • MAIN • Q5/5

539

QUESTION ID

S61-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is the image after a translation 4 units up; 4 units right gives y = 3x − 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 61 • MAIN • Q5/5

SET 61 • 8.EE.B.6

BACKWARD 1

Find horizontal and vertical distances between points • Grade 6 • The Number System • 6.NS.C.8

How far is it horizontally from (0, 3) to (4, 3)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 61 • BACKWARD 1 • Q1/5

541

QUESTION ID

S61-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 61 • BACKWARD 1 • Q1/5

How far is it vertically from (4, 3) to (4, 11)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 61 • BACKWARD 1 • Q2/5

542

QUESTION ID

S61-B1-Q2

TYPE

Short answer

CORRECT ANSWER

8 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 61 • BACKWARD 1 • Q2/5

What is the distance between (2, −1) and (2, 5)?

A. 3 units

B. 7 units

C. 4 units

D. 6 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 61 • BACKWARD 1 • Q3/5

543

QUESTION ID

S61-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 6 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 61 • BACKWARD 1 • Q3/5

True or false?

The distance between (−3, 2) and (4, 2) is 7 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 61 • BACKWARD 1 • Q4/5

544

QUESTION ID

S61-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 61 • BACKWARD 1 • Q4/5

True or false?

The distance between (1, −2) and (1, 6) is 4 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 61 • BACKWARD 1 • Q5/5

545

QUESTION ID

S61-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 8 units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 61 • BACKWARD 1 • Q5/5

SET 61 • 8.EE.B.6

BACKWARD 2

Solve equations of the form px + q = r • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Solve for b.

$11 = 3(4) + b$

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 61 • BACKWARD 2 • Q1/5

547

QUESTION ID

S61-B2-Q1

TYPE

Short answer

CORRECT ANSWER

−1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 61 • BACKWARD 2 • Q1/5

Solve for x.

$7 = 2x + 3$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 61 • BACKWARD 2 • Q2/5

548

QUESTION ID

S61-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 61 • BACKWARD 2 • Q2/5

Solve for b.

$5 = -2(3) + b$

A. b = 11

B. b = −1

C. b = 1

D. b = −11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 61 • BACKWARD 2 • Q3/5

549

QUESTION ID

S61-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. b = 11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 61 • BACKWARD 2 • Q3/5

True or false?

If 9 = 4(2) + b, then b = 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 61 • BACKWARD 2 • Q4/5

550

QUESTION ID

S61-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 61 • BACKWARD 2 • Q4/5

True or false?

If 2 = 5(0) + b, then b = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 61 • BACKWARD 2 • Q5/5

551

QUESTION ID

S61-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE b = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 61 • BACKWARD 2 • Q5/5

SET 61 • 8.EE.B.6

FORWARD 1

Construct linear functions from a point and a slope or from two points • High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

Write the equation of the line through (2, 5) with a slope of 3 in point-slope form, then in slope-intercept form.

Equations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q1

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 61 • FORWARD 1 • Q1/5

553

QUESTION ID

S61-F1-Q1

TYPE

Short answer

CORRECT ANSWER

y − 5 = 3(x − 2);
y = 3x − 1

GRADING NOTE Both forms are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q1

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 61 • FORWARD 1 • Q1/5

Explain why every point (x, y) on the line through (2, 5) with a slope of 3 satisfies (y − 5) ÷ (x − 2) = 3 when x ≠ 2.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q2

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 61 • FORWARD 1 • Q2/5

554

QUESTION ID

S61-F1-Q2

TYPE

Short answer

CORRECT ANSWER

The slope triangle from (2, 5) to (x, y) has run x − 2 and rise y − 5; every slope triangle on the line gives the same ratio, 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q2

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 61 • FORWARD 1 • Q2/5

Which equation describes the line through (2, 1) and (5, 10)?

A. $y = 9x - 17$

B. $y = 3x + 1$

C. $y = 3x - 5$

D. $y = \frac{1}{3}x + \frac{1}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q3

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 61 • FORWARD 1 • Q3/5

555

QUESTION ID

S61-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. y = 3x − 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q3

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 61 • FORWARD 1 • Q3/5

True or false?

The line $y - 2 = -4(x + 1)$ passes through $(-1, 2)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q4

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 61 • FORWARD 1 • Q4/5

556

QUESTION ID

S61-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q4

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 61 • FORWARD 1 • Q4/5

True or false?

The line through (0, 0) and (2, 6) is y = 2x + 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S61-F1-Q5

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 61 • FORWARD 1 • Q5/5

557

QUESTION ID

S61-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is y = 3x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S61-F1-Q5

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 61 • FORWARD 1 • Q5/5

SET 61 • 8.EE.B.6

FORWARD 2

Describe vertical and horizontal translations of linear functions • High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3

$f(x) = 2x$ and $g(x) = f(x) + 4$.

Write g(x), and describe how the graph of g compares with the graph of f.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 61 • FORWARD 2 • Q1/5

559

QUESTION ID

S61-F2-Q1

TYPE

Short answer

CORRECT ANSWER

g(x) = 2x + 4; the graph of f is translated 4 units up.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 61 • FORWARD 2 • Q1/5

$f(x) = 2x$ and $h(x) = f(x - 3)$.

Write h(x) in slope-intercept form, and describe the translation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 61 • FORWARD 2 • Q2/5

560

QUESTION ID

S61-F2-Q2

TYPE

Short answer

CORRECT ANSWER

h(x) = 2x − 6; the graph of f is translated 3 units right (which, for this line, is the same as 6 units down).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 61 • FORWARD 2 • Q2/5

f(x) = −x + 1. Which function translates the graph of f 5 units down?

A. g(x) = −x + 1 − 5x

B. g(x) = −(x − 5) + 1

C. g(x) = −x + 6

D. g(x) = −x − 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 61 • FORWARD 2 • Q3/5

561

QUESTION ID

S61-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. g(x) = −x − 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 61 • FORWARD 2 • Q3/5

True or false?

For a line f(x) = mx + b, f(x − 2) = f(x) − 2m for every x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 61 • FORWARD 2 • Q4/5

562

QUESTION ID

S61-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE m(x − 2) + b = mx + b − 2m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 61 • FORWARD 2 • Q4/5

True or false?

g(x) = f(x) − 5 moves the graph of f 5 units to the left.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 61 • FORWARD 2 • Q5/5

563

QUESTION ID

S61-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It moves the graph 5 units down.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 61 • FORWARD 2 • Q5/5

SET 16 • 8.EE.C.7.a • EXPRESSIONS & EQUATIONS

Decide whether a linear equation has one solution, no solution, or infinitely many

MAIN	Grade 8 main question	**8.EE.C.7.a**
BACKWARD 1	Expand linear expressions	**7.EE.A.1**
BACKWARD 2	Solve two-step equations	**7.EE.B.4.a**
FORWARD 1	Find parameter values that give a system one solution, no solution,	**HSA-REI.C.6**
FORWARD 2	Use the form $(x - p)^2 = q$ to find how many real solutions a quadra	**HSA-REI.B.4.a**

SET 16 • 8.EE.C.7.a

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.C.7.a

How many solutions does this equation have?

$4(x + 3) = 4x + 12$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q1/5

566

QUESTION ID

S16-M-Q1

TYPE

Short answer

CORRECT ANSWER

Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q1/5

How many solutions does this equation have?

$5x - 2 = 5x + 7$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q2/5

567

QUESTION ID

S16-M-Q2

TYPE

Short answer

CORRECT ANSWER

No solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q2/5

Which equation has exactly one solution?

A. $3(x + 1) = 3x + 3$

B. $2x - 2x = 0$

C. $3x + 1 = 2x + 9$

D. $3x + 1 = 3x + 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q3/5

568

QUESTION ID

S16-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3x + 1 = 2x + 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q3/5

What number in the blank makes the equation have no solution?

$6x + 5 = ___x + 1$

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q4/5

569

QUESTION ID

S16-M-Q4

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q4/5

True or false?

$2(x - 4) = 2x - 4$ has infinitely many solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q5/5

570

QUESTION ID

S16-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It simplifies to $-8 = -4$, so it has no solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q5/5

SET 16 • 8.EE.C.7.a

BACKWARD 1

Expand linear expressions • Grade 7 • Expressions & Equations • 7.EE.A.1

Expand.

$4(x + 3)$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q1/5

572

QUESTION ID

S16-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4x + 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q1/5

Expand.

$-2(3x - 5)$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q2/5

573

QUESTION ID

S16-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$-6x + 10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q2/5

Which expression is equal to 5(2x − 1)?

A. 10x + 5

B. 10x − 1

C. 10x − 5

D. 7x − 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q3/5

574

QUESTION ID

S16-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 10x − 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q3/5

True or false?

$3(x + 2) = 3x + 6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q4/5

575

QUESTION ID

S16-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q4/5

True or false?

$-(x - 4) = -x - 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q5/5

576

QUESTION ID

S16-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q5/5

SET 16 • 8.EE.C.7.a

BACKWARD 2

Solve two-step equations • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Solve for x.

3x + 7 = 22

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q1/5

578

QUESTION ID

S16-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q1/5

Solve for x.

$-2x + 4 = 10$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q2/5

579

QUESTION ID

S16-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q2/5

Solve for x.

$5x - 6 = 19$

A. $x = \frac{13}{5}$

B. $x = -5$

C. $x = 5$

D. $x = 25$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q3/5

580

QUESTION ID

S16-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q3/5

True or false?

$x = 2$ is the solution of $4x + 1 = 9$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q4/5

581

QUESTION ID

S16-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q4/5

True or false?

$x = 3$ is the solution of $2x - 5 = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q5/5

582

QUESTION ID

S16-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q5/5

SET 16 • 8.EE.C.7.a

FORWARD 1

Find parameter values that give a system one solution, no solution, or infinitely many • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6

For which value of k does the system have no solution?

$y = kx + 2$

$y = 3x - 5$

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q1/5

584

QUESTION ID

S16-F1-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F1-Q1**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q1/5

For which values of a and b does the system have infinitely many solutions?

$2x + 3y = 6$

$ax + 6y = b$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q2/5

585

QUESTION ID

S16-F1-Q2

TYPE

Short answer

CORRECT ANSWER

a = 4 and b = 12

GRADING NOTE Both values are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F1-Q2**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q2/5

For which value of c does the system have infinitely many solutions?

$4x - 2y = c$

$y = 2x - 3$

A. -6 **B.** 6

C. -3 **D.** 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q3/5

586

QUESTION ID

S16-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q3/5

True or false?

The system y = mx + 1 and y = 2x + 1 has exactly one solution for every value of m except 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q4/5

587

QUESTION ID

S16-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q4/5

True or false?

If two lines have different y-intercepts, the system they form has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q5/5

588

QUESTION ID

S16-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Lines with different slopes cross once, whatever their y-intercepts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q5/5

SET 16 • 8.EE.C.7.a

FORWARD 2

Use the form $(x - p)^2 = q$ to find how many real solutions a quadratic equation has • High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a

How many real solutions does $(x - 3)^2 = 0$ have? Give any solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q1/5

590

QUESTION ID

S16-F2-Q1

TYPE

Short answer

CORRECT ANSWER

One: x = 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q1/5

How many real solutions does $(x + 1)^2 = -4$ have? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q2/5

591

QUESTION ID

S16-F2-Q2

TYPE

Short answer

CORRECT ANSWER

None. A square of a real number cannot be negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q2/5

Completing the square on $x^2 + 6x + 13 = 0$ gives $(x + 3)^2 = q$.

What is q, and how many real solutions are there?

A. q = 4; two real solutions

B. q = −4; one real solution

C. q = 13; two real solutions

D. q = −4; no real solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q3/5

592

QUESTION ID

S16-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. q = −4; no real solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q3/5

True or false?

$(x - 2)^2 = 9$ has two real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q4/5

593

QUESTION ID

S16-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $x = 5$ and $x = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q4/5

True or false?

$x^2 - 6x + 9 = 0$ has two different real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q5/5

594

QUESTION ID

S16-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is $(x - 3)^2 = 0$, so $x = 3$ is the only solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q5/5

SET 62 • 8.EE.C.7.a • EXPRESSIONS & EQUATIONS

Give examples of linear equations with one, no, or infinitely many solutions and show which by transforming them

MAIN	Grade 8 main question	**8.EE.C.7.a**
BACKWARD 1	Simplify expressions with the distributive property	**7.EE.A.1**
BACKWARD 2	Decide whether a value makes an equation true	**6.EE.B.5**
FORWARD 1	Find the parameter values that give a linear equation one, no, or i…	**HSA-REI.B.3**
FORWARD 2	Justify each step and state the solution set of an equation	**HSA-REI.A.1**

SET 62 • 8.EE.C.7.a

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.C.7.a

ANSWER KEY Reduced copy of the question page, with its answer

Write a linear equation in x, with x on both sides, that has no solution. Transform it to show why.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 62 • MAIN • Q1/5

597

QUESTION ID

S62-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

Any correct example, such as 2x + 3 = 2x + 5 (3 = 5 is false)

GRADING NOTE REQUIRED WORK: Any equation that transforms to a false statement a = b with a ≠ b. Example: 2x + 3 = 2x + 5 becomes 3 = 5 after subtracting 2x. The steps must be shown.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 62 • MAIN • Q1/5

Write a linear equation in x, with x on both sides, that has infinitely many solutions. Transform it to show why.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 62 • MAIN • Q2/5

598

QUESTION ID

S62-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

Any correct example, such as 3(x + 1) = 3x + 3 (3 = 3 is always true)

GRADING NOTE REQUIRED WORK: Any equation that transforms to a true statement a = a. Example: 3(x + 1) = 3x + 3 becomes 3x + 3 = 3x + 3, then 3 = 3. The steps must be shown.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 62 • MAIN • Q2/5

Write a linear equation in x, with x on both sides, whose only solution is x = 4. Transform it to x = 4.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 62 • MAIN • Q3/5

599

QUESTION ID

S62-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

Any correct example, such as 5x − 2 = 3x + 6

GRADING NOTE REQUIRED WORK: Any equation with different coefficients of x that transforms to x = 4. Example: 5x − 2 = 3x + 6; 2x = 8; x = 4. The steps must be shown.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 62 • MAIN • Q3/5

Fill in the blanks so that the equation has infinitely many solutions.

$4(x - 2) = ___x - ___$

Blanks: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 62 • MAIN • Q4/5

600

QUESTION ID

S62-M-Q4

TYPE

Short answer

CORRECT ANSWER

4 and 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 62 • MAIN • Q4/5

Ana transformed an equation and got 7 = 7.

Which equation could she have started with?

A. 2x + 7 = 3x + 7

B. x + 7 = 0

C. 2x + 3 = 2x + 7

D. 2(x + 3) + 1 = 2x + 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 62 • MAIN • Q5/5

601

QUESTION ID

S62-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

D.

2(x + 3) + 1 = 2x + 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 62 • MAIN • Q5/5

SET 62 • 8.EE.C.7.a

BACKWARD 1

Simplify expressions with the distributive property • Grade 7 • Expressions & Equations • 7.EE.A.1

Simplify.

$2(x + 3) + 1$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 62 • BACKWARD 1 • Q1/5

603

QUESTION ID

S62-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2x + 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 62 • BACKWARD 1 • Q1/5

Simplify.

$5x - 2 - 3x$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 62 • BACKWARD 1 • Q2/5

604

QUESTION ID

S62-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2x − 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 62 • BACKWARD 1 • Q2/5

Which expression is equal to 4(x − 2)?

A. 4x − 2

B. 4x − 8

C. x − 8

D. 4x + 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 62 • BACKWARD 1 • Q3/5

605

QUESTION ID

S62-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4x − 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 62 • BACKWARD 1 • Q3/5

True or false?

$3x + 2x - 1 = 5x - 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 62 • BACKWARD 1 • Q4/5

606

QUESTION ID

S62-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 62 • BACKWARD 1 • Q4/5

True or false?

$-(2x - 3) = -2x - 3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 62 • BACKWARD 1 • Q5/5

607

QUESTION ID

S62-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $-(2x - 3) = -2x + 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 62 • BACKWARD 1 • Q5/5

SET 62 • 8.EE.C.7.a

BACKWARD 2

Decide whether a value makes an equation true • Grade 6 • Expressions & Equations • 6.EE.B.5

Is x = 3 a solution of 2x + 1 = x + 4? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 62 • BACKWARD 2 • Q1/5

609

QUESTION ID

S62-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. Both sides equal 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 62 • BACKWARD 2 • Q1/5

Which of the values 0, 1, and 2 makes $3x - 1 = x + 3$ true?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 62 • BACKWARD 2 • Q2/5

610

QUESTION ID

S62-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 62 • BACKWARD 2 • Q2/5

Which value makes 4x = 2x + 6 true?

A. 3

B. 4

C. 6

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 62 • BACKWARD 2 • Q3/5

611

QUESTION ID

S62-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 62 • BACKWARD 2 • Q3/5

True or false?

Every value of x makes x + x = 2x true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 62 • BACKWARD 2 • Q4/5

612

QUESTION ID

S62-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 62 • BACKWARD 2 • Q4/5

True or false?

$x = 5$ makes $x + 2 = 2x - 2$ true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 62 • BACKWARD 2 • Q5/5

613

QUESTION ID

S62-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $7 \neq 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 62 • BACKWARD 2 • Q5/5

SET 62 • 8.EE.C.7.a

FORWARD 1

Find the parameter values that give a linear equation one, no, or infinitely many solutions • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

For which value of a does the equation ax + 3 = 5x + 1 have no solution?

a = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 62 • FORWARD 1 • Q1/5

615

QUESTION ID

S62-F1-Q1

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 62 • FORWARD 1 • Q1/5

For which values of a and b does ax + b = 3x − 2 have infinitely many solutions?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 62 • FORWARD 1 • Q2/5

616

QUESTION ID

S62-F1-Q2

TYPE

Short answer

CORRECT ANSWER

a = 3 and b = −2

GRADING NOTE Both values are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 62 • FORWARD 1 • Q2/5

For which values of c does $cx - 4 = 2x + 4$ have exactly one solution?

A. Every value of c

B. No value of c

C. Every value except c = 2

D. Only c = 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 62 • FORWARD 1 • Q3/5

617

QUESTION ID

S62-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Every value except c = 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 62 • FORWARD 1 • Q3/5

True or false?

When $a \neq 4$, the equation $ax + 1 = 4x + 9$ has the one solution $x = 8 \div (a - 4)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 62 • FORWARD 1 • Q4/5

618

QUESTION ID

S62-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 62 • FORWARD 1 • Q4/5

True or false?

When k = 0, the equation kx = kx + 1 has infinitely many solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 62 • FORWARD 1 • Q5/5

619

QUESTION ID

S62-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It becomes 0 = 1, which has no solution (for every k).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 62 • FORWARD 1 • Q5/5

SET 62 • 8.EE.C.7.a

FORWARD 2

Justify each step and state the solution set of an equation • High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

Transform 2(x + 4) − 3 = 2x + 5, giving a reason for each step, and state the solution set.

Solution set: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 62 • FORWARD 2 • Q1/5

621

QUESTION ID

S62-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

All real numbers

GRADING NOTE REQUIRED WORK: Every step with a valid reason. Other valid orders of steps are accepted. A correct answer without reasons is not full credit. Example: distributive property (2x + 8 − 3 = 2x + 5); combine like terms (2x + 5 = 2x + 5); subtract 2x from both sides (5 = 5), which is always true.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 62 • FORWARD 2 • Q1/5

Transform 6x − 1 = 2(3x + 2), giving a reason for each step, and explain why the equation has no solution.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 62 • FORWARD 2 • Q2/5

622

QUESTION ID

S62-F2-Q2

TYPE

Answer with required work

CORRECT ANSWER

No solution

GRADING NOTE REQUIRED WORK: Every step with a valid reason. Other valid orders of steps are accepted. A correct answer without reasons is not full credit. Example: distributive property (6x − 1 = 6x + 4); subtract 6x from both sides (−1 = 4), which is never true.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 62 • FORWARD 2 • Q2/5

An equation transforms to 0 = 0.

What is its solution set?

A. All real numbers

B. Only x = 1

C. No solution

D. Only x = 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 62 • FORWARD 2 • Q3/5

623

QUESTION ID

S62-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. All real numbers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 62 • FORWARD 2 • Q3/5

True or false?

Adding the same expression, such as −3x, to both sides of an equation keeps the same solution set.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 62 • FORWARD 2 • Q4/5

624

QUESTION ID

S62-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 62 • FORWARD 2 • Q4/5

True or false?

Dividing both sides of $0 \cdot x = 0$ by 0 shows that its only solution is $x = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 62 • FORWARD 2 • Q5/5

625

QUESTION ID

S62-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Division by 0 is undefined; every value of x is a solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 62 • FORWARD 2 • Q5/5

SET 17 • 8.EE.C.7.b • EXPRESSIONS & EQUATIONS

Solve multistep linear equations by expanding and collecting like terms

MAIN	Grade 8 main question	**8.EE.C.7.b**
BACKWARD 1	Combine like terms	**7.EE.A.1**
BACKWARD 2	Solve two-step equations with rational numbers	**7.EE.B.4.a**
FORWARD 1	Solve linear equations with letter coefficients	**HSA-REI.B.3**
FORWARD 2	Justify each step of a solution and find an invalid step	**HSA-REI.A.1**

SET 17 • 8.EE.C.7.b

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$3(2x - 4) = 4x + 6$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q1/5

628

QUESTION ID

S17-M-Q1

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q1/5

Solve for x.

$5 - 2(x + 1) = 3x - 7$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q2/5

629

QUESTION ID

S17-M-Q2

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q2/5

Solve for x.

$4(x - 1) - (x + 2) = 2x + 3$

A. $x = 1$

B. $x = 5$

C. $x = 9$

D. $x = -9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q3/5

630

QUESTION ID

S17-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve. Show each step.

$-3(x + 5) + 2x = 4(x - 10)$

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q4/5

631

QUESTION ID

S17-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

5

GRADING NOTE REQUIRED WORK: Any valid sequence of steps. Example: $-3x - 15 + 2x = 4x - 40$; $-x - 15 = 4x - 40$; $25 = 5x$; $x = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q4/5

True or false?

$x = -1$ is the solution of $2(3x + 1) = 4x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q5/5

632

QUESTION ID

S17-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q5/5

SET 17 • 8.EE.C.7.b

BACKWARD 1

Combine like terms • Grade 7 • Expressions & Equations • 7.EE.A.1

Simplify.

$6x - 4 + 2x + 9$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q1/5

634

QUESTION ID

S17-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8x + 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q1/5

Simplify.

$-3a + 7 - a$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q2/5

635

QUESTION ID

S17-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$-4a + 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q2/5

Simplify.

$2(x + 3) + 5x$

A. $10x + 6$

B. $2x + 11$

C. $7x + 3$

D. $7x + 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q3/5

636

QUESTION ID

S17-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 7x + 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q3/5

True or false?

$4y - y = 3y$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q4/5

637

QUESTION ID

S17-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q4/5

True or false?

$5n + 2 - 2n = 3n$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q5/5

638

QUESTION ID

S17-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $5n + 2 - 2n = 3n + 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q5/5

SET 17 • 8.EE.C.7.b

BACKWARD 2

Solve two-step equations with rational numbers • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Solve for x.

$6x - 5 = 31$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q1/5

640

QUESTION ID

S17-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q1/5

Solve for x.

$\frac{x}{4} + 3 = 7$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q2/5

641

QUESTION ID

S17-B2-Q2

TYPE

Short answer

CORRECT ANSWER

16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q2/5

Solve for x.

$-3x + 8 = -10$

A. $x = -6$

B. $x = -\frac{2}{3}$

C. $x = \frac{2}{3}$

D. $x = 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q3/5

642

QUESTION ID

S17-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x = 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q3/5

True or false?

$x = -2$ is the solution of $5x + 12 = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q4/5

643

QUESTION ID

S17-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q4/5

True or false?

$x = 4$ is the solution of $3x - 2 = 14$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q5/5

644

QUESTION ID

S17-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q5/5

SET 17 • 8.EE.C.7.b

FORWARD 1

Solve linear equations with letter coefficients • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve for x. Assume a ≠ 3.

ax + 2 = 3x + 8

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q1/5

646

QUESTION ID

S17-F1-Q1

TYPE

Short answer

CORRECT ANSWER

6/(a − 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q1/5

Solve for x. Assume k ≠ 0.

$k(x - 4) = 2k$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q2/5

647

QUESTION ID

S17-F1-Q2

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q2/5

Solve for x. Assume b ≠ 0.

$bx - 5b = 2b$

A. $x = 7b$

B. $x = \frac{7}{b}$

C. $x = 7$

D. $x = 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q3/5

QUESTION ID

S17-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q3/5

True or false?

If $c \neq 1$, the solution of $cx + 1 = x + 1$ is $x = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q4/5

649

QUESTION ID

S17-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q4/5

True or false?

The solution of $mx = m + 4$ is $x = 5$ for every nonzero value of m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q5/5

650

QUESTION ID

S17-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x = (m + 4)/m$, which is 5 only when $m = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q5/5

SET 17 • 8.EE.C.7.b

FORWARD 2

Justify each step of a solution and find an invalid step • High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Solve 2(x − 3) + 4 = 5x − 11. Give a reason for each step.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q1/5

652

QUESTION ID

S17-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

3

GRADING NOTE REQUIRED WORK: Every step with a valid reason. Other valid orders of steps are accepted. A correct answer without reasons is not full credit. Example: distributive property (2x − 6 + 4 = 5x − 11); combine like terms (2x − 2 = 5x − 11); subtract 2x and add 11 to both sides (9 = 3x); divide both sides by 3 (x = 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q1/5

Dee solved $4 - 2(x - 1) = 8$ like this:

Step 1: $4 - 2x - 2 = 8$

Step 2: $2 - 2x = 8$

Step 3: $x = -3$

Which step is wrong? Correct it and give the solution.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q2/5

653

QUESTION ID

S17-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Step 1: $-2(x - 1) = -2x + 2$, so $6 - 2x = 8$ and $x = -1$.

GRADING NOTE Grade both: Step 1 with the correction, and $x = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q2/5

Which reason justifies rewriting $3x - 5 = 7$ as $3x = 12$?

A. Addition property of equality

B. Multiplication property of equality

C. Commutative property of addition

D. Distributive property

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q3/5

654

QUESTION ID

S17-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Addition property of equality

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-F2-Q3**

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q3/5

True or false?

Each step of a valid solution gives an equation with the same solutions as the original equation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q4/5

655

QUESTION ID

S17-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q4/5

True or false?

Multiplying both sides of an equation by 0 keeps the same solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q5/5

656

QUESTION ID

S17-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q5/5

SET 18 • 8.EE.C.7.b • EXPRESSIONS & EQUATIONS

Write and solve linear equations with rational coefficients for real-world problems

MAIN	Grade 8 main question	**8.EE.C.7.b**
BACKWARD 1	Solve word problems that lead to px + q = r	**7.EE.B.4.a**
BACKWARD 2	Multiply rational numbers in context	**7.NS.A.3**
FORWARD 1	Solve real-world systems, including mixtures and rates, and judge…	**HSA-REI.C.6**
FORWARD 2	Create equations and inequalities in one variable to solve proble…	**HSA-CED.A.1**

SET 18 • 8.EE.C.7.b

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Gym A charges $30 to join plus $12.50 per month. Gym B charges $55 to join plus $10 per month.

Write and solve an equation to find when the total costs are equal.

Equation and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q1/5

659

QUESTION ID

S18-M-Q1

TYPE

Short answer

CORRECT ANSWER

The equation 30 + 12.5m = 55 + 10m and the answer m = 10 months

GRADING NOTE Both parts are required: a valid equation (or system) and the correct solution. Any valid model is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Plant A is $6\frac{1}{2}$ inches tall and grows $\frac{3}{4}$ inch per week.

Plant B is 10 inches tall and grows $\frac{1}{4}$ inch per week.

Write and solve an equation to find when they are the same height.

Equation and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q2/5

660

QUESTION ID

S18-M-Q2

TYPE

Short answer

CORRECT ANSWER

The equation 6.5 + 0.75w = 10 + 0.25w and the answer w = 7 weeks

GRADING NOTE Both parts are required: a valid equation (or system) and the correct solution. Any valid model is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q2/5

Solve for x.

$\frac{2}{3}x - 4 = \frac{1}{2}x + 1$

A. x = 18

B. $x = \frac{30}{7}$

C. x = 30

D. x = 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q3/5

661

QUESTION ID

S18-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q3/5

Solve for x.

$0.4(x + 15) = 0.25x + 9$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q4/5

662

QUESTION ID

S18-M-Q4

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q4/5

True or false?

$x = 12$ is the solution of $\frac{3}{4}x + 2 = \frac{1}{2}x + 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q5/5

663

QUESTION ID

S18-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q5/5

SET 18 • 8.EE.C.7.b

BACKWARD 1

Solve word problems that lead to px + q = r • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Ana has $40. She buys some $6.50 tickets and has $1 left.

How many tickets did she buy?

Tickets: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q1/5

665

QUESTION ID

S18-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q1/5

A rental costs $20 plus $8 per hour. The total is $68.

For how many hours was it rented?

Hours: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q2/5

666

QUESTION ID

S18-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q2/5

Solve for x.

0.5x + 3 = 11

A. x = 7

B. x = 16

C. x = 28

D. x = 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q3/5

667

QUESTION ID

S18-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. x = 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q3/5

True or false?

$x = 9$ is the solution of $\frac{1}{3}x + 2 = 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q4/5

668

QUESTION ID

S18-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q4/5

True or false?

$x = 4$ is the solution of $2.5x - 1 = 11$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q5/5

669

QUESTION ID

S18-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = 4.8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q5/5

SET 18 • 8.EE.C.7.b

BACKWARD 2

Multiply rational numbers in context • Grade 7 • The Number System • 7.NS.A.3

A plant grows $\frac{3}{4}$ inch per week.

How much does it grow in 8 weeks?

Growth: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q1/5

671

QUESTION ID

S18-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q1/5

A gym charges $12.50 per month.

How much do 10 months cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q2/5

672

QUESTION ID

S18-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$125

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q2/5

A cyclist rides 0.4 mile per minute for 15 minutes.

How far does she ride?

A. 6 miles

B. 60 miles

C. 4 miles

D. 15.4 miles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q3/5

673

QUESTION ID

S18-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 6 miles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q3/5

True or false?

$\frac{2}{3}$ of 9 feet is 6 feet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q4/5

674

QUESTION ID

S18-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q4/5

True or false?

0.25 of a 12-hour day is 4 hours.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q5/5

675

QUESTION ID

S18-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 0.25 × 12 = 3 hours.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q5/5

SET 18 • 8.EE.C.7.b

FORWARD 1

Solve real-world systems, including mixtures and rates, and judge whether a solution makes sense • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6

A chemist mixes a 10% salt solution with a 40% salt solution to make 60 liters of a 20% salt solution.

Write and solve a system to find how many liters of each she uses.

System and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q1/5

677

QUESTION ID

S18-F1-Q1

TYPE

Short answer

CORRECT ANSWER

The system x + y = 60 and 0.10x + 0.40y = 12, and the answer 40 liters of the 10% solution and 20 liters of the 40% solution

GRADING NOTE Both parts are required: a valid system and the correct solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q1/5

A ticket system gives a = 37.5 adult tickets and s = 82.5 student tickets.

Does this solution make sense? Explain what it tells you.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q2/5

678

QUESTION ID

S18-F1-Q2

TYPE

Short answer

CORRECT ANSWER

No. Ticket counts must be whole numbers, so the given totals (or the model) cannot be right.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q2/5

Plan A costs $y = 12x + 40$ dollars and Plan B costs $y = 20x$ dollars for x months.

For which numbers of months is Plan A cheaper?

A. Exactly 5 months

B. Fewer than 5 months

C. More than 5 months

D. Never

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q3/5

679

QUESTION ID

S18-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. More than 5 months

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q3/5

True or false?

A plane flies 600 miles in 2 hours with the wind and 600 miles in 3 hours against the wind. Its speed in still air is 250 miles per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q4/5

680

QUESTION ID

S18-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q4/5

True or false?

A system for numbers of bikes b and cars c with the solution b = −4, c = 20 gives a sensible answer.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q5/5

681

QUESTION ID

S18-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A count of bikes cannot be negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q5/5

SET 18 • 8.EE.C.7.b

FORWARD 2

Create equations and inequalities in one variable to solve problems • High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1

Plan A costs $25 plus $0.15 per text. Plan B costs $40 with unlimited texts.

Write and solve an inequality to find when Plan A costs less.

Inequality and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q1

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q1/5

683

QUESTION ID

S18-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The inequality 25 + 0.15t < 40 and the answer t < 100 texts (0 to 99 texts)

GRADING NOTE Both parts are required: a valid inequality and the correct solution. Any valid model is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q1

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q1/5

Two trains leave a station at the same time in opposite directions, at 55 and 65 miles per hour.

Write and solve an equation to find when they are 300 miles apart.

Equation and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q2

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q2/5

684

QUESTION ID

S18-F2-Q2

TYPE

Short answer

CORRECT ANSWER

The equation 55t + 65t = 300 and the answer t = 2.5 hours

GRADING NOTE Both parts are required: a valid equation (or system) and the correct solution. Any valid model is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q2

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q2/5

A rectangle's length is 3 more than twice its width w. Its perimeter is 54.

Which equation can find w?

A. 2(2w) + 3 = 54

B. w + 2w + 3 = 54

C. 2w + 2(2w + 3) = 54

D. w(2w + 3) = 54

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q3

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q3/5

685

QUESTION ID

S18-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C.

2w + 2(2w + 3) = 54

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q3

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q3/5

True or false?

A rectangle has a length of 2w + 3 and a perimeter of 54. Its width w is 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q4

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q4/5

686

QUESTION ID

S18-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q4

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q4/5

True or false?

The inequality $25 + 0.15t < 40$ is true when $t = 120$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q5

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q5/5

687

QUESTION ID
S18-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $25 + 18 = 43$, which is not less than 40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q5

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q5/5

SET 19 • 8.EE.C.8.a • EXPRESSIONS & EQUATIONS

Solve a system of two linear equations by graphing

MAIN	Grade 8 main question	**8.EE.C.8.a**
BACKWARD 1	Make a table and graph an equation	**6.EE.C.9**
BACKWARD 2	Graph points in all four quadrants	**6.NS.C.8**
FORWARD 1	Solve f(x) = g(x) from the intersection of two graphs	**HSA-REI.D.11**
FORWARD 2	Graph the solution set of a system of linear inequalities	**HSA-REI.D.12**

SET 19 • 8.EE.C.8.a

MAIN

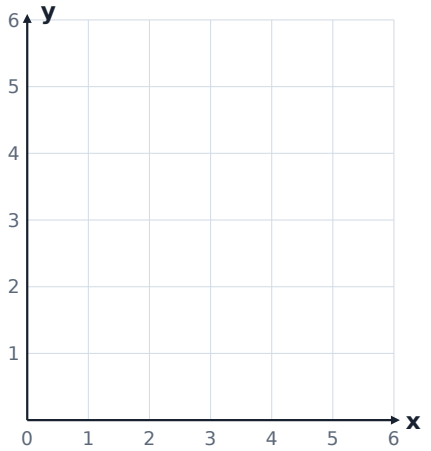
Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.C.8.a

ANSWER KEY Reduced copy of the question page, with its answer

Graph y = x + 1 and
y = −x + 5 on the grid.

What is the solution of the system?

Solution: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q1/5

690

QUESTION ID
S19-M-Q1

TYPE
Drawing + written answer

CORRECT ANSWER
Drawing: The line y = x + 1 through (0, 1) and (4, 5), and the line y = −x + 5 through (0, 5) and (5, 0), crossing at (2, 3)

Written: (2, 3)

GRADING NOTE Grade both: the two lines drawn correctly, and the solution (2, 3).

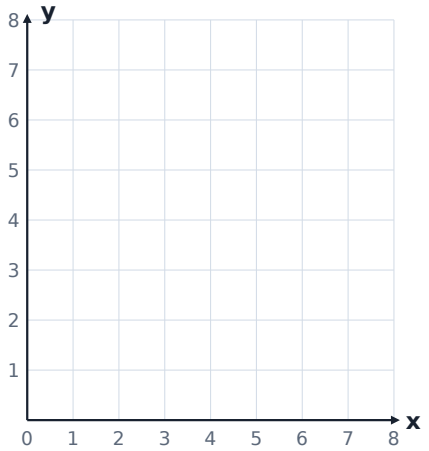
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q1/5

Graph y = 0.5x + 1 and y = −x + 7 on the grid.

What is the solution of the system?

Solution: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q2/5

691

QUESTION ID

S19-M-Q2

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: The line y = 0.5x + 1 through (0, 1) and (6, 4), and the line y = −x + 7 through (0, 7) and (7, 0), crossing at (4, 3)

Written: (4, 3)

GRADING NOTE Grade both: the two lines drawn correctly, and the solution (4, 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q2

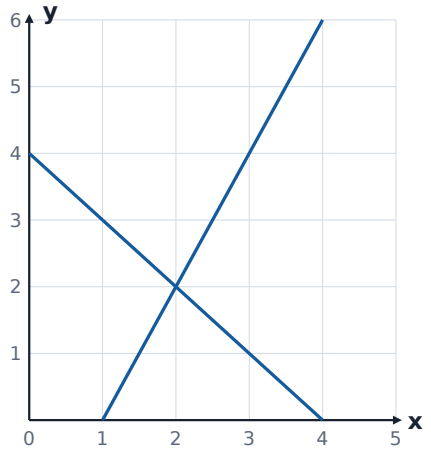
Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph shows the lines $y = -x + 4$ and $y = 2x - 2$.

What is the solution of the system? Check it in both equations.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q3/5

692

QUESTION ID

S19-M-Q3

TYPE

Short answer

CORRECT ANSWER

(2, 2): 2 = −2 + 4 and 2 = 2(2) − 2.

GRADING NOTE Grade both: (2, 2) and the check in both equations.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q3/5

The graphs of the two equations in a system are parallel lines.

What does this tell you about the system?

A. It has no solution.

B. It has infinitely many solutions.

C. It has one solution.

D. Its solution is the origin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q4/5

693

QUESTION ID

S19-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. It has no solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q4/5

True or false?

When the graphs of two linear equations are two different lines that cross, the crossing point is the only solution of the system.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q5/5

694

QUESTION ID

S19-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q5/5

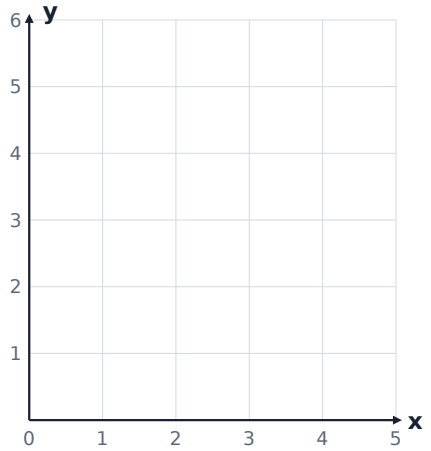
SET 19 • 8.EE.C.8.a

BACKWARD 1

Make a table and graph an equation • Grade 6 • Expressions & Equations • 6.EE.C.9

Make a table for y = x + 2 with x = 0, 1, 2, and 3. Write the table, then graph the points on the grid.

Table: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q1/5

696

QUESTION ID

S19-B1-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: Points at (0, 2), (1, 3), (2, 4), and (3, 5)

Written: x: 0, 1, 2, 3; y: 2, 3, 4, 5

GRADING NOTE Grade both: the table of all four pairs, and the four points graphed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q1/5

What is the value of y when x = 4 in y = 3x?

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q2/5

697

QUESTION ID

S19-B1-Q2

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q2/5

Which point is on the graph of y = x + 3?

A. (1, 3)

B. (3, 0)

C. (5, 2)

D. (2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q3/5

698

QUESTION ID

S19-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q3/5

True or false?

(4, 8) is on the graph of y = 2x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q4/5

699

QUESTION ID

S19-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q4/5

True or false?

(0, 1) is on the graph of y = x + 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q5/5

700

QUESTION ID

S19-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q5

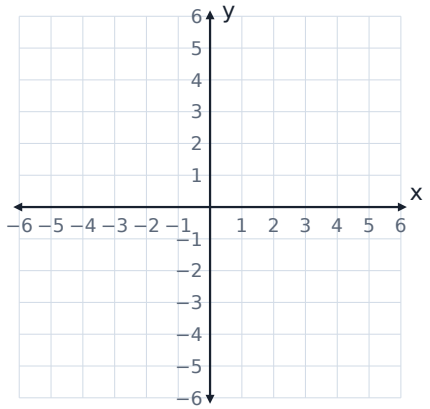
Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q5/5

SET 19 • 8.EE.C.8.a

BACKWARD 2

Graph points in all four quadrants • Grade 6 • The Number System • 6.NS.C.8

Plot A(2, −3) and B(−4, 1) on the coordinate plane.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q1/5

QUESTION ID

S19-B2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Point A at (2, −3) and point B at (−4, 1)

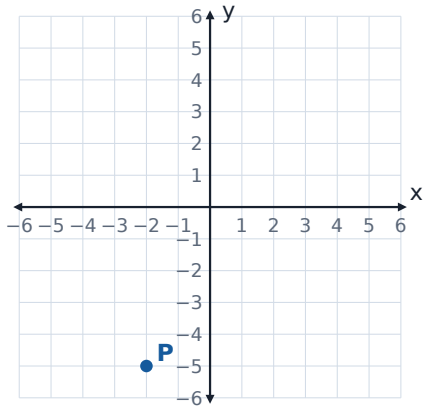
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q1/5

What are the coordinates of point P?

Coordinates: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q2/5

703

QUESTION ID

S19-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(−2, −5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q2/5

Which point is in Quadrant II?

A. (3, −4)

B. (−3, −4)

C. (3, 4)

D. (−3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q3/5

704

QUESTION ID

S19-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (−3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q3/5

True or false?

(0, −2) is on the y-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q4/5

705

QUESTION ID

S19-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q4/5

True or false?

(5, −1) is in Quadrant III.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q5/5

706

QUESTION ID

S19-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE (5, −1) is in Quadrant IV.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q5/5

SET 19 • 8.EE.C.8.a

FORWARD 1

Solve f(x) = g(x) from the intersection of two graphs • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

The graphs of f(x) = x + 1 and g(x) = 7 − x intersect at (3, 4).

What is the solution of x + 1 = 7 − x?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q1/5

708

QUESTION ID

S19-F1-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q1/5

Use a table of values for x = 0, 1, 2, and 3 to solve 2x + 1 = 7 − x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q2/5

709

QUESTION ID

S19-F1-Q2

TYPE

Answer with required work

CORRECT ANSWER

2

GRADING NOTE REQUIRED WORK: A table with both sides for each x: 2x + 1 is 1, 3, 5, 7 and 7 − x is 7, 6, 5, 4. The sides are equal at x = 2 (both 5). The table must be shown.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F1-Q2**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q2/5

The graphs of $f(x) = x^2$ and $g(x) = x + 2$ intersect at (2, 4) and (−1, 1).

What are the solutions of $x^2 = x + 2$?

A. x = 2 only

B. x = −1 only

C. x = 4 and x = 1

D. x = 2 and x = −1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q3/5

QUESTION ID

S19-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x = 2 and x = −1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q3/5

True or false?

If the graphs of y = f(x) and y = g(x) intersect at (5, 9), then f(5) = g(5) = 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q4/5

QUESTION ID

S19-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q4/5

True or false?

If the graphs of f and g never meet, then f(x) = g(x) has exactly one solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q5/5

712

QUESTION ID

S19-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has no solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q5/5

SET 19 • 8.EE.C.8.a

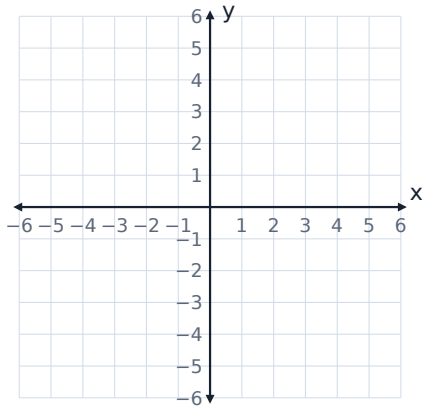
FORWARD 2

Graph the solution set of a system of linear inequalities • High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12

Graph the system on the coordinate plane.

$y \geq x$

$y < 4$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q1/5

714

QUESTION ID

S19-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

A solid line y = x and a dashed line y = 4, with the region above or on y = x and below y = 4 shaded

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q1/5

Is (1, 3) a solution of the system $y > 2x$ and $y < x + 3$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q2/5

715

QUESTION ID

S19-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. 3 > 2(1) and 3 < 1 + 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q2/5

Which point is a solution of the system $y \leq -x + 4$ and $y > 1$?

A. (1, 2)

B. (4, 3)

C. (3, 2)

D. (0, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q3/5

716

QUESTION ID

S19-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (1, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q3/5

True or false?

The boundary line of $y < 2x + 1$ is drawn dashed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q4/5

717

QUESTION ID

S19-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q4/5

True or false?

The solution set of a system of two inequalities is every point that satisfies at least one of them.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q5/5

718

QUESTION ID
S19-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE A solution must satisfy both inequalities (the overlap of the two regions).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q5/5

SET 20 • 8.EE.C.8.b • EXPRESSIONS & EQUATIONS

Solve a system of two linear equations algebraically

MAIN	Grade 8 main question	**8.EE.C.8.b**
BACKWARD 1	Evaluate expressions for given values	**6.EE.A.2.c**
BACKWARD 2	Solve equations with the distributive property	**7.EE.B.4.a**
FORWARD 1	Prove that the elimination step keeps the same solutions	**HSA-REI.C.5**
FORWARD 2	Solve a system of a linear equation and a quadratic equation by gr.	**HSA-REI.C.7**

SET 20 • 8.EE.C.8.b

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.C.8.b

ANSWER KEY Reduced copy of the question page, with its answer

Solve the system. Show your work.

$y = 3x - 2$

$2x + y = 13$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q1/5

721

QUESTION ID

S20-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

x = 3, y = 7

GRADING NOTE REQUIRED WORK: Any valid method (substitution, elimination, or another), with the steps shown. Example: substitute $y = 3x - 2$ into $2x + y = 13$ to get $5x - 2 = 13$, so $x = 3$ and $y = 7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve the system. Show your work.

$x + 2y = 10$

$3x - 2y = 6$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q2/5

722

QUESTION ID

S20-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

x = 4, y = 3

GRADING NOTE REQUIRED WORK: Any valid method, with the steps shown. Example: add the equations to get $4x = 16$, so $x = 4$; then $4 + 2y = 10$, so $y = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q2/5

Solve the system.

$y = x + 4$

$y = -2x + 1$

A. $x = 1, y = 5$

B. $x = 3, y = 7$

C. $x = -1, y = 5$

D. $x = -1, y = 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q3/5

723

QUESTION ID

S20-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x = −1, y = 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q3/5

Solve the system.

$4x + 3y = 18$

$2x - y = 4$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q4/5

724

QUESTION ID

S20-M-Q4

TYPE

Short answer

CORRECT ANSWER

x = 3, y = 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q4/5

True or false?

$x = 2, y = -1$ is a solution of both $3x + y = 5$ and $x - y = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q5/5

725

QUESTION ID

S20-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q5/5

SET 20 • 8.EE.C.8.b

BACKWARD 1

Evaluate expressions for given values • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Evaluate 2x + y when x = 3 and y = 7.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q1/5

727

QUESTION ID

S20-B1-Q1

TYPE

Short answer

CORRECT ANSWER

13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q1/5

Evaluate 3x − 2 when x = 4.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q2/5

728

QUESTION ID

S20-B1-Q2

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q2/5

Evaluate x + 2y when x = 4 and y = 3.

A. 9

B. 10

C. 11

D. 14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q3/5

729

QUESTION ID

S20-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q3/5

True or false?

When x = 2, the value of 5x − 1 is 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q4/5

730

QUESTION ID

S20-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q4/5

True or false?

When y = 3, the value of 2y + 4 is 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q5/5

731

QUESTION ID

S20-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q5/5

SET 20 • 8.EE.C.8.b

BACKWARD 2

Solve equations with the distributive property • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Solve for x.

$5x - 2 = 13$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q1/5

733

QUESTION ID

S20-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q1/5

Solve for x.

$2x + 3(x - 2) = 9$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q2/5

734

QUESTION ID

S20-B2-Q2

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q2/5

Solve for x.

$4x + 2 = -10$

A. $x = 3$

B. $x = -2$

C. $x = -3$

D. $x = 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q3/5

735

QUESTION ID

S20-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = −3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q3/5

True or false?

$x = 4$ is the solution of $3x - 5 = 7$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q4/5

736

QUESTION ID

S20-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q4/5

True or false?

$x = -1$ is the solution of $2x + 9 = 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q5/5

737

QUESTION ID

S20-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = -2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q5/5

SET 20 • 8.EE.C.8.b

FORWARD 1

Prove that the elimination step keeps the same solutions • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5

System A: $x + y = 9$ and $x - y = 3$.

System B: $x + y = 9$ and $2x = 12$ (its second equation is the sum of the two equations of A).

Show that every solution of A is a solution of B, and every solution of B is a solution of A.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q1/5

739

QUESTION ID

S20-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

The systems have the same solutions; both have only (6, 3).

GRADING NOTE REQUIRED WORK: Both directions are required. A to B: if (x, y) makes both equations of A true, adding them gives $2x = 12$, so (x, y) solves B. B to A: if (x, y) solves B, subtracting $x + y = 9$ from $2x = 12$ gives $x - y = 3$, so (x, y) solves A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q1/5

System A: $x + 2y = 7$ and $-3x + y = 0$. Replace the second equation by its sum with 3 times the first equation.

Write the new system, and show how to get the original second equation back from it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q2/5

740

QUESTION ID
S20-F1-Q2

TYPE
Short answer

CORRECT ANSWER
New system:
$x + 2y = 7$ and
$7y = 21$. Subtracting
$3(x + 2y) = 21$ from
$7y = 21$ gives
$-3x + y = 0$ again.

GRADING NOTE Grade both: the new system and the reverse step.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q2/5

Why does replacing one equation by its sum with a multiple of the other keep the same solutions?

A. Adding equations always gives the solution directly.

B. The new equation always describes the same line as the old one.

C. Multiplying an equation by a number changes its graph.

D. The step can be undone by subtracting the same multiple, so no solution is gained or lost.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q3/5

741

QUESTION ID

S20-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The step can be undone by subtracting the same multiple, so no solution is gained or lost.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q3/5

True or false?

The system $x + y = 5$ and $x - y = 1$ has the same solutions as the system $x + y = 5$ and $2x = 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q4/5

742

QUESTION ID

S20-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q4/5

True or false?

The system x + y = 5 and x − y = 1 has the same solutions as the single equation 2x = 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q5/5

743

QUESTION ID

S20-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 2x = 6 alone is true for (3, y) with any y; the system has only (3, 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q5/5

SET 20 • 8.EE.C.8.b

FORWARD 2

Solve a system of a linear equation and a quadratic equation by graphing and algebraically • High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7

ANSWER KEY Reduced copy of the question page, with its answer

Solve the system. Give all solutions.

$y = x^2$

$y = 4$

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q1/5

745

QUESTION ID

S20-F2-Q1

TYPE

Short answer

CORRECT ANSWER

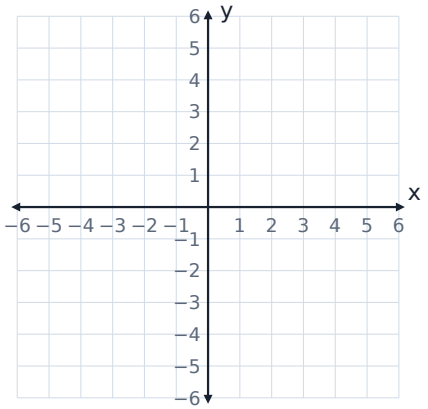
(2, 4) and (−2, 4)

GRADING NOTE Both solutions are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q1/5

Graph $y = x^2 - 1$ and $y = x + 1$ on the coordinate plane. Mark every intersection point, then check each one in both equations.



Solutions and check: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q2/5

746

QUESTION ID

S20-F2-Q2

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: The parabola $y = x^2 - 1$ through $(-2, 3)$, $(-1, 0)$, $(0, -1)$, $(1, 0)$, $(2, 3)$, and the line $y = x + 1$, crossing at $(-1, 0)$ and $(2, 3)$

Written: $(2, 3)$ and $(-1, 0)$: $3 = 2^2 - 1 = 2 + 1$ and $0 = (-1)^2 - 1 = -1 + 1$.

GRADING NOTE Grade all three: both graphs, both intersection points, and the algebraic check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q2/5

How many solutions does the system have?

$y = x^2 + 3$

$y = 1$

A. Infinitely many

B. Two

C. One

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q3/5

747

QUESTION ID

S20-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q3/5

True or false?

The line $y = 2x$ and the parabola $y = x^2$ meet at (0, 0) and (2, 4).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q4/5

748

QUESTION ID

S20-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q4/5

True or false?

A line and a parabola always meet at exactly two points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q5/5

749

QUESTION ID

S20-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q5/5

SET 21 • 8.EE.C.8.b • EXPRESSIONS & EQUATIONS

Solve simple systems by inspection

MAIN	Grade 8 main question	**8.EE.C.8.b**
BACKWARD 1	Recognize equivalent expressions	**7.EE.A.1**
BACKWARD 2	Decide whether a value makes two equations true	**6.EE.B.5**
FORWARD 1	Use elimination to confirm the number of solutions and solve syst..	**HSA-REI.C.6**
FORWARD 2	Classify lines as parallel, perpendicular, or neither from their equ..	**HSG-GPE.B.5**

SET 21 • 8.EE.C.8.b

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.C.8.b

Without solving, explain why this system has no solution.

3x + 2y = 5

3x + 2y = 6

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q1/5

752

QUESTION ID

S21-M-Q1

TYPE

Short answer

CORRECT ANSWER

The same expression, 3x + 2y, cannot equal both 5 and 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q1/5

Without solving, how many solutions does this system have? Explain.

$y = 4x - 1$

$y = 4x + 3$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q2/5

753

QUESTION ID

S21-M-Q2

TYPE

Short answer

CORRECT ANSWER

None. The lines have the same slope and different y-intercepts, so they are parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q2/5

Which system has infinitely many solutions?

A. $y = x$ and $y = -x$

B. $x + y = 4$ and $x + y = 8$

C. $x + y = 4$ and $x - y = 4$

D. $x + y = 4$ and $2x + 2y = 8$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q3/5

754

QUESTION ID

S21-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $x + y = 4$ and $2x + 2y = 8$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q3/5

Without solving, how many solutions does this system have? Explain.

$y = -2x + 5$

$y = 3x + 5$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q4/5

755

QUESTION ID

S21-M-Q4

TYPE

Short answer

CORRECT ANSWER

One. The slopes are different, so the lines cross once (at (0, 5)).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q4/5

True or false?

The system $2x - y = 7$ and $6x - 3y = 21$ has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q5/5

QUESTION ID

S21-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The second equation is 3 times the first, so there are infinitely many solutions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q5/5

SET 21 • 8.EE.C.8.b

BACKWARD 1

Recognize equivalent expressions • Grade 7 • Expressions & Equations • 7.EE.A.1

Is 2x + 2y equivalent to 2(x + y)? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q1/5

758

QUESTION ID

S21-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. By the distributive property, 2(x + y) = 2x + 2y.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q1/5

Write 3(2x + y) without parentheses.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q2/5

759

QUESTION ID

S21-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6x + 3y

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q2/5

Which expression is equivalent to 4x + 6y?

A. 6(4x + y)

B. 4(x + 6y)

C. 2(4x + 6y)

D. 2(2x + 3y)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q3/5

760

QUESTION ID

S21-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2(2x + 3y)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q3/5

True or false?

$5x - 5y = 5(x - y)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q4/5

761

QUESTION ID

S21-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q4/5

True or false?

$2(x + 3) = 2x + 3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q5/5

762

QUESTION ID

S21-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q5/5

SET 21 • 8.EE.C.8.b

BACKWARD 2

Decide whether a value makes two equations true • Grade 6 • Expressions & Equations • 6.EE.B.5

Is x = 2 a solution of both x + 3 = 5 and 2x = 4?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q1/5

764

QUESTION ID

S21-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q1/5

Which of the values 1, 2, and 3 makes both 2x = 6 and x + 1 = 4 true?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q2/5

765

QUESTION ID

S21-B2-Q2

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q2/5

Which value makes both x + 2 = 7 and 3x = 15 true?

A. 2

B. 7

C. 3

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q3/5

766

QUESTION ID

S21-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q3/5

True or false?

No value of x makes both x + 1 = 3 and x + 1 = 4 true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q4/5

767

QUESTION ID

S21-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q4/5

True or false?

$x = 4$ makes both $x - 1 = 3$ and $2x = 6$ true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q5/5

768

QUESTION ID

S21-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q5/5

SET 21 • 8.EE.C.8.b

FORWARD 1

Use elimination to confirm the number of solutions and solve systems with letter coefficients • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6

Add 2 times the first equation to the second equation. What equation results, and what does it tell you about the system?

$x - 3y = 4$

$-2x + 6y = 1$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q1/5

770

QUESTION ID

S21-F1-Q1

TYPE

Short answer

CORRECT ANSWER

0 = 9, which is false, so the system has no solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q1/5

Solve the system for x and y in terms of a.

$x + y = 3a$

$x - y = a$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q2/5

771

QUESTION ID

S21-F1-Q2

TYPE

Short answer

CORRECT ANSWER

x = 2a and y = a

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q2/5

Elimination on a system of two linear equations gives 0 = 0.

What does this tell you?

A. There is exactly one solution.

B. The only solution is (0, 0).

C. The equations describe the same line, so there are infinitely many solutions.

D. There is no solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q3/5

772

QUESTION ID

S21-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. The equations describe the same line, so there are infinitely many solutions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q3/5

True or false?

The system $x + y = 1.5$ and $2x - y = 0$ has the solution $x = 0.5$, $y = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q4/5

773

QUESTION ID

S21-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q4/5

True or false?

If elimination gives 0 = 5, the system has infinitely many solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q5/5

774

QUESTION ID

S21-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 0 = 5 is false, so there is no solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q5/5

SET 21 • 8.EE.C.8.b

FORWARD 2

Classify lines as parallel, perpendicular, or neither from their equations • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Are the lines y = 2x + 1 and 4x − 2y = 6 parallel, perpendicular, or neither? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q1/5

776

QUESTION ID

S21-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Parallel; both have slope 2 (and different y-intercepts).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q1/5

Are the lines 3x + y = 2 and x − 3y = 9 parallel, perpendicular, or neither? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q2/5

777

QUESTION ID

S21-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Perpendicular; the slopes are −3 and 1/3, and their product is −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q2/5

One line passes through (0, 0) and (2, 4). Another passes through (1, 1) and (3, 5).

How are the lines related?

A. Perpendicular

B. The same line

C. Parallel

D. Neither

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q3/5

778

QUESTION ID

S21-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Parallel

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q3/5

True or false?

Lines with slopes $\frac{2}{3}$ and $-\frac{3}{2}$ are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q4/5

779

QUESTION ID

S21-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q4/5

True or false?

The lines y = x and y = −x are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q5/5

780

QUESTION ID

S21-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are perpendicular.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q5/5

SET 22 • 8.EE.C.8.c • EXPRESSIONS & EQUATIONS

Write and solve systems of linear equations for real-world problems

MAIN	Grade 8 main question	**8.EE.C.8.c**
BACKWARD 1	Solve word problems with one unknown	**7.EE.B.4.a**
BACKWARD 2	Use variables to represent quantities in a situation	**6.EE.B.6**
FORWARD 1	Create cost, revenue, and profit equations in two variables and gr.	**HSA-CED.A.2**
FORWARD 2	Represent constraints with a system of inequalities	**HSA-CED.A.3**

SET 22 • 8.EE.C.8.c

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.C.8.c

A school sells 200 tickets. Adult tickets cost $6 and student tickets cost $4. The sales total $940.

Write a system and solve it to find how many of each ticket were sold.

System and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q1/5

783

QUESTION ID

S22-M-Q1

TYPE

Short answer

CORRECT ANSWER

The system a + s = 200 and 6a + 4s = 940, and the answer 70 adult and 130 student tickets

GRADING NOTE Both parts are required: a valid equation (or system) and the correct solution. Any valid model is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Plan A costs $15 plus $0.05 per minute. Plan B costs $0.10 per minute.

Write a system and solve it to find when the plans cost the same.

System and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q2/5

784

QUESTION ID

S22-M-Q2

TYPE

Short answer

CORRECT ANSWER

The system c = 15 + 0.05m and c = 0.10m, and the answer 300 minutes (both cost $30)

GRADING NOTE Both parts are required: a valid equation (or system) and the correct solution. Any valid model is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q2/5

Jo has 25 coins, all dimes and quarters, worth $4.00 in all.

Which system models this?

A. $d - q = 25$ and $0.10d + 0.25q = 4$

B. $d + q = 25$ and $0.10d + 0.25q = 4$

C. $d + q = 4$ and $0.10d + 0.25q = 25$

D. $d + q = 25$ and $0.25d + 0.10q = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q3/5

QUESTION ID

S22-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $d + q = 25$ and $0.10d + 0.25q = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q3/5

Jo has 25 coins, all dimes and quarters, worth $4.00 in all: d + q = 25 and 0.10d + 0.25q = 4.

How many of each coin does Jo have?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q4/5

786

QUESTION ID

S22-M-Q4

TYPE

Short answer

CORRECT ANSWER

15 dimes and 10 quarters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q4/5

True or false?

Two numbers have a sum of 40 and a difference of 12. The larger number is 26.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q5/5

787

QUESTION ID

S22-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q5/5

SET 22 • 8.EE.C.8.c

BACKWARD 1

Solve word problems with one unknown • Grade 7 • Expressions & Equations • 7.EE.B.4.a

A taxi charges $3 plus $2 per mile. A ride costs $19.

How many miles long was the ride?

Miles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q1/5

789

QUESTION ID

S22-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q1/5

Concert tickets cost $9 each, plus a $5 fee for the whole order. An order costs $50.

How many tickets were bought?

Tickets: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q2/5

790

QUESTION ID

S22-B1-Q2

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q2/5

The sum of a number and 3 times the number is 48.

What is the number?

A. 16

B. 45

C. 12

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q3/5

791

QUESTION ID

S22-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q3/5

True or false?

If 2x + 7 = 25, then x = 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q4/5

792

QUESTION ID

S22-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q4/5

True or false?

Items cost $4 each, and shipping is $6. A $30 order has 7 items.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q5/5

793

QUESTION ID

S22-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 4n + 6 = 30 gives n = 6 items.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q5/5

SET 22 • 8.EE.C.8.c

BACKWARD 2

Use variables to represent quantities in a situation • Grade 6 • Expressions & Equations • 6.EE.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Write an expression for the value, in dollars, of d dimes.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q1/5

795

QUESTION ID

S22-B2-Q1

TYPE

Short answer

CORRECT ANSWER

0.10d

GRADING NOTE 0.1d is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q1/5

Adult tickets cost $6.

Write an expression for the cost of a adult tickets.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q2/5

796

QUESTION ID

S22-B2-Q2

TYPE

Short answer

CORRECT ANSWER

6a

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q2/5

There are 25 coins: d dimes and the rest quarters.

Which expression gives the number of quarters?

A. d − 25

B. 25d

C. 25 − d

D. d + 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q3/5

797

QUESTION ID

S22-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 25 − d

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q3/5

True or false?

q quarters are worth 0.25q dollars.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q4/5

798

QUESTION ID

S22-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q4/5

True or false?

There are s students and 3 more teachers than students. There are 3s teachers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q5/5

799

QUESTION ID

S22-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE There are s + 3 teachers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q5/5

SET 22 • 8.EE.C.8.c

FORWARD 1

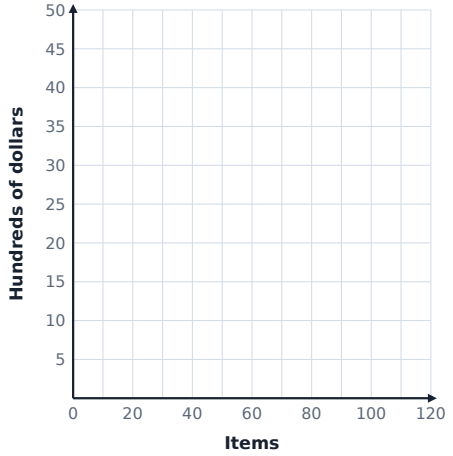
Create cost, revenue, and profit equations in two variables and graph them • High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A company's cost is $C = 2{,}000 + 15x$ dollars and its revenue is $R = 40x$ dollars for x items.

Graph both equations for $0 \le x \le 120$ (the vertical axis is in hundreds of dollars), label each line, and mark the break-even point.

Break-even point: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q1

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q1/5

801

QUESTION ID

S22-F1-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: On the grid (in hundreds of dollars): the cost line from (0, 20) to (120, 38) and the revenue line from (0, 0) to (120, 48), labeled, meeting at (80, 32)

Written: (80, 3,200): 80 items, where cost and revenue are both $3,200

GRADING NOTE Grade both: the two labeled lines, and the break-even point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q1

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q1/5

A theater sells tickets at p dollars each. The number of tickets sold is n = 600 − 20p.

Write an equation for the revenue R in terms of p.

R = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q2

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q2/5

802

QUESTION ID

S22-F1-Q2

TYPE

Short answer

CORRECT ANSWER

p(600 − 20p)

GRADING NOTE 600p − 20p² is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q2

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q2/5

The revenue is $R = p(600 - 20p)$ dollars for a ticket price of p dollars.

Which price gives the greatest revenue?

A. $15

B. $20

C. $30

D. $600

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q3

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q3/5

803

QUESTION ID

S22-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q3

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q3/5

True or false?

At the break-even point of C = 2,000 + 15x and R = 40x, cost and revenue are both $3,200.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q4

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q4/5

804

QUESTION ID

S22-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q4

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q4/5

True or false?

$R = p(600 - 20p)$ is a linear equation in p.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q5

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q5/5

805

QUESTION ID

S22-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has a p^2 term, so it is quadratic.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q5

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q5/5

SET 22 • 8.EE.C.8.c

FORWARD 2

Represent constraints with a system of inequalities • High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3

A bakery has 30 pounds of flour. Each batch of bread uses 2 pounds and each batch of muffins uses 0.5 pound.

Write an inequality for b batches of bread and m batches of muffins.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q1

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q1/5

807

QUESTION ID

S22-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$2b + 0.5m \leq 30$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q1

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q1/5

A bakery has at most 20 hours of oven time. Each batch of bread takes 1 hour and each batch of muffins takes 2 hours.

Write an inequality for b batches of bread and m batches of muffins.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q2

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q2/5

808

QUESTION ID

S22-F2-Q2

TYPE

Short answer

CORRECT ANSWER

b + 2m ≤ 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q2

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q2/5

A bakery's limits are $2b + 0.5m \leq 30$ (flour) and $b + 2m \leq 20$ (oven hours).

Can it make 10 batches of bread and 5 batches of muffins?

A. No; both limits are broken.

B. No; the flour runs out.

C. Yes; both limits are met.

D. No; the oven time runs out.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q3

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q3/5

809

QUESTION ID

S22-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Yes; both limits are met.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q3

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q3/5

True or false?

In a model of batches of bread b and muffins m, the constraints $b \geq 0$ and $m \geq 0$ make sense because batches cannot be negative.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q4

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q4/5

QUESTION ID

S22-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q4

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q4/5

True or false?

The point (12, 6) meets both limits $2b + 0.5m \leq 30$ and $b + 2m \leq 20$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q5

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q5/5

811

QUESTION ID
S22-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $12 + 2(6) = 24$, which is more than 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q5

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q5/5

SET 23 • 8.F.A.1 • FUNCTIONS

Decide whether a relation is a function

MAIN	Grade 8 main question	**8.F.A.1**
BACKWARD 1	Find the output of a rule	**6.EE.C.9**
BACKWARD 2	Read points in the first quadrant	**5.G.A.2**
FORWARD 1	Use function notation and find the domain and range	**HSF-IF.A.1**
FORWARD 2	Find the inverse of a linear function	**HSF-BF.B.4.a**

SET 23 • 8.F.A.1

MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.A.1

True or false?

The table shows y as a function of x.

x	1	2	3	4
y	5	5	7	9

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q1

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q1/5

814

QUESTION ID

S23-M-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q1

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q1/5

Is the relation in the table a function? Explain.

x	2	3	3	5
y	4	6	8	10

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q2

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q2/5

815

QUESTION ID

S23-M-Q2

TYPE

Short answer

CORRECT ANSWER

No. The input 3 has two different outputs (6 and 8).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q2

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q2/5

Which set of ordered pairs is a function?

A. (5, 1), (6, 2), (5, 3)

B. (0, 0), (0, 1), (1, 1)

C. (1, 2), (1, 3), (2, 4)

D. (1, 4), (2, 4), (3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q3

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q3/5

QUESTION ID

S23-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (1, 4), (2, 4), (3, 4)

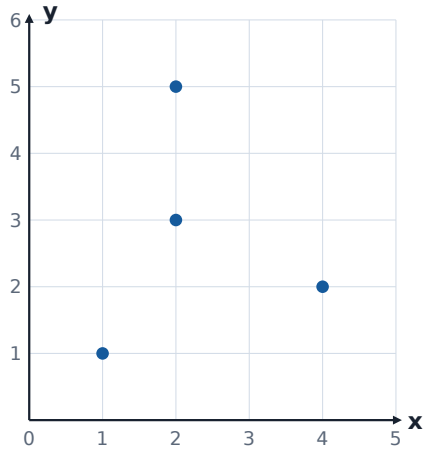
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q3

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q3/5

Do the points on the graph show y as a function of x? Explain.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q4

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q4/5

817

QUESTION ID

S23-M-Q4

TYPE

Short answer

CORRECT ANSWER

No. The input x = 2 has two outputs (3 and 5).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q4

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q4/5

True or false?

A rule takes a birth month as its input and gives one student born in that month as its output. Ana and Ben were both born in January. This rule is a function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q5

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q5/5

818

QUESTION ID

S23-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The input January is paired with two different outputs, Ana and Ben, so the rule is not a function.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q5

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q5/5

SET 23 • 8.F.A.1

BACKWARD 1

Find the output of a rule • Grade 6 • Expressions & Equations • 6.EE.C.9

The rule is y = 2x + 1.

What is y when x = 4?

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q1/5

820

QUESTION ID

S23-B1-Q1

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q1/5

The table follows a pattern.

What is y when x = 4?

x	1	2	3
y	4	7	10

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q2/5

821

QUESTION ID

S23-B1-Q2

TYPE

Short answer

CORRECT ANSWER

13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q2/5

The rule is: output = input × 3.

What is the output for an input of 7?

A. 4

B. 73

C. 21

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q3/5

822

QUESTION ID

S23-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q3/5

True or false?

For $y = x - 5$, an input of $x = 9$ gives $y = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q4/5

823

QUESTION ID

S23-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q4/5

True or false?

For y = 4x, an input of x = 3 gives y = 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q5/5

824

QUESTION ID

S23-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q5

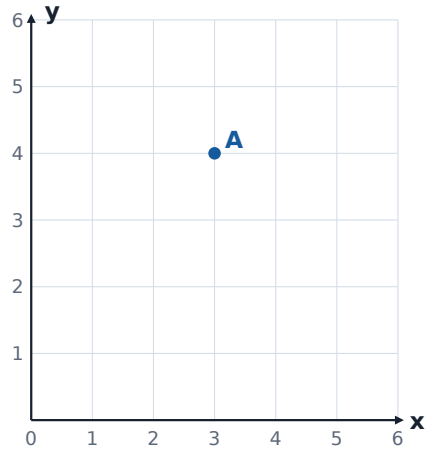
Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q5/5

SET 23 • 8.F.A.1

BACKWARD 2

Read points in the first quadrant • Grade 5 • Geometry • 5.G.A.2

What are the coordinates of point A?



Coordinates: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q1/5

826

QUESTION ID

S23-B2-Q1

TYPE

Short answer

CORRECT ANSWER

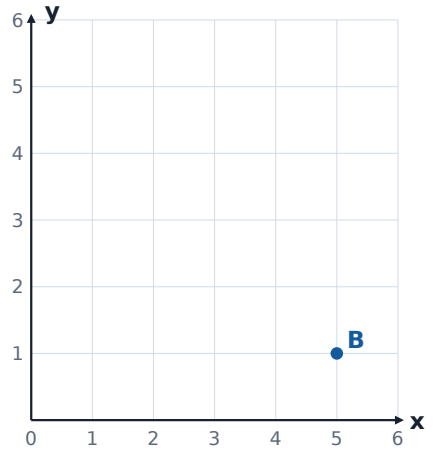
(3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q1/5

What are the coordinates of point B?



Coordinates: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q2/5

827

QUESTION ID

S23-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(5, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q2/5

Which point is 2 units to the right of the origin and 5 units up?

A. (5, 5)

B. (5, 2)

C. (2, 2)

D. (2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q3/5

828

QUESTION ID

S23-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q3/5

True or false?

The point (0, 3) is on the y-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q4/5

829

QUESTION ID

S23-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q4/5

True or false?

The point (4, 0) is 4 units up from the origin.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q5/5

QUESTION ID

S23-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q5/5

SET 23 • 8.F.A.1

FORWARD 1

Use function notation and find the domain and range • High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1

A function f is made of the pairs (1, 3), (2, 5), and (4, 9).

What is its domain?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q1/5

832

QUESTION ID

S23-F1-Q1

TYPE

Short answer

CORRECT ANSWER

1, 2, and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q1/5

A function f is made of the pairs (1, 3), (2, 5), and (4, 9).

What is its range?

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q2/5

833

QUESTION ID

S23-F1-Q2

TYPE

Short answer

CORRECT ANSWER

3, 5, and 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q2/5

The output of f is 7 when the input is 2.

Which statement says this in function notation?

A. $f = 2 \times 7$

B. $f(7) = 2$

C. $2f = 7$

D. $f(2) = 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q3/5

834

QUESTION ID

S23-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. f(2) = 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q3/5

True or false?

If $f(x) = x^2$, then $f(-3) = 9$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q4/5

835

QUESTION ID

S23-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q4/5

True or false?

A function can assign two different outputs to the same input.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q5/5

836

QUESTION ID

S23-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q5/5

SET 23 • 8.F.A.1

FORWARD 2

Find the inverse of a linear function • High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a

$f(x) = 2x + 3$

Write the inverse function.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q1/5

838

QUESTION ID

S23-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(x − 3)/2

GRADING NOTE $\frac{x}{2} - \frac{3}{2}$ is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-F2-Q1**

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q1/5

$f(x) = 2x + 3$

Solve $f(x) = 11$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q2/5

839

QUESTION ID

S23-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q2/5

$g(x) = \frac{x}{5}$

What is the inverse function?

A. $g^{-1}(x) = x + 5$

B. $g^{-1}(x) = x - 5$

C. $g^{-1}(x) = 5x$

D. $g^{-1}(x) = \frac{5}{x}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q3/5

840

QUESTION ID

S23-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $g^{-1}(x) = 5x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q3/5

True or false?

If $f(2) = 7$ and f has an inverse, then $f^{-1}(7) = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q4/5

841

QUESTION ID

S23-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q4/5

True or false?

Every function has an inverse function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q5/5

842

QUESTION ID

S23-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $f(x) = x^2$ gives the same output for 2 and -2, so it has no inverse on all real numbers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q5/5

SET 24 • 8.F.A.1 • FUNCTIONS

Graph a function as the set of ordered pairs of inputs and outputs

MAIN	Grade 8 main question	**8.F.A.1**
BACKWARD 1	Make ordered pairs from an equation	**6.EE.C.9**
BACKWARD 2	Graph points from a pattern	**5.G.A.2**
FORWARD 1	Choose a domain that fits a function's situation	**HSF-IF.B.5**
FORWARD 2	Graph quadratic functions and show their key features	**HSF-IF.C.7.a**

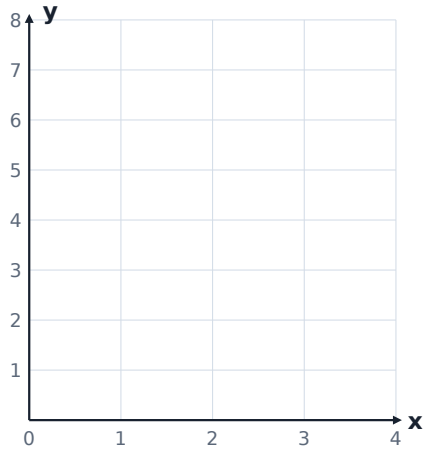
SET 24 • 8.F.A.1

MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.A.1

The function y = 2x + 1 has the inputs 0, 1, 2, and 3.

Plot the ordered pairs (input, output) on the grid.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q1

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q1/5

845

QUESTION ID

S24-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Points at (0, 1), (1, 3), (2, 5), and (3, 7)

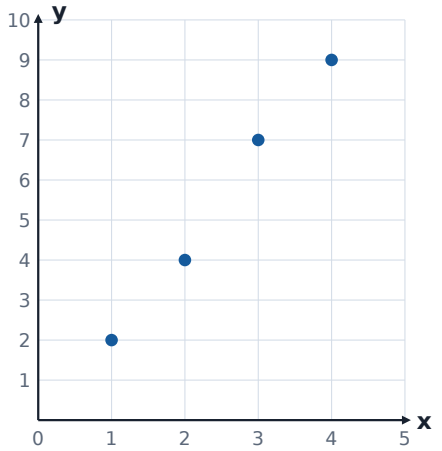
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q1

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q1/5

The points show a function.

What is the output when the input is 3?



Output: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q2

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q2/5

846

QUESTION ID

S24-M-Q2

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q2

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q2/5

The graph of a function contains the point (4, 10).

Which statement is true?

A. When the input is 4, the output is 10.

B. When the input is 10, the output is 4.

C. The graph has only one point.

D. The function adds 4 to every input.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q3

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q3/5

847

QUESTION ID

S24-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. When the input is 4, the output is 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q3

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q3/5

The rule is "output = input squared."

Write the ordered pairs (input, output) for the inputs −2, −1, 0, 1, and 2.

Ordered pairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q4

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q4/5

848

QUESTION ID

S24-M-Q4

TYPE

Short answer

CORRECT ANSWER

(−2, 4), (−1, 1), (0, 0), (1, 1), (2, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q4

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q4/5

True or false?

The graph of a function can contain both (2, 5) and (2, 8).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q5

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q5/5

849

QUESTION ID

S24-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The input 2 cannot have two outputs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q5

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q5/5

SET 24 • 8.F.A.1

BACKWARD 1

Make ordered pairs from an equation • Grade 6 • Expressions & Equations • 6.EE.C.9

For y = 3x, what are the values of y when x = 0, 1, and 2?

Values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q1/5

851

QUESTION ID

S24-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0, 3, and 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

For y = x + 4, what is y when x = 5?

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q2/5

852

QUESTION ID

S24-B1-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q2/5

Which pair (x, y) fits y = 5x?

A. (2, 7)

B. (10, 2)

C. (2, 10)

D. (5, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q3/5

853

QUESTION ID

S24-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (2, 10)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q3/5

True or false?

The pair (3, 9) fits y = 3x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q4/5

854

QUESTION ID

S24-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q4/5

True or false?

The pair (4, 6) fits y = x + 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q5/5

855

QUESTION ID

S24-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q5

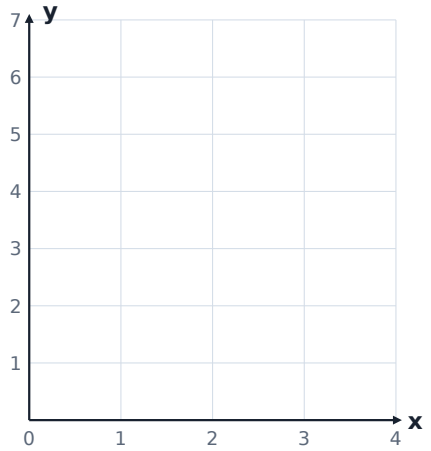
Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q5/5

SET 24 • 8.F.A.1

BACKWARD 2

Graph points from a pattern • Grade 5 • Geometry • 5.G.A.2

Plot the points (1, 2), (2, 4), and (3, 6) on the grid.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q1/5

857

QUESTION ID

S24-B2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Points at (1, 2), (2, 4), and (3, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q1/5

The y-values start at 1 and increase by 2: (1, 1), (2, 3), (3, 5), ...

What is the point for x = 4?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q2/5

858

QUESTION ID

S24-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(4, 7)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q2/5

Which point is 3 units to the right of the origin and 1 unit up?

A. (3, 1)

B. (3, 3)

C. (1, 3)

D. (1, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q3/5

859

QUESTION ID

S24-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (3, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q3/5

True or false?

The point (2, 6) is 2 units to the right of the origin and 6 units up.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q4/5

860

QUESTION ID

S24-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q4/5

True or false?

To plot (5, 2), you move 5 units up and 2 units to the right.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q5/5

861

QUESTION ID

S24-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q5/5

SET 24 • 8.F.A.1

FORWARD 1

Choose a domain that fits a function's situation • High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5

A function gives the cost of n concert tickets at $12 each. A buyer may buy at most 8 tickets.

What is a sensible domain?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q1/5

863

QUESTION ID

S24-F1-Q1

TYPE

Short answer

CORRECT ANSWER

The whole numbers 0 through 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q1/5

h(t) gives the height of a ball t seconds after it is thrown, until it lands at t = 3.

What is a sensible domain?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q2/5

QUESTION ID

S24-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$0 \leq t \leq 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q2/5

p is the number of people in a car that seats 5.

Which domain makes sense for p?

A. $-5 \leq p \leq 5$

B. 0, 1, 2, 3, 4, 5

C. All real numbers

D. $p \geq 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q3/5

865

QUESTION ID

S24-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 0, 1, 2, 3, 4, 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q3/5

True or false?

For the area of a square as a function of its side length, the domain contains only positive numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q4/5

866

QUESTION ID

S24-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q4/5

True or false?

A function that gives the number of buses needed for n students should have all real numbers as its domain.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q5/5

867

QUESTION ID

S24-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q5

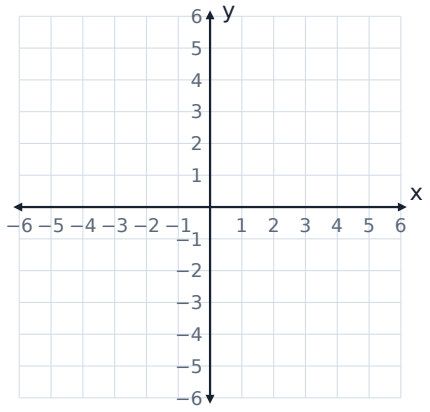
High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q5/5

SET 24 • 8.F.A.1

FORWARD 2

Graph quadratic functions and show their key features • High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a

Graph $y = x^2 - 4$ on the coordinate plane. Mark the vertex and the x-intercepts.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q1/5

869

QUESTION ID

S24-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

A U-shaped parabola with its vertex at (0, −4), crossing the x-axis at (−2, 0) and (2, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q1/5

What is the vertex of the graph of $y = (x - 1)^2 + 2$?

Vertex: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q2/5

870

QUESTION ID

S24-F2-Q2

TYPE

Short answer

CORRECT ANSWER

(1, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q2/5

Where does the graph of $y = x^2 - 9$ cross the x-axis?

A. At (3, 0) and (−3, 0)

B. At (9, 0) and (−9, 0)

C. At (0, −9)

D. At (3, 0) only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q3/5

871

QUESTION ID

S24-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. At (3, 0) and (−3, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q3/5

True or false?

The graph of $y = -x^2$ opens downward.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q4/5

872

QUESTION ID

S24-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q4/5

True or false?

The graph of $y = 2x + 3$ has a vertex.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q5/5

873

QUESTION ID

S24-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q5/5

SET 25 • 8.F.A.2 • FUNCTIONS

Compare two functions represented in different ways

MAIN	Grade 8 main question	**8.F.A.2**
BACKWARD 1	Find a unit rate from a table, graph, or equation	**7.RP.A.2.b**
BACKWARD 2	Find the starting value of a relationship	**6.EE.C.9**
FORWARD 1	Compare average rates of change of linear and nonlinear function…	**HSF-IF.B.6**
FORWARD 2	Describe the effect of transformations on the graph of a function	**HSF-BF.B.3**

SET 25 • 8.F.A.2

MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.A.2

f(x) = 3x + 2. The table shows g.

Which function has the greater rate of change?

x	0	2	4
g(x)	1	9	17

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q1

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q1/5

876

QUESTION ID

S25-M-Q1

TYPE

Short answer

CORRECT ANSWER

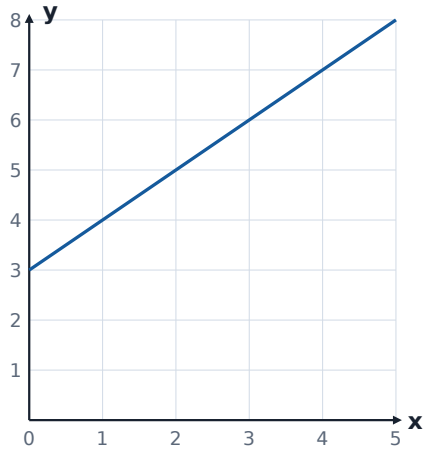
g (its rate is 4; f's is 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q1

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q1/5

The graph shows f.
$g(x) = -2x + 5$.

Which function has the greater initial value?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q2

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q2/5

877

QUESTION ID

S25-M-Q2

TYPE

Short answer

CORRECT ANSWER

g (5 is greater than 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q2

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q2/5

Function A is y = 0.5x + 10. Function B is a line through (0, 4) and (2, 8).

Which statement is true?

A. B has the greater rate of change; A has the greater initial value.

B. B has the greater rate of change and the greater initial value.

C. A has the greater rate of change and the greater initial value.

D. They have the same rate of change.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q3

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q3/5

878

QUESTION ID

S25-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. B has the greater rate of change; A has the greater initial value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q3

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q3/5

Car A travels d = 45t miles in t hours. The table shows Car B.

Which car is faster, and by how much?

Hours	1	2	3
Miles	50	100	150

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q4

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q4/5

879

QUESTION ID

S25-M-Q4

TYPE

Short answer

CORRECT ANSWER

Car B, by 5 miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q4

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q4/5

True or false?

A function with x-values 0, 1, 2 and y-values 6, 4, 2 decreases faster than $y = -x + 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q5

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q5/5

880

QUESTION ID

S25-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q5

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q5/5

SET 25 • 8.F.A.2

BACKWARD 1

Find a unit rate from a table, graph, or equation • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b

The table shows a proportional relationship.

What is the unit rate?

x	2	4	6
y	5	10	15

Unit rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q1/5

882

QUESTION ID

S25-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q1/5

What is the unit rate in y = 7x?

Unit rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q2/5

883

QUESTION ID

S25-B1-Q2

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q2/5

A proportional graph passes through (0, 0) and (3, 12).

What is the unit rate?

A. 36

B. 12

C. 3

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q3/5

884

QUESTION ID

S25-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q3/5

True or false?

A proportional table with the pairs (1, 9) and (2, 18) has a unit rate of 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q4/5

885

QUESTION ID

S25-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q4/5

True or false?

The unit rate in y = 0.5x is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q5/5

886

QUESTION ID

S25-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q5/5

SET 25 • 8.F.A.2

BACKWARD 2

Find the starting value of a relationship • Grade 6 • Expressions & Equations • 6.EE.C.9

For y = 4x + 7, what is y when x = 0?

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q1/5

888

QUESTION ID

S25-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q1/5

For the table, what is y when x = 0?

x	0	1	2
y	9	11	13

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q2/5

889

QUESTION ID

S25-B2-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q2/5

Which equation gives y = 5 when x = 0?

A. y = 5x + 2

B. y = 2x + 5

C. y = 5x

D. y = x − 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q3/5

890

QUESTION ID

S25-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. y = 2x + 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q3/5

True or false?

For y = 3x + 1, y = 1 when x = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q4/5

891

QUESTION ID

S25-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q4/5

True or false?

For y = 6x, y = 6 when x = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q5/5

892

QUESTION ID

S25-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q5/5

SET 25 • 8.F.A.2

FORWARD 1

Compare average rates of change of linear and nonlinear functions over intervals • High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6

$f(x) = x^2$. The table shows g.

Find the average rate of change of each function from $x = 0$ to $x = 3$.

x	0	1	2	3
g(x)	0	3	6	9

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q1/5

894

QUESTION ID

S25-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Both are 3: f goes from 0 to 9 and g goes from 0 to 9 over 3 units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q1/5

f(x) = 2^x and g(x) = 3x + 5.

Find the average rate of change of each function from x = 0 to x = 4. Which is greater?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q2/5

895

QUESTION ID

S25-F1-Q2

TYPE

Short answer

CORRECT ANSWER

f: (16 − 1) ÷ 4 = 3.75; g: 3. f has the greater average rate of change.

GRADING NOTE Both rates and the comparison are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q2/5

$f(x) = x^2$ and $g(x) = 4x$.

Which statement about the average rates of change from $x = 1$ to $x = 3$ is true?

A. f's rate is greater.

B. f's rate is 9 and g's rate is 12.

C. g's rate is greater.

D. They are equal (both are 4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q3/5

896

QUESTION ID

S25-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. They are equal (both are 4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q3/5

True or false?

The average rate of change of $f(x) = -x^2 + 4$ from $x = 0$ to $x = 2$ is -2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q4/5

897

QUESTION ID

S25-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q4/5

True or false?

The average rate of change of a linear function depends on which interval you use.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q5/5

898

QUESTION ID

S25-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A linear function has the same rate over every interval.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q5/5

SET 25 • 8.F.A.2

FORWARD 2

Describe the effect of transformations on the graph of a function • High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3

How does the graph of g(x) = f(x) + 3 compare with the graph of f?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q1/5

900

QUESTION ID

S25-F2-Q1

TYPE

Short answer

CORRECT ANSWER

It is the graph of f shifted up 3 units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q1/5

How does the graph of $g(x) = (x - 2)^2$ compare with the graph of $y = x^2$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q2/5

901

QUESTION ID

S25-F2-Q2

TYPE

Short answer

CORRECT ANSWER

It is shifted 2 units to the right.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q2/5

Which function shifts the graph of y = |x| down 4 units?

A. y = |x| − 4

B. y = |x + 4|

C. y = 4|x|

D. y = |x − 4|

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q3/5

902

QUESTION ID

S25-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. y = |x| − 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q3/5

True or false?

The graph of $y = -f(x)$ is the reflection of the graph of $y = f(x)$ across the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q4/5

903

QUESTION ID

S25-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q4/5

True or false?

The graph of y = f(x + 5) is the graph of y = f(x) shifted 5 units to the right.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q5/5

904

QUESTION ID

S25-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is shifted 5 units to the left.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q5/5

SET 26 • 8.F.A.3 • FUNCTIONS

Decide whether a function is linear or nonlinear

MAIN	Grade 8 main question	**8.F.A.3**
BACKWARD 1	Decide whether a table shows a proportional relationship	**7.RP.A.2.a**
BACKWARD 2	Find how much y changes for each change in x	**6.EE.C.9**
FORWARD 1	Show that linear functions grow by equal differences and expone..	**HSF-LE.A.1.a**
FORWARD 2	Use factoring to find the zeros and maximum of a quadratic functi..	**HSF-IF.C.8.a**

SET 26 • 8.F.A.3

MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.A.3

Which equation is a linear function?

A. $y = 4x - 7$

B. $y = x^2 + 1$

C. $y = 2^x$

D. $y = \frac{3}{x}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q1

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q1/5

907

QUESTION ID

S26-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. y = 4x − 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q1

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q1/5

Is the function in the table linear? Explain.

x	0	1	2	3
y	1	2	5	10

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q2

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q2/5

908

QUESTION ID

S26-M-Q2

TYPE

Short answer

CORRECT ANSWER

No. y increases by 1, then 3, then 5, so the rate of change is not constant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q2

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q2/5

When n people each shake hands once with every other person, the number of handshakes is h = n(n − 1) ÷ 2.

Is h a linear function of n? Use the values for n = 2, 3, and 4 to support your answer.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q3

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q3/5

909

QUESTION ID

S26-M-Q3

TYPE

Short answer

CORRECT ANSWER

No. The values are 1, 3, and 6, which increase by 2 and then by 3, so the rate of change is not constant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q3

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q3/5

Is the function in the table linear? Explain.

x	0	2	4	6
y	7	4	1	−2

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q4

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q4/5

910

QUESTION ID

S26-M-Q4

TYPE

Short answer

CORRECT ANSWER

Yes. y decreases by 3 each time x increases by 2, a constant rate of change of −1.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q4

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q4/5

True or false?

y = 3(x + 2) is a linear function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q5

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q5/5

911

QUESTION ID

S26-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q5

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q5/5

SET 26 • 8.F.A.3

BACKWARD 1

Decide whether a table shows a proportional relationship • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

True or false?

The table shows a proportional relationship.

x	1	2	3
y	4	8	12

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q1/5

913

QUESTION ID

S26-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q1/5

Is the relationship in the table proportional? Explain.

x	1	2	3
y	3	5	7

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q2/5

914

QUESTION ID

S26-B1-Q2

TYPE

Short answer

CORRECT ANSWER

No. The ratios 3, 2.5, and about 2.33 are not equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q2/5

Which table shows a proportional relationship?

A. x: 0, 1, 2 and y: 1, 2, 3

B. x: 1, 2, 3 and y: 1, 4, 9

C. x: 1, 2, 3 and y: 2, 3, 4

D. x: 2, 4, 6 and y: 3, 6, 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q3/5

915

QUESTION ID

S26-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x: 2, 4, 6 and y: 3, 6, 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q3/5

The pairs (5, 2) and (10, 4) are from a proportional relationship.

What is the constant of proportionality (y per x)?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q4/5

916

QUESTION ID

S26-B1-Q4

TYPE

Short answer

CORRECT ANSWER

0.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q4/5

True or false?

y = x + 1 is a proportional relationship.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q5/5

917

QUESTION ID

S26-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q5/5

SET 26 • 8.F.A.3

BACKWARD 2

Find how much y changes for each change in x • Grade 6 • Expressions & Equations • 6.EE.C.9

How much does y increase each time x increases by 1?

x	0	1	2	3
y	5	8	11	14

Increase: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q1/5

919

QUESTION ID

S26-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q1/5

How much does y increase each time x increases by 2?

x	0	2	4
y	1	5	9

Increase: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q2/5

920

QUESTION ID

S26-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q2/5

In y = 6x, how much does y increase when x increases by 1?

A. 1

B. 7

C. 6

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q3/5

921

QUESTION ID

S26-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q3/5

True or false?

In the table with x-values 1, 2, 3 and y-values 10, 20, 30, y increases by 10 each time x increases by 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q4/5

922

QUESTION ID

S26-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q4/5

True or false?

In y = x + 9, y increases by 9 each time x increases by 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q5/5

923

QUESTION ID

S26-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q5/5

SET 26 • 8.F.A.3

FORWARD 1

Show that linear functions grow by equal differences and exponential functions by equal factors • High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a

Show that f(x) = 4x + 1 grows by the same amount over every interval of length 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q1

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q1/5

925

QUESTION ID

S26-F1-Q1

TYPE

Short answer

CORRECT ANSWER

f(x + 2) − f(x) = 4(x + 2) + 1 − (4x + 1) = 8, for every x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q1

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q1/5

Show that g(x) = 3^x grows by the same factor over every interval of length 1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q2

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q2/5

926

QUESTION ID

S26-F1-Q2

TYPE

Short answer

CORRECT ANSWER

g(x + 1) ÷ g(x)
= 3^(x + 1) ÷ 3^x
= 3, for every x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q2

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q2/5

Which function grows by equal factors over equal intervals?

A. $y = 2x^2$

B. $y = x + 5$

C. $y = 5x + 2$

D. $y = 5(2)^x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q3

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q3/5

927

QUESTION ID

S26-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. y = 5(2)^x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q3

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q3/5

True or false?

The linear function $f(x) = mx + b$ changes by $m(x_2 - x_1)$ from x_1 to x_2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q4

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q4/5

928

QUESTION ID

S26-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q4

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q4/5

True or false?

$y = x^2$ grows by equal differences over equal intervals.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q5

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q5/5

929

QUESTION ID

S26-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q5

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q5/5

SET 26 • 8.F.A.3

FORWARD 2

Use factoring to find the zeros and maximum of a quadratic function • High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a

ANSWER KEY Reduced copy of the question page, with its answer

A ball's height is $h(t) = -16t^2 + 32t = -16t(t - 2)$ feet after t seconds.

Use the factored form to find when the ball is at height 0.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q1/5

931

QUESTION ID

S26-F2-Q1

TYPE

Short answer

CORRECT ANSWER

t = 0 and t = 2 seconds

GRADING NOTE Both times are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q1/5

A ball's height is h(t) = −16t(t − 2) feet after t seconds. Its greatest height is halfway between the zeros.

When does it reach its greatest height, and what is that height?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q2/5

932

QUESTION ID

S26-F2-Q2

TYPE

Short answer

CORRECT ANSWER

At t = 1 second; 16 feet

GRADING NOTE Both the time and the height are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q2/5

$f(x) = x^2 - 6x + 8 = (x - 2)(x - 4)$

What are the zeros of f?

A. 2 and 4

B. −2 and −4

C. 8 only

D. 6 and 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q3/5

933

QUESTION ID

S26-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2 and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q3/5

True or false?

The graph of a quadratic function is symmetric about a vertical line through its vertex.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q4/5

934

QUESTION ID

S26-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q4/5

True or false?

The zeros of f(x) = (x − 3)(x + 5) are 3 and 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q5/5

935

QUESTION ID

S26-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The zeros are 3 and −5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q5/5

SET 27 • 8.F.B.4 • FUNCTIONS

Find the rate of change and initial value from two points, a table, or a graph

MAIN	Grade 8 main question	**8.F.B.4**
BACKWARD 1	Find a constant rate from a table	**7.RP.A.2.b**
BACKWARD 2	Subtract integers	**7.NS.A.1.c**
FORWARD 1	Construct linear and exponential functions from two points	**HSF-LE.A.2**
FORWARD 2	Write arithmetic and geometric sequences recursively and explicit..	**HSF-BF.A.2**

SET 27 • 8.F.B.4

MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.B.4

A linear function passes through (2, 7) and (5, 16).

What are its rate of change and its initial value?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q1

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q1/5

938

QUESTION ID

S27-M-Q1

TYPE

Short answer

CORRECT ANSWER

Rate of change 3; initial value 1

GRADING NOTE Grade both values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q1

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q1/5

The table shows a linear function.

What are its rate of change and its initial value?

x	1	3	5
y	10	16	22

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q2

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q2/5

939

QUESTION ID

S27-M-Q2

TYPE

Short answer

CORRECT ANSWER

Rate of change 3; initial value 7

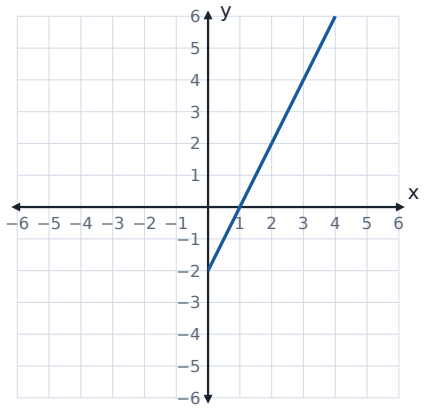
GRADING NOTE Grade both values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q2

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q2/5

The graph shows a linear function.

What are its rate of change and its initial value?



A. Rate 4, initial value −2

B. Rate −2, initial value 2

C. Rate 2, initial value −2

D. Rate 2, initial value 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q3

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q3/5

940

QUESTION ID

S27-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Rate 2, initial value −2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q3

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q3/5

A line passes through (−1, 8) and (3, 0).

Write its equation in the form y = mx + b.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q4

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q4/5

941

QUESTION ID

S27-M-Q4

TYPE

Short answer

CORRECT ANSWER

y = −2x + 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q4

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q4/5

True or false?

A linear function with f(0) = 5 and f(2) = 1 has a rate of change of −2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q5

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q5/5

942

QUESTION ID

S27-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q5

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q5/5

SET 27 • 8.F.B.4

BACKWARD 1

Find a constant rate from a table • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b

The table shows a proportional relationship.

What is the constant rate (y per x)?

x	3	6	9
y	12	24	36

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q1/5

944

QUESTION ID

S27-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q1/5

The table shows a proportional relationship.

What is the constant rate (y per x)?

Hours	2	5	8
Miles	7	17.5	28

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q2/5

945

QUESTION ID

S27-B1-Q2

TYPE

Short answer

CORRECT ANSWER

3.5 miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q2/5

A proportional table contains the pair (4, 10).

What is the constant rate (y per x)?

A. 0.4

B. 40

C. 2.5

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q3/5

946

QUESTION ID

S27-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q3/5

True or false?

A proportional table with the pairs (2, 1) and (6, 3) has a constant rate of 0.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q4/5

947

QUESTION ID

S27-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q4/5

True or false?

A proportional table with the pairs (5, 15) and (8, 24) has a constant rate of 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q5/5

948

QUESTION ID

S27-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q5/5

SET 27 • 8.F.B.4

BACKWARD 2

Subtract integers • Grade 7 • The Number System • 7.NS.A.1.c

Subtract.

$6 - (-2)$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q1/5

950

QUESTION ID

S27-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q1/5

Subtract.

$-3 - 5$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q2/5

951

QUESTION ID

S27-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q2/5

Subtract.

$-4 - (-9)$

A. 5

B. 13

C. -13

D. -5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q3/5

952

QUESTION ID

S27-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q3/5

True or false?

$2 - 7 = -5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q4/5

953

QUESTION ID

S27-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q4/5

True or false?

$-6 - (-6) = -12$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q5/5

954

QUESTION ID

S27-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q5/5

SET 27 • 8.F.B.4

FORWARD 1

Construct linear and exponential functions from two points • High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

Write the linear function whose graph passes through (1, 4) and (3, 10).

f(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q1

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q1/5

956

QUESTION ID

S27-F1-Q1

TYPE

Short answer

CORRECT ANSWER

3x + 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q1

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q1/5

Write the exponential function f(x) = a · b^x whose graph passes through (0, 5) and (1, 15).

f(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q2

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q2/5

957

QUESTION ID

S27-F1-Q2

TYPE

Short answer

CORRECT ANSWER

5 · 3^x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-F1-Q2**

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q2/5

Which exponential function passes through (0, 2) and (2, 18)?

A. f(x) = 8x + 2

B. f(x) = 2 · 3^x

C. f(x) = 18 · 2^x

D. f(x) = 2 · 9^x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q3

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q3/5

958

QUESTION ID

S27-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. f(x) = 2 · 3^x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q3

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q3/5

True or false?

The linear function through (0, −1) and (4, 7) is f(x) = 2x − 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q4

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q4/5

959

QUESTION ID

S27-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q4

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q4/5

True or false?

The exponential function through (0, 4) and (1, 2) is f(x) = 4 · 2^x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q5

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q5/5

960

QUESTION ID

S27-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The function is f(x) = 4 · (0.5)^x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q5

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q5/5

SET 27 • 8.F.B.4

FORWARD 2

Write arithmetic and geometric sequences recursively and explicitly • High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2

Write an explicit formula for the arithmetic sequence 7, 11, 15, ..., starting with n = 1.

a(n) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q1/5

962

QUESTION ID

S27-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4n + 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q1/5

Write a recursive formula for the sequence 7, 11, 15, ...

Formula: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q2/5

963

QUESTION ID

S27-F2-Q2

TYPE

Short answer

CORRECT ANSWER

a(1) = 7;

a(n) = a(n − 1) + 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q2/5

Which explicit formula gives 3, 6, 12, 24, ..., starting with n = 1?

A. a(n) = 2 · 3^(n − 1)

B. a(n) = 3 · 2^(n − 1)

C. a(n) = 3 + 2n

D. a(n) = 3n

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q3/5

964

QUESTION ID

S27-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. a(n) = 3 · 2^(n − 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q3/5

True or false?

An arithmetic sequence is a linear function whose domain is the counting numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q4/5

965

QUESTION ID

S27-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q4/5

True or false?

The sequence 2, 5, 8, 11, ... has a common ratio of 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q5/5

966

QUESTION ID

S27-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has a common difference of 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q5/5

SET 28 • 8.F.B.4 • FUNCTIONS

Construct a linear function to model a situation

MAIN	Grade 8 main question	**8.F.B.4**
BACKWARD 1	Write an equation for a situation	**6.EE.C.9**
BACKWARD 2	Solve a two-step equation from a model	**7.EE.B.4.a**
FORWARD 1	Write functions for situations with fees, thresholds, and restricted.	**HSF-BF.A.1.a**
FORWARD 2	Rearrange a formula to solve for a chosen quantity	**HSA-CED.A.4**

SET 28 • 8.F.B.4

MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.B.4

A plumber charges $65 for a visit plus $40 per hour.

Write a function for the cost C after h hours.

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q1

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q1/5

969

QUESTION ID

S28-M-Q1

TYPE

Short answer

CORRECT ANSWER

40h + 65

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q1

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A candle is 30 cm tall and burns down 2.5 cm each hour.

Write a function for its height h after t hours.

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q2

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q2/5

970

QUESTION ID

S28-M-Q2

TYPE

Short answer

CORRECT ANSWER

30 − 2.5t

GRADING NOTE −2.5t + 30 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q2

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q2/5

A phone plan costs $20 per month after a $50 activation fee.

Which function gives the total cost C after m months?

A. C = 20m − 50

B. C = 20m + 50

C. C = 50m + 20

D. C = 70m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q3

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q3/5

971

QUESTION ID

S28-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. C = 20m + 50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q3

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q3/5

The cost of a taxi ride is a linear function of the miles m. A 3-mile ride costs $11 and a 7-mile ride costs $21.

Write the function.

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q4

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q4/5

QUESTION ID

S28-M-Q4

TYPE

Short answer

CORRECT ANSWER

2.5m + 3.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q4

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q4/5

True or false?

A tank holds 500 liters and drains 20 liters each minute. The function V = 500 − 20t gives the liters V after t minutes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q5

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q5/5

973

QUESTION ID

S28-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q5

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q5/5

SET 28 • 8.F.B.4

BACKWARD 1

Write an equation for a situation • Grade 6 • Expressions & Equations • 6.EE.C.9

Tickets cost $9 each.

Write an equation for the cost c of t tickets.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q1/5

975

QUESTION ID

S28-B1-Q1

TYPE

Short answer

CORRECT ANSWER

c = 9t

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q1/5

Mia has $15 and saves $5 each week.

Write an equation for her savings s after w weeks.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q2/5

976

QUESTION ID

S28-B1-Q2

TYPE

Short answer

CORRECT ANSWER

s = 5w + 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q2/5

A pool holds 100 gallons and fills at 10 gallons per minute.

Which equation gives the gallons g after m minutes?

A. g = 110m

B. g = 100m + 10

C. g = 10m + 100

D. g = 10m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q3/5

977

QUESTION ID

S28-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. g = 10m + 100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q3/5

True or false?

Apples cost $3 per pound. The equation c = 3p gives the cost c of p pounds.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q4/5

978

QUESTION ID

S28-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q4/5

True or false?

A game costs a $10 fee plus $2 per game. The equation c = 10g + 2 gives the cost c of g games.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q5/5

979

QUESTION ID

S28-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The equation is c = 2g + 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q5/5

SET 28 • 8.F.B.4

BACKWARD 2

Solve a two-step equation from a model • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Solve for h.

40h + 65 = 185

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q1/5

981

QUESTION ID

S28-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q1/5

Solve for t.

$30 - 2.5t = 10$

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q2/5

982

QUESTION ID

S28-B2-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q2/5

Solve for m.

20m + 50 = 170

A. m = 3

B. m = 11

C. m = 8.5

D. m = 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q3/5

983

QUESTION ID

S28-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. m = 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q3/5

True or false?

t = 4 is the solution of 500 − 20t = 420.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q4/5

984

QUESTION ID

S28-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q4/5

True or false?

h = 5 is the solution of 12h + 8 = 50.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q5/5

985

QUESTION ID

S28-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is h = 3.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q5/5

SET 28 • 8.F.B.4

FORWARD 1

Write functions for situations with fees, thresholds, and restricted domains • High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a

ANSWER KEY Reduced copy of the question page, with its answer

A phone plan costs $30 per month for up to 5 GB of data, plus $8 for each GB over 5.

Write a function C(g) for the monthly cost when g > 5 GB, and find C(9).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q1

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q1/5

987

QUESTION ID

S28-F1-Q1

TYPE

Short answer

CORRECT ANSWER

C(g) = 30 + 8(g − 5);
C(9) = $62

GRADING NOTE Equivalent forms such as C(g) = 8g − 10 are correct. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q1

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q1/5

A pool holds 12,000 gallons and drains at 400 gallons per hour.

Write a function V(t) for the water left after t hours, and give the domain that makes sense.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q2

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q2/5

988

QUESTION ID
S28-F1-Q2

TYPE
Short answer

CORRECT ANSWER
V(t) = 12,000 − 400t, for 0 ≤ t ≤ 30

GRADING NOTE Both the function and the domain are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q2

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q2/5

A taxi charges $3 for the first mile and $2 for each mile after the first.

Which function gives the cost for m miles, where m ≥ 1?

A. C(m) = 3 + 2(m − 1)

B. C(m) = 3 + 2m

C. C(m) = 5m

D. C(m) = 2 + 3(m − 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q3

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q3/5

989

QUESTION ID

S28-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A.

C(m) = 3 + 2(m − 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q3

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q3/5

True or false?

A gym charges a $50 fee and then $20 per month. For $C(m) = 50 + 20m$, $C(m + 1) - C(m) = 20$ for every m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q4

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q4/5

990

QUESTION ID

S28-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q4

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q4/5

True or false?

A store takes $10 off any purchase over $50. For a purchase of p dollars, where p > 50, the cost is c(p) = 10 − p.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q5

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q5/5

991

QUESTION ID

S28-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The cost is c(p) = p − 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q5

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q5/5

SET 28 • 8.F.B.4

FORWARD 2

Rearrange a formula to solve for a chosen quantity • High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Solve C = 40h + 65 for h.

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q1

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q1/5

993

QUESTION ID

S28-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(C − 65)/40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q1

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $F = \frac{9}{5}C + 32$ for C.

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q2

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q2/5

994

QUESTION ID

S28-F2-Q2

TYPE

Short answer

CORRECT ANSWER

(5/9)(F − 32)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q2

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q2/5

Solve $y = mx + b$ for b.

A. $b = \frac{y}{mx}$

B. $b = mx - y$

C. $b = y + mx$

D. $b = y - mx$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q3

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q3/5

995

QUESTION ID

S28-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. b = y − mx

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q3

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q3/5

True or false?

For $t \neq 0$, solving $d = rt$ for r gives $r = \frac{d}{t}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q4

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q4/5

996

QUESTION ID

S28-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q4

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

For $l \neq 0$, solving $A = lw$ for w gives $w = A - l$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q5

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q5/5

997

QUESTION ID
S28-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $w = A/l$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q5

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q5/5

SET 29 • 8.F.B.4 • FUNCTIONS

Interpret the rate of change and initial value of a linear function in context

MAIN	Grade 8 main question	**8.F.B.4**
BACKWARD 1	Explain points on the graph of a proportional relationship	**7.RP.A.2.d**
BACKWARD 2	Identify the independent and dependent variables	**6.EE.C.9**
FORWARD 1	Interpret the slope and intercept of a linear model when the units…	**HSS-ID.C.7**
FORWARD 2	Interpret the parameters of linear and exponential functions	**HSF-LE.B.5**

SET 29 • 8.F.B.4

MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Renting a bike costs C = 4h + 10 dollars for h hours.

What do the 4 and the 10 mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q1

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q1/5

1000

QUESTION ID

S29-M-Q1

TYPE

Short answer

CORRECT ANSWER

4: the cost goes up $4 for each hour. 10: a $10 fee (the cost at 0 hours).

GRADING NOTE Grade both numbers.

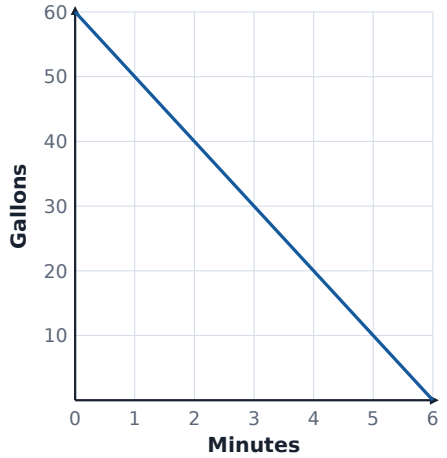
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q1

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q1/5

The graph shows the water in a tank.

What do the rate of change and the initial value mean?

Meaning: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q2

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q2/5

1001

QUESTION ID

S29-M-Q2

TYPE

Short answer

CORRECT ANSWER

The tank starts with 60 gallons (initial value) and drains 10 gallons per minute (rate of change −10).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q2

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q2/5

A plant's height is h = 0.5w + 12 centimeters after w weeks.

What does 0.5 mean?

A. The plant grows for 0.5 week.

B. The plant grows 0.5 cm each week.

C. The plant was 0.5 cm tall at the start.

D. The plant grows 12 cm each week.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q3

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q3/5

1002

QUESTION ID

S29-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. The plant grows 0.5 cm each week.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q3

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The table shows the money in a savings account.

What does the rate of change mean?

Weeks	0	4	8
Dollars	150	210	270

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q4

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q4/5

1003

QUESTION ID

S29-M-Q4

TYPE

Short answer

CORRECT ANSWER

$15 is added each week.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q4

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q4/5

True or false?

For C = 4h + 10, the initial value means the bike costs $4 before any hours.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q5

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q5/5

1004

QUESTION ID

S29-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The initial value is $10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q5

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q5/5

SET 29 • 8.F.B.4

BACKWARD 1

Explain points on the graph of a proportional relationship • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d

A proportional graph of gallons (x) and miles (y) passes through (1, 28).

What does this point mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q1/5

1006

QUESTION ID

S29-B1-Q1

TYPE

Short answer

CORRECT ANSWER

The car travels 28 miles on 1 gallon.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q1/5

A proportional graph shows hours worked (x) and pay (y).

What does the point (0, 0) mean?

A. 0 hours of work earn $0.

B. The worker starts with $0 per hour.

C. The pay is $0 per hour.

D. There is no graph.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q2/5

1007

QUESTION ID

S29-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. 0 hours of work earn $0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q2/5

A proportional graph of minutes (x) and laps (y) passes through (6, 3).

What does this point mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q3/5

1008

QUESTION ID

S29-B1-Q3

TYPE

Short answer

CORRECT ANSWER

3 laps are swum in 6 minutes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q3/5

True or false?

On the graph of a proportional relationship, the point (1, r) shows the unit rate r.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q4/5

1009

QUESTION ID

S29-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q4/5

True or false?

On a graph of tickets (x) and cost (y), the point (2, 18) means each ticket costs $18.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q5/5

1010

QUESTION ID

S29-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It means 2 tickets cost $18 ($9 each).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q5/5

SET 29 • 8.F.B.4

BACKWARD 2

Identify the independent and dependent variables • Grade 6 • Expressions & Equations • 6.EE.C.9

ANSWER KEY Reduced copy of the question page, with its answer

In C = 4h + 10, which variable is the independent variable?

Variable: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q1/5

1012

QUESTION ID

S29-B2-Q1

TYPE

Short answer

CORRECT ANSWER

h

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

"Pay depends on the hours worked." Which quantity is the dependent variable?

Quantity: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q2/5

1013

QUESTION ID

S29-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Pay

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q2/5

In d = 60t, what is d?

A. The rate

B. The independent variable

C. A constant

D. The dependent variable

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q3/5

1014

QUESTION ID

S29-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The dependent variable

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q3/5

True or false?

In a table of hours worked and pay, the hours are usually the independent variable.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q4/5

1015

QUESTION ID

S29-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q4/5

True or false?

In y = 3x + 2, x depends on y.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q5/5

1016

QUESTION ID

S29-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q5/5

SET 29 • 8.F.B.4

FORWARD 1

Interpret the slope and intercept of a linear model when the units or the starting point change • High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7

ANSWER KEY Reduced copy of the question page, with its answer

A model gives a city's population, in thousands, as $P = 2.4t + 85$, where t is the number of years after 2010.

What do 2.4 and 85 mean? Include units.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q1/5

1018

QUESTION ID

S29-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2.4: the population grows by about 2,400 people per year. 85: the model gives about 85,000 people in 2010.

GRADING NOTE Both meanings, with units, are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q1/5

A phone plan is modeled by C = 0.05m + 20 dollars for m minutes of calls.

Rewrite the model for h hours of calls, and say what the new slope means.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q2/5

1019

QUESTION ID

S29-F1-Q2

TYPE

Short answer

CORRECT ANSWER

C = 3h + 20; the plan costs $3 for each hour of calls.

GRADING NOTE Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-F1-Q2**

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q2/5

A model gives the air temperature as $T = -6.5h + 15$ °C at a height of h kilometers.

What does -6.5 mean?

A. The air is 6.5 km high.

B. The temperature drops about 6.5°C for each kilometer of height.

C. The temperature at the ground is −6.5°C.

D. The temperature rises 6.5°C for each kilometer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q3

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q3/5

1020

QUESTION ID

S29-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. The temperature drops about 6.5°C for each kilometer of height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q3

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q3/5

True or false?

In the model $y = 4(x - 2000) + 150$ for the sales y in year x, 150 is the predicted sales in the year 2000.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q4/5

1021

QUESTION ID

S29-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q4/5

True or false?

A model for x years since 2000 has the intercept 40. This means the model predicts 40 in the year 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q5/5

1022

QUESTION ID

S29-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE x = 0 is the year 2000, so 40 is the prediction for 2000.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q5/5

SET 29 • 8.F.B.4

FORWARD 2

Interpret the parameters of linear and exponential functions • High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5

P(t) = 2,000(1.05)^t gives an account balance after t years.

What do 2,000 and 1.05 mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q1

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q1/5

1024

QUESTION ID

S29-F2-Q1

TYPE

Short answer

CORRECT ANSWER

2,000 is the starting amount; 1.05 means the balance grows 5% each year.

GRADING NOTE Grade both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q1

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q1/5

V(t) = 25,000(0.85)^t gives a car's value after t years.

What is the yearly percent change?

Change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q2

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q2/5

1025

QUESTION ID

S29-F2-Q2

TYPE

Short answer

CORRECT ANSWER

A 15% decrease each year

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q2

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q2/5

A phone plan costs C(m) = 0.12m + 9.99 dollars for m minutes.

What does 9.99 represent?

A. A fixed monthly charge

B. The number of minutes

C. The total cost

D. The cost per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q3

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q3/5

1026

QUESTION ID

S29-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. A fixed monthly charge

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-F2-Q3**

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q3/5

True or false?

In f(t) = 100(2)^t, the quantity doubles each time t increases by 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q4

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q4/5

1027

QUESTION ID

S29-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q4

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q4/5

True or false?

In f(t) = 50(0.9)^t, the quantity grows by 90% each period.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q5

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q5/5

1028

QUESTION ID

S29-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It decreases by 10% each period.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q5

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q5/5

SET 30 • 8.F.B.5 • FUNCTIONS

Describe the relationship shown by a graph

MAIN	Grade 8 main question	**8.F.B.5**
BACKWARD 1	Interpret points on a graph of a situation	**5.G.A.2**
BACKWARD 2	Compare speeds as unit rates	**6.RP.A.3.b**
FORWARD 1	Interpret the key features of a nonlinear graph in context	**HSF-IF.B.4**
FORWARD 2	Graph and evaluate piecewise-defined functions	**HSF-IF.C.7.b**

SET 30 • 8.F.B.5

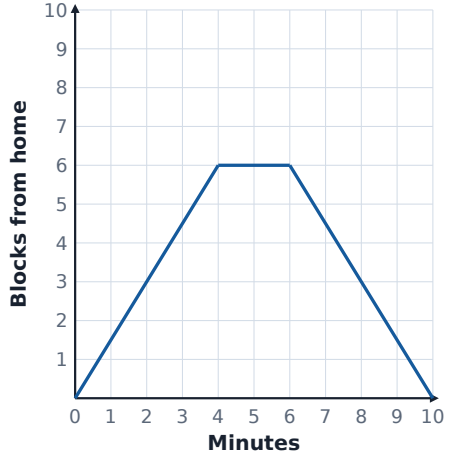
MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.B.5

The graph shows Ana's walk.

Describe what happens from 4 to 6 minutes.

Description: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q1

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q1/5

1031

QUESTION ID

S30-M-Q1

TYPE

Short answer

CORRECT ANSWER

She stays 6 blocks from home; she stops.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q1

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q1/5

The graph shows Ana's walk.

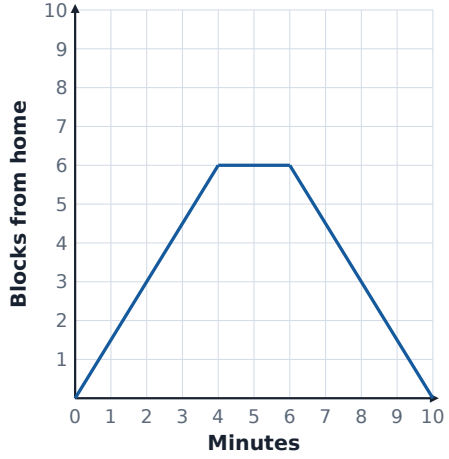
During which interval is Ana moving toward home?

A. 0 to 4 minutes

B. 6 to 10 minutes

C. 4 to 6 minutes

D. 0 to 6 minutes



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q2

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q2/5

1032

QUESTION ID

S30-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 6 to 10 minutes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

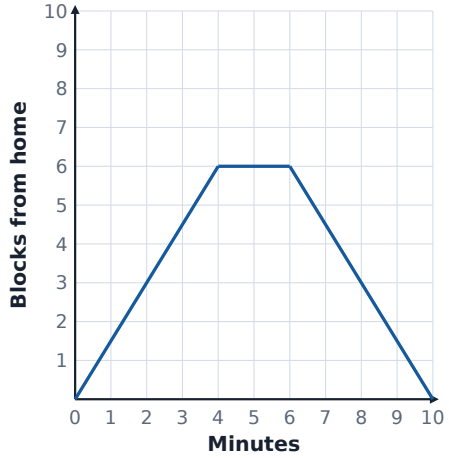
ID S30-M-Q2

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q2/5

The graph shows Ana's walk.

Is the relationship linear or nonlinear over the whole walk? Explain.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q3

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q3/5

1033

QUESTION ID

S30-M-Q3

TYPE

Short answer

CORRECT ANSWER

Nonlinear. The graph is not one straight line; the rate of change is different on different parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

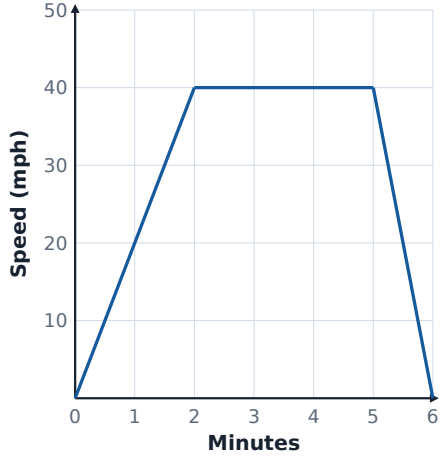
ID S30-M-Q3

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q3/5

The graph shows a car's speed.

Describe how the speed changes over the 6 minutes.

Description: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q4

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q4/5

1034

QUESTION ID

S30-M-Q4

TYPE

Short answer

CORRECT ANSWER

The car speeds up for 2 minutes, travels at a steady 40 mph until minute 5, then slows to a stop.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q4

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q4/5

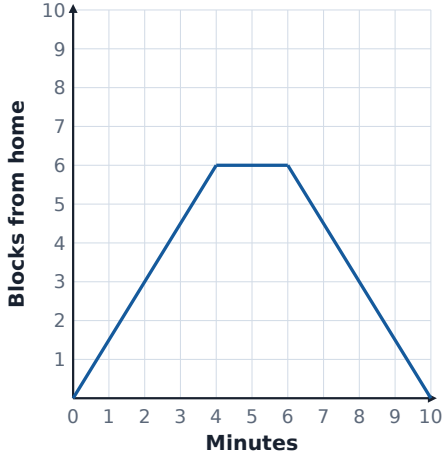
ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In the graph of Ana's walk, she walks faster going home than going away from home.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q5

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q5/5

1035

QUESTION ID
S30-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Both trips cover 6 blocks in 4 minutes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q5

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q5/5

SET 30 • 8.F.B.5

BACKWARD 1

Interpret points on a graph of a situation • Grade 5 • Geometry • 5.G.A.2

A graph shows minutes (x) and blocks from home (y).
What does the point (3, 5) mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q1/5

1037

QUESTION ID

S30-B1-Q1

TYPE

Short answer

CORRECT ANSWER

After 3 minutes, the walker is 5 blocks from home.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q1/5

A graph shows days (x) and plant height in centimeters (y). What does the point (10, 4) mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q2/5

1038

QUESTION ID

S30-B1-Q2

TYPE

Short answer

CORRECT ANSWER

After 10 days, the plant is 4 cm tall.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q2/5

A graph shows hours (x) and miles driven (y).

Which point shows 120 miles after 2 hours?

A. (2, 120)

B. (60, 2)

C. (2, 60)

D. (120, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q3/5

1039

QUESTION ID

S30-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (2, 120)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q3/5

True or false?

On a graph of time (x) and distance (y), the point (0, 0) can mean the start of a trip.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q4/5

1040

QUESTION ID

S30-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q4/5

True or false?

The point (4, 1) on a graph of hours (x) and miles (y) means 4 miles in 1 hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q5/5

1041

QUESTION ID

S30-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q5/5

SET 30 • 8.F.B.5

BACKWARD 2

Compare speeds as unit rates • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b

Ana walks 6 blocks in 4 minutes at a steady pace.

What is her speed in blocks per minute?

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q1/5

1043

QUESTION ID

S30-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1.5 blocks per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q1/5

Which is faster: 12 miles in 3 hours, or 10 miles in 2 hours?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q2/5

1044

QUESTION ID

S30-B2-Q2

TYPE

Short answer

CORRECT ANSWER

10 miles in 2 hours (5 mph, compared with 4 mph)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q2/5

Which rate is the greatest?

A. 60 miles in 1 hour

B. 100 miles in 2 hours

C. 20 miles in 0.5 hour

D. 150 miles in 3 hours

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q3/5

1045

QUESTION ID

S30-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 60 miles in 1 hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q3/5

True or false?

Walking 3 miles in 1 hour is faster than walking 5 miles in 2 hours.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q4/5

1046

QUESTION ID

S30-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q4/5

True or false?

Running 8 km in 40 minutes is the same rate as running 2 km in 20 minutes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q5/5

1047

QUESTION ID

S30-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q5/5

SET 30 • 8.F.B.5

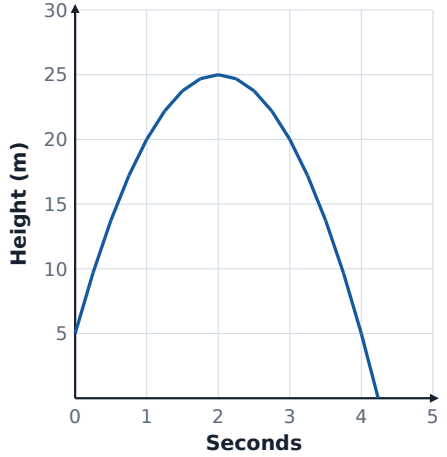
FORWARD 1

Interpret the key features of a nonlinear graph in context • High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4

ANSWER KEY Reduced copy of the question page, with its answer

The graph shows a ball's height after it is thrown.

What is the greatest height, and when does it happen?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q1/5

1049

QUESTION ID

S30-F1-Q1

TYPE

Short answer

CORRECT ANSWER

25 m, at 2 seconds

GRADING NOTE Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q1

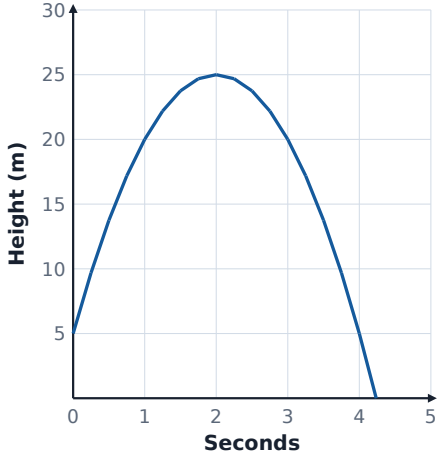
High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph shows a ball's height after it is thrown.

On what interval is the height increasing, and on what interval is it decreasing?

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q2/5

1050

QUESTION ID

S30-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Increasing from 0 to 2 seconds; decreasing from 2 seconds until it lands (a little after 4 seconds).

GRADING NOTE Both intervals are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph shows a ball's height after it is thrown.

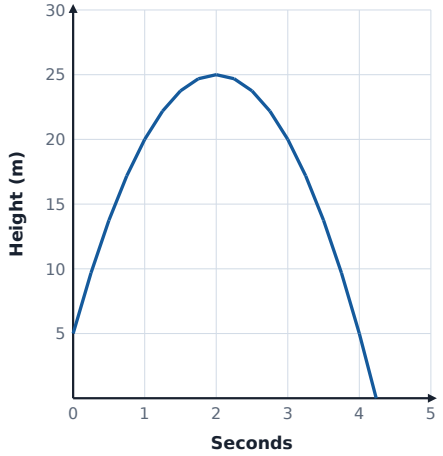
What does the point where the graph meets the vertical axis mean?

A. The ball lands after 5 seconds.

B. The ball was 5 m above the ground when it was thrown.

C. The ball rises 5 m each second.

D. The ball's greatest height is 5 m.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q3/5

1051

QUESTION ID

S30-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. The ball was 5 m above the ground when it was thrown.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q3

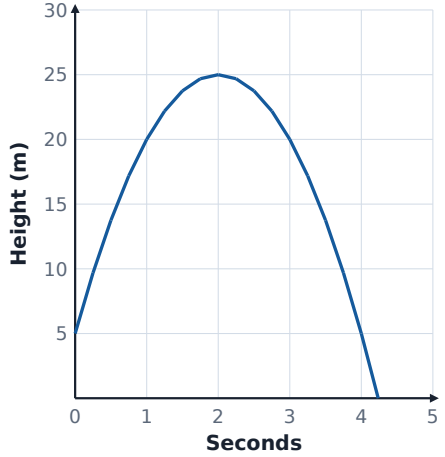
High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q3/5

True or false?

In the graph, the ball hits the ground a little after 4 seconds.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q4/5

1052

QUESTION ID

S30-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q4/5

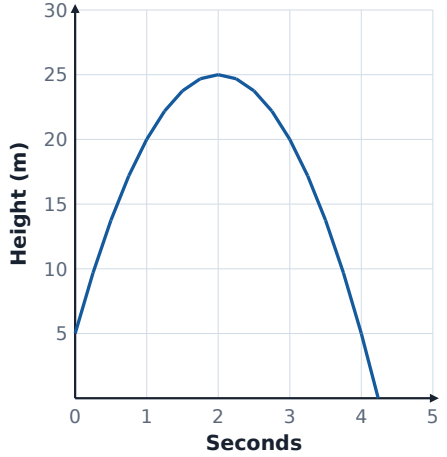
ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In the graph, the ball's height changes by the same amount each second.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q5/5

1053

QUESTION ID
S30-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The graph is curved, so the rate of change is not constant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q5/5

SET 30 • 8.F.B.5

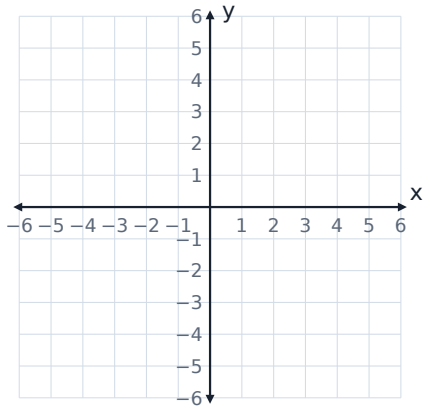
FORWARD 2

Graph and evaluate piecewise-defined functions • High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b

Graph the function on the coordinate plane.

$f(x) = x + 2$ for $x < 0$

$f(x) = 2$ for $x \geq 0$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q1/5

1055

QUESTION ID

S30-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

The line $y = x + 2$ for $x < 0$ (an open end at (0, 2) is acceptable) and the horizontal line $y = 2$ for $x \geq 0$, with a closed point at (0, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q1/5

$f(x) = 3$ for $x \leq 2$, and $f(x) = x + 1$ for $x > 2$.

Find $f(5)$.

$f(5) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q2/5

1056

QUESTION ID

S30-F2-Q2

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q2/5

A taxi charges $3 for the first mile and $2 for each mile after that.

Which function gives the cost for m miles?

A. C(m) = 2m + 3 for all m

B. C(m) = 3m for all m

C. C(m) = 3 for 0 < m ≤ 1; C(m) = 3 + 2(m − 1) for m > 1

D. C(m) = 3 + 2m for m ≤ 1 only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q3/5

1057

QUESTION ID

S30-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. C(m) = 3 for 0 < m ≤ 1; C(m) = 3 + 2(m − 1) for m > 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q3/5

True or false?

For f(x) = |x|, f(−4) = 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q4/5

1058

QUESTION ID

S30-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q4/5

True or false?

A piecewise-defined function can never have an unbroken graph.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q5/5

1059

QUESTION ID

S30-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE For example, f(x) = |x| is piecewise-defined and unbroken.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q5/5

SET 31 • 8.F.B.5 • FUNCTIONS

Sketch a graph from a description of a relationship

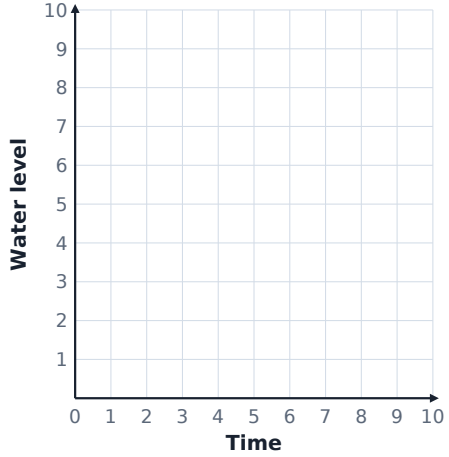
MAIN	Grade 8 main question	**8.F.B.5**
BACKWARD 1	Graph a relationship from a table	**6.EE.C.9**
BACKWARD 2	Plot points that represent a situation	**5.G.A.2**
FORWARD 1	Sketch graphs that show given key features	**HSF-IF.B.4**
FORWARD 2	Graph exponential functions and describe their end behavior	**HSF-IF.C.7.e**

SET 31 • 8.F.B.5

MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.B.5

Sketch a graph of the water level in a bathtub over time: it fills at a steady rate, stays the same while someone bathes, and then drains quickly.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q1

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q1/5

1062

QUESTION ID

S31-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

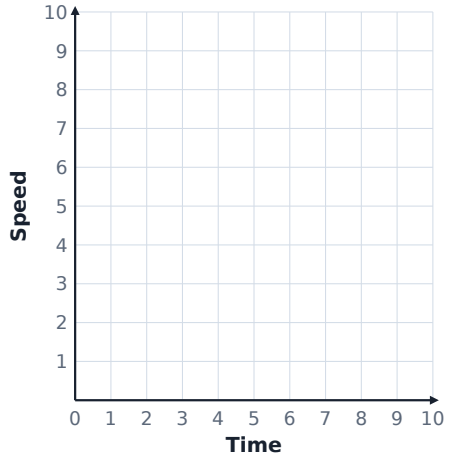
A rising straight segment from 0, then a horizontal segment, then a steeper falling segment back to 0. Any lengths are accepted if the order and steepness are right.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q1

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q1/5

Sketch a graph of a car's speed over time: it speeds up, travels at a steady speed, and then slows down to a stop.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q2

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q2/5

1063

QUESTION ID

S31-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

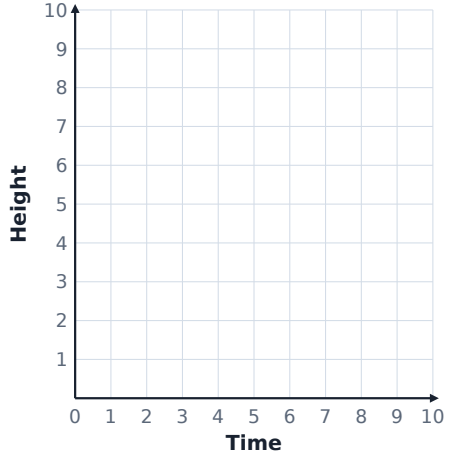
A rising segment from 0, a horizontal segment, then a falling segment back to 0. Any lengths are accepted if the order is right.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q2

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q2/5

Sketch the height of a ball thrown straight up over time: it rises, slows down, reaches a highest point, and falls back to the ground.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q3

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q3/5

1064

QUESTION ID

S31-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

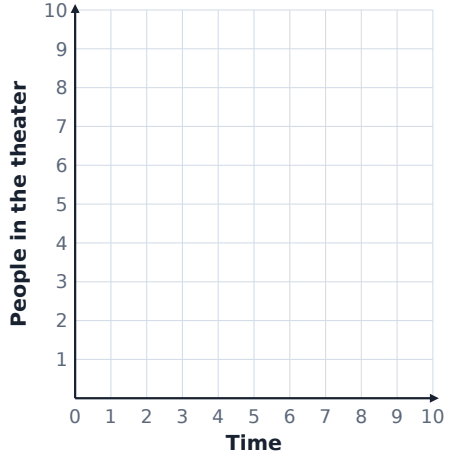
A curve that rises, levels off at one highest point, and falls back to height 0, like an upside-down U. Starting at ground level or at the throwing height is accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q3

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q3/5

Sketch the number of people in a theater from 6:00 to 10:00. It fills quickly before a 7:00 show, stays full, and empties after the show ends at 9:30.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q4

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q4/5

1065

QUESTION ID

S31-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

A steep rise ending at about 7:00, a horizontal segment from 7:00 to 9:30, then a steep fall after 9:30 back to about 0 by 10:00. The rise, level part, and fall must be placed at these clock times; exact heights are not required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q4

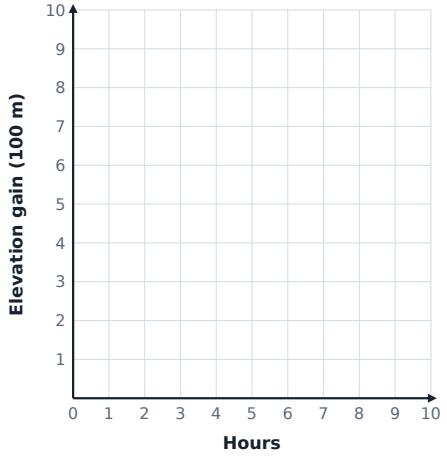
Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Sketch a graph of a hiker's elevation gain over time: she gains elevation at a steady 200 meters per hour for 2 hours, then at a steady 80 meters per hour for 1 hour, then rests for 1 hour.

Is elevation gain a linear function of time for the whole hike? Explain.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q5

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q5/5

1066

QUESTION ID

S31-M-Q5

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A rising segment from (0, 0) to (2, 4) (400 m), a less steep rising segment to (3, 4.8) (480 m), and a horizontal segment to (4, 4.8)

Written: No. The rate of change is different on each part (200, then 80, then 0 meters per hour), so the whole graph is not one line.

GRADING NOTE Grade both: the three-part graph with the right order and steepness, and "no" with a reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q5

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q5/5

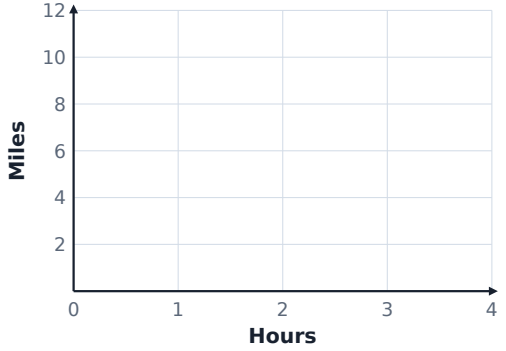
SET 31 • 8.F.B.5

BACKWARD 1

Graph a relationship from a table • Grade 6 • Expressions & Equations • 6.EE.C.9

Graph the pairs in the table.

Hours	0	1	2	3
Miles	0	4	8	12



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q1/5

1068

QUESTION ID

S31-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Points at (0, 0), (1, 4), (2, 8), and (3, 12)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q1/5

A hiker walks 4 miles each hour, so miles = 4 × hours.

How many miles does she walk in 2.5 hours?

Miles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q2/5

1069

QUESTION ID

S31-B1-Q2

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q2/5

Which table shows y increasing by the same amount each time?

A. x: 0, 1, 2 and y: 0, 3, 4

B. x: 0, 1, 2 and y: 5, 5, 6

C. x: 0, 1, 2 and y: 6, 3, 9

D. x: 0, 1, 2 and y: 0, 3, 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q3/5

1070

QUESTION ID

S31-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x: 0, 1, 2 and y: 0, 3, 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q3/5

True or false?

A table where y is 5 for every x shows y not changing.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q4/5

1071

QUESTION ID

S31-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q4/5

True or false?

A table with y-values 2, 4, 3 shows y always increasing.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q5/5

1072

QUESTION ID

S31-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q5/5

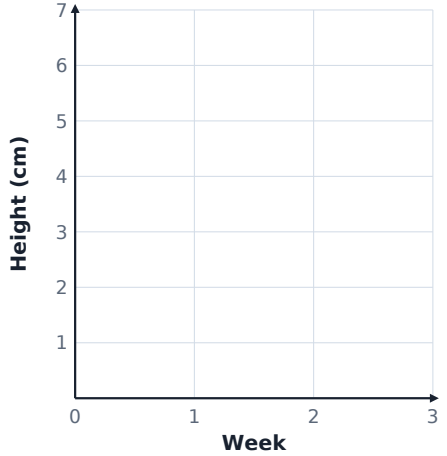
SET 31 • 8.F.B.5

BACKWARD 2

Plot points that represent a situation • Grade 5 • Geometry • 5.G.A.2

A plant is 2 cm tall at week 0, 4 cm at week 1, and 6 cm at week 2.

Plot the points (week, height).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q1/5

1074

QUESTION ID

S31-B2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Points at (0, 2), (1, 4), and (2, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q1/5

A plant is 2 cm tall at week 0 and grows 2 cm each week.

Which point shows its height at week 3?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q2/5

1075

QUESTION ID

S31-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(3, 8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q2/5

On a graph of weeks (x) and plant height in cm (y), what does the point (2, 6) mean?

A. The plant grows 2 cm.

B. At week 6 the plant is 2 cm tall.

C. The plant grows 6 cm each week.

D. At week 2 the plant is 6 cm tall.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q3/5

1076

QUESTION ID

S31-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. At week 2 the plant is 6 cm tall.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q3/5

True or false?

On a graph of weeks (x) and height (y), the point (0, 2) shows the height at the start.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q4/5

1077

QUESTION ID

S31-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q4/5

True or false?

On a graph of weeks (x) and height in cm (y), the point (1, 4) means the plant is 1 cm tall at week 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q5/5

1078

QUESTION ID

S31-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q5/5

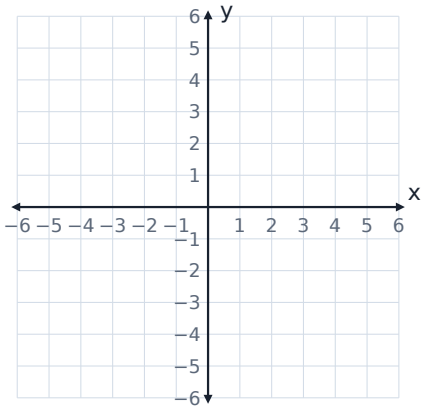
SET 31 • 8.F.B.5

FORWARD 1

Sketch graphs that show given key features • High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Sketch a function that increases for $0 < x < 3$, has a maximum value of 5 at $x = 3$, and decreases for $x > 3$.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q1/5

1080

QUESTION ID

S31-F1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

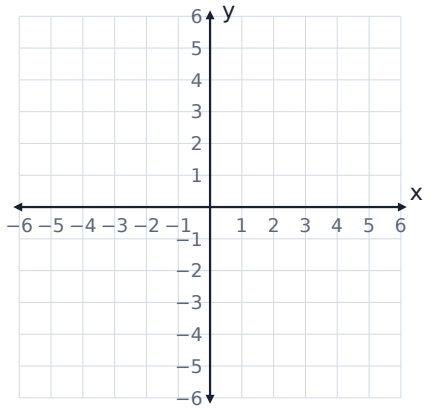
Any graph that rises up to the point (3, 5) and falls after it, with 5 as its highest value

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Sketch a function whose graph crosses the y-axis at 2 and that increases for all x shown.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q2/5

1081

QUESTION ID

S31-F1-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Any always-increasing graph through (0, 2); a line or a curve is accepted

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q2/5

Which description matches $y = -(x - 2)^2 + 4$?

A. A maximum of 4 at $x = 2$; increasing for $x < 2$

B. A minimum of 4 at $x = 2$

C. A maximum of 2 at $x = 4$

D. Increasing for every x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q3/5

1082

QUESTION ID

S31-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. A maximum of 4 at x = 2; increasing for x < 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q3/5

True or false?

The function $f(x) = -(x - 3)^2 + 5$ has its maximum at $x = 3$ and decreases for every x greater than 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q4/5

1083

QUESTION ID

S31-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q4/5

True or false?

A function that is always decreasing can have its maximum in the middle of its domain.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q5/5

1084

QUESTION ID

S31-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q5

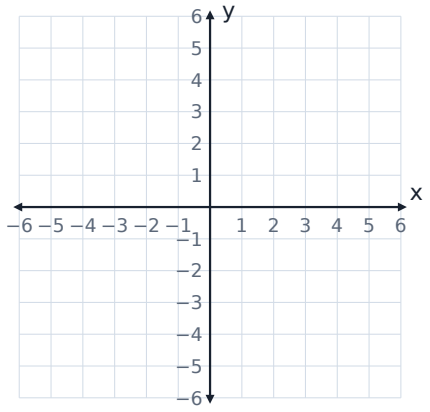
High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q5/5

SET 31 • 8.F.B.5

FORWARD 2

Graph exponential functions and describe their end behavior • High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e

Graph y = 2^x for x from −2 to 2 on the coordinate plane.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q1/5

1086

QUESTION ID

S31-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

The points (−2, 1/4), (−1, 1/2), (0, 1), (1, 2), and (2, 4) joined by a smooth increasing curve that gets close to the x-axis on the left

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q1/5

What is the y-intercept of the graph of y = 5(0.5)^x?

y-intercept: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q2/5

1087

QUESTION ID

S31-F2-Q2

TYPE

Short answer

CORRECT ANSWER

(0, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q2/5

For y = 3^x, what happens to y as x decreases without bound?

A. y gets closer to 3.

B. y decreases without bound.

C. y increases.

D. y gets closer to 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q3/5

1088

QUESTION ID

S31-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. y gets closer to 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q3/5

True or false?

The graph of y = 2^x never touches the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q4/5

1089

QUESTION ID

S31-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q4/5

True or false?

The function y = (0.5)^x is increasing.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q5/5

1090

QUESTION ID

S31-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q5/5

SET 32 • 8.G.A.1.a • GEOMETRY

Verify that rigid motions take lines to lines and segments to segments of the same length

MAIN	Grade 8 main question	**8.G.A.1.a**
BACKWARD 1	Reflect points across the axes	**6.NS.C.6.b**
BACKWARD 2	Find the distance between points with the same x- or y-coordinate	**6.NS.C.8**
FORWARD 1	Define rotations, reflections, and translations precisely	**HSG-CO.A.4**
FORWARD 2	Use rigid motions to decide whether figures are congruent	**HSG-CO.B.6**

SET 32 • 8.G.A.1.a

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.1.a

Segment AB has endpoints A(1, 2) and B(5, 2). It is translated 3 units down.

What are the endpoints of the image, and how long is the image?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q1

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q1/5

1093

QUESTION ID

S32-M-Q1

TYPE

Short answer

CORRECT ANSWER

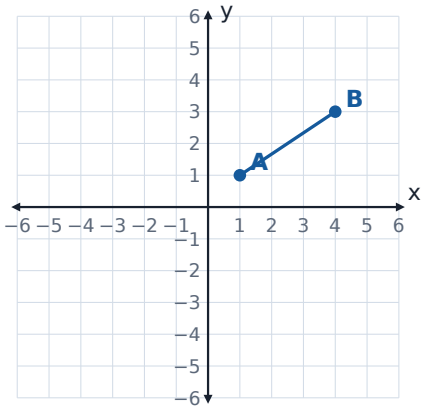
A'(1, −1) and B'(5, −1); it is 4 units long, the same as AB.

GRADING NOTE Grade both the endpoints and the length.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q1

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q1/5

Reflect segment AB across the y-axis. Draw the image A'B'. Then find the lengths of AB and A'B' and compare them.



Lengths: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q2

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q2/5

1094

QUESTION ID

S32-M-Q2

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: The segment from A'(−1, 1) to B'(−4, 3)

Written: AB = A'B' = √13 units (about 3.6 units); the reflection kept the length.

GRADING NOTE Grade both: the image segment, and the two equal lengths found from the grid (run 3, rise 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q2

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q2/5

A(1, 2) and B(4, 6) are reflected across the x-axis to A' and B'.

How long is A'B'?

A. 4 units

B. 5 units

C. 3 units

D. 8 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q3

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q3/5

1095

QUESTION ID

S32-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 5 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q3

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q3/5

True or false?

The segment from (0, 0) to (3, 4) is rotated 90° counterclockwise about the origin to the segment from (0, 0) to (−4, 3). Both segments are 5 units long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q4

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q4/5

1096

QUESTION ID

S32-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q4

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q4/5

True or false?

Segment PQ from P(2, 1) to Q(2, 5) is translated 3 units left. Its image is 7 units long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q5

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q5/5

1097

QUESTION ID

S32-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The image runs from (−1, 1) to (−1, 5), so it is 4 units long, the same as PQ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q5

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q5/5

SET 32 • 8.G.A.1.a

BACKWARD 1

Reflect points across the axes • Grade 6 • The Number System • 6.NS.C.6.b

The point (3, −4) is reflected across the x-axis.

What are the coordinates of the image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q1/5

1099

QUESTION ID

S32-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q1/5

The point (−2, 5) is reflected across the y-axis.

What are the coordinates of the image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q2/5

1100

QUESTION ID

S32-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q2/5

Which point is the reflection of (6, 1) across the y-axis?

A. (1, 6)

B. (6, −1)

C. (−6, −1)

D. (−6, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q3/5

1101

QUESTION ID

S32-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (−6, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q3/5

True or false?

The reflection of (−3, −3) across the x-axis is (−3, 3).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q4/5

1102

QUESTION ID

S32-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q4/5

True or false?

The reflection of (4, 2) across the x-axis is (−4, 2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q5/5

1103

QUESTION ID

S32-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q5/5

SET 32 • 8.G.A.1.a

BACKWARD 2

Find the distance between points with the same x- or y-coordinate • Grade 6 • The Number System • 6.NS.C.8

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance between (1, 2) and (5, 2).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q1/5

1105

QUESTION ID

S32-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q1/5

Find the distance between (1, −1) and (5, −1).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q2/5

1106

QUESTION ID

S32-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q2/5

Which two points are 6 units apart?

A. (−2, 3) and (2, 3)

B. (0, 1) and (0, 5)

C. (3, −2) and (3, 2)

D. (−2, 3) and (4, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q3/5

1107

QUESTION ID

S32-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (−2, 3) and (4, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q3/5

True or false?

(0, −3) and (0, 4) are 7 units apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q4/5

1108

QUESTION ID

S32-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

(−5, 1) and (2, 1) are 3 units apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q5/5

1109

QUESTION ID

S32-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are 7 units apart.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q5/5

SET 32 • 8.G.A.1.a

FORWARD 1

Define rotations, reflections, and translations precisely • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4

A reflection across line m maps a point P that is not on m to the point P'.

How is line m related to segment PP'?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q1/5

1111

QUESTION ID

S32-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Line m is the perpendicular bisector of PP'.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q1/5

A rotation of 90° about point O maps a point P (P ≠ O) to P'.

What is true about OP, OP', and angle POP'?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q2/5

1112

QUESTION ID

S32-F1-Q2

TYPE

Short answer

CORRECT ANSWER

OP = OP', and angle POP' measures 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q2/5

In a translation by a nonzero distance, the segments that join each point to its image are...

A. perpendicular to each other

B. all through one point

C. different lengths

D. parallel and the same length

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q3/5

1113

QUESTION ID

S32-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. parallel and the same length

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q3/5

True or false?

A rotation keeps every point the same distance from the center of rotation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q4/5

1114

QUESTION ID

S32-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q4/5

True or false?

In a reflection, a point (not on the line) and its image are on the same side of the line of reflection.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q5/5

1115

QUESTION ID

S32-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q5/5

SET 32 • 8.G.A.1.a

FORWARD 2

Use rigid motions to decide whether figures are congruent • High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6

Triangle ABC is mapped onto triangle DEF by a rotation followed by a translation.

What can you conclude about the triangles?

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q1/5

1117

QUESTION ID

S32-F2-Q1

TYPE

Short answer

CORRECT ANSWER

They are congruent; their corresponding sides and angles are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q1/5

Explain why two figures are congruent when a sequence of rigid motions maps one onto the other.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q2/5

1118

QUESTION ID

S32-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Rigid motions keep all lengths and angle measures, so every part of one figure matches the corresponding part of the other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q2/5

Which transformation does NOT always map a figure onto a congruent figure?

A. A translation

B. A rotation

C. A dilation with a scale factor of 2

D. A reflection

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q3/5

1119

QUESTION ID

S32-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. A dilation with a scale factor of 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q3/5

True or false?

If a translation maps segment AB onto segment CD, then AB = CD.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q4/5

1120

QUESTION ID

S32-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q4/5

True or false?

Two triangles with the same area must be congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q5/5

1121

QUESTION ID

S32-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q5/5

SET 33 • 8.G.A.1.b • GEOMETRY

Verify that rigid motions take angles to angles of the same measure

MAIN	Grade 8 main question	**8.G.A.1.b**
BACKWARD 1	Find unknown angles with supplementary, complementary, and v…	**7.G.B.5**
BACKWARD 2	Classify triangles by their angles	**4.G.A.2**
FORWARD 1	Specify a sequence of transformations that maps a figure onto a…	**HSG-CO.A.5**
FORWARD 2	Prove theorems about parallelograms — nearest related (no direc…	**HSG-CO.C.11**

SET 33 • 8.G.A.1.b

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.1.b

Angle ABC has A(4, 0), B(0, 0), and C(3, 3). It is reflected across the y-axis.

Find A', B', and C', and explain why angle A'B'C' has the same measure as angle ABC.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q1

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q1/5

1124

QUESTION ID

S33-M-Q1

TYPE

Short answer

CORRECT ANSWER

A'(−4, 0), B'(0, 0), C'(−3, 3). Ray BC and ray B'C' each run along a diagonal of the grid squares, so each angle is half of a right angle: both measure 45°.

GRADING NOTE Grade both: the three image points, and a check that both angles measure 45°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q1

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The right triangle with vertices (0, 0), (4, 0), and (0, 4) is translated 2 units right and 1 unit up.

Find the image vertices, and explain why the angle at the image of (0, 0) is still a right angle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q2

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q2/5

1125

QUESTION ID

S33-M-Q2

TYPE

Short answer

CORRECT ANSWER

(2, 1), (6, 1), and (2, 5). The sides from (2, 1) are horizontal and vertical, so they are perpendicular: the angle is 90°.

GRADING NOTE Grade both: the vertices and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q2

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q2/5

Angle DEF has D(0, 3), E(0, 0), and F(3, 0). It is reflected across the line y = x.

What does the image angle measure?

A. 90°

B. 30°

C. 180°

D. 45°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q3

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q3/5

1126

QUESTION ID

S33-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 90°

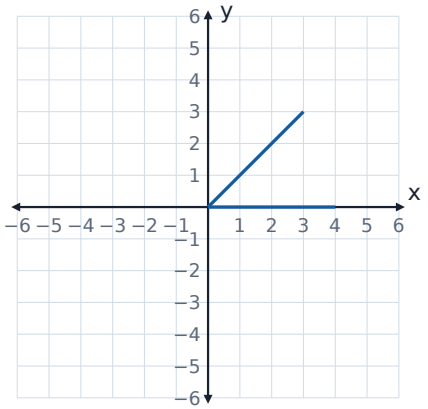
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q3

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The angle has its vertex at the origin. Rotate it 90° counterclockwise about the origin and draw the image. Then compare the measures of the two angles.



Measures: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q4

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q4/5

1127

QUESTION ID

S33-M-Q4

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: An angle with its vertex at the origin and rays through (0, 4) and (−3, 3)

Written: Both angles measure 45°.

GRADING NOTE Grade both: the rotated angle, and the two equal measures (from a protractor or the grid diagonals).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q4

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q4/5

True or false?

Angle ABC with A(2, 0), B(0, 0), and C(0, 2) is translated 4 units right. The image angle measures 45°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q5

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q5/5

1128

QUESTION ID

S33-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The image is still a right angle (90°).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q5

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q5/5

SET 33 • 8.G.A.1.b

BACKWARD 1

Find unknown angles with supplementary, complementary, and vertical angles • Grade 7 • Geometry • 7.G.B.5

Two angles form a straight line. One measures 65°.

What does the other measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q1/5

1130

QUESTION ID

S33-B1-Q1

TYPE

Short answer

CORRECT ANSWER

115°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q1/5

Two lines intersect. One angle measures 40°.

What does the vertical angle measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q2/5

1131

QUESTION ID

S33-B1-Q2

TYPE

Short answer

CORRECT ANSWER

40°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q2/5

What is the complement of a 25° angle?

A. 65°

B. 75°

C. 155°

D. 25°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q3/5

1132

QUESTION ID

S33-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 65°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q3/5

True or false?

Vertical angles have equal measures.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q4/5

1133

QUESTION ID

S33-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q4/5

True or false?

Supplementary angles add to 90°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q5/5

1134

QUESTION ID

S33-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q5/5

SET 33 • 8.G.A.1.b

BACKWARD 2

Classify triangles by their angles • Grade 4 • Geometry • 4.G.A.2

A triangle has angles of 40°, 60°, and 80°.

Classify it by its angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q1/5

1136

QUESTION ID

S33-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Acute triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q1/5

A triangle has a 90° angle.

Classify it by its angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q2/5

1137

QUESTION ID

S33-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Right triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q2/5

Which angle measures make an obtuse triangle?

A. 60°, 60°, 60°

B. 30°, 40°, 110°

C. 45°, 45°, 90°

D. 50°, 60°, 70°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q3/5

1138

QUESTION ID

S33-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 30°, 40°, 110°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q3/5

True or false?

An acute triangle has three angles that each measure less than 90°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q4/5

1139

QUESTION ID

S33-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q4/5

True or false?

A right triangle can have an obtuse angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q5/5

1140

QUESTION ID

S33-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q5/5

SET 33 • 8.G.A.1.b

FORWARD 1

Specify a sequence of transformations that maps a figure onto another • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle A(1, 1), B(3, 1), C(1, 4) is mapped onto A'(−1, −1), B'(−3, −1), C'(−1, −4).

Describe a transformation or sequence of transformations that does this.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q1/5

1142

QUESTION ID

S33-F1-Q1

TYPE

Short answer

CORRECT ANSWER

A 180° rotation about the origin.

GRADING NOTE A reflection across the x-axis followed by a reflection across the y-axis (or the reverse) is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q1/5

Which single transformation maps (2, 3) to (2, −3) and (5, 1) to (5, −1)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q2/5

1143

QUESTION ID

S33-F1-Q2

TYPE

Short answer

CORRECT ANSWER

A reflection across the x-axis

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q2/5

Which rotation about the origin maps (4, 0) to (0, 4)?

A. 270° counterclockwise

B. 180°

C. 90° clockwise

D. 90° counterclockwise

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q3/5

1144

QUESTION ID

S33-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 90° counterclockwise

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q3/5

True or false?

Reflecting a figure across a line and then reflecting the image across the same line gives back the original figure.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q4/5

1145

QUESTION ID

S33-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q4/5

True or false?

A translation 3 units right followed by a translation 3 units left moves the figure.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q5/5

1146

QUESTION ID
S33-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The figure returns to where it started.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q5/5

SET 33 • 8.G.A.1.b

FORWARD 2

Prove theorems about parallelograms • High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11

⚑ No high-school standard directly continues this skill at this step. This branch uses the nearest related high-school standard (placed at Grade 10 in this bank).

ANSWER KEY Reduced copy of the question page, with its answer

ABCD is a parallelogram. Prove that angle A and angle B are supplementary. Give a reason for each step.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q1/5 • NEAREST RELATED

1148

QUESTION ID

S33-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

Angle A + angle B = 180°

GRADING NOTE REQUIRED WORK: A valid chain of reasons. Other valid orders are accepted. A conclusion without reasons is not full credit. Example: AD ∥ BC (definition of a parallelogram), and AB is a transversal; angles A and B are same-side interior angles, so they are supplementary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q1/5 • NEAREST RELATED

In parallelogram ABCD, angle A = 70°.

Find the other three angles and give a reason for each.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q2/5 • NEAREST RELATED

1149

QUESTION ID

S33-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Angle B = 110° and angle D = 110° (consecutive angles are supplementary); angle C = 70° (opposite angles are congruent).

GRADING NOTE Grade the angles and the reasons.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F2-Q2**

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q2/5 • NEAREST RELATED

Which statement about every parallelogram can be proved?

A. All of its angles are right angles.

B. Its diagonals are congruent.

C. Its diagonals bisect each other.

D. Its diagonals are perpendicular.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q3/5 • NEAREST RELATED

1150

QUESTION ID

S33-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Its diagonals bisect each other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

Opposite sides of a parallelogram are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q4/5 • NEAREST RELATED

1151

QUESTION ID

S33-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

A quadrilateral with one pair of parallel sides must be a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q5/5 • NEAREST RELATED

1152

QUESTION ID

S33-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q5/5 • NEAREST RELATED

SET 34 • 8.G.A.1.c • GEOMETRY

Verify that rigid motions take parallel lines to parallel lines

MAIN	Grade 8 main question	**8.G.A.1.c**
BACKWARD 1	Identify parallel lines and segments	**4.G.A.1**
BACKWARD 2	Draw polygons in the coordinate plane and find side lengths	**6.G.A.3**
FORWARD 1	Prove theorems about angles formed by parallel lines	**HSG-CO.C.9**
FORWARD 2	Use coordinates to prove simple geometric theorems	**HSG-GPE.B.4**

SET 34 • 8.G.A.1.c

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.1.c

The parallel lines y = 2x and y = 2x + 3 are reflected across the x-axis.

Write the equations of the images and check whether the images are parallel.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q1

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q1/5

1155

QUESTION ID

S34-M-Q1

TYPE

Short answer

CORRECT ANSWER

y = −2x and y = −2x − 3; both have slope −2 (and different y-intercepts), so the images are parallel.

GRADING NOTE Grade both: the two equations and the slope check.

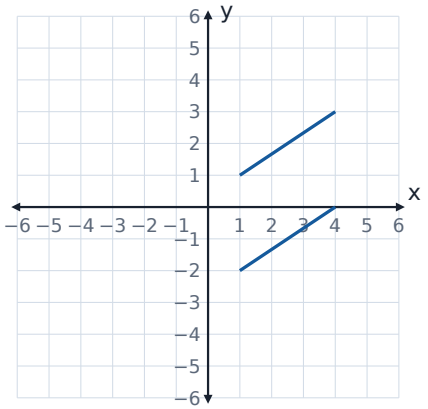
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q1

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The two segments are parallel. Translate both segments 4 units to the left and draw the images. Then find the slope of each image and compare.

Slopes: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q2

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q2/5

1156

GRADING NOTE Grade both: the two image segments, and the two equal slopes.

QUESTION ID

S34-M-Q2

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: Segments from (−3, 1) to (0, 3) and from (−3, −2) to (0, 0)

Written: Both images have slope 2/3, so they are still parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q2

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q2/5

The parallel segments from (0, 0) to (2, 1) and from (0, 3) to (2, 4) are rotated 90° counterclockwise about the origin.

What are the slopes of the images?

A. -2 and $\frac{1}{2}$

B. Both 2

C. Both -2

D. Both $\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q3

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q3/5

1157

QUESTION ID

S34-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Both −2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q3

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q3/5

The line y = 2x + 1 is translated 3 units up.

What is the equation of the image? Is the image parallel to the original line?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q4

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q4/5

1158

QUESTION ID

S34-M-Q4

TYPE

Short answer

CORRECT ANSWER

y = 2x + 4; yes, the slope is still 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q4

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q4/5

True or false?

The parallel segments from (1, 1) to (3, 2) and from (1, 4) to (3, 5) are translated 2 units down. The images have different slopes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q5

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q5/5

1159

QUESTION ID

S34-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Both images have slope 1/2, so they are still parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q5

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q5/5

SET 34 • 8.G.A.1.c

BACKWARD 1

Identify parallel lines and segments • Grade 4 • Geometry • 4.G.A.1

Name a quadrilateral that has two pairs of parallel sides.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q1/5

1161

QUESTION ID

S34-B1-Q1

TYPE

Short answer

CORRECT ANSWER

A rectangle

GRADING NOTE Any parallelogram (square, rhombus, rectangle) is correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q1/5

How many pairs of parallel sides does a trapezoid that is not a parallelogram have?

Pairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q2/5

1162

QUESTION ID

S34-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q2/5

Which capital letter contains a pair of parallel segments?

A. V

B. H

C. T

D. L

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q3/5

1163

QUESTION ID

S34-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. H

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q3/5

True or false?

Parallel lines never meet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q4/5

1164

QUESTION ID

S34-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q4/5

True or false?

Perpendicular lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q5/5

1165

QUESTION ID

S34-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q5

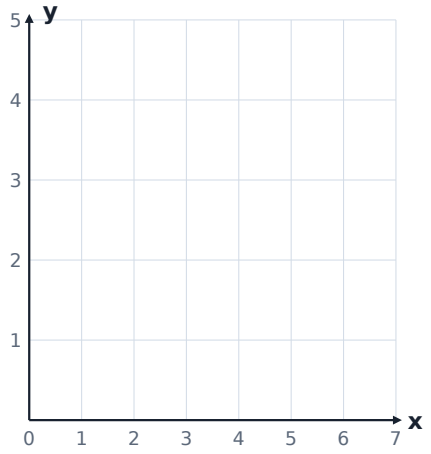
Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q5/5

SET 34 • 8.G.A.1.c

BACKWARD 2

Draw polygons in the coordinate plane and find side lengths • Grade 6 • Geometry • 6.G.A.3

Draw the quadrilateral with vertices (1, 1), (5, 1), (6, 4), and (2, 4).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q1/5

1167

QUESTION ID

S34-B2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

The quadrilateral with those four vertices (a parallelogram)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q1/5

A quadrilateral has vertices (1, 1), (5, 1), (6, 4), and (2, 4).

How long is the side from (1, 1) to (5, 1)?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q2/5

1168

QUESTION ID

S34-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q2/5

A quadrilateral has vertices (1, 1), (5, 1), (6, 4), and (2, 4).

Which side is parallel to the side from (1, 1) to (5, 1)?

A. From (1, 1) to (6, 4)

B. From (2, 4) to (6, 4)

C. From (1, 1) to (2, 4)

D. From (5, 1) to (6, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q3/5

1169

QUESTION ID

S34-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. From (2, 4) to (6, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q3/5

True or false?

The side from (2, 4) to (6, 4) is 4 units long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q4/5

1170

QUESTION ID

S34-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The segment from (1, 1) to (1, 4) is 4 units long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q5/5

1171

QUESTION ID

S34-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 3 units long.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q5/5

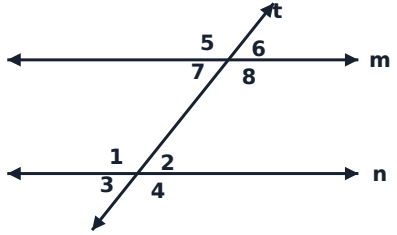
SET 34 • 8.G.A.1.c

FORWARD 1

Prove theorems about angles formed by parallel lines • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9

ANSWER KEY Reduced copy of the question page, with its answer

In the figure, lines m and n are parallel. Prove that angle 2 = angle 7. Give a reason for each step.



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q1/5

1173

QUESTION ID

S34-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

Angle 2 = angle 7

GRADING NOTE REQUIRED WORK: A valid chain of reasons. Other valid orders are accepted. A conclusion without reasons is not full credit. Example: angle 2 = angle 6 (corresponding angles, m ∥ n); angle 6 = angle 7 (vertical angles); so angle 2 = angle 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q1/5

Two parallel lines are cut by a transversal. One of the interior angles measures 72°.

What does the other interior angle on the same side of the transversal measure? Give the reason.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q2/5

1174

QUESTION ID

S34-F1-Q2

TYPE

Short answer

CORRECT ANSWER

108°; same-side interior angles are supplementary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q2/5

Which statement can be proved from "corresponding angles formed by parallel lines are congruent" and "vertical angles are congruent"?

A. Alternate interior angles are congruent.

B. The transversal is parallel to the lines.

C. The lines are perpendicular.

D. All the angles are right angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q3/5

1175

QUESTION ID

S34-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Alternate interior angles are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q3/5

True or false?

If alternate interior angles formed by a transversal are congruent, then the two lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q4/5

1176

QUESTION ID

S34-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q4/5

True or false?

Same-side interior angles formed by parallel lines are always congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q5/5

1177

QUESTION ID

S34-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are supplementary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q5/5

SET 34 • 8.G.A.1.c

FORWARD 2

Use coordinates to prove simple geometric theorems • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Show that the quadrilateral with vertices (0, 0), (4, 0), (5, 3), and (1, 3) is a parallelogram.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q1/5

1179

QUESTION ID

S34-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The sides from (0, 0) to (4, 0) and from (1, 3) to (5, 3) both have slope 0; the sides from (0, 0) to (1, 3) and from (4, 0) to (5, 3) both have slope 3. Both pairs of opposite sides are parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q1/5

Is the triangle with vertices (0, 0), (4, 0), and (0, 3) a right triangle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q2/5

1180

QUESTION ID

S34-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. One side is horizontal and one is vertical, so they are perpendicular.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-F2-Q2**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q2/5

In order, (0, 0), (2, 0), (3, 2), and a fourth point form a parallelogram.

Which is the fourth point?

A. (0, 3)

B. (3, 0)

C. (1, 2)

D. (2, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q3/5

1181

QUESTION ID

S34-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (1, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q3/5

True or false?

A quadrilateral whose opposite sides have equal slopes is a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q4/5

1182

QUESTION ID

S34-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q4/5

True or false?

The points (0, 0), (1, 1), and (2, 3) lie on one line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q5/5

1183

QUESTION ID

S34-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q5/5

SET 35 • 8.G.A.2 • GEOMETRY

Describe a sequence of rigid motions that shows two figures are congruent

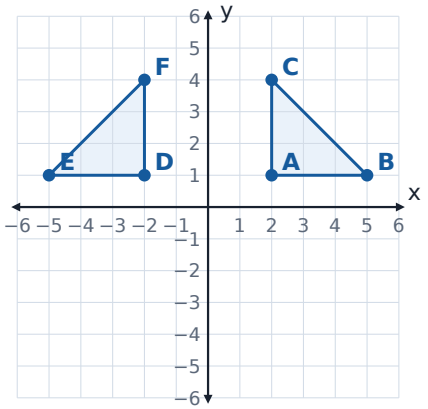
MAIN	Grade 8 main question	**8.G.A.2**
BACKWARD 1	Draw a polygon from its vertices	**6.G.A.3**
BACKWARD 2	Use signs to locate points across the axes	**6.NS.C.6.b**
FORWARD 1	Use corresponding parts to explain triangle congruence	**HSG-CO.B.7**
FORWARD 2	Explain the SAS, ASA, and SSS criteria with rigid motions	**HSG-CO.B.8**

SET 35 • 8.G.A.2

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.2

Describe a rigid motion that maps triangle ABC onto triangle DEF.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q1/5

1186

QUESTION ID

S35-M-Q1

TYPE

Short answer

CORRECT ANSWER

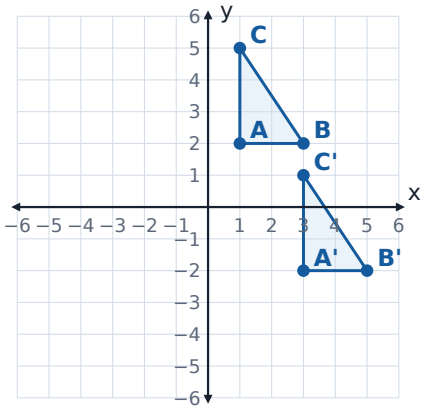
A reflection across the y-axis

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q1/5

Describe a rigid motion that maps triangle ABC onto triangle A'B'C'.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q2/5

1187

QUESTION ID

S35-M-Q2

TYPE

Short answer

CORRECT ANSWER

A translation 2 units right and 4 units down

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q2/5

Figure P is mapped onto figure Q by a 90° rotation followed by a translation.

Which statement is true?

A. P and Q are congruent.

B. Q is larger than P.

C. P and Q are not congruent because they face different directions.

D. P and Q have different angle measures.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q3/5

1188

QUESTION ID

S35-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. P and Q are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q3/5

Explain why a triangle and its image after a dilation with a scale factor of 2 are not congruent.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q4/5

1189

QUESTION ID

S35-M-Q4

TYPE

Short answer

CORRECT ANSWER

The dilation doubles every side length, but rigid motions (which show congruence) keep lengths the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q4/5

True or false?

Two figures are congruent if a sequence of rotations, reflections, and translations maps one onto the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q5/5

1190

QUESTION ID

S35-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q5

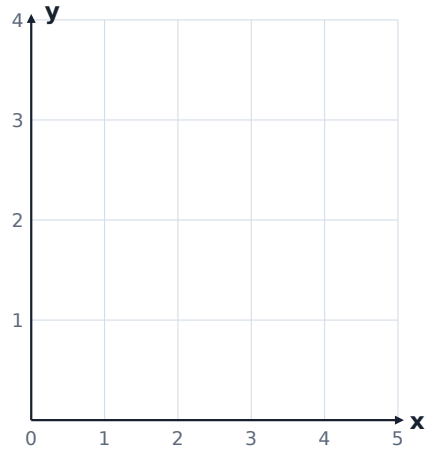
Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q5/5

SET 35 • 8.G.A.2

BACKWARD 1

Draw a polygon from its vertices • Grade 6 • Geometry • 6.G.A.3

Draw the triangle with vertices (1, 1), (4, 1), and (1, 3).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q1/5

1192

QUESTION ID

S35-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

The triangle with those three vertices

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q1/5

A triangle has vertices (1, 1), (4, 1), and (1, 3).

How long is the side from (1, 1) to (4, 1)?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q2/5

1193

QUESTION ID

S35-B1-Q2

TYPE

Short answer

CORRECT ANSWER

3 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q2/5

A rectangle has vertices (0, 0), (5, 0), and (5, 2).

What is the fourth vertex?

A. (0, 2)

B. (2, 0)

C. (5, 5)

D. (2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q3/5

1194

QUESTION ID

S35-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (0, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q3/5

True or false?

The side from (1, 1) to (1, 3) is 2 units long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q4/5

1195

QUESTION ID

S35-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q4/5

True or false?

The points (2, 1), (2, 5), and (6, 1) form a triangle with a side 3 units long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q5/5

1196

QUESTION ID

S35-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its sides along the grid lines are 4 units long.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q5/5

SET 35 • 8.G.A.2

BACKWARD 2

Use signs to locate points across the axes • Grade 6 • The Number System • 6.NS.C.6.b

The point (5, 2) is reflected across the y-axis.

What is its image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q1/5

1198

QUESTION ID

S35-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(−5, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q1/5

The point $(-1, -4)$ is reflected across the x-axis.

What is its image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q2/5

1199

QUESTION ID

S35-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$(-1, 4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q2/5

Points (3, 2) and (3, −2) differ only in the sign of the y-coordinate.

How are they related?

A. They are reflections across the x-axis.

B. They are reflections across the y-axis.

C. They are the same point.

D. They are 3 units apart.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q3/5

1200

QUESTION ID

S35-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. They are reflections across the x-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q3/5

True or false?

(−4, 1) and (4, 1) are reflections of each other across the y-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q4/5

1201

QUESTION ID

S35-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q4/5

True or false?

(2, −6) and (−2, 6) are reflections of each other across the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q5/5

1202

QUESTION ID

S35-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q5/5

SET 35 • 8.G.A.2

FORWARD 1

Use corresponding parts to explain triangle congruence • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7

A sequence of rigid motions maps triangle ABC onto triangle DEF, taking A to D, B to E, and C to F. AB = 5 and angle B = 40°.

Which side and which angle of DEF must equal these? Explain using rigid motions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q1/5

1204

QUESTION ID

S35-F1-Q1

TYPE

Short answer

CORRECT ANSWER

DE = 5 and angle E = 40°: the rigid motions map segment AB onto DE and angle B onto angle E, and rigid motions keep lengths and angle measures.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q1/5

Two triangles have all corresponding sides and all corresponding angles congruent.

What can you conclude? Explain with rigid motions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q2/5

1205

QUESTION ID

S35-F1-Q2

TYPE

Short answer

CORRECT ANSWER

The triangles are congruent; a sequence of rigid motions maps one onto the other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q2/5

Triangle PQR is congruent to triangle XYZ.

Which statement must be true?

A. PQ = YZ

B. QR = YZ

C. Angle P = angle Z

D. PR = XY

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q3/5

1206

QUESTION ID

S35-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. QR = YZ

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q3/5

True or false?

Corresponding parts of congruent triangles are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q4/5

1207

QUESTION ID

S35-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If triangle ABC is congruent to triangle DEF, then AC = EF.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q5/5

1208

QUESTION ID

S35-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE AC = DF.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q5/5

SET 35 • 8.G.A.2

FORWARD 2

Explain the SAS, ASA, and SSS criteria with rigid motions • High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Triangles ABC and DEF have AB = DE, angle A = angle D, and AC = DF (SAS).

Explain the rigid motions that map triangle ABC onto triangle DEF, and why C must land on F.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q1/5

1210

QUESTION ID

S35-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

A rigid motion maps A to D, B to E, and C to F, so the triangles are congruent.

GRADING NOTE REQUIRED WORK: A complete argument: translate A to D; rotate so ray AB lies along ray DE, and B lands on E because AB = DE; if C is on the other side of DE from F, reflect across DE. Since angle A = angle D, ray AC lies along ray DF, and since AC = DF, C lands on F. Any equivalent order is accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangles ABC and DEF have angle A = angle D, AB = DE, and angle B = angle E (ASA).

After rigid motions map segment AB onto segment DE (with C and F on the same side), explain why C must land on F.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q2/5

1211

QUESTION ID

S35-F2-Q2

TYPE

Answer with required work

CORRECT ANSWER

C lands on F, so the triangles are congruent.

GRADING NOTE REQUIRED WORK: Because angle A = angle D, the image of ray AC lies along ray DF; because angle B = angle E, the image of ray BC lies along ray EF. C is on both rays, and the two rays meet at only one point, F.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q2/5

Triangles ABC and DEF have AB = DE, BC = EF, and AC = DF (SSS). Rigid motions map AB onto DE, with C and F on the same side of DE.

Explain why C must land on F.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q3/5

1212

QUESTION ID

S35-F2-Q3

TYPE

Short answer

CORRECT ANSWER

C is at distance AC from D and BC from E. On that side of DE, only one point (F) is at those two distances (the two circles meet there once), so C lands on F.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q3/5

Why do three pairs of equal angles (AAA) NOT prove that two triangles are congruent?

A. A triangle has only two angles that matter.

B. Angles cannot be measured exactly.

C. A dilation keeps every angle but changes the side lengths.

D. AAA always proves congruence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q4/5

1213

QUESTION ID

S35-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. A dilation keeps every angle but changes the side lengths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q4/5

True or false?

In the rigid-motion explanation of SAS, a reflection may be needed to put C on the same side of DE as F.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q5/5

1214

QUESTION ID

S35-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q5/5

SET 36 • 8.G.A.3 • GEOMETRY

Translate figures on the coordinate plane

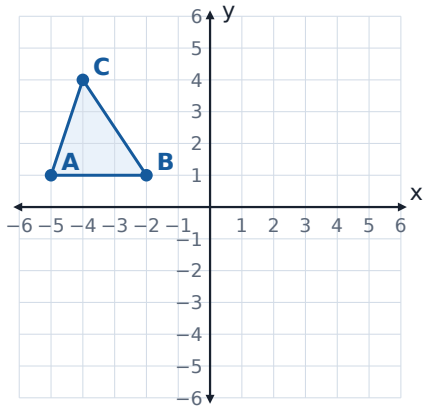
MAIN	Grade 8 main question	**8.G.A.3**
BACKWARD 1	Find and plot points in all four quadrants	**6.NS.C.6.c**
BACKWARD 2	Add integers	**7.NS.A.1.b**
FORWARD 1	Describe transformations as functions and decide which preserve…	**HSG-CO.A.2**
FORWARD 2	Find the point that divides a segment in a given ratio — nearest r…	**HSG-GPE.B.6**

SET 36 • 8.G.A.3

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.3

Translate triangle ABC 4 units right and 2 units down. Draw the image.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q1/5

1217

QUESTION ID

S36-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

The triangle with vertices A'(−1, −1), B'(2, −1), and C'(0, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q1/5

The point P(3, −2) is translated 5 units left and 4 units up.

What are the coordinates of the image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q2/5

1218

QUESTION ID

S36-M-Q2

TYPE

Short answer

CORRECT ANSWER

(−2, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q2/5

Which rule describes a translation 6 units down?

A. $(x, y) \to (x, y - 6)$

B. $(x, y) \to (x, y + 6)$

C. $(x, y) \to (x + 6, y)$

D. $(x, y) \to (x - 6, y)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q3/5

1219

QUESTION ID

S36-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $(x, y) \to (x, y - 6)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q3/5

A translation maps A(2, 5) to A'(−1, 7).

Where does it map B(4, 0)?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q4/5

1220

QUESTION ID

S36-M-Q4

TYPE

Short answer

CORRECT ANSWER

(1, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q4/5

True or false?

The rule (x, y) → (x + 3, y − 1) moves every point 3 units right and 1 unit down.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q5/5

1221

QUESTION ID

S36-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q5

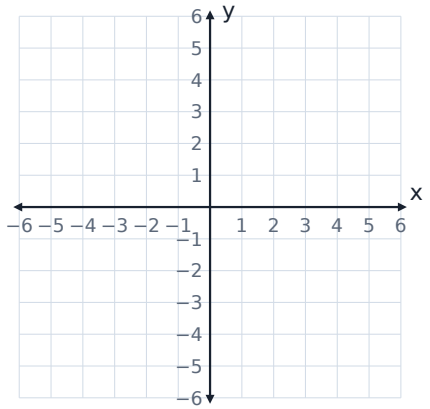
Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q5/5

SET 36 • 8.G.A.3

BACKWARD 1

Find and plot points in all four quadrants • Grade 6 • The Number System • 6.NS.C.6.c

Plot the points (−3, 2) and (4, −5).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q1/5

1223

QUESTION ID

S36-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

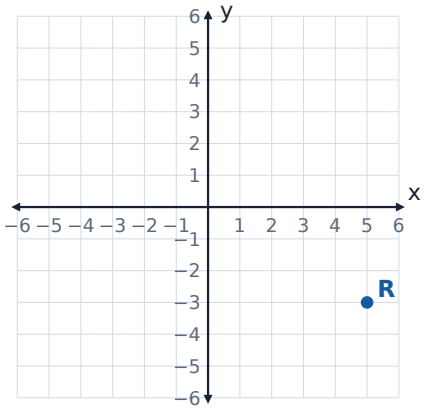
Points at (−3, 2) and (4, −5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q1/5

What are the coordinates of point R?



Coordinates: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q2/5

1224

QUESTION ID

S36-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(5, −3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q2/5

Which point is 2 units to the left of (1, 4)?

A. (1, 6)

B. (−1, 4)

C. (1, 2)

D. (3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q3/5

1225

QUESTION ID

S36-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. (−1, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q3/5

True or false?

Moving 3 units up from (2, −1) gives (2, 2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q4/5

1226

QUESTION ID

S36-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q4/5

True or false?

Moving 4 units left from (0, 0) gives (0, −4).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q5/5

1227

QUESTION ID

S36-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q5/5

SET 36 • 8.G.A.3

BACKWARD 2

Add integers • Grade 7 • The Number System • 7.NS.A.1.b

ANSWER KEY Reduced copy of the question page, with its answer

Add.

$-4 + 4$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q1/5

1229

QUESTION ID

S36-B2-Q1

TYPE

Short answer

CORRECT ANSWER

0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q1/5

Add.

$3 + (-5)$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q2

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q2/5

1230

QUESTION ID

S36-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q2

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Add.

$-1 + 6$

A. 5

B. -5

C. 7

D. -7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q3/5

1231

QUESTION ID

S36-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q3/5

True or false?

$-2 + (-3) = -5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q4/5

1232

QUESTION ID

S36-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q4/5

True or false?

$5 + (-8) = 3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q5

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q5/5

1233

QUESTION ID

S36-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q5

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q5/5

SET 36 • 8.G.A.3

FORWARD 1

Describe transformations as functions and decide which preserve distance • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2

Write the rule for a translation 2 units left and 7 units up.

Rule: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q1/5

1235

QUESTION ID

S36-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$(x, y) \rightarrow (x - 2, y + 7)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q1/5

Does the rule (x, y) → (2x, y) keep distances the same? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q2/5

1236

QUESTION ID

S36-F1-Q2

TYPE

Short answer

CORRECT ANSWER

No. It doubles horizontal distances; the segment from (0, 0) to (1, 0) becomes 2 units long.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q2/5

Which transformation keeps all distances and angles the same?

A. $(x, y) \to (x, 2y)$

B. $(x, y) \to (x^2, y)$

C. $(x, y) \to (3x, 3y)$

D. $(x, y) \to (x + 4, y - 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q3/5

1237

QUESTION ID

S36-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (x, y) → (x + 4, y − 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q3/5

True or false?

A translation is a function whose inputs and outputs are points of the plane.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q4/5

1238

QUESTION ID

S36-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q4/5

True or false?

The rule (x, y) → (x + 1, y + 1) changes the length of every segment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q5/5

1239

QUESTION ID

S36-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q5/5

SET 36 • 8.G.A.3

FORWARD 2

Find the point that divides a segment in a given ratio • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6

⚑ No high-school standard directly continues this skill at this step. This branch uses the nearest related high-school standard (placed at Grade 10 in this bank).

Find the midpoint of the segment from (2, 4) to (8, 10).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q1/5 • NEAREST RELATED

1241

QUESTION ID

S36-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(5, 7)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-F2-Q1**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q1/5 • NEAREST RELATED

Find the point that is $\frac{1}{3}$ of the way from A(0, 0) to B(9, 6).

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q2/5 • NEAREST RELATED

1242

QUESTION ID

S36-F2-Q2

TYPE

Short answer

CORRECT ANSWER

(3, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-F2-Q2**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q2/5 • NEAREST RELATED

Point P is on segment AB from A(1, 1) to B(7, 4), with AP : PB = 2 : 1.

What are the coordinates of P?

A. (3, 2)

B. (4, 2.5)

C. (6, 3)

D. (5, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q3/5 • NEAREST RELATED

1243

QUESTION ID

S36-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (5, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

The midpoint of the segment from (−4, 2) to (6, −2) is (1, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q4/5 • NEAREST RELATED

1244

QUESTION ID

S36-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

The point halfway from (0, 0) to (10, 4) is (10, 2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q5/5 • NEAREST RELATED

1245

QUESTION ID

S36-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The midpoint is (5, 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-F2-Q5**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q5/5 • NEAREST RELATED

SET 37 • 8.G.A.3 • GEOMETRY

Reflect figures across the x-axis and the y-axis

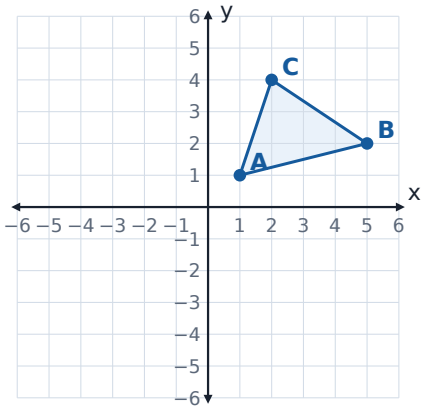
MAIN	Grade 8 main question	**8.G.A.3**
BACKWARD 1	Locate points that are reflections across the axes	**6.NS.C.6.b**
BACKWARD 2	Find the opposite of a number	**6.NS.C.6.a**
FORWARD 1	Describe the rotations and reflections that carry a polygon onto it…	**HSG-CO.A.3**
FORWARD 2	Draw the image of a figure after a sequence of transformations	**HSG-CO.B.6**

SET 37 • 8.G.A.3

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.3

Reflect triangle ABC across the x-axis. Draw the image.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q1/5

1248

QUESTION ID

S37-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

The triangle with vertices A'(1, −1), B'(5, −2), and C'(2, −4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q1/5

The point $(-3, 5)$ is reflected across the y-axis.

What are the coordinates of the image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q2/5

1249

QUESTION ID

S37-M-Q2

TYPE

Short answer

CORRECT ANSWER

(3, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q2/5

Which rule describes a reflection across the x-axis?

A. (x, y) → (−x, y)

B. (x, y) → (−x, −y)

C. (x, y) → (y, x)

D. (x, y) → (x, −y)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q3/5

1250

QUESTION ID

S37-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (x, y) → (x, −y)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q3/5

A segment has endpoints (2, 3) and (5, 1). It is reflected across the y-axis.

What are the endpoints of the image?

Endpoints: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q4/5

1251

QUESTION ID

S37-M-Q4

TYPE

Short answer

CORRECT ANSWER

(−2, 3) and (−5, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q4/5

True or false?

Reflecting (4, −6) across the x-axis gives (−4, −6).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q5/5

1252

QUESTION ID

S37-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The image is (4, 6).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q5/5

SET 37 • 8.G.A.3

BACKWARD 1

Locate points that are reflections across the axes • Grade 6 • The Number System • 6.NS.C.6.b

The point (7, −2) is reflected across the x-axis.

What is its image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q1/5

1254

QUESTION ID

S37-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(7, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q1/5

The point (−6, −3) is reflected across the y-axis.

What is its image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q2/5

1255

QUESTION ID

S37-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(6, −3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q2/5

Reflecting a point across the y-axis changes...

A. the sign of the y-coordinate

B. neither sign

C. the sign of the x-coordinate

D. both signs

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q3/5

1256

QUESTION ID

S37-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. the sign of the x-coordinate

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q3/5

True or false?

(0, 5) is its own reflection across the y-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q4/5

1257

QUESTION ID

S37-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q4/5

True or false?

The reflection of (−1, 8) across the y-axis is (−1, −8).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q5/5

1258

QUESTION ID

S37-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q5/5

SET 37 • 8.G.A.3

BACKWARD 2

Find the opposite of a number • Grade 6 • The Number System • 6.NS.C.6.a

What is the opposite of −7?

Opposite: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q1/5

1260

QUESTION ID

S37-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the opposite of 2.5?

Opposite: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q2/5

1261

QUESTION ID

S37-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q2/5

Which pair are opposites?

A. −4 and 4

B. 4 and $\frac{1}{4}$

C. 0 and 1

D. −4 and −4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q3/5

1262

QUESTION ID

S37-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. −4 and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q3/5

True or false?

The opposite of the opposite of 3 is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q4/5

1263

QUESTION ID

S37-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q4/5

True or false?

The opposite of 0 is −1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q5/5

1264

QUESTION ID

S37-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q5/5

SET 37 • 8.G.A.3

FORWARD 1

Describe the rotations and reflections that carry a polygon onto itself • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3

How many lines of symmetry does a square have?

Lines: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q1/5

1266

QUESTION ID

S37-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q1/5

Which rotations (between 0° and 360°) about its center carry a regular hexagon onto itself?

Rotations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q2/5

1267

QUESTION ID

S37-F1-Q2

TYPE

Short answer

CORRECT ANSWER

60°, 120°, 180°, 240°, and 300°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q2/5

Which figure is carried onto itself by a 180° rotation but has no line of symmetry?

A. A parallelogram that is not a rectangle or a rhombus

B. A square

C. An equilateral triangle

D. An isosceles trapezoid

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q3/5

1268

QUESTION ID

S37-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. A parallelogram that is not a rectangle or a rhombus

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q3/5

True or false?

A rectangle that is not a square has exactly 2 lines of symmetry.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q4/5

1269

QUESTION ID

S37-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q4/5

True or false?

An isosceles trapezoid is carried onto itself by a 90° rotation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q5/5

1270

QUESTION ID

S37-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q5

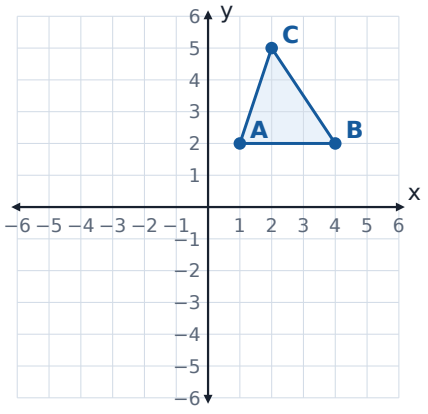
High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q5/5

SET 37 • 8.G.A.3

FORWARD 2

Draw the image of a figure after a sequence of transformations • High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6

Reflect triangle ABC across the x-axis, then reflect that image across the y-axis. Draw the final image.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q1/5

1272

QUESTION ID

S37-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

The triangle with vertices A''(−1, −2), B''(−4, −2), and C''(−2, −5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q1/5

What single transformation has the same effect as a reflection across the x-axis followed by a reflection across the y-axis?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q2/5

1273

QUESTION ID

S37-F2-Q2

TYPE

Short answer

CORRECT ANSWER

A 180° rotation about the origin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q2/5

The point (3, 1) is reflected across the line y = x.

What is its image?

A. (3, −1)

B. (−1, −3)

C. (−3, 1)

D. (1, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q3/5

1274

QUESTION ID

S37-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (1, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q3/5

True or false?

Reflecting a figure across one line and then across a parallel line has the same effect as a translation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q4/5

1275

QUESTION ID

S37-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q4/5

True or false?

Doing a translation and a reflection in the other order always gives the same image.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q5/5

1276

QUESTION ID

S37-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q5/5

SET 38 • 8.G.A.3 • GEOMETRY

Rotate figures 90° and 180° about the origin

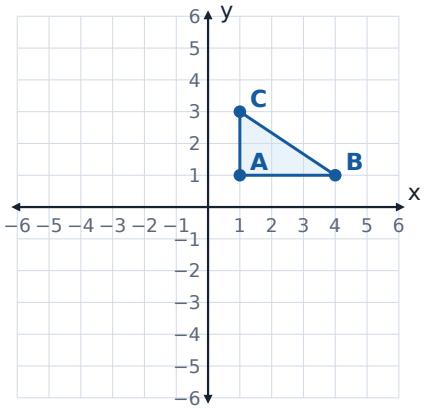
MAIN	Grade 8 main question	**8.G.A.3**
BACKWARD 1	Relate turns to angle measures	**4.MD.C.5.a**
BACKWARD 2	Locate a point reflected across both axes	**6.NS.C.6.b**
FORWARD 1	Describe a rotation in terms of distance and angle	**HSG-CO.A.4**
FORWARD 2	Use relationships among central angles, inscribed angles, and rad…	**HSG-C.A.2**

SET 38 • 8.G.A.3

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.3

Rotate triangle ABC 90° counterclockwise about the origin. Draw the image.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q1/5

1279

QUESTION ID

S38-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

The triangle with vertices A'(−1, 1), B'(−1, 4), and C'(−3, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q1/5

The point (5, −2) is rotated 180° about the origin.

What are the coordinates of the image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q2/5

1280

QUESTION ID

S38-M-Q2

TYPE

Short answer

CORRECT ANSWER

(−5, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q2/5

Which rule describes a 90° clockwise rotation about the origin?

A. (x, y) → (y, x)

B. (x, y) → (y, −x)

C. (x, y) → (−y, x)

D. (x, y) → (−x, −y)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q3/5

1281

QUESTION ID

S38-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. (x, y) → (y, −x)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q3/5

The point (2, 3) is rotated 90° counterclockwise about the origin.

What are the coordinates of the image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q4/5

1282

QUESTION ID

S38-M-Q4

TYPE

Short answer

CORRECT ANSWER

(−3, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q4/5

True or false?

Rotating $(-4, 1)$ by $180°$ about the origin gives $(4, -1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q5/5

1283

QUESTION ID

S38-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q5/5

SET 38 • 8.G.A.3

BACKWARD 1

Relate turns to angle measures • Grade 4 • Measurement & Data • 4.MD.C.5.a

ANSWER KEY Reduced copy of the question page, with its answer

How many degrees are in a quarter turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q1/5

1285

QUESTION ID

S38-B1-Q1

TYPE

Short answer

CORRECT ANSWER

90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q1/5

How many degrees are in a half turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q2/5

1286

QUESTION ID

S38-B1-Q2

TYPE

Short answer

CORRECT ANSWER

180°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q2/5

How many degrees are in a three-quarter turn?

A. 360°

B. 180°

C. 270°

D. 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q3/5

1287

QUESTION ID

S38-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 270°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q3/5

True or false?

A full turn is 360°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q4/5

1288

QUESTION ID

S38-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q4/5

True or false?

A half turn is 90°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q5/5

1289

QUESTION ID

S38-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q5/5

SET 38 • 8.G.A.3

BACKWARD 2

Locate a point reflected across both axes • Grade 6 • The Number System • 6.NS.C.6.b

The point (3, 4) is reflected across the x-axis and then across the y-axis.

What is the final image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q1/5

1291

QUESTION ID

S38-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(−3, −4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q1/5

The point (−2, 5) is reflected across both axes.

What is the final image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q2/5

1292

QUESTION ID

S38-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(2, −5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q2/5

Reflecting a point across both axes changes...

A. the signs of both coordinates

B. only the y-coordinate

C. neither coordinate

D. only the x-coordinate

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q3/5

1293

QUESTION ID

S38-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. the signs of both coordinates

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q3/5

True or false?

Reflecting (1, −6) across both axes gives (−1, 6).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q4/5

1294

QUESTION ID

S38-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q4/5

True or false?

Reflecting (4, 4) across both axes gives (4, 4).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q5/5

1295

QUESTION ID

S38-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q5/5

SET 38 • 8.G.A.3

FORWARD 1

Describe a rotation in terms of distance and angle • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4

A rotation of 120° about point O maps a point A (A ≠ O) to A'.

Which lengths and which angle are determined by the rotation?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q1/5

1297

QUESTION ID

S38-F1-Q1

TYPE

Short answer

CORRECT ANSWER

OA = OA', and angle AOA' = 120°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q1/5

P(6, 0) is rotated 60° counterclockwise about the origin.

How far is the image from the origin?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q2/5

1298

QUESTION ID

S38-F1-Q2

TYPE

Short answer

CORRECT ANSWER

6 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q2/5

A rotation about point O through an angle θ, with $0° < θ ≤ 180°$, maps each point P other than O to a point P' so that...

A. P' is on the line of reflection

B. P' is twice as far from O as P

C. OP = OP' and angle POP' = θ

D. PP' is parallel to one fixed line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q3/5

1299

QUESTION ID

S38-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. OP = OP' and angle POP' = θ

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q3/5

True or false?

In a rotation by a nonzero angle less than 360°, the center is the only point that does not move.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q4/5

1300

QUESTION ID

S38-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q4/5

True or false?

A 90° rotation maps every horizontal segment to another horizontal segment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q5/5

1301

QUESTION ID
S38-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It maps horizontal segments to vertical segments.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q5/5

SET 38 • 8.G.A.3

FORWARD 2

Use relationships among central angles, inscribed angles, and radii • High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2

⚑ No high-school standard directly continues this skill at this step. This branch uses the nearest related high-school standard (placed at Grade 10 in this bank).

A central angle of a circle measures 80°.

What does an inscribed angle that intercepts the same arc measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q1

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q1/5 • NEAREST RELATED

1303

QUESTION ID

S38-F2-Q1

TYPE

Short answer

CORRECT ANSWER

40°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-F2-Q1**

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q1/5 • NEAREST RELATED

An inscribed angle intercepts a semicircle.

What is its measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q2

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q2/5 • NEAREST RELATED

1304

QUESTION ID

S38-F2-Q2

TYPE

Short answer

CORRECT ANSWER

90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-F2-Q2**

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q2/5 • NEAREST RELATED

A radius is drawn to the point where a tangent line touches a circle.

What angle do the radius and the tangent line form?

A. 90°

B. 45°

C. 180°

D. 60°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q3

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q3/5 • NEAREST RELATED

1305

QUESTION ID

S38-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q3

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

Rotating a circle about its center by any angle carries the circle onto itself.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q4

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q4/5 • NEAREST RELATED

1306

QUESTION ID

S38-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q4

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

An inscribed angle measures twice its intercepted arc.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q5

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q5/5 • NEAREST RELATED

1307

QUESTION ID

S38-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It measures half of its intercepted arc.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q5

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q5/5 • NEAREST RELATED

SET 39 • 8.G.A.3 • GEOMETRY

Dilate figures on the coordinate plane with the origin as the center

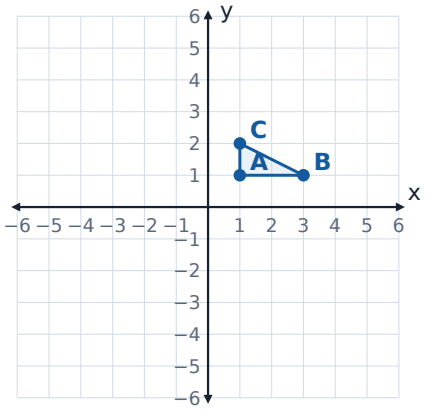
MAIN	Grade 8 main question	**8.G.A.3**
BACKWARD 1	Use a scale factor to find lengths in a scale drawing	**7.G.A.1**
BACKWARD 2	Multiply rational numbers	**7.NS.A.2.a**
FORWARD 1	Verify the properties of dilations	**HSG-SRT.A.1**
FORWARD 2	Use similar triangles to solve problems	**HSG-SRT.B.5**

SET 39 • 8.G.A.3

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.3

Dilate triangle ABC by a scale factor of 2 with the origin as the center. Draw the image.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q1/5

1310

QUESTION ID

S39-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

The triangle with vertices A'(2, 2), B'(6, 2), and C'(2, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q1/5

The point $(6, -4)$ is dilated by a scale factor of $\frac{1}{2}$ with the origin as the center.

What are the coordinates of the image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q2/5

1311

QUESTION ID

S39-M-Q2

TYPE

Short answer

CORRECT ANSWER

(3, −2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q2/5

Which rule describes a dilation with a scale factor of 3 and the origin as the center?

A. $(x, y) \rightarrow (3x, y)$

B. $(x, y) \rightarrow (x + 3, y + 3)$

C. $(x, y) \rightarrow (\frac{x}{3}, \frac{y}{3})$

D. $(x, y) \rightarrow (3x, 3y)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q3/5

1312

QUESTION ID

S39-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (x, y) → (3x, 3y)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q3/5

A dilation with the origin as the center maps (4, 2) to (10, 5).

What is the scale factor?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q4/5

1313

GRADING NOTE 5/2 is also correct.

QUESTION ID

S39-M-Q4

TYPE

Short answer

CORRECT ANSWER

2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q4/5

True or false?

A dilation with a scale factor of $\frac{1}{3}$ and the origin as the center maps (9, 6) to (3, 2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q5/5

1314

QUESTION ID

S39-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q5/5

SET 39 • 8.G.A.3

BACKWARD 1

Use a scale factor to find lengths in a scale drawing • Grade 7 • Geometry • 7.G.A.1

A 4 cm segment is enlarged by a scale factor of 3.

How long is the new segment?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q1/5

1316

QUESTION ID

S39-B1-Q1

TYPE

Short answer

CORRECT ANSWER

12 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q1/5

A 10 cm segment is reduced by a scale factor of $\frac{1}{2}$.

How long is the new segment?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q2/5

1317

QUESTION ID

S39-B1-Q2

TYPE

Short answer

CORRECT ANSWER

5 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q2/5

A rectangle 2 cm by 5 cm is enlarged by a scale factor of 4.

What are the new dimensions?

A. 8 cm by 20 cm

B. 8 cm by 5 cm

C. 6 cm by 9 cm

D. 2 cm by 20 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q3/5

1318

QUESTION ID

S39-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 8 cm by 20 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q3/5

True or false?

A scale factor greater than 1 makes a figure larger.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q4/5

1319

QUESTION ID

S39-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q4/5

True or false?

A scale factor of $\frac{1}{4}$ makes each length 4 times as long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q5/5

1320

QUESTION ID

S39-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q5/5

SET 39 • 8.G.A.3

BACKWARD 2

Multiply rational numbers • Grade 7 • The Number System • 7.NS.A.2.a

ANSWER KEY Reduced copy of the question page, with its answer

Multiply.

$2 \times (-4)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q1/5

1322

QUESTION ID

S39-B2-Q1

TYPE

Short answer

CORRECT ANSWER

−8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply.

$\frac{1}{2} \times (-6)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q2/5

1323

QUESTION ID

S39-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q2/5

Multiply.

3.5×4

A. 12

B. 0.875

C. 14

D. 7.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q3/5

1324

QUESTION ID

S39-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q3/5

True or false?

$3 \times (-2) = -6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q4/5

1325

QUESTION ID

S39-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q4/5

True or false?

$\frac{1}{3} \times 9 = 27$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q5/5

1326

QUESTION ID

S39-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q5/5

SET 39 • 8.G.A.3

FORWARD 1

Verify the properties of dilations • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1

A dilation with the origin as the center and a scale factor of 3 is applied to the line y = x + 1.

What is the equation of the image?

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q1/5

1328

QUESTION ID

S39-F1-Q1

TYPE

Short answer

CORRECT ANSWER

y = x + 3 (a line parallel to the original)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q1/5

A segment 4 units long is dilated by a scale factor of 2.5.

How long is the image?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q2/5

1329

QUESTION ID

S39-F1-Q2

TYPE

Short answer

CORRECT ANSWER

10 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F1-Q2**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q2/5

A dilation is applied to a line that passes through the center of dilation.

What is the image?

A. A parallel line

B. A single point

C. A perpendicular line

D. The same line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q3/5

1330

QUESTION ID

S39-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The same line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q3/5

True or false?

A dilation maps a line that does not pass through the center to a parallel line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q4/5

1331

QUESTION ID

S39-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q4/5

True or false?

A dilation with a scale factor of 2 doubles every angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q5/5

1332

QUESTION ID

S39-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Dilations keep angle measures the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q5/5

SET 39 • 8.G.A.3

FORWARD 2

Use similar triangles to solve problems • High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

A 6-foot person casts a 4-foot shadow. At the same time, a tree casts a 22-foot shadow.

How tall is the tree?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q1

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q1/5

1334

QUESTION ID

S39-F2-Q1

TYPE

Short answer

CORRECT ANSWER

33 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q1

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q1/5

Triangle ABC is similar to triangle DEF, with AB = 8, DE = 12, and BC = 10.

Find EF.

EF = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q2

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q2/5

1335

QUESTION ID

S39-F2-Q2

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F2-Q2**

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q2/5

In triangle ABC, D is on AB and E is on AC, and segment DE is parallel to BC. AD = 3, DB = 6, and AE = 4.

How long is EC?

A. 2

B. 12

C. 8

D. 4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q3

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q3/5

1336

QUESTION ID

S39-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q3/5

True or false?

In similar triangles, the ratio of the perimeters equals the scale factor.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q4

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q4/5

1337

QUESTION ID

S39-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q4

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q4/5

True or false?

Triangles with sides 3, 4, 5 and 6, 8, 12 are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q5

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q5/5

1338

QUESTION ID

S39-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q5

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q5/5

SET 40 • 8.G.A.4 • GEOMETRY

Describe a sequence of transformations that shows two figures are similar

MAIN	Grade 8 main question	**8.G.A.4**
BACKWARD 1	Find a scale factor between scale copies	**7.G.A.1**
BACKWARD 2	Decide from a table whether two quantities are in a constant ratio	**7.RP.A.2.a**
FORWARD 1	Use the definition of similarity transformations to prove equal ang…	**HSG-SRT.A.2**
FORWARD 2	Use a translation and a dilation to show that any two circles are s…	**HSG-C.A.1**

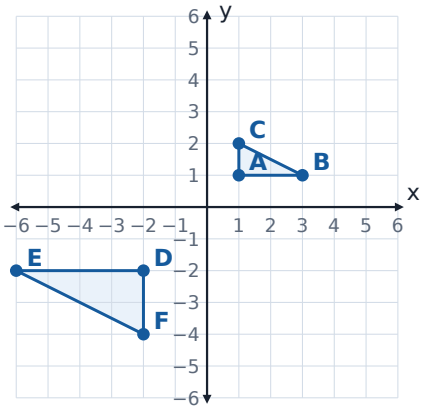
SET 40 • 8.G.A.4

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Describe a sequence of transformations that maps triangle ABC onto triangle DEF.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q1

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q1/5

1341

QUESTION ID

S40-M-Q1

TYPE

Short answer

CORRECT ANSWER

A dilation with a scale factor of 2 and the origin as the center, followed by a 180° rotation about the origin.

GRADING NOTE Any correct sequence is accepted, for example the rotation first, then the dilation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q1

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q1/5

Explain why a 2-by-3 rectangle and a 4-by-6 rectangle are similar.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q2

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q2/5

1342

QUESTION ID

S40-M-Q2

TYPE

Short answer

CORRECT ANSWER

A dilation with a scale factor of 2 (followed by rigid motions to move it into place) maps the first rectangle onto the second.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q2

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q2/5

Which sequence of transformations maps a figure onto a figure that is similar but not congruent?

A. A rotation, then a translation

B. A translation only

C. Two reflections

D. A dilation with a scale factor of 3, then a translation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q3

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q3/5

1343

QUESTION ID

S40-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A dilation with a scale factor of 3, then a translation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q3

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q3/5

Are a 3-4-5 triangle and a 5-12-13 triangle similar? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q4

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q4/5

1344

QUESTION ID

S40-M-Q4

TYPE

Short answer

CORRECT ANSWER

No. The ratios of corresponding sides (3/5, 4/12, 5/13) are not equal, so no dilation and rigid motions can map one onto the other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q4

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q4/5

True or false?

Two figures are similar if a sequence of rotations, reflections, translations, and dilations maps one onto the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q5

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q5/5

1345

QUESTION ID

S40-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q5

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q5/5

SET 40 • 8.G.A.4

BACKWARD 1

Find a scale factor between scale copies • Grade 7 • Geometry • 7.G.A.1

A 3 cm side becomes 12 cm in a scale copy.

What is the scale factor?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q1/5

1347

QUESTION ID

S40-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q1/5

A 15 m side becomes 5 m in a scale copy.

What is the scale factor?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q2/5

1348

QUESTION ID

S40-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q2/5

A triangle's sides are 2, 3, and 4. Which side lengths are a scale copy?

A. 6, 9, 12

B. 4, 5, 6

C. 2, 6, 8

D. 3, 4, 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q3/5

1349

QUESTION ID

S40-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 6, 9, 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q3/5

True or false?

A scale copy with a scale factor of 1 is the same size as the original.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q4/5

1350

QUESTION ID

S40-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q4/5

True or false?

A rectangle 2 by 3 and a rectangle 3 by 4 are scale copies of each other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q5/5

1351

QUESTION ID

S40-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q5/5

SET 40 • 8.G.A.4

BACKWARD 2

Decide from a table whether two quantities are in a constant ratio • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

The table shows side lengths of a figure and of its copy.

Are the pairs in a constant ratio? Explain.

Figure	4	6	8
Copy	10	15	20

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q1/5

1353

QUESTION ID

S40-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. Each copy length is 2.5 times the figure length.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q1/5

The table shows side lengths of a figure and of another figure.

Are the pairs in a constant ratio? Explain.

Figure	3	5
Other	9	12

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q2/5

1354

QUESTION ID

S40-B2-Q2

TYPE

Short answer

CORRECT ANSWER

No. 9 ÷ 3 = 3, but 12 ÷ 5 = 2.4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q2/5

Which table shows y in a constant ratio to x?

A. x: 2, 4, 6; y: 1, 3, 5

B. x: 2, 4, 6; y: 5, 10, 15

C. x: 2, 4, 6; y: 4, 16, 36

D. x: 2, 4, 6; y: 5, 7, 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q3/5

1355

QUESTION ID

S40-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. x: 2, 4, 6; y: 5, 10, 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q3/5

True or false?

For the pairs (2, 7), (4, 14), and (6, 21), y is always 3.5 times x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q4/5

1356

QUESTION ID

S40-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q4/5

True or false?

The pairs (4, 5) and (5, 6) are in the same ratio.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q5/5

1357

QUESTION ID

S40-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $4 \div 5 = 0.8$, but $5 \div 6 \approx 0.83$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q5/5

SET 40 • 8.G.A.4

FORWARD 1

Use the definition of similarity transformations to prove equal angles and proportional sides • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC is dilated by a scale factor of 3 about point O, then rotated 90° about O, giving triangle A'B'C'. AB = 4 and angle A = 50°.

Find A'B' and angle A'. Explain using the properties of the transformations.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q1/5

1359

QUESTION ID

S40-F1-Q1

TYPE

Short answer

CORRECT ANSWER

A'B' = 12 and angle A' = 50°: the dilation multiplies lengths by 3 and the rotation keeps lengths; both keep angle measures.

GRADING NOTE Both values and the reasons are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Use the definition of similarity (a sequence of rigid motions and dilations) to explain why similar triangles have equal corresponding angles and proportional corresponding sides.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q2/5

1360

QUESTION ID

S40-F1-Q2

TYPE

Answer with required work

CORRECT ANSWER

Corresponding angles are equal, and corresponding sides are proportional.

GRADING NOTE REQUIRED WORK: Each rigid motion and each dilation keeps angle measures, so corresponding angles are equal. Rigid motions keep lengths, and a dilation multiplies every length by the same scale factor, so every pair of corresponding sides has the same ratio.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q2/5

Rectangle P is 2 by 6. Rectangle Q is 3 by 8.

Which statement is true?

A. They are not similar, because $3 \div 2 \neq 8 \div 6$.

B. They are similar, because all their angles are 90°.

C. They are similar with a scale factor of 1.5.

D. They are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q3/5

1361

QUESTION ID

S40-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. They are not similar, because $3 \div 2 \neq 8 \div 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q3/5

True or false?

If triangle DEF is the image of triangle ABC under a similarity transformation, then
DE ÷ AB = EF ÷ BC = DF ÷ AC.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q4/5

1362

QUESTION ID
S40-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q4/5

True or false?

Two quadrilaterals whose four pairs of corresponding angles are equal must be similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q5/5

1363

QUESTION ID

S40-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A square and a long rectangle have equal angles but are not similar.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q5/5

SET 40 • 8.G.A.4

FORWARD 2

Use a translation and a dilation to show that any two circles are similar • High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1

Circle A has center (0, 0) and radius 2. Circle B has center (5, 1) and radius 6.

Describe a translation followed by a dilation that maps circle A onto circle B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q1

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q1/5

1365

QUESTION ID

S40-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Translate 5 units right and 1 unit up, then dilate about (5, 1) with a scale factor of 3.

GRADING NOTE Any correct sequence is accepted, for example a dilation about (0, 0) with scale factor 3 followed by a translation 5 units right and 1 unit up.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q1

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q1/5

What scale factor of a dilation about the center maps a circle with a radius of 4 onto a circle with the same center and a radius of 10?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q2

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q2/5

1366

QUESTION ID

S40-F2-Q2

TYPE

Short answer

CORRECT ANSWER

2.5

GRADING NOTE $\frac{5}{2}$ is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q2

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q2/5

Which statement explains why any two circles are similar?

A. A rotation alone maps any circle onto any other circle.

B. All circles have the same radius.

C. A translation moves one center onto the other, and then a dilation about that center makes the radii equal.

D. All circles have the same area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q3

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q3/5

1367

QUESTION ID

S40-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. A translation moves one center onto the other, and then a dilation about that center makes the radii equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q3

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q3/5

True or false?

Any two circles are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q4

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q4/5

1368

QUESTION ID

S40-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q4

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q4/5

True or false?

A circle with a radius of 3 can be mapped onto a circle with a radius of 6 by a translation alone.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q5

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q5/5

1369

QUESTION ID

S40-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A translation keeps the radius 3; a dilation is also needed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q5

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q5/5

SET 41 • 8.G.A.5 • GEOMETRY

Use the angle sum and exterior angles of a triangle

MAIN	Grade 8 main question	**8.G.A.5**
BACKWARD 1	Write and solve equations for supplementary and complementary…	**7.G.B.5**
BACKWARD 2	Find an unknown angle in a sum of angles	**4.MD.C.7**
FORWARD 1	Prove theorems about triangles	**HSG-CO.C.10**
FORWARD 2	Define trigonometric ratios from similar right triangles — nearest r	**HSG-SRT.C.6**

SET 41 • 8.G.A.5

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.5

Two angles of a triangle measure 52° and 71°.

What does the third angle measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q1/5

1372

QUESTION ID

S41-M-Q1

TYPE

Short answer

CORRECT ANSWER

57°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q1/5

The angles of a triangle measure x°, (x + 10)°, and (2x − 30)°.

Write and solve an equation to find x.

Equation and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q2/5

1373

QUESTION ID

S41-M-Q2

TYPE

Short answer

CORRECT ANSWER

The equation x + (x + 10) + (2x − 30) = 180 and the answer x = 50

GRADING NOTE Both parts are required: a valid equation and the correct solution. Any valid equation is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q2/5

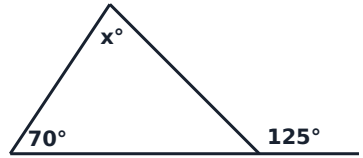
In the figure, what is the value of x?

A. 55

B. 70

C. 125

D. 35



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q3/5

1374

QUESTION ID

S41-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 55

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q3/5

Explain why an exterior angle of a triangle equals the sum of the two remote interior angles. Give a reason for each step.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q4/5

1375

QUESTION ID

S41-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

The exterior angle = the sum of the two remote interior angles

GRADING NOTE REQUIRED WORK: A valid chain of reasons. Other valid orders are accepted. A conclusion without reasons is not full credit. Example: the exterior angle and the adjacent interior angle form a linear pair, so they add to 180°; the three interior angles also add to 180°; so the exterior angle equals the sum of the other two interior angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q4/5

True or false?

A triangle can have two obtuse angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q5/5

1376

QUESTION ID

S41-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Two obtuse angles alone add to more than 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q5/5

SET 41 • 8.G.A.5

BACKWARD 1

Write and solve equations for supplementary and complementary angles • Grade 7 • Geometry • 7.G.B.5

Two supplementary angles measure x° and 130°.

What is x?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q1/5

1378

QUESTION ID

S41-B1-Q1

TYPE

Short answer

CORRECT ANSWER

50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q1/5

Two complementary angles measure 2x° and 40°.

What is x?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q2/5

1379

QUESTION ID

S41-B1-Q2

TYPE

Short answer

CORRECT ANSWER

25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q2/5

Angles of x° and (x + 30)° are supplementary.

What is x?

A. 75

B. 150

C. 105

D. 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q3/5

1380

QUESTION ID

S41-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q3/5

True or false?

Angles of 47° and 43° are complementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q4/5

1381

QUESTION ID

S41-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q4/5

True or false?

Angles of 100° and 90° are supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q5/5

1382

QUESTION ID

S41-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q5/5

SET 41 • 8.G.A.5

BACKWARD 2

Find an unknown angle in a sum of angles • Grade 4 • Measurement & Data • 4.MD.C.7

A 90° angle is split into two angles. One is 35°.

What is the other?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q1/5

1384

QUESTION ID

S41-B2-Q1

TYPE

Short answer

CORRECT ANSWER

55°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q1/5

A straight angle (180°) is split into two angles. One is 110°.

What is the other?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q2/5

1385

QUESTION ID

S41-B2-Q2

TYPE

Short answer

CORRECT ANSWER

70°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q2/5

An angle of 120° is split into angles of 75° and x°.

What is x?

A. 195

B. 55

C. 45

D. 35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q3/5

1386

QUESTION ID

S41-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q3/5

True or false?

Angles of 30° and 60° together make a 90° angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q4/5

1387

QUESTION ID

S41-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q4/5

True or false?

Angles of 80° and 80° together make a 180° angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q5/5

1388

QUESTION ID

S41-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q5/5

SET 41 • 8.G.A.5

FORWARD 1

Prove theorems about triangles • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10

ANSWER KEY Reduced copy of the question page, with its answer

Prove that the angles of triangle ABC add to 180°. Use a line through C that is parallel to AB. Give a reason for each step.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q1/5

1390

QUESTION ID

S41-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

Angle A + angle B + angle C = 180°

GRADING NOTE REQUIRED WORK: A valid chain of reasons. Other valid orders are accepted. A conclusion without reasons is not full credit. Example: draw line l through C parallel to AB. The angles at C along l add to 180° (straight angle). The two outer ones equal angles A and B (alternate interior angles, l ∥ AB). So angle A + angle B + angle C = 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q1/5

The angles of a triangle are 2x°, 3x°, and 4x°.

What is x?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q2/5

1391

QUESTION ID

S41-F1-Q2

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q2/5

An isosceles triangle has base angles of 40°.

What does its third angle measure?

A. 100°

B. 40°

C. 80°

D. 140°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q3/5

1392

QUESTION ID

S41-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 100°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q3/5

True or false?

The base angles of an isosceles triangle are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q4/5

1393

QUESTION ID

S41-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q4/5

True or false?

An exterior angle of a triangle equals the sum of all three interior angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q5/5

1394

QUESTION ID

S41-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q5/5

SET 41 • 8.G.A.5

FORWARD 2

Define trigonometric ratios from similar right triangles • High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

⚑ No high-school standard directly continues this skill at this step. This branch uses the nearest related high-school standard (placed at Grade 10 in this bank).

A right triangle has a leg of 3 opposite angle A, a leg of 4 next to angle A, and a hypotenuse of 5.

What is sin A?

sin A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q1

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q1/5 • NEAREST RELATED

1396

QUESTION ID

S41-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q1

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q1/5 • NEAREST RELATED

A right triangle has a leg of 3 opposite angle A, a leg of 4 next to angle A, and a hypotenuse of 5.

What is tan A?

tan A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q2

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q2/5 • NEAREST RELATED

1397

QUESTION ID

S41-F2-Q2

TYPE

Short answer

CORRECT ANSWER

3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q2

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q2/5 • NEAREST RELATED

A right triangle has a leg of 3 opposite angle A, a leg of 4 next to angle A, and a hypotenuse of 5.

What is cos A?

A. $\frac{3}{5}$

B. $\frac{5}{4}$

C. $\frac{4}{5}$

D. $\frac{3}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S41-F2-Q3

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q3/5 • NEAREST RELATED

1398

QUESTION ID

S41-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 4/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S41-F2-Q3

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

In every right triangle with a 30° angle, the side opposite the 30° angle is half the hypotenuse, so $\sin 30° = \frac{1}{2}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q4

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q4/5 • NEAREST RELATED

1399

QUESTION ID

S41-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q4

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q4/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

sin A changes when a right triangle is enlarged but its angles stay the same.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q5

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q5/5 • NEAREST RELATED

1400

QUESTION ID
S41-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Similar right triangles have equal ratios.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q5

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q5/5 • NEAREST RELATED

SET 63 • 8.G.A.5 • GEOMETRY

Give informal arguments for the triangle angle sum and the exterior angle of a triangle

MAIN	Grade 8 main question	**8.G.A.5**
BACKWARD 1	Use linear pairs and straight angles to find unknown angles	**7.G.B.5**
BACKWARD 2	Find an unknown part of an angle that is split into parts	**4.MD.C.7**
FORWARD 1	Prove theorems about isosceles and right triangles	**HSG-CO.C.10**
FORWARD 2	Use the triangle angle sum to prove angle facts about parallelogra	**HSG-CO.C.11**

SET 63 • 8.G.A.5

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.5

Three copies of the same triangle are placed side by side so that its three different angles meet at one point, with no gaps or overlaps, along a straight line.

Explain why this shows that the angles of the triangle add to 180°.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 63 • MAIN • Q1/5

1403

QUESTION ID

S63-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

The three angles fill a straight angle, so they add to 180°.

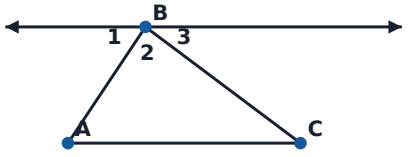
GRADING NOTE REQUIRED WORK: The three angles, one of each kind, together fill a straight angle, which measures 180°. So angle A + angle B + angle C = 180°. Any equivalent explanation is accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 63 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In the figure, the line through B is parallel to AC. Use angles 1, 2, and 3 to explain why the angles of triangle ABC add to 180°.



Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 63 • MAIN • Q2/5

1404

QUESTION ID

S63-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

Angle A + angle B + angle C = 180°

GRADING NOTE REQUIRED WORK: Angle 1 = angle A and angle 3 = angle C (alternate interior angles, because the line is parallel to AC). Angles 1, 2, and 3 together form a straight angle, so angle A + angle B + angle C = 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 63 • MAIN • Q2/5

Use the triangle angle sum to explain why the exterior angle at C of triangle ABC equals angle A + angle B.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 63 • MAIN • Q3/5

1405

QUESTION ID

S63-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

The exterior angle at C = angle A + angle B

GRADING NOTE REQUIRED WORK: The exterior angle and angle C form a linear pair, so exterior angle = 180° − angle C. The triangle's angles add to 180°, so angle A + angle B = 180° − angle C. Therefore exterior angle = angle A + angle B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 63 • MAIN • Q3/5

A student tears off the three corners of a paper triangle and places them together, point to point, with no gaps.

What does she see?

A. The three angles form a full turn (360°).

B. The three angles form a straight angle (180°).

C. The angles cannot fit together.

D. The three angles form a right angle (90°).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 63 • MAIN • Q4/5

1406

QUESTION ID

S63-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

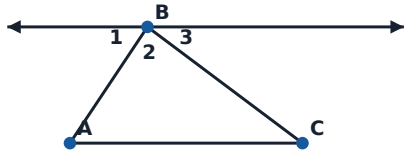
B. The three angles form a straight angle (180°).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 63 • MAIN • Q4/5

True or false?

In the figure, the argument for the triangle angle sum uses the fact that alternate interior angles formed by parallel lines are equal.



A. True **B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 63 • MAIN • Q5/5

1407

QUESTION ID

S63-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 63 • MAIN • Q5/5

SET 63 • 8.G.A.5

BACKWARD 1

Use linear pairs and straight angles to find unknown angles • Grade 7 • Geometry • 7.G.B.5

Two angles form a straight line. One measures 47°.

What does the other measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 63 • BACKWARD 1 • Q1/5

1409

QUESTION ID

S63-B1-Q1

TYPE

Short answer

CORRECT ANSWER

133°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 63 • BACKWARD 1 • Q1/5

Three angles together form a straight line. Two of them measure 50° and 60°.

What does the third measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 63 • BACKWARD 1 • Q2/5

1410

QUESTION ID

S63-B1-Q2

TYPE

Short answer

CORRECT ANSWER

70°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 63 • BACKWARD 1 • Q2/5

Angles of 2x° and 3x° together form a straight angle.

What is x?

A. 45

B. 18

C. 72

D. 36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 63 • BACKWARD 1 • Q3/5

1411

QUESTION ID

S63-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 63 • BACKWARD 1 • Q3/5

True or false?

Two angles that form a linear pair are supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 63 • BACKWARD 1 • Q4/5

1412

QUESTION ID

S63-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 63 • BACKWARD 1 • Q4/5

True or false?

Angles of 95° and 95° together form a straight angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 63 • BACKWARD 1 • Q5/5

1413

QUESTION ID

S63-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They add to 190°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 63 • BACKWARD 1 • Q5/5

SET 63 • 8.G.A.5

BACKWARD 2

Find an unknown part of an angle that is split into parts • Grade 4 • Measurement & Data • 4.MD.C.7

A 180° angle is split into three angles of 40°, 75°, and x°.

What is x?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 63 • BACKWARD 2 • Q1/5

1415

QUESTION ID

S63-B2-Q1

TYPE

Short answer

CORRECT ANSWER

65

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 63 • BACKWARD 2 • Q1/5

A 90° angle is split into angles of 28° and x°.

What is x?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 63 • BACKWARD 2 • Q2/5

1416

QUESTION ID

S63-B2-Q2

TYPE

Short answer

CORRECT ANSWER

62

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 63 • BACKWARD 2 • Q2/5

A 150° angle is split into angles of 90° and x°.

What is x?

A. 240

B. 70

C. 60

D. 50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 63 • BACKWARD 2 • Q3/5

1417

QUESTION ID

S63-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 63 • BACKWARD 2 • Q3/5

True or false?

Angles of 45° and 45° together make a 90° angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 63 • BACKWARD 2 • Q4/5

1418

QUESTION ID

S63-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 63 • BACKWARD 2 • Q4/5

True or false?

Angles of 70°, 50°, and 50° together make a 180° angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 63 • BACKWARD 2 • Q5/5

1419

QUESTION ID

S63-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They make 170°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 63 • BACKWARD 2 • Q5/5

SET 63 • 8.G.A.5

FORWARD 1

Prove theorems about isosceles and right triangles • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC has AB = AC. Prove that angle B = angle C. (Use the reflection across the line that bisects angle A.)

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 63 • FORWARD 1 • Q1/5

1421

QUESTION ID

S63-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

Angle B = angle C

GRADING NOTE REQUIRED WORK: A valid chain of reasons. Other valid orders are accepted. A conclusion without reasons is not full credit. Example: the reflection across the bisector of angle A maps ray AB onto ray AC; since AB = AC, it maps B to C (and C to B). So it maps triangle ABC onto itself and angle B onto angle C; reflections keep angle measures, so angle B = angle C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 63 • FORWARD 1 • Q1/5

An isosceles triangle has a vertex angle of 50°.

Find each base angle, and name the theorems you used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 63 • FORWARD 1 • Q2/5

1422

QUESTION ID

S63-F1-Q2

TYPE

Short answer

CORRECT ANSWER

65° each: the base angles are congruent, and the three angles add to 180°.

GRADING NOTE Both the measure and the theorems are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 63 • FORWARD 1 • Q2/5

Which statement can be proved from the triangle angle sum theorem and the fact that an equilateral triangle's angles are equal?

A. Opposite sides of a parallelogram are congruent.

B. The diagonals of a rectangle are congruent.

C. Vertical angles are congruent.

D. Each angle of an equilateral triangle measures 60°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S63-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 63 • FORWARD 1 • Q3/5

1423

QUESTION ID

S63-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Each angle of an equilateral triangle measures 60°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S63-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 63 • FORWARD 1 • Q3/5

True or false?

In a right triangle, the two acute angles are complementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 63 • FORWARD 1 • Q4/5

1424

QUESTION ID

S63-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 63 • FORWARD 1 • Q4/5

True or false?

A triangle can have two right angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 63 • FORWARD 1 • Q5/5

1425

QUESTION ID

S63-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Two right angles alone add to 180°, leaving 0° for the third angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 63 • FORWARD 1 • Q5/5

SET 63 • 8.G.A.5

FORWARD 2

Use the triangle angle sum to prove angle facts about parallelograms • High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11

In parallelogram ABCD, diagonal AC is drawn. Prove that angle B = angle D. Give a reason for each step.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 63 • FORWARD 2 • Q1/5

1427

QUESTION ID

S63-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

Angle B = angle D

GRADING NOTE REQUIRED WORK: A valid chain of reasons. Other valid orders are accepted. A conclusion without reasons is not full credit. Example: angle BAC = angle DCA (alternate interior angles, AB ∥ DC); angle BCA = angle DAC (alternate interior angles, BC ∥ AD); in each triangle the angles add to 180°, so angle B = 180° − angle BAC − angle BCA = 180° − angle DCA − angle DAC = angle D.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 63 • FORWARD 2 • Q1/5

In parallelogram ABCD, angle A = 3x° and angle B = (x + 20)°.

Find x and all four angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 63 • FORWARD 2 • Q2/5

1428

QUESTION ID

S63-F2-Q2

TYPE

Short answer

CORRECT ANSWER

x = 40; angle A = angle C = 120° and angle B = angle D = 60°.

GRADING NOTE Consecutive angles are supplementary: 3x + x + 20 = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 63 • FORWARD 2 • Q2/5

Which fact, with the triangle angle sum, shows that the four angles of a parallelogram add to 360°?

A. Its diagonals are congruent.

B. A diagonal splits it into two triangles whose angles together make up its four angles.

C. It has four right angles.

D. All of its sides are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 63 • FORWARD 2 • Q3/5

1429

QUESTION ID

S63-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. A diagonal splits it into two triangles whose angles together make up its four angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 63 • FORWARD 2 • Q3/5

True or false?

If one angle of a parallelogram is 90°, all four of its angles are 90°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 63 • FORWARD 2 • Q4/5

1430

QUESTION ID

S63-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 63 • FORWARD 2 • Q4/5

True or false?

The diagonals of every parallelogram are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 63 • FORWARD 2 • Q5/5

1431

QUESTION ID

S63-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only a rectangle's diagonals are always congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 63 • FORWARD 2 • Q5/5

SET 42 • 8.G.A.5 • GEOMETRY

Write and solve equations for angles formed by parallel lines cut by a transversal

MAIN	Grade 8 main question	**8.G.A.5**
BACKWARD 1	Write and solve equations for vertical and adjacent angles	**7.G.B.5**
BACKWARD 2	Solve two-step equations	**7.EE.B.4.a**
FORWARD 1	Prove that alternate interior angles are congruent	**HSG-CO.C.9**
FORWARD 2	Use slopes to show that lines are parallel or perpendicular	**HSG-GPE.B.5**

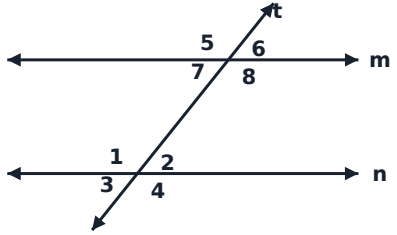
SET 42 • 8.G.A.5

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.5

In the figure, lines m and n are parallel and angle 2 = 52°.

What does angle 6 measure? Give the reason.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q1/5

1434

QUESTION ID

S42-M-Q1

TYPE

Short answer

CORRECT ANSWER

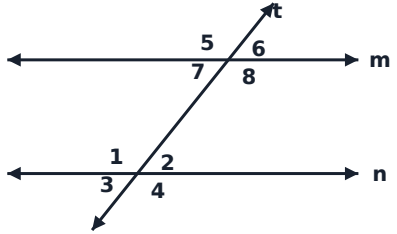
52°; corresponding angles are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q1/5

In the figure, lines m and n are parallel. Angle 1 = (3x + 10)° and angle 8 = 115°.

Write and solve an equation for x.



Equation and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q2/5

1435

QUESTION ID

S42-M-Q2

TYPE

Short answer

CORRECT ANSWER

The equation 3x + 10 = 115 (alternate interior angles are congruent) and the answer x = 35

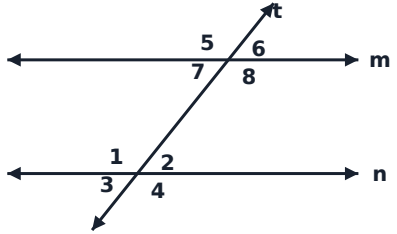
GRADING NOTE Both parts are required: a valid equation and the correct solution. Any valid equation is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q2/5

In the figure, lines m and n are parallel. Angle 2 = (2x + 4)° and angle 8 = (4x − 10)°.

What is x?



A. 93 **B.** 31

C. 43 **D.** 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q3/5

1436

QUESTION ID

S42-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

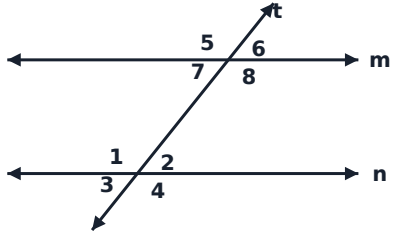
B. 31

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q3/5

In the figure, lines m and n are parallel and angle 4 = 128°.

What does angle 5 measure? Give the reasons.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q4/5

1437

QUESTION ID

S42-M-Q4

TYPE

Short answer

CORRECT ANSWER

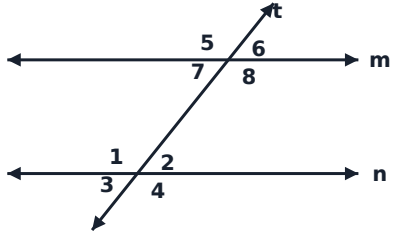
128°: angle 1 = angle 4 (vertical angles), and angle 5 = angle 1 (corresponding angles).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q4/5

True or false?

In the figure, angles 3 and 6 are alternate exterior angles, so they are congruent when m and n are parallel.



A. True **B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q5/5

1438

QUESTION ID

S42-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q5/5

SET 42 • 8.G.A.5

BACKWARD 1

Write and solve equations for vertical and adjacent angles • Grade 7 • Geometry • 7.G.B.5

Two lines intersect. Vertical angles measure (2x + 20)° and 80°.

What is x?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q1/5

1440

QUESTION ID

S42-B1-Q1

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q1/5

Two lines intersect. Adjacent angles measure x° and 3x°.

What is x?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q2/5

1441

QUESTION ID

S42-B1-Q2

TYPE

Short answer

CORRECT ANSWER

45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q2/5

Two lines intersect. Adjacent angles measure (x + 40)° and 2x°.

What is x?

A. $\frac{140}{3}$

B. 50

C. 40

D. 70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q3/5

1442

QUESTION ID

S42-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 140/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q3/5

True or false?

Vertical angles of 5x° and 75° give x = 15.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q4/5

1443

QUESTION ID

S42-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q4/5

True or false?

Adjacent angles that form a straight line measure (x + 10)° and x°. Then x = 90.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q5/5

1444

QUESTION ID

S42-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE x = 85.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q5/5

SET 42 • 8.G.A.5

BACKWARD 2

Solve two-step equations • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Solve for x.

3x + 10 = 115

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q1/5

1446

QUESTION ID

S42-B2-Q1

TYPE

Short answer

CORRECT ANSWER

35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q1/5

Solve for x.

$2x - 15 = 75$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q2/5

1447

QUESTION ID

S42-B2-Q2

TYPE

Short answer

CORRECT ANSWER

45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q2/5

Solve for x.

$6x - 6 = 180$

A. x = 174

B. x = 29

C. x = 31

D. x = 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q3/5

1448

QUESTION ID

S42-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 31

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q3/5

True or false?

x = 20 is the solution of 4x + 20 = 100.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q4/5

1449

QUESTION ID

S42-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$x = 25$ is the solution of $5x - 5 = 115$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q5/5

1450

QUESTION ID

S42-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = 24$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q5/5

1450

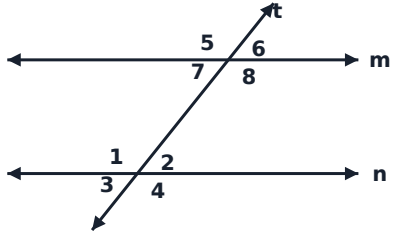
SET 42 • 8.G.A.5

FORWARD 1

Prove that alternate interior angles are congruent • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9

ANSWER KEY Reduced copy of the question page, with its answer

In the figure, lines m and n are parallel. Prove that angle 1 = angle 8. Give a reason for each step.



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q1/5

1452

QUESTION ID

S42-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

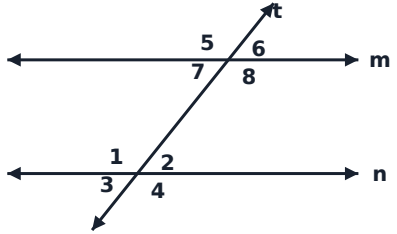
Angle 1 = angle 8

GRADING NOTE REQUIRED WORK: A valid chain of reasons. Other valid orders are accepted. A conclusion without reasons is not full credit. Example: angle 1 = angle 5 (corresponding angles, m ∥ n); angle 5 = angle 8 (vertical angles); so angle 1 = angle 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q1/5

In the figure, lines m and n are parallel. Prove that angle 2 + angle 8 = 180°. Give a reason for each step.



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q2/5

1453

QUESTION ID

S42-F1-Q2

TYPE

Answer with required work

CORRECT ANSWER

Angle 2 + angle 8 = 180°

GRADING NOTE REQUIRED WORK: A valid chain of reasons. Other valid orders are accepted. A conclusion without reasons is not full credit. Example: angle 6 + angle 8 = 180° (linear pair); angle 2 = angle 6 (corresponding angles); so angle 2 + angle 8 = 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q2/5

A proof says: angle 1 = angle 5 (corresponding angles). Angle 5 = angle 8 (?). So angle 1 = angle 8.

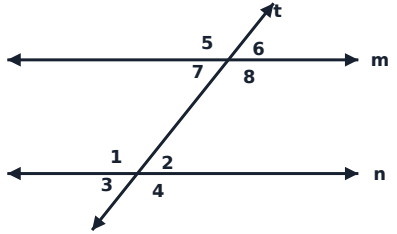
What is the missing reason?

A. Linear pairs are supplementary.

B. Vertical angles are congruent.

C. Alternate interior angles are congruent.

D. The triangle angle sum is 180°.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q3/5

1454

QUESTION ID

S42-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Vertical angles are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q3/5

True or false?

If corresponding angles formed by a transversal are congruent, then the two lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q4/5

1455

QUESTION ID

S42-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q4/5

True or false?

"The angles look equal in the figure" is a valid reason in a proof.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q5/5

1456

QUESTION ID

S42-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q5/5

SET 42 • 8.G.A.5

FORWARD 2

Use slopes to show that lines are parallel or perpendicular • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

A transversal crosses the lines $y = 2x + 1$ and $y = 2x - 3$.

Are the two lines parallel? Explain with slopes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q1/5

1458

QUESTION ID

S42-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes; both lines have slope 2 and different y-intercepts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q1/5

Write an equation of the line through (1, 4) that is perpendicular to $y = -\frac{1}{2}x + 3$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q2/5

1459

QUESTION ID

S42-F2-Q2

TYPE

Short answer

CORRECT ANSWER

y = 2x + 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-F2-Q2**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q2/5

Which pair of lines is perpendicular?

A. $y = x$ and $y = x + 2$

B. $y = 3x$ and $y = -3x$

C. $y = 3x + 1$ and $y = 3x - 1$

D. $y = 3x + 1$ and $y = -\frac{1}{3}x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q3/5

1460

QUESTION ID

S42-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $y = 3x + 1$ and $y = -(1/3)x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q3/5

True or false?

Two distinct non-vertical lines are parallel exactly when their slopes are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q4/5

1461

QUESTION ID

S42-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q4/5

True or false?

The lines $y = \frac{2}{5}x$ and $y = \frac{5}{2}x$ are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q5/5

1462

QUESTION ID

S42-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Perpendicular slopes multiply to −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q5/5

SET 64 • 8.G.A.5 • GEOMETRY

Give informal arguments for the angles formed when parallel lines are cut by a transversal

MAIN	Grade 8 main question	**8.G.A.5**
BACKWARD 1	Use vertical angles and linear pairs to find unknown angles	**7.G.B.5**
BACKWARD 2	Recognize parallel and perpendicular lines	**4.G.A.1**
FORWARD 1	Prove that lines are parallel from congruent angles	**HSG-CO.C.9**
FORWARD 2	Construct a line through a point parallel to a given line	**HSG-CO.D.12**

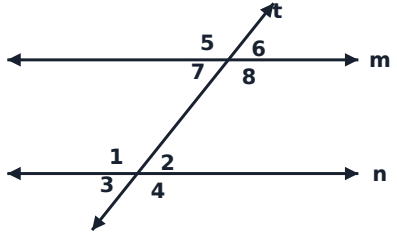
SET 64 • 8.G.A.5

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.5

In the figure, lines m and n are parallel. A translation along the transversal t carries line n onto line m.

Use this translation to explain why angle 2 = angle 6.



Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 64 • MAIN • Q1/5

1465

QUESTION ID

S64-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

Angle 2 = angle 6

GRADING NOTE REQUIRED WORK: The translation moves the point where t meets n to the point where t meets m, keeps t on itself, and carries n onto m (they are parallel). So it carries angle 2 onto angle 6. A translation keeps angle measures, so angle 2 = angle 6.

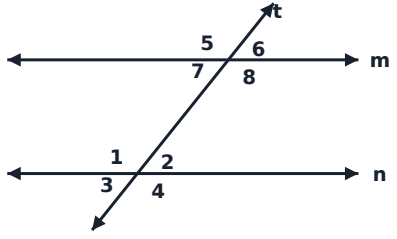
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 64 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In the figure, lines m and n are parallel. A 180° rotation about the midpoint of the part of t between m and n swaps the two lines.

Use this rotation to explain why angle 2 = angle 7.



Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 64 • MAIN • Q2/5

1466

QUESTION ID

S64-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

Angle 2 = angle 7

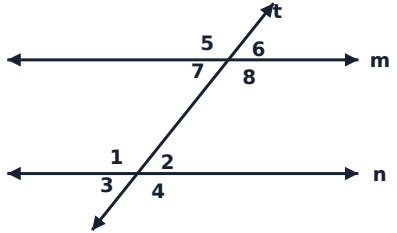
GRADING NOTE REQUIRED WORK: The rotation maps t onto itself and swaps the two intersection points; it maps n onto the parallel line through the other point, which is m. So it carries angle 2 onto angle 7. A rotation keeps angle measures, so angle 2 = angle 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 64 • MAIN • Q2/5

In the figure, lines m and n are parallel and angle 3 = 115°.

Find all eight angles, using only vertical angles, linear pairs, and corresponding angles.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 64 • MAIN • Q3/5

1467

QUESTION ID

S64-M-Q3

TYPE

Short answer

CORRECT ANSWER

Angles 2, 3, 6, and 7 measure 115°; angles 1, 4, 5, and 8 measure 65°.

GRADING NOTE Any correct chain of reasons is accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 64 • MAIN • Q3/5

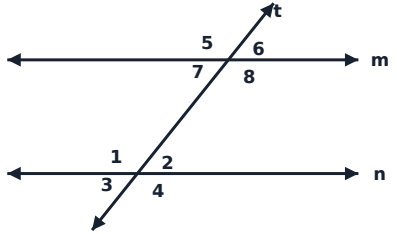
In the figure, which rigid motion carries angle 1 onto angle 5?

A. A translation along t that carries line n onto line m

B. A dilation centered where t meets m

C. A reflection across line m

D. A 90° rotation about the point where t meets n



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 64 • MAIN • Q4/5

1468

QUESTION ID

S64-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. A translation along t that carries line n onto line m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 64 • MAIN • Q4/5

True or false?

If a transversal cuts two lines that are NOT parallel, its corresponding angles are still equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 64 • MAIN • Q5/5

1469

QUESTION ID

S64-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Corresponding angles are equal only when the lines are parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 64 • MAIN • Q5/5

SET 64 • 8.G.A.5

BACKWARD 1

Use vertical angles and linear pairs to find unknown angles • Grade 7 • Geometry • 7.G.B.5

Two lines intersect, and one of the four angles measures 70°.

Find the other three angles.

Angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 64 • BACKWARD 1 • Q1/5

1471

QUESTION ID

S64-B1-Q1

TYPE

Short answer

CORRECT ANSWER

110°, 70°, and 110°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 64 • BACKWARD 1 • Q1/5

Two angles form a linear pair. One is 3 times as large as the other.

What are the two angles?

Angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 64 • BACKWARD 1 • Q2/5

1472

QUESTION ID

S64-B1-Q2

TYPE

Short answer

CORRECT ANSWER

45° and 135°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 64 • BACKWARD 1 • Q2/5

Two lines intersect. One angle measures 125°.

What does its vertical angle measure?

A. 235°

B. 55°

C. 35°

D. 125°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 64 • BACKWARD 1 • Q3/5

1473

QUESTION ID

S64-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 125°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 64 • BACKWARD 1 • Q3/5

True or false?

Vertical angles are the opposite angles formed by two intersecting lines.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 64 • BACKWARD 1 • Q4/5

1474

QUESTION ID

S64-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 64 • BACKWARD 1 • Q4/5

True or false?

The two angles of a linear pair are always equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 64 • BACKWARD 1 • Q5/5

1475

QUESTION ID

S64-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They add to 180°; they are equal only when both are 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 64 • BACKWARD 1 • Q5/5

SET 64 • 8.G.A.5

BACKWARD 2

Recognize parallel and perpendicular lines • Grade 4 • Geometry • 4.G.A.1

What are lines in a plane called that never meet, however far they are extended?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 64 • BACKWARD 2 • Q1/5

1477

QUESTION ID

S64-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Parallel lines

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 64 • BACKWARD 2 • Q1/5

What are lines called that meet at a right angle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 64 • BACKWARD 2 • Q2/5

1478

QUESTION ID

S64-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Perpendicular lines

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 64 • BACKWARD 2 • Q2/5

Which shape always has two pairs of parallel sides?

A. Kite

B. Triangle

C. Parallelogram

D. Trapezoid

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 64 • BACKWARD 2 • Q3/5

1479

QUESTION ID

S64-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Parallelogram

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 64 • BACKWARD 2 • Q3/5

True or false?

The two rails of a straight railroad track are a model of parallel lines.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 64 • BACKWARD 2 • Q4/5

1480

QUESTION ID

S64-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 64 • BACKWARD 2 • Q4/5

True or false?

Two lines that cross each other can be parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 64 • BACKWARD 2 • Q5/5

1481

QUESTION ID

S64-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 64 • BACKWARD 2 • Q5/5

SET 64 • 8.G.A.5

FORWARD 1

Prove that lines are parallel from congruent angles • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9

ANSWER KEY Reduced copy of the question page, with its answer

Lines m and n are cut by transversal t, and a pair of alternate interior angles is congruent.

Explain why m and n must be parallel. (Suppose they meet at a point P, and look at the triangle formed with t.)

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 64 • FORWARD 1 • Q1/5

1483

QUESTION ID

S64-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

The lines are parallel.

GRADING NOTE REQUIRED WORK: A complete argument: if m and n met at P, they would form a triangle with t. One of the congruent angles would be an exterior angle of that triangle and the other a remote interior angle. But an exterior angle equals the sum of the two remote interior angles, so it is greater than either one. That is a contradiction, so m and n do not meet: they are parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 64 • FORWARD 1 • Q1/5

Lines a and b are cut by a transversal. A pair of corresponding angles measure (4x + 6)° and (5x − 9)°.

For what value of x are the lines parallel?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 64 • FORWARD 1 • Q2/5

1484

QUESTION ID

S64-F1-Q2

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 64 • FORWARD 1 • Q2/5

Two lines are cut by a transversal. Which angle pair, if congruent, proves that the lines are parallel?

A. A pair of alternate interior angles

B. A pair of vertical angles

C. A linear pair

D. Two adjacent angles on the same line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 64 • FORWARD 1 • Q3/5

1485

QUESTION ID

S64-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. A pair of alternate interior angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 64 • FORWARD 1 • Q3/5

True or false?

If the same-side interior angles formed by a transversal add to 180°, the two lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 64 • FORWARD 1 • Q4/5

1486

QUESTION ID

S64-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 64 • FORWARD 1 • Q4/5

True or false?

If the vertical angles at one intersection are congruent, the two lines cut by the transversal are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 64 • FORWARD 1 • Q5/5

1487

QUESTION ID

S64-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Vertical angles are always congruent, so they prove nothing about the lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 64 • FORWARD 1 • Q5/5

SET 64 • 8.G.A.5

FORWARD 2

Construct a line through a point parallel to a given line • High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.D.12

Use a compass and straightedge to construct the line through P that is parallel to line k. Draw a transversal through P, then copy the angle it makes with k so that it is a corresponding angle at P. Leave your construction marks.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.D.12 • Set 64 • FORWARD 2 • Q1/5

1489

QUESTION ID

S64-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

A transversal through P and k, the arcs used to copy the angle at P, and a line through P parallel to k. Any correct compass-and-straightedge construction with visible marks is accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.D.12 • Set 64 • FORWARD 2 • Q1/5

Why does copying the angle in that construction give a line that is parallel to k?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.D.12 • Set 64 • FORWARD 2 • Q2/5

1490

QUESTION ID

S64-F2-Q2

TYPE

Short answer

CORRECT ANSWER

The copied angle and the angle at k are congruent corresponding angles, so the lines are parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.D.12 • Set 64 • FORWARD 2 • Q2/5

When a parallel line is constructed by copying an angle, which angles are made congruent?

A. Vertical angles

B. Corresponding angles formed by the transversal

C. A linear pair

D. Two angles of a triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.D.12 • Set 64 • FORWARD 2 • Q3/5

1491

QUESTION ID

S64-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Corresponding angles formed by the transversal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.D.12 • Set 64 • FORWARD 2 • Q3/5

True or false?

In a compass-and-straightedge construction, the compass is used to copy distances.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.D.12 • Set 64 • FORWARD 2 • Q4/5

1492

QUESTION ID

S64-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.D.12 • Set 64 • FORWARD 2 • Q4/5

True or false?

A compass-and-straightedge construction may use a protractor to measure the angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.D.12 • Set 64 • FORWARD 2 • Q5/5

1493

QUESTION ID

S64-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only a compass and a straightedge are allowed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.D.12 • Set 64 • FORWARD 2 • Q5/5

SET 43 • 8.G.A.5 • GEOMETRY

Use the angle-angle criterion to show that triangles are similar

MAIN	Grade 8 main question	**8.G.A.5**
BACKWARD 1	Find a missing length in a scale copy	**7.G.A.1**
BACKWARD 3	Decide whether three angle measures can make a triangle and w…	**7.G.A.2**
FORWARD 1	Establish the angle-angle criterion with similarity transformations	**HSG-SRT.A.3**
FORWARD 2	Use similar triangles to solve multistep problems, including altitud…	**HSG-SRT.B.5**

SET 43 • 8.G.A.5

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.5

Triangle ABC has angles of 35° and 85°. Triangle DEF has angles of 35° and 60°.

Are the triangles similar? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q1/5

1496

QUESTION ID

S43-M-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. The third angle of DEF is 180° − 35° − 60° = 85°, so two pairs of angles are equal (AA).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q1/5

A 5-foot post casts a 3-foot shadow. At the same time, a tree casts a 12-foot shadow.

Explain why the two triangles are similar, and find the height of the tree.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q2/5

1497

QUESTION ID

S43-M-Q2

TYPE

Short answer

CORRECT ANSWER

Both triangles have a right angle and the same angle of the sun's rays, so they are similar by AA. The tree is 20 feet tall.

GRADING NOTE Grade both: the AA reason and 20 feet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q2/5

Which information shows that two triangles are similar by the angle-angle criterion?

A. One pair of sides is proportional.

B. All the sides are equal.

C. Two pairs of corresponding angles are equal.

D. One pair of corresponding angles is equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q3/5

1498

QUESTION ID

S43-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Two pairs of corresponding angles are equal.

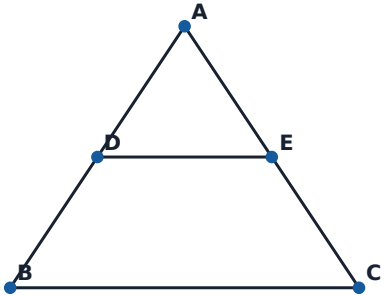
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q3/5

In the figure, segment DE is parallel to BC.

Explain why triangle ADE is similar to triangle ABC.



Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q4/5

1499

QUESTION ID

S43-M-Q4

TYPE

Short answer

CORRECT ANSWER

The triangles share angle A, and angle ADE = angle ABC (corresponding angles, because DE ∥ BC). So they are similar by AA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q4/5

True or false?

Two triangles that each have a 90° angle and a 40° angle are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q5/5

1500

QUESTION ID

S43-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q5/5

SET 43 • 8.G.A.5

BACKWARD 1

Find a missing length in a scale copy • Grade 7 • Geometry • 7.G.A.1

A triangle has sides 4, 6, and 8. In a scale copy, the side of 4 becomes 10.

What does the side of 6 become?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q1/5

1502

QUESTION ID

S43-B1-Q1

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q1/5

A drawing with a scale factor of 3 shows a 2 cm side.

How long is that side in the scale copy?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q2/5

1503

QUESTION ID

S43-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q2/5

A 5-by-8 rectangle is scaled so the 5 becomes 15.

What does the 8 become?

A. 40

B. 24

C. 13

D. 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q3/5

1504

QUESTION ID

S43-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q3/5

True or false?

In a scale copy, every length is multiplied by the same scale factor.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q4/5

1505

QUESTION ID

S43-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q4/5

True or false?

In a scale copy with a scale factor of 2, every angle is doubled.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q5/5

1506

QUESTION ID

S43-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q5/5

SET 43 • 8.G.A.5

BACKWARD 3

Decide whether three angle measures can make a triangle and whether they determine one triangle • Grade 7 • Geometry • 7.G.A.2

Can angles of 45°, 55°, and 80° be the three angles of a triangle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 43 • BACKWARD 3 • Q1/5

1508

QUESTION ID

S43-B3-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. They add to 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 43 • BACKWARD 3 • Q1/5

A triangle must have angles of 30°, 60°, and 90°.

Do these conditions determine one triangle, more than one triangle, or no triangle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 43 • BACKWARD 3 • Q2/5

1509

QUESTION ID

S43-B3-Q2

TYPE

Short answer

CORRECT ANSWER

More than one triangle: triangles of different sizes have these angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 43 • BACKWARD 3 • Q2/5

Which three angle measures can be the angles of a triangle?

A. 50°, 60°, 70°

B. 90°, 90°, 10°

C. 100°, 50°, 40°

D. 50°, 60°, 80°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 43 • BACKWARD 3 • Q3/5

1510

QUESTION ID

S43-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 50°, 60°, 70°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 43 • BACKWARD 3 • Q3/5

True or false?

Triangles with the same three angle measures can have different side lengths.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 43 • BACKWARD 3 • Q4/5

1511

QUESTION ID

S43-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 43 • BACKWARD 3 • Q4/5

True or false?

Angles of 90°, 60°, and 40° can be the angles of one triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 43 • BACKWARD 3 • Q5/5

1512

QUESTION ID

S43-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They add to 190°, not 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 43 • BACKWARD 3 • Q5/5

SET 43 • 8.G.A.5

FORWARD 1

Establish the angle-angle criterion with similarity transformations • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

Triangles ABC and DEF have angle A = angle D and angle B = angle E.

Which transformations show that the triangles are similar?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q1/5

1514

QUESTION ID

S43-F1-Q1

TYPE

Short answer

CORRECT ANSWER

A dilation that makes AB the same length as DE, followed by rigid motions that move the image onto DEF; the equal angles make the third vertices match.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangles ABC and DEF have angle A = angle D and angle B = angle E.

Explain why a similarity transformation maps triangle ABC onto triangle DEF.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q2/5

1515

QUESTION ID

S43-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Dilate ABC by the scale factor DE ÷ AB, so the image of AB has the same length as DE. Rigid motions then move that image onto DE (A onto D, B onto E), with the third vertex on the same side as F. Because angle A = angle D and angle B = angle E, the image's other two sides lie along rays DF and EF, and two rays meet in only one point, so the third vertex lands on F. (The third angles are then equal too.)

GRADING NOTE Full credit needs the dilation, the rigid motions, and why the third vertex must land on F.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F1-Q2**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q2/5

Triangles with angles 30°, 70°, 80° and 70°, 30°, 80° are...

A. congruent in every case

B. similar

C. not similar

D. impossible to compare

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q3/5

1516

QUESTION ID

S43-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. similar

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F1-Q3**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q3/5

True or false?

Two equilateral triangles are always similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q4/5

1517

QUESTION ID

S43-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q4/5

True or false?

Two isosceles triangles are always similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q5/5

1518

QUESTION ID

S43-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q5/5

SET 43 • 8.G.A.5

FORWARD 2

Use similar triangles to solve multistep problems, including altitudes and parallel segments • High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

In right triangle ABC, the right angle is at C, and D is the foot of the altitude from C to hypotenuse AB. AD = 4 and DB = 9.

Find CD, and name the similar triangles you used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q1

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q1/5

1520

QUESTION ID

S43-F2-Q1

TYPE

Short answer

CORRECT ANSWER

CD = 6. Triangles ACD and CBD are similar, so AD ÷ CD = CD ÷ DB and $CD^2 = 4 \cdot 9 = 36$.

GRADING NOTE Both the length and the similar triangles are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q1

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q1/5

In triangle ABC, D is on AB and E is on AC, and DE is parallel to BC. AD = 5, AB = 15, and DE = 4.

Find BC.

BC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q2

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q2/5

1521

QUESTION ID

S43-F2-Q2

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F2-Q2**

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q2/5

A mirror lies on level ground 6 m from the base of a tree. A person stands 2 m from the mirror, on the other side, and sees the top of the tree in it. Her eyes are 1.5 m above the ground. The angles at the mirror are equal.

How tall is the tree?

A. 8 m

B. 0.5 m

C. 4.5 m

D. 9 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q3

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q3/5

1522

QUESTION ID

S43-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 4.5 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q3/5

True or false?

In triangle ABC, D is on AB and E is on AC, and DE is parallel to BC. If AD = 2, DB = 6, and AE = 3, then EC = 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q4

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q4/5

1523

QUESTION ID

S43-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q4

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q4/5

True or false?

In right triangle ABC with the right angle at C, and D the foot of the altitude from C to hypotenuse AB, $AC^2 = AD \cdot DB$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q5

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q5/5

1524

QUESTION ID

S43-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $AC^2 = AD \cdot AB$; it is CD^2 that equals $AD \cdot DB$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q5

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q5/5

SET 65 • 8.G.A.5 • GEOMETRY

Give an informal argument for the angle-angle criterion for similar triangles

MAIN	Grade 8 main question	**8.G.A.5**
BACKWARD 1	Use scale drawings: lengths change, angles stay the same	**7.G.A.1**
BACKWARD 2	Use tables of equivalent ratios	**6.RP.A.3.a**
FORWARD 1	Prove the triangle proportionality theorem with similar triangles	**HSG-SRT.B.4**
FORWARD 2	Explain why trigonometric ratios depend only on the angle (simila.	**HSG-SRT.C.6**

SET 65 • 8.G.A.5

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.5

Triangle ABC has angles of 40° at A and 75° at B. Triangle DEF has angles of 40° at D and 75° at E, and DE = 2 · AB.

Explain, using a dilation and rigid motions, why the triangles are similar.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 65 • MAIN • Q1/5

1527

QUESTION ID

S65-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

The triangles are similar.

GRADING NOTE REQUIRED WORK: Dilate ABC by a scale factor of 2, so the image of AB has the same length as DE. Rigid motions then move that image onto DE, with the third vertex on the same side as F. The 40° and 75° angles make the other two sides lie along rays DF and EF, which meet at only one point, so the third vertex lands on F. A dilation followed by rigid motions maps ABC onto DEF.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 65 • MAIN • Q1/5

Explain why two triangles with two pairs of equal angles must also have the third pair of angles equal.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 65 • MAIN • Q2/5

1528

QUESTION ID

S65-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

The third angles are equal.

GRADING NOTE REQUIRED WORK: In each triangle the three angles add to 180°, so each third angle equals 180° minus the same two measures. The third angles are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 65 • MAIN • Q2/5

Use a ruler and a protractor to draw two triangles of different sizes that each have angles of 50° and 60°. Measure their sides, and compare the ratios of corresponding sides.

Comparison: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 65 • MAIN • Q3/5

1529

QUESTION ID

S65-M-Q3

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: Two triangles of different sizes, each with angles of 50°, 60°, and 70°

Written: The ratios of corresponding sides are all about the same (equal up to measuring error), so the triangles are similar.

GRADING NOTE Grade the two drawings and the ratios computed from the student's own measurements; there is no preset size.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 65 • MAIN • Q3/5

Which argument shows that two triangles with two pairs of equal angles are similar?

A. Rotate one triangle 90°.

B. Dilate one so a pair of corresponding sides match, then use rigid motions; the equal angles force the third vertices to match.

C. Check that the triangles have equal areas.

D. Measure all three sides of each and check that they are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 65 • MAIN • Q4/5

1530

QUESTION ID

S65-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Dilate one so a pair of corresponding sides match, then use rigid motions; the equal angles force the third vertices to match.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 65 • MAIN • Q4/5

True or false?

Knowing that just one pair of angles is equal is enough to show two triangles are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 65 • MAIN • Q5/5

1531

QUESTION ID

S65-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Two pairs of equal angles are needed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 65 • MAIN • Q5/5

SET 65 • 8.G.A.5

BACKWARD 1

Use scale drawings: lengths change, angles stay the same • Grade 7 • Geometry • 7.G.A.1

A triangle has angles of 30°, 60°, and 90°, and its longest side is 6 cm. A scale copy has a scale factor of 2.

What are the angles and the longest side of the copy?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 65 • BACKWARD 1 • Q1/5

1533

QUESTION ID

S65-B1-Q1

TYPE

Short answer

CORRECT ANSWER

30°, 60°, and 90°; the longest side is 12 cm.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 65 • BACKWARD 1 • Q1/5

In a scale copy, a 4 cm side becomes 10 cm.

What is the scale factor, and what does a 70° angle become?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 65 • BACKWARD 1 • Q2/5

1534

QUESTION ID

S65-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2.5; the angle is still 70°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 65 • BACKWARD 1 • Q2/5

A scale drawing of a triangle uses a scale factor of $\frac{1}{2}$.

Which statement is true?

A. Each side stays the same length.

B. Each angle is half as large.

C. Each angle doubles.

D. Each angle stays the same, and each side is half as long.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 65 • BACKWARD 1 • Q3/5

1535

QUESTION ID

S65-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Each angle stays the same, and each side is half as long.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 65 • BACKWARD 1 • Q3/5

True or false?

A scale copy of a triangle has the same angle measures as the original.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 65 • BACKWARD 1 • Q4/5

1536

QUESTION ID

S65-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 65 • BACKWARD 1 • Q4/5

True or false?

In a scale copy with a scale factor of 3, a side of 5 cm becomes 8 cm.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 65 • BACKWARD 1 • Q5/5

1537

QUESTION ID

S65-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It becomes 15 cm.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 65 • BACKWARD 1 • Q5/5

SET 65 • 8.G.A.5

BACKWARD 2

Use tables of equivalent ratios • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a

Complete the equivalent ratios.

3 : 5 = 6 : ___ = 9 : ___

Missing values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 65 • BACKWARD 2 • Q1/5

1539

QUESTION ID

S65-B2-Q1

TYPE

Short answer

CORRECT ANSWER

10 and 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 65 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Are 4 : 6 and 6 : 9 equivalent ratios? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 65 • BACKWARD 2 • Q2/5

1540

QUESTION ID

S65-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. Both equal 2 : 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 65 • BACKWARD 2 • Q2/5

Which ratio is equivalent to 8 : 12?

A. 3 : 2

B. 8 : 4

C. 2 : 3

D. 4 : 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 65 • BACKWARD 2 • Q3/5

1541

QUESTION ID

S65-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2 : 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 65 • BACKWARD 2 • Q3/5

True or false?

5 : 7 and 15 : 21 are equivalent ratios.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 65 • BACKWARD 2 • Q4/5

1542

QUESTION ID

S65-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 65 • BACKWARD 2 • Q4/5

True or false?

2 : 5 and 4 : 7 are equivalent ratios.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 65 • BACKWARD 2 • Q5/5

1543

QUESTION ID

S65-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 65 • BACKWARD 2 • Q5/5

SET 65 • 8.G.A.5

FORWARD 1

Prove the triangle proportionality theorem with similar triangles • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, D is on AB and E is on AC, and DE is parallel to BC.

Prove that AD ÷ AB = AE ÷ AC. Give a reason for each step.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 65 • FORWARD 1 • Q1/5

1545

QUESTION ID

S65-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

AD ÷ AB = AE ÷ AC

GRADING NOTE REQUIRED WORK: A valid chain of reasons. Other valid orders are accepted. A conclusion without reasons is not full credit. Example: angle ADE = angle ABC (corresponding angles, DE ∥ BC); angle A is shared; so triangle ADE is similar to triangle ABC (AA), and corresponding sides are proportional: AD ÷ AB = AE ÷ AC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-F1-Q1**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 65 • FORWARD 1 • Q1/5

In triangle ABC, D is on AB and E is on AC, and DE is parallel to BC. AD = 4, DB = 6, and AE = 6.

Find EC.

EC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 65 • FORWARD 1 • Q2/5

1546

QUESTION ID

S65-F1-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-F1-Q2**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 65 • FORWARD 1 • Q2/5

In triangle ABC, D is on AB and E is on AC, with AD ÷ DB = AE ÷ EC.

What does the converse of the triangle proportionality theorem say?

A. DE is parallel to BC.

B. D and E must be midpoints.

C. DE is perpendicular to BC.

D. DE is as long as BC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S65-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 65 • FORWARD 1 • Q3/5

1547

QUESTION ID

S65-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. DE is parallel to BC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S65-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 65 • FORWARD 1 • Q3/5

True or false?

The segment joining the midpoints of two sides of a triangle is parallel to the third side and half as long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 65 • FORWARD 1 • Q4/5

1548

QUESTION ID

S65-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 65 • FORWARD 1 • Q4/5

True or false?

If DE is parallel to BC in triangle ABC, with D on AB and E on AC, then DE = BC.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 65 • FORWARD 1 • Q5/5

1549

QUESTION ID

S65-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE DE is shorter: DE ÷ BC = AD ÷ AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 65 • FORWARD 1 • Q5/5

SET 65 • 8.G.A.5

FORWARD 2

Explain why trigonometric ratios depend only on the angle (similar right triangles) • High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

Two right triangles each have a 35° angle.

Explain why the ratio (leg opposite 35°) ÷ (hypotenuse) is the same in both.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q1

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 65 • FORWARD 2 • Q1/5

1551

QUESTION ID

S65-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Both triangles have a 90° angle and a 35° angle, so they are similar by AA, and corresponding sides of similar triangles have equal ratios.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-F2-Q1**

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 65 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A right triangle with a 35° angle has a hypotenuse of 10 and a leg of 5.74 opposite the 35° angle. A larger right triangle with a 35° angle has a hypotenuse of 25.

How long is its leg opposite the 35° angle?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q2

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 65 • FORWARD 2 • Q2/5

1552

QUESTION ID

S65-F2-Q2

TYPE

Short answer

CORRECT ANSWER

14.35

GRADING NOTE 5.74 ÷ 10 = x ÷ 25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-F2-Q2**

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 65 • FORWARD 2 • Q2/5

Why does sin 35° have only one value?

A. All right triangles are congruent.

B. Every hypotenuse has length 1.

C. The opposite leg is always 35 units long.

D. All right triangles with a 35° angle are similar (AA).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q3

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 65 • FORWARD 2 • Q3/5

1553

QUESTION ID

S65-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. All right triangles with a 35° angle are similar (AA).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q3

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 65 • FORWARD 2 • Q3/5

True or false?

In every right triangle with a 20° angle, the ratio (leg next to 20°) ÷ (hypotenuse) is the same.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q4

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 65 • FORWARD 2 • Q4/5

1554

QUESTION ID

S65-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q4

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 65 • FORWARD 2 • Q4/5

True or false?

Of two right triangles with a 50° angle, the one with the longer hypotenuse has the larger sine of 50°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q5

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 65 • FORWARD 2 • Q5/5

1555

QUESTION ID

S65-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The triangles are similar, so sin 50° is the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q5

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 65 • FORWARD 2 • Q5/5

SET 44 • 8.G.B.6 • GEOMETRY

Explain a proof of the Pythagorean theorem and its converse

MAIN	Grade 8 main question	**8.G.B.6**
BACKWARD 1	Find the area of a right triangle	**6.G.A.1**
BACKWARD 2	Find the area of a square	**3.MD.C.7.b**
FORWARD 1	Prove the Pythagorean theorem with similar triangles	**HSG-SRT.B.4**
FORWARD 2	Derive the equation of a circle from the Pythagorean theorem	**HSG-GPE.A.1**

SET 44 • 8.G.B.6

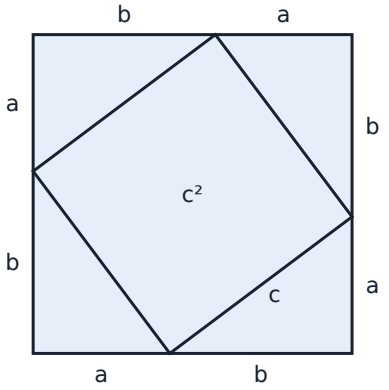
MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.B.6

ANSWER KEY Reduced copy of the question page, with its answer

The figure shows a square with sides a + b, made of four right triangles and a tilted square with side c.

Explain how the figure proves that $a^2 + b^2 = c^2$.



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q1/5

1558

QUESTION ID

S44-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

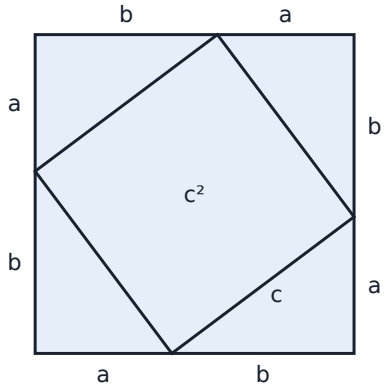
$a^2 + b^2 = c^2$

GRADING NOTE REQUIRED WORK: A complete argument: the large square's area is $(a + b)^2 = a^2 + 2ab + b^2$; it is also 4 triangles ($4 \cdot \frac{1}{2}ab = 2ab$) plus the inner square c^2; so $a^2 + 2ab + b^2 = 2ab + c^2$, and subtracting 2ab gives $a^2 + b^2 = c^2$. Any equivalent wording is accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q1/5

In the figure, write the area of the large square as the sum of the areas of the four triangles and the inner square.



Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q2/5

1559

QUESTION ID

S44-M-Q2

TYPE

Short answer

CORRECT ANSWER

4 ·

$$\frac{1}{2}ab + c^2 = 2ab + c^2$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q2/5

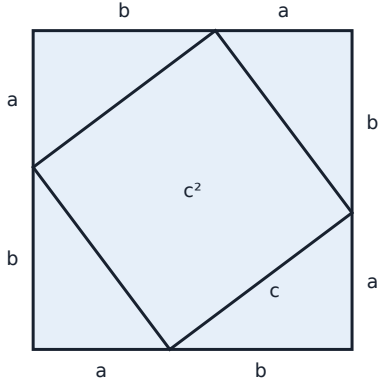
In the proof with the figure, which step comes after writing $(a + b)^2 = 2ab + c^2$?

A. Divide both sides by c

B. Expand to $a^2 + 2ab + b^2 = 2ab + c^2$, then subtract 2ab from both sides

C. Add 2ab to both sides

D. Take the square root of each term



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q3/5

1560

QUESTION ID

S44-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Expand to $a^2 + 2ab + b^2 = 2ab + c^2$, then subtract 2ab from both sides

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q3/5

A triangle has sides a, b, and c with $a^2 + b^2 = c^2$. Explain why it must be a right triangle.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q4/5

1561

QUESTION ID

S44-M-Q4

TYPE

Short answer

CORRECT ANSWER

A right triangle with legs a and b has a hypotenuse of $\sqrt{(a^2 + b^2)} = c$. So the two triangles have the same three sides and are congruent (SSS), and the given triangle has a right angle too.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q4/5

True or false?

The Pythagorean theorem $a^2 + b^2 = c^2$ is true for every triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q5/5

1562

QUESTION ID

S44-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is true only for right triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q5/5

SET 44 • 8.G.B.6

BACKWARD 1

Find the area of a right triangle • Grade 6 • Geometry • 6.G.A.1

A right triangle has legs of 3 and 4.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q1/5

1564

QUESTION ID

S44-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q1/5

A right triangle has legs of 5 and 12.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q2/5

1565

QUESTION ID

S44-B1-Q2

TYPE

Short answer

CORRECT ANSWER

30 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q2/5

What is the total area of four right triangles that each have legs of 2 and 6?

A. 24

B. 12

C. 48

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q3/5

1566

QUESTION ID

S44-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q3/5

True or false?

A right triangle with legs of 8 and 3 has an area of 12 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q4/5

1567

QUESTION ID

S44-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q4/5

True or false?

A right triangle with legs of 6 and 6 has an area of 36 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q5/5

1568

QUESTION ID

S44-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q5/5

SET 44 • 8.G.B.6

BACKWARD 2

Find the area of a square • Grade 3 • Measurement & Data • 3.MD.C.7.b

A square has sides of 7 units.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q1/5

1570

QUESTION ID

S44-B2-Q1

TYPE

Short answer

CORRECT ANSWER

49 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q1/5

A square has sides of 9 units.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q2/5

1571

QUESTION ID

S44-B2-Q2

TYPE

Short answer

CORRECT ANSWER

81 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q2/5

A square has sides of 5 units.

What is its area?

A. 15 square units

B. 25 square units

C. 10 square units

D. 20 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q3/5

1572

QUESTION ID

S44-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 25 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q3/5

True or false?

A square with 4-unit sides has an area of 16 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q4/5

1573

QUESTION ID

S44-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q4/5

True or false?

A square with 6-unit sides has an area of 24 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q5/5

1574

QUESTION ID

S44-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q5/5

SET 44 • 8.G.B.6

FORWARD 1

Prove the Pythagorean theorem with similar triangles • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

In right triangle ABC with the right angle at C, altitude CD is drawn to hypotenuse AB.

Why is $AC^2 = AD \cdot AB$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q1/5

1576

QUESTION ID

S44-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Triangle ACD is similar to triangle ABC (AA), so $AC/AB = AD/AC$, which gives $AC^2 = AD \cdot AB$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q1/5

In right triangle ABC, the right angle is at C, and D is the foot of the altitude from C to hypotenuse AB. Similar triangles give $AC^2 = AD \cdot AB$ and $BC^2 = BD \cdot AB$.

Add these two equations. What do you get?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q2/5

1577

QUESTION ID

S44-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$AC^2 + BC^2 = AD \cdot AB + BD \cdot AB = AB(AD + BD) = AB \cdot AB = AB^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q2/5

In right triangle ABC, the right angle is at C, and D is the foot of the altitude from C to hypotenuse AB.

Which criterion shows that triangle ACD is similar to triangle ABC?

A. ASA

B. AA

C. SAS

D. SSS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q3/5

1578

QUESTION ID

S44-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. AA

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F1-Q3**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q3/5

True or false?

The altitude to the hypotenuse divides a right triangle into two triangles similar to it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q4/5

1579

QUESTION ID

S44-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q4/5

True or false?

The altitude to the hypotenuse always cuts the hypotenuse into two equal parts.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q5/5

1580

QUESTION ID

S44-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q5/5

SET 44 • 8.G.B.6

FORWARD 2

Derive the equation of a circle from the Pythagorean theorem • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

Write the equation of the circle with center (0, 0) and radius 5.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q1/5

1582

QUESTION ID

S44-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2 + y^2 = 25$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F2-Q1**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A point (x, y) is on the circle with center (2, −1) and radius 3. Use a right triangle with legs |x − 2| and |y + 1| and the Pythagorean theorem to derive the equation of the circle.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q2/5

1583

QUESTION ID

S44-F2-Q2

TYPE

Answer with required work

CORRECT ANSWER

$(x - 2)^2 + (y + 1)^2 = 9$

GRADING NOTE REQUIRED WORK: The horizontal leg is |x − 2|, the vertical leg is |y + 1|, and the hypotenuse is the radius 3, so $(x - 2)^2 + (y + 1)^2 = 3^2 = 9$. The right triangle and the use of the Pythagorean theorem must be shown.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q2/5

Which point is on the circle $x^2 + y^2 = 100$?

A. (5, 5)

B. (10, 10)

C. (6, 8)

D. (6, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q3/5

1584

QUESTION ID

S44-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (6, 8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q3/5

True or false?

The equation $(x - h)^2 + (y - k)^2 = r^2$ comes from the Pythagorean theorem.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q4/5

1585

QUESTION ID

S44-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q4/5

True or false?

The circle $x^2 + y^2 = 16$ has a radius of 16.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q5/5

1586

QUESTION ID
S44-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Its radius is 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q5/5

SET 45 • 8.G.B.6 • GEOMETRY

Use the converse of the Pythagorean theorem to decide whether a triangle is a right triangle

MAIN	Grade 8 main question	**8.G.B.6**
BACKWARD 1	Find the squares of whole numbers	**6.EE.A.1**
BACKWARD 2	Evaluate an expression with exponents	**6.EE.A.2.c**
FORWARD 1	Prove the converse of the Pythagorean theorem with congruent tri	**HSG-CO.C.10**
FORWARD 2	Use coordinates to show that a triangle is a right triangle	**HSG-GPE.B.4**

SET 45 • 8.G.B.6

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.B.6

Is a triangle with sides 9, 12, and 15 a right triangle? Show your check.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q1/5

1589

QUESTION ID

S45-M-Q1

TYPE

Short answer

CORRECT ANSWER

Yes.

$9^2 + 12^2 = 81 + 144 = 225 = 15^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q1/5

Is a triangle with sides 7, 8, and 11 a right triangle? Show your check.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q2/5

1590

QUESTION ID

S45-M-Q2

TYPE

Short answer

CORRECT ANSWER

No. $7^2 + 8^2 = 49 + 64 = 113$, but $11^2 = 121$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q2/5

Which side lengths form a right triangle?

A. 6, 8, 12

B. 20, 21, 29

C. 5, 7, 9

D. 10, 12, 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q3/5

1591

QUESTION ID

S45-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 20, 21, 29

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q3/5

A carpenter checks a corner. Points 3 feet and 4 feet from the corner along the two edges are 5 feet apart.

Is the corner a right angle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q4/5

1592

QUESTION ID

S45-M-Q4

TYPE

Short answer

CORRECT ANSWER

Yes. $3^2 + 4^2 = 5^2$, so the angle is 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q4/5

True or false?

A triangle with sides 2, 3, and 4 is a right triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q5/5

1593

QUESTION ID

S45-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $4 + 9 = 13$, but $4^2 = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q5/5

SET 45 • 8.G.B.6

BACKWARD 1

Find the squares of whole numbers • Grade 6 • Expressions & Equations • 6.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Find the value.

15^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q1/5

1595

QUESTION ID

S45-B1-Q1

TYPE

Short answer

CORRECT ANSWER

225

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q1/5

Find the value.

11^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q2/5

1596

QUESTION ID

S45-B1-Q2

TYPE

Short answer

CORRECT ANSWER

121

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q2/5

Find the value.

13^2

A. 26

B. 169

C. 139

D. 196

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q3/5

1597

QUESTION ID

S45-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 169

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$20^2 = 400$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q4/5

1598

QUESTION ID

S45-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q4/5

True or false?

$9^2 = 18$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q5/5

1599

QUESTION ID

S45-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q5/5

SET 45 • 8.G.B.6

BACKWARD 2

Evaluate an expression with exponents • Grade 6 • Expressions & Equations • 6.EE.A.2.c

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $a^2 + b^2$ when a = 6 and b = 8.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 45 • BACKWARD 2 • Q1/5

1601

QUESTION ID

S45-B2-Q1

TYPE

Short answer

CORRECT ANSWER

100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 45 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $a^2 + b^2$ when a = 5 and b = 12.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 45 • BACKWARD 2 • Q2/5

1602

QUESTION ID

S45-B2-Q2

TYPE

Short answer

CORRECT ANSWER

169

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 45 • BACKWARD 2 • Q2/5

Evaluate $c^2 - a^2$ when c = 10 and a = 6.

A. 16

B. 136

C. 64

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 45 • BACKWARD 2 • Q3/5

1603

QUESTION ID

S45-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 45 • BACKWARD 2 • Q3/5

True or false?

When a = 3 and b = 4, $a^2 + b^2 = 25$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 45 • BACKWARD 2 • Q4/5

1604

QUESTION ID

S45-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 45 • BACKWARD 2 • Q4/5

True or false?

When a = 2 and b = 3, $a^2 + b^2 = 25$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 45 • BACKWARD 2 • Q5/5

1605

QUESTION ID

S45-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 45 • BACKWARD 2 • Q5/5

SET 45 • 8.G.B.6

FORWARD 1

Prove the converse of the Pythagorean theorem with congruent triangles • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC has sides BC = a, AC = b, and AB = c, with $a^2 + b^2 = c^2$.

Prove that angle C is a right angle. (Build a right triangle with legs a and b.)

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 45 • FORWARD 1 • Q1/5

1607

QUESTION ID

S45-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

Angle C = 90°

GRADING NOTE REQUIRED WORK: A complete proof: build right triangle DEF with legs EF = a and DF = b and its right angle at F. By the Pythagorean theorem, $DE^2 = a^2 + b^2 = c^2$, so DE = c. Then triangles ABC and DEF have all three pairs of sides equal, so they are congruent (SSS), and angle C = angle F = 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 45 • FORWARD 1 • Q1/5

In that proof, which congruence criterion is used, and why does it apply?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 45 • FORWARD 1 • Q2/5

1608

QUESTION ID

S45-F1-Q2

TYPE

Short answer

CORRECT ANSWER

SSS: BC = EF = a, AC = DF = b, and AB = DE = c, so all three pairs of sides are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 45 • FORWARD 1 • Q2/5

In that proof, which theorem gives the length of the hypotenuse of the triangle that was built?

A. The triangle angle sum theorem

B. The converse of the Pythagorean theorem

C. The Pythagorean theorem

D. The exterior angle theorem

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 45 • FORWARD 1 • Q3/5

1609

QUESTION ID

S45-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. The Pythagorean theorem

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 45 • FORWARD 1 • Q3/5

True or false?

If $a^2 + b^2 > c^2$ for the longest side c of a triangle, the angle opposite c is acute.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 45 • FORWARD 1 • Q4/5

1610

QUESTION ID

S45-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 45 • FORWARD 1 • Q4/5

True or false?

Measuring angle C with a protractor once is a proof that a triangle with $a^2 + b^2 = c^2$ has a right angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 45 • FORWARD 1 • Q5/5

1611

QUESTION ID

S45-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A measurement is evidence, not a proof; the proof uses congruent triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 45 • FORWARD 1 • Q5/5

SET 45 • 8.G.B.6

FORWARD 2

Use coordinates to show that a triangle is a right triangle • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Is the triangle with vertices (1, 1), (4, 1), and (4, 5) a right triangle? Use side lengths.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 45 • FORWARD 2 • Q1/5

1613

QUESTION ID

S45-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. The sides are 3, 4, and 5, and $3^2 + 4^2 = 5^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 45 • FORWARD 2 • Q1/5

Use slopes to decide whether the triangle with vertices (0, 0), (2, 4), and (6, 2) has a right angle at (2, 4).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 45 • FORWARD 2 • Q2/5

1614

QUESTION ID

S45-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. The slope from (2, 4) to (0, 0) is 2 and the slope from (2, 4) to (6, 2) is −1/2; their product is −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 45 • FORWARD 2 • Q2/5

Which triangle is a right triangle?

A. (0, 0), (2, 2), (4, 1)

B. (0, 0), (3, 1), (1, 3)

C. (1, 1), (2, 3), (3, 1)

D. (0, 0), (3, 0), (0, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 45 • FORWARD 2 • Q3/5

1615

QUESTION ID

S45-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (0, 0), (3, 0), (0, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 45 • FORWARD 2 • Q3/5

True or false?

The triangle with vertices (0, 0), (5, 0), and (0, 5) is a right triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 45 • FORWARD 2 • Q4/5

1616

QUESTION ID

S45-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 45 • FORWARD 2 • Q4/5

True or false?

The triangle with vertices (0, 0), (4, 0), and (2, 1) is a right triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 45 • FORWARD 2 • Q5/5

1617

QUESTION ID

S45-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 45 • FORWARD 2 • Q5/5

SET 46 • 8.G.B.7 • GEOMETRY

Use the Pythagorean theorem to find unknown lengths in two dimensions

MAIN	Grade 8 main question	**8.G.B.7**
BACKWARD 1	Find the squares of decimals and whole numbers	**6.EE.A.1**
BACKWARD 2	Solve one-step equations	**6.EE.B.7**
FORWARD 1	Solve right triangles with the Pythagorean theorem and trigonom…	**HSG-SRT.C.8**
FORWARD 2	Use right-triangle reasoning to make design decisions under const…	**HSG-MG.A.3**

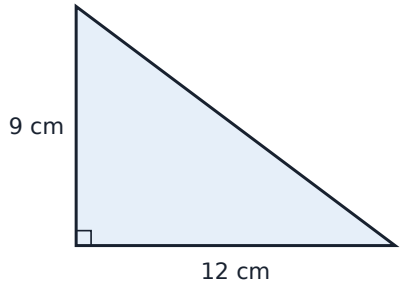
SET 46 • 8.G.B.7

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.B.7

Find the length of the hypotenuse.

Hypotenuse: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 46 • MAIN • Q1/5

1620

QUESTION ID

S46-M-Q1

TYPE

Short answer

CORRECT ANSWER

15 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 46 • MAIN • Q1/5

A 13-foot ladder leans against a wall. Its base is 5 feet from the wall.

How high up the wall does it reach?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 46 • MAIN • Q2/5

1621

QUESTION ID

S46-M-Q2

TYPE

Short answer

CORRECT ANSWER

12 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 46 • MAIN • Q2/5

A rectangular screen is 48 inches wide and 36 inches tall.

How long is its diagonal?

A. 60 inches

B. 84 inches

C. 72 inches

D. 42 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 46 • MAIN • Q3/5

1622

QUESTION ID

S46-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 60 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 46 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A right triangle has a hypotenuse of 17 and a leg of 8.

How long is the other leg?

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 46 • MAIN • Q4/5

1623

QUESTION ID

S46-M-Q4

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 46 • MAIN • Q4/5

True or false?

A square with 10 cm sides has a diagonal of about 14.1 cm.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 46 • MAIN • Q5/5

1624

QUESTION ID

S46-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 46 • MAIN • Q5/5

SET 46 • 8.G.B.7

BACKWARD 1

Find the squares of decimals and whole numbers • Grade 6 • Expressions & Equations • 6.EE.A.1

Find the value.

14^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 46 • BACKWARD 1 • Q1/5

1626

QUESTION ID

S46-B1-Q1

TYPE

Short answer

CORRECT ANSWER

196

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 46 • BACKWARD 1 • Q1/5

Find the value.

2.5^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 46 • BACKWARD 1 • Q2/5

1627

QUESTION ID

S46-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 46 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the value.

17^2

A. 189

B. 279

C. 289

D. 34

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 46 • BACKWARD 1 • Q3/5

1628

QUESTION ID

S46-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 289

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 46 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$9^2 = 81$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 46 • BACKWARD 1 • Q4/5

1629

QUESTION ID

S46-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 46 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$1.5^2 = 3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 46 • BACKWARD 1 • Q5/5

1630

QUESTION ID

S46-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 46 • BACKWARD 1 • Q5/5

SET 46 • 8.G.B.7

BACKWARD 2

Solve one-step equations • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

x + 64 = 289

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 46 • BACKWARD 2 • Q1/5

1632

QUESTION ID

S46-B2-Q1

TYPE

Short answer

CORRECT ANSWER

225

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 46 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve for x.

25 + x = 169

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 46 • BACKWARD 2 • Q2/5

1633

QUESTION ID

S46-B2-Q2

TYPE

Short answer

CORRECT ANSWER

144

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 46 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve for x.

81 + x = 100

A. x = 19

B. x = 181

C. x = 9

D. x = 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 46 • BACKWARD 2 • Q3/5

1634

QUESTION ID

S46-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. x = 19

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 46 • BACKWARD 2 • Q3/5

True or false?

x = 15 is the solution of x + 49 = 64.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 46 • BACKWARD 2 • Q4/5

1635

QUESTION ID

S46-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 46 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

x = 46 is the solution of 64 + x = 100.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 46 • BACKWARD 2 • Q5/5

1636

QUESTION ID

S46-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is x = 36.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 46 • BACKWARD 2 • Q5/5

SET 46 • 8.G.B.7

FORWARD 1

Solve right triangles with the Pythagorean theorem and trigonometric ratios • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

A right triangle has a hypotenuse of 10 and a 30° angle. sin 30° = 0.5.

How long is the side opposite the 30° angle?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 46 • FORWARD 1 • Q1/5

1638

QUESTION ID

S46-F1-Q1

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F1-Q1**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 46 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A ramp makes a 5° angle with the ground and rises 2 feet. Use sin 5° ≈ 0.087.

How long is the ramp, to the nearest foot?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 46 • FORWARD 1 • Q2/5

1639

QUESTION ID

S46-F1-Q2

TYPE

Short answer

CORRECT ANSWER

23 feet

GRADING NOTE 2 ÷ 0.087 ≈ 23.0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F1-Q2**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 46 • FORWARD 1 • Q2/5

A kite string is 50 m long and makes a 37° angle with the level ground. Use sin 37° ≈ 0.60 and cos 37° ≈ 0.80.

About how high is the kite?

A. 20 m

B. 30 m

C. 40 m

D. 83 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 46 • FORWARD 1 • Q3/5

1640

QUESTION ID

S46-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 30 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 46 • FORWARD 1 • Q3/5

True or false?

A right triangle with two 45° angles and legs of 7 has a hypotenuse of $7\sqrt{2}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 46 • FORWARD 1 • Q4/5

1641

QUESTION ID

S46-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 46 • FORWARD 1 • Q4/5

True or false?

tan 45° = 2

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 46 • FORWARD 1 • Q5/5

1642

QUESTION ID

S46-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE tan 45° = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 46 • FORWARD 1 • Q5/5

SET 46 • 8.G.B.7

FORWARD 2

Use right-triangle reasoning to make design decisions under constraints • High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3

A ramp must rise 2 feet. Building rules allow at most 1 foot of rise for every 12 feet of horizontal run.

What is the shortest allowed run, and how long is the ramp surface, to the nearest tenth of a foot?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q1

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 46 • FORWARD 2 • Q1/5

1644

QUESTION ID

S46-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The run is 24 feet; the surface is $\sqrt{(24^2 + 2^2)} = \sqrt{580} \approx 24.1$ feet.

GRADING NOTE Both values are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q1

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 46 • FORWARD 2 • Q1/5

A 50-inch TV (measured on the diagonal) is 40 inches wide. A cabinet opening is 32 inches tall and 42 inches wide.

Does the TV fit? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q2

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 46 • FORWARD 2 • Q2/5

1645

QUESTION ID

S46-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. The TV is 30 inches tall ($30^2 + 40^2 = 50^2$), which is less than 32 inches, and 40 inches wide, which is less than 42 inches.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q2

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 46 • FORWARD 2 • Q2/5

A cable runs from the top of a 12 m pole to a stake on level ground. Safety rules say the stake must be at least 5 m from the pole, and a path makes 6 m to 8 m from the pole off-limits. The company wants the shortest cable.

Where should the stake go, and how long is the cable?

A. 5 m from the pole; 13 m of cable

B. 9 m from the pole; 15 m of cable

C. At the base of the pole; 12 m of cable

D. 12 m from the pole; about 17 m of cable

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q3

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 46 • FORWARD 2 • Q3/5

1646

QUESTION ID

S46-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5 m from the pole; 13 m of cable

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q3

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 46 • FORWARD 2 • Q3/5

True or false?

A rectangular field is 120 m by 50 m. A diagonal path is 40 m shorter than walking along two sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q4

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 46 • FORWARD 2 • Q4/5

1647

QUESTION ID

S46-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q4

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 46 • FORWARD 2 • Q4/5

True or false?

A 10-foot ladder must reach a window 9 feet up, and its base must be at least 5 feet from the wall. This ladder meets both rules.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q5

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 46 • FORWARD 2 • Q5/5

1648

QUESTION ID
S46-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE To reach 9 feet, the base is $\sqrt{(100 - 81)} \approx 4.4$ feet from the wall, which is less than 5 feet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q5

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 46 • FORWARD 2 • Q5/5

SET 47 • 8.G.B.7 • GEOMETRY

Use the Pythagorean theorem to find unknown lengths in three dimensions

MAIN	Grade 8 main question	**8.G.B.7**
BACKWARD 1	Identify the faces, edges, and vertices of a prism	**6.G.A.4**
BACKWARD 2	Describe cross sections of right prisms	**7.G.A.3**
FORWARD 1	Use trigonometric ratios and the Pythagorean theorem to solve rig	**HSG-SRT.C.8**
FORWARD 2	Identify solids made by rotating two-dimensional figures — neare	**HSG-GMD.B.4**

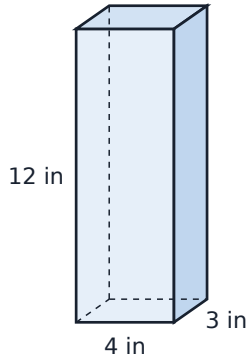
SET 47 • 8.G.B.7

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.B.7

A box is 4 inches long, 3 inches wide, and 12 inches tall.

How long is the diagonal from a bottom corner to the opposite top corner?



Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 47 • MAIN • Q1/5

1651

QUESTION ID

S47-M-Q1

TYPE

Short answer

CORRECT ANSWER

13 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 47 • MAIN • Q1/5

A cube has 6 cm edges.

Find the diagonal of one face, and then the diagonal from one corner of the cube to the opposite corner. Give exact values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 47 • MAIN • Q2/5

1652

QUESTION ID

S47-M-Q2

TYPE

Short answer

CORRECT ANSWER

Face diagonal 6√2 cm; cube diagonal 6√3 cm

GRADING NOTE Grade both values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 47 • MAIN • Q2/5

Will a 26-inch stick fit inside a box that is 24 by 6 by 8 inches, from one corner to the opposite corner?

A. No; the box is too short.

B. Yes; the face diagonal is 30 inches.

C. Yes; the corner-to-corner diagonal is exactly 26 inches.

D. No; the diagonal is only 24 inches.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 47 • MAIN • Q3/5

1653

QUESTION ID

S47-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Yes; the corner-to-corner diagonal is exactly 26 inches.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 47 • MAIN • Q3/5

A right square pyramid has its top directly above the center of its base. Its base edges are 10 m and its height is 12 m.

What is the slant height (from the top to the midpoint of a base edge)?

Slant height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 47 • MAIN • Q4/5

1654

QUESTION ID

S47-M-Q4

TYPE

Short answer

CORRECT ANSWER

13 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 47 • MAIN • Q4/5

True or false?

The corner-to-corner diagonal of a 2 by 3 by 6 box is 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 47 • MAIN • Q5/5

1655

QUESTION ID

S47-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 47 • MAIN • Q5/5

SET 47 • 8.G.B.7

BACKWARD 1

Identify the faces, edges, and vertices of a prism • Grade 6 • Geometry • 6.G.A.4

How many faces does a rectangular prism have?

Faces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 47 • BACKWARD 1 • Q1/5

1657

QUESTION ID

S47-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 47 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

How many edges does a rectangular prism have?

Edges: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 47 • BACKWARD 1 • Q2/5

1658

QUESTION ID

S47-B1-Q2

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 47 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

How many vertices does a cube have?

A. 6

B. 4

C. 12

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 47 • BACKWARD 1 • Q3/5

1659

QUESTION ID

S47-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 47 • BACKWARD 1 • Q3/5

True or false?

Each face of a rectangular prism is a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 47 • BACKWARD 1 • Q4/5

1660

QUESTION ID

S47-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 47 • BACKWARD 1 • Q4/5

True or false?

A triangular prism has 6 faces.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 47 • BACKWARD 1 • Q5/5

1661

QUESTION ID

S47-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has 5 faces.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 47 • BACKWARD 1 • Q5/5

SET 47 • 8.G.B.7

BACKWARD 2

Describe cross sections of right prisms • Grade 7 • Geometry • 7.G.A.3

A right rectangular prism is sliced by a plane through two opposite vertical edges.

What is the shape of the cross section?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q1

Grade 7 • Geometry • 7.G.A.3 • Set 47 • BACKWARD 2 • Q1/5

1663

QUESTION ID

S47-B2-Q1

TYPE

Short answer

CORRECT ANSWER

A rectangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q1

Grade 7 • Geometry • 7.G.A.3 • Set 47 • BACKWARD 2 • Q1/5

A cube is sliced by a plane parallel to one face, through its interior.

What is the shape of the cross section?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q2

Grade 7 • Geometry • 7.G.A.3 • Set 47 • BACKWARD 2 • Q2/5

1664

QUESTION ID

S47-B2-Q2

TYPE

Short answer

CORRECT ANSWER

A square

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q2

Grade 7 • Geometry • 7.G.A.3 • Set 47 • BACKWARD 2 • Q2/5

A right rectangular prism is sliced parallel to its base, through its interior.

What is the cross section?

A. A circle

B. A triangle

C. A pentagon

D. A rectangle the same size as the base

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q3

Grade 7 • Geometry • 7.G.A.3 • Set 47 • BACKWARD 2 • Q3/5

1665

QUESTION ID

S47-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A rectangle the same size as the base

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q3

Grade 7 • Geometry • 7.G.A.3 • Set 47 • BACKWARD 2 • Q3/5

True or false?

A diagonal slice through a box can make a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q4

Grade 7 • Geometry • 7.G.A.3 • Set 47 • BACKWARD 2 • Q4/5

1666

QUESTION ID

S47-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q4

Grade 7 • Geometry • 7.G.A.3 • Set 47 • BACKWARD 2 • Q4/5

True or false?

A slice of a cube parallel to a face can be a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q5

Grade 7 • Geometry • 7.G.A.3 • Set 47 • BACKWARD 2 • Q5/5

1667

QUESTION ID

S47-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q5

Grade 7 • Geometry • 7.G.A.3 • Set 47 • BACKWARD 2 • Q5/5

SET 47 • 8.G.B.7

FORWARD 1

Use trigonometric ratios and the Pythagorean theorem to solve right-triangle problems in three dimensions • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

ANSWER KEY Reduced copy of the question page, with its answer

A room is 12 ft long, 9 ft wide, and 8 ft high.

Find the diagonal from a floor corner to the opposite ceiling corner, and the angle it makes with the floor diagonal. Use tan θ = 8 ÷ 15 and tan 28° ≈ 0.53.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 47 • FORWARD 1 • Q1/5

1669

QUESTION ID

S47-F1-Q1

TYPE

Short answer

CORRECT ANSWER

The floor diagonal is 15 ft, the room diagonal is 17 ft, and the angle is about 28°.

GRADING NOTE Both the length and the angle are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 47 • FORWARD 1 • Q1/5

A right square pyramid has base edges of 10 m and a height of 12 m.

Find the angle between a face's slant height and the base, to the nearest degree. Use tan 67° ≈ 2.36 and tan 68° ≈ 2.48.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 47 • FORWARD 1 • Q2/5

1670

QUESTION ID

S47-F1-Q2

TYPE

Short answer

CORRECT ANSWER

About 67° (tan θ = 12 ÷ 5 = 2.4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F1-Q2**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 47 • FORWARD 1 • Q2/5

A 6 m ladder leans against a wall and makes a 70° angle with the level ground. Use sin 70° ≈ 0.94 and cos 70° ≈ 0.34.

About how high up the wall does it reach?

A. About 5.6 m

B. About 17.6 m

C. About 6.4 m

D. About 2.0 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 47 • FORWARD 1 • Q3/5

1671

QUESTION ID

S47-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. About 5.6 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 47 • FORWARD 1 • Q3/5

True or false?

In a right triangle, if tan A = 1, then angle A = 45°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 47 • FORWARD 1 • Q4/5

1672

QUESTION ID

S47-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 47 • FORWARD 1 • Q4/5

True or false?

The corner-to-corner diagonal of a 3 by 4 by 12 box makes an angle with the floor whose tangent is 12 ÷ 13.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 47 • FORWARD 1 • Q5/5

1673

QUESTION ID
S47-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The floor diagonal is 5, so the tangent is 12 ÷ 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F1-Q5**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 47 • FORWARD 1 • Q5/5

SET 47 • 8.G.B.7

FORWARD 2

Identify solids made by rotating two-dimensional figures • High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4

⚑ No high-school standard directly continues this skill at this step. This branch uses the nearest related high-school standard (placed at Grade 10 in this bank).

A right triangle with legs 3 and 4 is rotated around the leg of length 4.

What solid does it make, and what is its slant height?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q1

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 47 • FORWARD 2 • Q1/5 • NEAREST RELATED

1675

QUESTION ID

S47-F2-Q1

TYPE

Short answer

CORRECT ANSWER

A cone with radius 3 and height 4; its slant height is 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q1

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 47 • FORWARD 2 • Q1/5 • NEAREST RELATED

A 2-by-5 rectangle is rotated around one of its 5-unit sides.

What solid does it make?

Solid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q2

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 47 • FORWARD 2 • Q2/5 • NEAREST RELATED

1676

QUESTION ID

S47-F2-Q2

TYPE

Short answer

CORRECT ANSWER

A cylinder with radius 2 and height 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q2

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 47 • FORWARD 2 • Q2/5 • NEAREST RELATED

A sphere is sliced by a plane through its center.

What is the cross section?

A. A circle with the same radius as the sphere

B. An oval that is not a circle

C. A square

D. A smaller circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q3

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 47 • FORWARD 2 • Q3/5 • NEAREST RELATED

1677

QUESTION ID

S47-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. A circle with the same radius as the sphere

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 47 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

Rotating a semicircle about its diameter makes a sphere.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q4

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 47 • FORWARD 2 • Q4/5 • NEAREST RELATED

1678

QUESTION ID

S47-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q4

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 47 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

A cube can never have a triangular cross section.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q5

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 47 • FORWARD 2 • Q5/5 • NEAREST RELATED

1679

QUESTION ID
S47-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE A slice through three corners next to one vertex is a triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F2-Q5**

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 47 • FORWARD 2 • Q5/5 • NEAREST RELATED

SET 48 • 8.G.B.8 • GEOMETRY

Find the distance between two points on the coordinate plane

MAIN	Grade 8 main question	**8.G.B.8**
BACKWARD 1	Find distances along grid lines	**6.NS.C.8**
BACKWARD 2	Find a distance as the absolute value of a difference	**7.NS.A.1.c**
FORWARD 1	Use the Pythagorean theorem and trigonometric ratios with coord.	**HSG-SRT.C.8**
FORWARD 2	Use coordinates to find perimeters and areas of polygons	**HSG-GPE.B.7**

SET 48 • 8.G.B.8

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.B.8

Find the distance between (1, 2) and (7, 10).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 48 • MAIN • Q1/5

1682

QUESTION ID

S48-M-Q1

TYPE

Short answer

CORRECT ANSWER

10 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 48 • MAIN • Q1/5

Find the distance between (−3, 4) and (2, −8).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 48 • MAIN • Q2/5

1683

QUESTION ID

S48-M-Q2

TYPE

Short answer

CORRECT ANSWER

13 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 48 • MAIN • Q2/5

What is the distance between (0, 0) and (4, 4), to the nearest tenth?

A. 16.0 units

B. 4.0 units

C. 5.7 units

D. 8.0 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 48 • MAIN • Q3/5

1684

QUESTION ID

S48-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 5.7 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

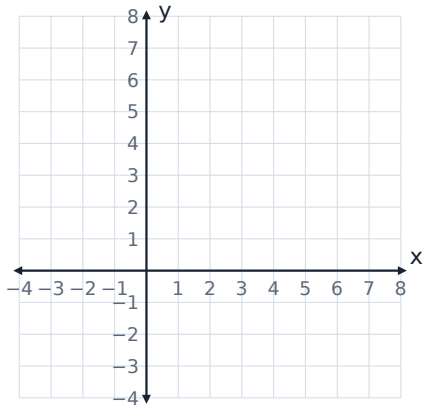
ID S48-M-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 48 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Plot A(−2, −1) and B(4, 7). Draw a right triangle with AB as its hypotenuse, and find AB.

AB = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 48 • MAIN • Q4/5

1685

QUESTION ID

S48-M-Q4

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: Points A and B joined by segment AB, and any right triangle with AB as its hypotenuse. Examples: the right angle at (4, −1) or at (−2, 7) (legs 6 and 8), or at (5, 0) (legs √50 and √50).

Written: 10 units

GRADING NOTE Grade both: any correct right triangle with AB as its hypotenuse, and AB = 10 units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 48 • MAIN • Q4/5

True or false?

The distance between (−1, −1) and (2, 3) is 5 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 48 • MAIN • Q5/5

1686

QUESTION ID

S48-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 48 • MAIN • Q5/5

SET 48 • 8.G.B.8

BACKWARD 1

Find distances along grid lines • Grade 6 • The Number System • 6.NS.C.8

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance between (−4, 2) and (5, 2).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 48 • BACKWARD 1 • Q1/5

1688

QUESTION ID

S48-B1-Q1

TYPE

Short answer

CORRECT ANSWER

9 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 48 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance between (3, −6) and (3, 1).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 48 • BACKWARD 1 • Q2/5

1689

QUESTION ID

S48-B1-Q2

TYPE

Short answer

CORRECT ANSWER

7 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 48 • BACKWARD 1 • Q2/5

Which two points are 8 units apart?

A. (1, 1) and (1, 8)

B. (−3, 0) and (3, 0)

C. (−3, 0) and (5, 0)

D. (0, −4) and (0, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 48 • BACKWARD 1 • Q3/5

1690

QUESTION ID

S48-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (−3, 0) and (5, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 48 • BACKWARD 1 • Q3/5

True or false?

(2, −5) and (2, 5) are 10 units apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 48 • BACKWARD 1 • Q4/5

1691

QUESTION ID

S48-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 48 • BACKWARD 1 • Q4/5

True or false?

(−6, 3) and (−1, 3) are 7 units apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 48 • BACKWARD 1 • Q5/5

1692

QUESTION ID

S48-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 48 • BACKWARD 1 • Q5/5

SET 48 • 8.G.B.8

BACKWARD 2

Find a distance as the absolute value of a difference • Grade 7 • The Number System • 7.NS.A.1.c

Find $|7 - (-2)|$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 48 • BACKWARD 2 • Q1/5

1694

QUESTION ID

S48-B2-Q1

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 48 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

How far apart are −4 and 6 on a number line?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 48 • BACKWARD 2 • Q2/5

1695

QUESTION ID

S48-B2-Q2

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 48 • BACKWARD 2 • Q2/5

Which expression gives the distance between −8 and −3?

A. |−8 − (−3)|

B. −8 − 3

C. |−8 + 3| − 1

D. −8 + (−3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 48 • BACKWARD 2 • Q3/5

1696

QUESTION ID

S48-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. |−8 − (−3)|

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 48 • BACKWARD 2 • Q3/5

True or false?

The distance between −5 and 1 is 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 48 • BACKWARD 2 • Q4/5

1697

QUESTION ID

S48-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 48 • BACKWARD 2 • Q4/5

True or false?

The distance between −7 and −2 is 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 48 • BACKWARD 2 • Q5/5

1698

QUESTION ID

S48-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 48 • BACKWARD 2 • Q5/5

SET 48 • 8.G.B.8

FORWARD 1

Use the Pythagorean theorem and trigonometric ratios with coordinates and distances • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

ANSWER KEY Reduced copy of the question page, with its answer

A drone flies in a straight line from (0, 0) to (12, 5) on a grid in meters.

How far does it fly, and what angle does its path make with the x-axis, to the nearest degree? Use tan 22° ≈ 0.40 and tan 23° ≈ 0.42.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 48 • FORWARD 1 • Q1/5

1700

QUESTION ID

S48-F1-Q1

TYPE

Short answer

CORRECT ANSWER

13 m; about 23° (tan θ = 5 ÷ 12 ≈ 0.417)

GRADING NOTE Both values are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 48 • FORWARD 1 • Q1/5

From a point on level ground 40 m from a building, the angle up to the top of the building is 35°. Use tan 35° ≈ 0.70.

How tall is the building?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 48 • FORWARD 1 • Q2/5

1701

QUESTION ID

S48-F1-Q2

TYPE

Short answer

CORRECT ANSWER

About 28 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F1-Q2**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 48 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which point is exactly 5 units from (1, 1)?

A. (7, 1)

B. (5, 5)

C. (1, 7)

D. (4, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 48 • FORWARD 1 • Q3/5

1702

QUESTION ID

S48-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (4, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 48 • FORWARD 1 • Q3/5

True or false?

The segment from (0, 0) to (3, 3) makes a 45° angle with the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 48 • FORWARD 1 • Q4/5

1703

QUESTION ID

S48-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 48 • FORWARD 1 • Q4/5

True or false?

The points (0, 0) and (3, 3) are 6 units apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 48 • FORWARD 1 • Q5/5

1704

QUESTION ID

S48-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are $3\sqrt{2} \approx 4.2$ units apart.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 48 • FORWARD 1 • Q5/5

SET 48 • 8.G.B.8

FORWARD 2

Use coordinates to find perimeters and areas of polygons • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

Find the perimeter of the triangle with vertices (0, 0), (6, 0), and (6, 8).

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 48 • FORWARD 2 • Q1/5

1706

QUESTION ID

S48-F2-Q1

TYPE

Short answer

CORRECT ANSWER

24 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F2-Q1**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 48 • FORWARD 2 • Q1/5

Find the area of the triangle with vertices (0, 0), (6, 0), and (6, 8).

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 48 • FORWARD 2 • Q2/5

1707

QUESTION ID

S48-F2-Q2

TYPE

Short answer

CORRECT ANSWER

24 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F2-Q2**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 48 • FORWARD 2 • Q2/5

A rhombus has vertices (0, 0), (3, 4), (8, 4), and (5, 0).

What is its perimeter?

A. 20 units

B. 18 units

C. 16 units

D. 25 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 48 • FORWARD 2 • Q3/5

1708

QUESTION ID

S48-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 20 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 48 • FORWARD 2 • Q3/5

True or false?

A square with vertices (0, 0), (3, 1), (2, 4), and (−1, 3) has an area of 10 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 48 • FORWARD 2 • Q4/5

1709

QUESTION ID

S48-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 48 • FORWARD 2 • Q4/5

True or false?

The rectangle with vertices (0, 0), (4, 0), (4, 3), and (0, 3) has a perimeter of 12 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 48 • FORWARD 2 • Q5/5

1710

QUESTION ID

S48-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its perimeter is 14 units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 48 • FORWARD 2 • Q5/5

SET 49 • 8.G.C.9 • GEOMETRY

Find the volume of a cylinder

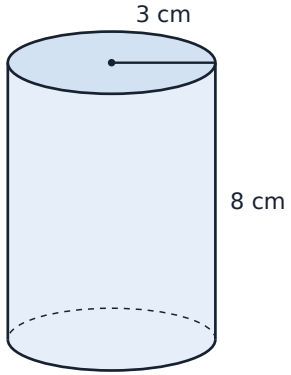
MAIN	Grade 8 main question	**8.G.C.9**
BACKWARD 1	Find the area of a circle	**7.G.B.4**
BACKWARD 2	Find the volume of a right prism as base area times height	**7.G.B.6**
FORWARD 1	Give an informal argument for volume formulas	**HSG-GMD.A.1**
FORWARD 2	Use density with volume in a model	**HSG-MG.A.2**

SET 49 • 8.G.C.9

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.C.9

Find the volume of the cylinder in terms of π.



Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 49 • MAIN • Q1/5

1713

QUESTION ID

S49-M-Q1

TYPE

Short answer

CORRECT ANSWER

72π cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 49 • MAIN • Q1/5

A cylinder has a diameter of 10 inches and a height of 6 inches.

Find its volume. Use 3.14 for π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 49 • MAIN • Q2/5

1714

QUESTION ID

S49-M-Q2

TYPE

Short answer

CORRECT ANSWER

471 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 49 • MAIN • Q2/5

A can holds 128π cubic centimeters and has a radius of 4 cm.

What is its height?

A. 8 cm

B. 2 cm

C. 16 cm

D. 32 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 49 • MAIN • Q3/5

1715

QUESTION ID

S49-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 8 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 49 • MAIN • Q3/5

A cylinder-shaped tank has a radius of 2 m and a height of 5 m.

How many liters does it hold? Use 3.14 for π. (1 cubic meter = 1,000 liters)

Liters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 49 • MAIN • Q4/5

1716

QUESTION ID

S49-M-Q4

TYPE

Short answer

CORRECT ANSWER

62,800 liters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 49 • MAIN • Q4/5

True or false?

Doubling the radius of a cylinder while keeping its height the same doubles its volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 49 • MAIN • Q5/5

1717

QUESTION ID

S49-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It multiplies the volume by 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 49 • MAIN • Q5/5

SET 49 • 8.G.C.9

BACKWARD 1

Find the area of a circle • Grade 7 • Geometry • 7.G.B.4

A circle has a radius of 3.

What is its area in terms of π?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 49 • BACKWARD 1 • Q1/5

1719

QUESTION ID

S49-B1-Q1

TYPE

Short answer

CORRECT ANSWER

9π square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 49 • BACKWARD 1 • Q1/5

A circle has a diameter of 10 cm.

What is its area? Use 3.14 for π.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 49 • BACKWARD 1 • Q2/5

1720

QUESTION ID

S49-B1-Q2

TYPE

Short answer

CORRECT ANSWER

78.5 square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 49 • BACKWARD 1 • Q2/5

A circle has a radius of 4 m.

What is its area in terms of π?

A. 64π square meters

B. 4π square meters

C. 16π square meters

D. 8π square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 49 • BACKWARD 1 • Q3/5

1721

QUESTION ID

S49-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 16π square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 49 • BACKWARD 1 • Q3/5

True or false?

A circle with a radius of 1 has an area of π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 49 • BACKWARD 1 • Q4/5

1722

QUESTION ID

S49-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 49 • BACKWARD 1 • Q4/5

True or false?

A circle with a diameter of 6 has an area of 36π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 49 • BACKWARD 1 • Q5/5

1723

QUESTION ID

S49-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The radius is 3, so the area is 9π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 49 • BACKWARD 1 • Q5/5

SET 49 • 8.G.C.9

BACKWARD 2

Find the volume of a right prism as base area times height • Grade 7 • Geometry • 7.G.B.6

A right prism has a base area of 20 square cm and a height of 7 cm.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 49 • BACKWARD 2 • Q1/5

1725

QUESTION ID

S49-B2-Q1

TYPE

Short answer

CORRECT ANSWER

140 cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 49 • BACKWARD 2 • Q1/5

A right triangular prism has a triangle base with area 6 square inches and a length of 10 inches.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 49 • BACKWARD 2 • Q2/5

1726

QUESTION ID

S49-B2-Q2

TYPE

Short answer

CORRECT ANSWER

60 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 49 • BACKWARD 2 • Q2/5

A right prism has a base area of 15 square m and a height of 4 m.

What is its volume?

A. 19 cubic meters

B. 30 cubic meters

C. 225 cubic meters

D. 60 cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 49 • BACKWARD 2 • Q3/5

1727

QUESTION ID

S49-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 60 cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 49 • BACKWARD 2 • Q3/5

True or false?

The volume of any right prism is the area of its base times its height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 49 • BACKWARD 2 • Q4/5

1728

QUESTION ID

S49-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 49 • BACKWARD 2 • Q4/5

True or false?

A prism with a base area of 9 and a height of 2 has a volume of 11.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 49 • BACKWARD 2 • Q5/5

1729

QUESTION ID

S49-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 49 • BACKWARD 2 • Q5/5

SET 49 • 8.G.C.9

FORWARD 1

Give an informal argument for volume formulas • High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

Explain why the volume of a cylinder is the area of its base times its height.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q1

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 49 • FORWARD 1 • Q1/5

1731

QUESTION ID

S49-F1-Q1

TYPE

Short answer

CORRECT ANSWER

The cylinder is a stack of equal circles (cross sections) with area πr^2; stacking them to height h gives $\pi r^2 \cdot h$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q1

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 49 • FORWARD 1 • Q1/5

A straight stack of coins and a slanted stack have the same coins and the same height.

Do they have the same volume? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q2

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 49 • FORWARD 1 • Q2/5

1732

QUESTION ID

S49-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. At every height the cross sections have the same area (Cavalieri's principle).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-F1-Q2**

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 49 • FORWARD 1 • Q2/5

An oblique cylinder and a right cylinder have the same radius and height.

How do their volumes compare?

A. The oblique one is larger.

B. It cannot be known.

C. The right one is larger.

D. They are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q3

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 49 • FORWARD 1 • Q3/5

1733

QUESTION ID

S49-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. They are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q3

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 49 • FORWARD 1 • Q3/5

True or false?

Cavalieri's principle compares the areas of cross sections at every height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q4

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 49 • FORWARD 1 • Q4/5

1734

QUESTION ID

S49-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q4

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 49 • FORWARD 1 • Q4/5

True or false?

Leaning a stack of paper to one side increases its volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q5

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 49 • FORWARD 1 • Q5/5

1735

QUESTION ID

S49-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q5

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 49 • FORWARD 1 • Q5/5

SET 49 • 8.G.C.9

FORWARD 2

Use density with volume in a model • High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2

An aluminum cylinder has a radius of 2 cm and a height of 10 cm. Aluminum has a density of 2.7 grams per cubic centimeter.

What is its mass? Use 3.14 for π.

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q1

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 49 • FORWARD 2 • Q1/5

1737

QUESTION ID

S49-F2-Q1

TYPE

Short answer

CORRECT ANSWER

339.12 grams

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q1

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 49 • FORWARD 2 • Q1/5

A town has 12,000 people and covers 30 square miles.

What is its population density?

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q2

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 49 • FORWARD 2 • Q2/5

1738

QUESTION ID

S49-F2-Q2

TYPE

Short answer

CORRECT ANSWER

400 people per square mile

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q2

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 49 • FORWARD 2 • Q2/5

A 500 cubic-centimeter block has a mass of 4,000 grams.

What is its density?

A. 8 grams per cubic centimeter

B. 2 grams per cubic centimeter

C. 0.125 gram per cubic centimeter

D. 4,500 grams

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q3

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 49 • FORWARD 2 • Q3/5

1739

QUESTION ID

S49-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 8 grams per cubic centimeter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q3

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 49 • FORWARD 2 • Q3/5

True or false?

If two objects have the same volume, the one with the greater density has the greater mass.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q4

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 49 • FORWARD 2 • Q4/5

1740

QUESTION ID

S49-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q4

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 49 • FORWARD 2 • Q4/5

True or false?

Density is volume divided by mass.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q5

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 49 • FORWARD 2 • Q5/5

1741

QUESTION ID

S49-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Density is mass divided by volume.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q5

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 49 • FORWARD 2 • Q5/5

SET 50 • 8.G.C.9 • GEOMETRY

Find the volume of a cone

MAIN	Grade 8 main question	**8.G.C.9**
BACKWARD 1	Find the area of a circle from its radius or diameter	**7.G.B.4**
BACKWARD 2	Multiply a fraction by a whole number	**5.NF.B.4.a**
FORWARD 1	Explain why a cone's volume is one-third of a cylinder's	**HSG-GMD.A.1**
FORWARD 2	Use volume formulas to solve problems	**HSG-GMD.A.3**

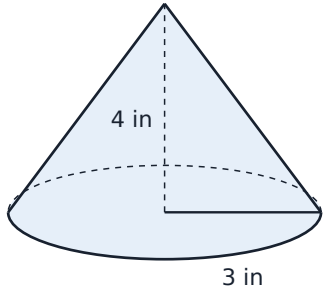
SET 50 • 8.G.C.9

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.C.9

Find the volume of the cone in terms of π.

Volume: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 50 • MAIN • Q1/5

1744

QUESTION ID

S50-M-Q1

TYPE

Short answer

CORRECT ANSWER

12π cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 50 • MAIN • Q1/5

A cone has a diameter of 12 feet and a height of 10 feet.

Find its volume. Use 3.14 for π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 50 • MAIN • Q2/5

1745

QUESTION ID

S50-M-Q2

TYPE

Short answer

CORRECT ANSWER

376.8 cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 50 • MAIN • Q2/5

A cone and a cylinder have the same base and the same height. The cylinder holds 90 cubic cm.

How much does the cone hold?

A. 30 cubic cm

B. 90 cubic cm

C. 270 cubic cm

D. 45 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 50 • MAIN • Q3/5

1746

QUESTION ID

S50-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 30 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 50 • MAIN • Q3/5

A cone has a volume of 50π cubic units and a radius of 5 units.

What is its height?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 50 • MAIN • Q4/5

1747

QUESTION ID

S50-M-Q4

TYPE

Short answer

CORRECT ANSWER

6 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 50 • MAIN • Q4/5

True or false?

A cone with a radius of 6 and a height of 2 has a volume of 24π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 50 • MAIN • Q5/5

1748

QUESTION ID

S50-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 50 • MAIN • Q5/5

SET 50 • 8.G.C.9

BACKWARD 1

Find the area of a circle from its radius or diameter • Grade 7 • Geometry • 7.G.B.4

A circle has a radius of 6.

What is its area in terms of π?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 50 • BACKWARD 1 • Q1/5

1750

QUESTION ID

S50-B1-Q1

TYPE

Short answer

CORRECT ANSWER

36π square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 50 • BACKWARD 1 • Q1/5

A circle has a diameter of 4 m.

What is its area in terms of π?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 50 • BACKWARD 1 • Q2/5

1751

QUESTION ID

S50-B1-Q2

TYPE

Short answer

CORRECT ANSWER

4π square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 50 • BACKWARD 1 • Q2/5

A circle has a radius of 10 cm.

What is its area? Use 3.14 for π.

A. 314 square cm

B. 31.4 square cm

C. 62.8 square cm

D. 628 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 50 • BACKWARD 1 • Q3/5

1752

QUESTION ID

S50-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 314 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 50 • BACKWARD 1 • Q3/5

True or false?

A circle with a radius of 2 has an area of 4π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 50 • BACKWARD 1 • Q4/5

1753

QUESTION ID

S50-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 50 • BACKWARD 1 • Q4/5

True or false?

A circle with a radius of 5 has an area of 10π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 50 • BACKWARD 1 • Q5/5

1754

QUESTION ID

S50-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 50 • BACKWARD 1 • Q5/5

SET 50 • 8.G.C.9

BACKWARD 2

Multiply a fraction by a whole number • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

Multiply.

$\frac{1}{3} \times 36$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 50 • BACKWARD 2 • Q1/5

1756

QUESTION ID

S50-B2-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 50 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply.

$\frac{1}{3} \times 90$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 50 • BACKWARD 2 • Q2/5

1757

QUESTION ID

S50-B2-Q2

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 50 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply.

$\frac{1}{3} \times 48$

A. 144

B. 12

C. 16

D. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 50 • BACKWARD 2 • Q3/5

1758

QUESTION ID

S50-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 50 • BACKWARD 2 • Q3/5

True or false?

$\frac{1}{3} \times 27 = 9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 50 • BACKWARD 2 • Q4/5

1759

QUESTION ID

S50-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 50 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\frac{1}{3} \times 15 = 45$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 50 • BACKWARD 2 • Q5/5

1760

QUESTION ID

S50-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 50 • BACKWARD 2 • Q5/5

SET 50 • 8.G.C.9

FORWARD 1

Explain why a cone's volume is one-third of a cylinder's • High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

A cone and a cylinder have the same base and height. Water from 3 full cones exactly fills the cylinder.

What formula does this suggest for the cone's volume?

Formula: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q1

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 50 • FORWARD 1 • Q1/5

1762

QUESTION ID

S50-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$$V = \frac{1}{3}\pi r^2 h$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q1

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 50 • FORWARD 1 • Q1/5

Explain why an oblique cone and a right cone with the same base and height have the same volume.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q2

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 50 • FORWARD 1 • Q2/5

1763

QUESTION ID

S50-F1-Q2

TYPE

Short answer

CORRECT ANSWER

At every height their cross sections are circles with the same area (Cavalieri's principle).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q2

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 50 • FORWARD 1 • Q2/5

A pyramid and a prism have the same base and height.

How do their volumes compare?

A. The pyramid's is $\frac{1}{3}$ of the prism's.

B. The prism's is $\frac{1}{3}$ of the pyramid's.

C. The pyramid's is $\frac{1}{2}$ of the prism's.

D. They are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q3

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 50 • FORWARD 1 • Q3/5

1764

QUESTION ID

S50-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. The pyramid's is 1/3 of the prism's.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q3

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 50 • FORWARD 1 • Q3/5

True or false?

A cone's volume formula has the same form as a pyramid's: $\frac{1}{3}$ × base area × height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q4

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 50 • FORWARD 1 • Q4/5

1765

QUESTION ID

S50-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q4

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 50 • FORWARD 1 • Q4/5

True or false?

A cone has half the volume of a cylinder with the same base and height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q5

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 50 • FORWARD 1 • Q5/5

1766

QUESTION ID

S50-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q5

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 50 • FORWARD 1 • Q5/5

SET 50 • 8.G.C.9

FORWARD 2

Use volume formulas to solve problems • High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A paper cone cup has a radius of 4 cm and a height of 9 cm.

How many full cups fill a cylinder-shaped jug with a radius of 8 cm and a height of 18 cm?

Cups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q1

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 50 • FORWARD 2 • Q1/5

1768

QUESTION ID

S50-F2-Q1

TYPE

Short answer

CORRECT ANSWER

24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q1

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 50 • FORWARD 2 • Q1/5

A cone-shaped pile of sand has a radius of 3 m and holds 30π cubic meters.

How tall is the pile?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q2

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 50 • FORWARD 2 • Q2/5

1769

QUESTION ID

S50-F2-Q2

TYPE

Short answer

CORRECT ANSWER

10 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F2-Q2**

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 50 • FORWARD 2 • Q2/5

A right circular cone has a radius of 6 and a slant height of 10.

What is its volume? (Find its height first.)

A. 120π

B. 96π

C. 32π

D. 288π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q3

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 50 • FORWARD 2 • Q3/5

1770

QUESTION ID

S50-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 96π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q3

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 50 • FORWARD 2 • Q3/5

True or false?

Doubling the height of a cone while keeping its radius the same doubles its volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q4

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 50 • FORWARD 2 • Q4/5

1771

QUESTION ID

S50-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q4

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 50 • FORWARD 2 • Q4/5

True or false?

Doubling the radius of a cone while keeping its height the same doubles its volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q5

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 50 • FORWARD 2 • Q5/5

1772

QUESTION ID
S50-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It multiplies the volume by 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q5

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 50 • FORWARD 2 • Q5/5

SET 51 • 8.G.C.9 • GEOMETRY

Find the volume of a sphere

MAIN	Grade 8 main question	**8.G.C.9**
BACKWARD 1	Find the volume of a cube	**6.G.A.2**
BACKWARD 2	Evaluate a formula with a fraction and an exponent	**6.EE.A.2.c**
FORWARD 1	Model objects with spheres, cylinders, and cones	**HSG-MG.A.1**
FORWARD 2	Use volume formulas to solve problems with spheres	**HSG-GMD.A.3**

SET 51 • 8.G.C.9

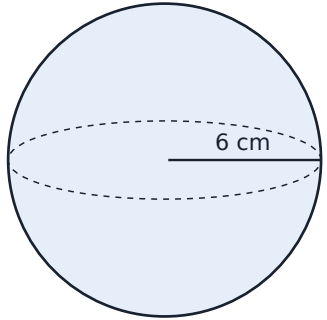
MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.C.9

ANSWER KEY Reduced copy of the question page, with its answer

Find the volume of the sphere in terms of π.

Volume: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 51 • MAIN • Q1/5

1775

QUESTION ID

S51-M-Q1

TYPE

Short answer

CORRECT ANSWER

288π cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 51 • MAIN • Q1/5

A ball has a diameter of 10 cm.

Find its volume to the nearest tenth. Use 3.14 for π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 51 • MAIN • Q2/5

1776

QUESTION ID

S51-M-Q2

TYPE

Short answer

CORRECT ANSWER

523.3 cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 51 • MAIN • Q2/5

A hemisphere (half of a sphere) has a radius of 3.

What is its volume in terms of π?

A. 36π

B. 18π

C. 9π

D. 27π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 51 • MAIN • Q3/5

1777

QUESTION ID

S51-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 18π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 51 • MAIN • Q3/5

A sphere has a volume of 36π cubic units.

What is its radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 51 • MAIN • Q4/5

1778

QUESTION ID

S51-M-Q4

TYPE

Short answer

CORRECT ANSWER

3 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 51 • MAIN • Q4/5

True or false?

A sphere with a radius of 1 has a volume of $\frac{4}{3}\pi$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 51 • MAIN • Q5/5

1779

QUESTION ID

S51-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 51 • MAIN • Q5/5

SET 51 • 8.G.C.9

BACKWARD 1

Find the volume of a cube • Grade 6 • Geometry • 6.G.A.2

A cube has edges of 3 units.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 51 • BACKWARD 1 • Q1/5

1781

QUESTION ID

S51-B1-Q1

TYPE

Short answer

CORRECT ANSWER

27 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 51 • BACKWARD 1 • Q1/5

A cube has edges of $\frac{1}{2}$ foot.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 51 • BACKWARD 1 • Q2/5

1782

QUESTION ID

S51-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1/8 cubic foot

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 51 • BACKWARD 1 • Q2/5

A cube has edges of 6 cm.

What is its volume?

A. 18 cubic cm

B. 216 cubic cm

C. 72 cubic cm

D. 36 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 51 • BACKWARD 1 • Q3/5

1783

QUESTION ID

S51-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 216 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 51 • BACKWARD 1 • Q3/5

True or false?

A cube with 4-inch edges has a volume of 64 cubic inches.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 51 • BACKWARD 1 • Q4/5

1784

QUESTION ID

S51-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 51 • BACKWARD 1 • Q4/5

True or false?

A cube with 5-meter edges has a volume of 15 cubic meters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 51 • BACKWARD 1 • Q5/5

1785

QUESTION ID

S51-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 51 • BACKWARD 1 • Q5/5

SET 51 • 8.G.C.9

BACKWARD 2

Evaluate a formula with a fraction and an exponent • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Use $V = \frac{4}{3}r^3$. Find V when r = 3.

V = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 51 • BACKWARD 2 • Q1/5

1787

QUESTION ID

S51-B2-Q1

TYPE

Short answer

CORRECT ANSWER

36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 51 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Use $V = \frac{4}{3}r^3$. Find V when r = 6.

V = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 51 • BACKWARD 2 • Q2/5

1788

QUESTION ID

S51-B2-Q2

TYPE

Short answer

CORRECT ANSWER

288

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 51 • BACKWARD 2 • Q2/5

Use $V = \frac{1}{2}r^3$. Find V when r = 2.

A. 6

B. 4

C. 3

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 51 • BACKWARD 2 • Q3/5

1789

QUESTION ID

S51-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 51 • BACKWARD 2 • Q3/5

True or false?

When r = 1, $\frac{4}{3}r^3 = \frac{4}{3}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 51 • BACKWARD 2 • Q4/5

1790

QUESTION ID

S51-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 51 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

When r = 2, $\frac{4}{3}r^3 = 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 51 • BACKWARD 2 • Q5/5

1791

QUESTION ID

S51-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 32/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 51 • BACKWARD 2 • Q5/5

SET 51 • 8.G.C.9

FORWARD 1

Model objects with spheres, cylinders, and cones • High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1

ANSWER KEY Reduced copy of the question page, with its answer

An orange is modeled as a sphere with a diameter of 8 cm.

Find its volume in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q1

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 51 • FORWARD 1 • Q1/5

1793

QUESTION ID

S51-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$\frac{256}{3}$π cubic centimeters

GRADING NOTE The exact answer in terms of π is required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q1

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 51 • FORWARD 1 • Q1/5

A tennis ball can holds 3 balls in a cylinder whose radius equals the ball's radius r and whose height is 6r.

What fraction of the can is filled by the balls?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q2

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 51 • FORWARD 1 • Q2/5

1794

QUESTION ID

S51-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q2

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 51 • FORWARD 1 • Q2/5

Which shape is the best model for a scoop of ice cream?

A. A cylinder

B. A sphere

C. A cone

D. A cube

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q3

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 51 • FORWARD 1 • Q3/5

1795

QUESTION ID

S51-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. A sphere

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q3

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 51 • FORWARD 1 • Q3/5

True or false?

Modeling Earth as a sphere is a useful approximation even though Earth is slightly flattened.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q4

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 51 • FORWARD 1 • Q4/5

1796

QUESTION ID

S51-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q4

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 51 • FORWARD 1 • Q4/5

True or false?

A soccer ball is best modeled by a cylinder.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q5

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 51 • FORWARD 1 • Q5/5

1797

QUESTION ID

S51-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q5

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 51 • FORWARD 1 • Q5/5

SET 51 • 8.G.C.9

FORWARD 2

Use volume formulas to solve problems with spheres • High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A spherical tank has a radius of 3 m.

How many liters does it hold? Use 3.14 for π. (1 cubic meter = 1,000 liters)

Liters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q1

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 51 • FORWARD 2 • Q1/5

1799

QUESTION ID

S51-F2-Q1

TYPE

Short answer

CORRECT ANSWER

113,040 liters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-F2-Q1**

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 51 • FORWARD 2 • Q1/5

A sphere and a cylinder have the same radius r, and the cylinder's height is 2r.

What is the ratio of the sphere's volume to the cylinder's volume?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q2

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 51 • FORWARD 2 • Q2/5

1800

QUESTION ID

S51-F2-Q2

TYPE

Short answer

CORRECT ANSWER

2 : 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-F2-Q2**

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 51 • FORWARD 2 • Q2/5

Doubling a sphere's radius multiplies its volume by...

A. 6

B. 2

C. 8

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q3

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 51 • FORWARD 2 • Q3/5

1801

QUESTION ID

S51-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q3

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 51 • FORWARD 2 • Q3/5

True or false?

A sphere with volume 288π has a radius of 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q4

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 51 • FORWARD 2 • Q4/5

1802

QUESTION ID

S51-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q4

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 51 • FORWARD 2 • Q4/5

True or false?

A hemisphere with radius 6 has a volume of 288π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q5

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 51 • FORWARD 2 • Q5/5

1803

QUESTION ID

S51-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has a volume of 144π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q5

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 51 • FORWARD 2 • Q5/5

SET 52 • 8.G.C.9 • GEOMETRY

Solve real-world problems with the volumes of cylinders, cones, and spheres

MAIN	Grade 8 main question	**8.G.C.9**
BACKWARD 1	Find the volume of a solid made of two prisms	**5.MD.C.5.c**
BACKWARD 2	Find the volume of a right prism	**7.G.B.6**
FORWARD 1	Model real objects with combinations of solids	**HSG-MG.A.1**
FORWARD 2	Apply volume to design and capacity problems	**HSG-MG.A.3**

SET 52 • 8.G.C.9

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.C.9

A grain silo is a cylinder with a radius of 6 m and a height of 10 m, topped with half of a sphere with a radius of 6 m.

Find its volume in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 52 • MAIN • Q1/5

1806

QUESTION ID

S52-M-Q1

TYPE

Short answer

CORRECT ANSWER

504π cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 52 • MAIN • Q1/5

An ice cream cone (radius 3 cm, height 10 cm) is filled and topped with half of a sphere of ice cream with a radius of 3 cm.

Find the total volume of ice cream in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 52 • MAIN • Q2/5

1807

QUESTION ID

S52-M-Q2

TYPE

Short answer

CORRECT ANSWER

48π cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 52 • MAIN • Q2/5

A cone and a cylinder each have a radius of 2 and a height of 6.

How many full cones of water fill the cylinder?

A. 6

B. 3

C. 1

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 52 • MAIN • Q3/5

1808

QUESTION ID

S52-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 52 • MAIN • Q3/5

A straight-sided cylindrical container has an inside radius of 5 cm and holds water. It is tall enough that no water spills. A ball with a radius of 3 cm is dropped in and sinks completely under the water.

How many centimeters does the water level rise?

Rise: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 52 • MAIN • Q4/5

1809

QUESTION ID
S52-M-Q4

TYPE
Short answer

CORRECT ANSWER
1.44 cm

GRADING NOTE The rise is the ball's volume, 36π, divided by the base area, 25π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 52 • MAIN • Q4/5

True or false?

A sphere with radius r fits exactly in a cylinder with radius r and height 2r. The sphere's volume is $\frac{2}{3}$ of the cylinder's.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 52 • MAIN • Q5/5

1810

QUESTION ID

S52-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 52 • MAIN • Q5/5

SET 52 • 8.G.C.9

BACKWARD 1

Find the volume of a solid made of two prisms • Grade 5 • Measurement & Data • 5.MD.C.5.c

A 4 by 3 by 2 prism and a 2 by 3 by 2 prism are joined with no overlap.

What is the total volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 52 • BACKWARD 1 • Q1/5

1812

QUESTION ID

S52-B1-Q1

TYPE

Short answer

CORRECT ANSWER

36 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 52 • BACKWARD 1 • Q1/5

Two cubes with 2-unit edges are joined face to face.

What is the total volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 52 • BACKWARD 1 • Q2/5

1813

QUESTION ID

S52-B1-Q2

TYPE

Short answer

CORRECT ANSWER

16 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 52 • BACKWARD 1 • Q2/5

A 5 by 2 by 1 prism and a 1 by 2 by 1 prism are joined with no overlap.

What is the total volume?

A. 12 cubic units

B. 7 cubic units

C. 20 cubic units

D. 10 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 52 • BACKWARD 1 • Q3/5

1814

QUESTION ID

S52-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 12 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 52 • BACKWARD 1 • Q3/5

True or false?

Volumes of parts that do not overlap can be added to find a total volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 52 • BACKWARD 1 • Q4/5

1815

QUESTION ID

S52-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 52 • BACKWARD 1 • Q4/5

True or false?

A 3 by 3 by 3 cube and a 1 by 1 by 1 cube joined with no overlap have a volume of 30 cubic units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 52 • BACKWARD 1 • Q5/5

1816

QUESTION ID

S52-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 27 + 1 = 28.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 52 • BACKWARD 1 • Q5/5

SET 52 • 8.G.C.9

BACKWARD 2

Find the volume of a right prism • Grade 7 • Geometry • 7.G.B.6

A right rectangular prism is 6 cm by 4 cm by 5 cm.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 52 • BACKWARD 2 • Q1/5

1818

QUESTION ID

S52-B2-Q1

TYPE

Short answer

CORRECT ANSWER

120 cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 52 • BACKWARD 2 • Q1/5

A right prism has a base area of 12 square feet and a height of 3 feet.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 52 • BACKWARD 2 • Q2/5

1819

QUESTION ID

S52-B2-Q2

TYPE

Short answer

CORRECT ANSWER

36 cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 52 • BACKWARD 2 • Q2/5

A right triangular prism has a base area of 10 square inches and a length of 8 inches.

What is its volume?

A. 40 cubic inches

B. 80 cubic inches

C. 18 cubic inches

D. 800 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 52 • BACKWARD 2 • Q3/5

1820

QUESTION ID

S52-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 80 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 52 • BACKWARD 2 • Q3/5

True or false?

A cube with edges of 10 m has a volume of 1,000 cubic meters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 52 • BACKWARD 2 • Q4/5

1821

QUESTION ID

S52-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 52 • BACKWARD 2 • Q4/5

True or false?

A 2 by 5 by 10 box has a volume of 17 cubic units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 52 • BACKWARD 2 • Q5/5

1822

QUESTION ID

S52-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 52 • BACKWARD 2 • Q5/5

SET 52 • 8.G.C.9

FORWARD 1

Model real objects with combinations of solids • High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1

A capsule-shaped vitamin is a cylinder 1 cm long with a radius of 0.3 cm, with half of a sphere of radius 0.3 cm on each end.

What solids make up the model?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q1

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 52 • FORWARD 1 • Q1/5

1824

QUESTION ID

S52-F1-Q1

TYPE

Short answer

CORRECT ANSWER

A cylinder and two half-spheres (together, one whole sphere)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q1

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 52 • FORWARD 1 • Q1/5

A pencil is modeled as a cylinder with a cone at one end.

Which measurements are needed to find its volume?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q2

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 52 • FORWARD 1 • Q2/5

1825

QUESTION ID

S52-F1-Q2

TYPE

Short answer

CORRECT ANSWER

The radius, the length of the cylinder part, and the height of the cone part.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q2

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 52 • FORWARD 1 • Q2/5

Which solids best model a traffic cone sitting on a square base?

A. A cone on a square prism

B. A sphere on a cylinder

C. Two cylinders

D. A cube on a pyramid

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q3

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 52 • FORWARD 1 • Q3/5

1826

QUESTION ID

S52-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. A cone on a square prism

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q3

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 52 • FORWARD 1 • Q3/5

True or false?

A model can use simpler shapes even when the real object is not exactly that shape.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q4

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 52 • FORWARD 1 • Q4/5

1827

QUESTION ID

S52-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q4

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 52 • FORWARD 1 • Q4/5

True or false?

A water bottle is best modeled as one sphere.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q5

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 52 • FORWARD 1 • Q5/5

1828

QUESTION ID

S52-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q5

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 52 • FORWARD 1 • Q5/5

SET 52 • 8.G.C.9

FORWARD 2

Apply volume to design and capacity problems • High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3

A company wants a cylinder-shaped can that holds 500π cubic cm with a radius of 5 cm.

How tall must the can be?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q1

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 52 • FORWARD 2 • Q1/5

1830

QUESTION ID

S52-F2-Q1

TYPE

Short answer

CORRECT ANSWER

20 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q1

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 52 • FORWARD 2 • Q1/5

A cone-shaped funnel has a radius of 6 cm and a height of 10 cm. Water drains at 6π cubic cm per second.

How long does a full funnel take to drain?

Time: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q2

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 52 • FORWARD 2 • Q2/5

1831

QUESTION ID

S52-F2-Q2

TYPE

Short answer

CORRECT ANSWER

20 seconds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q2

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 52 • FORWARD 2 • Q2/5

Each can is a cylinder holding the same volume.

Which design uses the most metal for the curved side?

A. Radius 4 cm, height 6.25 cm

B. Radius 2.5 cm, height 16 cm

C. Radius 5 cm, height 4 cm

D. Radius 2 cm, height 25 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q3

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 52 • FORWARD 2 • Q3/5

1832

QUESTION ID

S52-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Radius 2 cm, height 25 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q3

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 52 • FORWARD 2 • Q3/5

True or false?

Two cans with the same volume can have different surface areas.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q4

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 52 • FORWARD 2 • Q4/5

1833

QUESTION ID

S52-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q4

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 52 • FORWARD 2 • Q4/5

True or false?

A taller can always holds more than a shorter can.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q5

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 52 • FORWARD 2 • Q5/5

1834

QUESTION ID

S52-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q5

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 52 • FORWARD 2 • Q5/5

SET 53 • 8.SP.A.1 • STATISTICS & PROBABILITY

Construct a scatter plot for bivariate measurement data

MAIN	Grade 8 main question	**8.SP.A.1**
BACKWARD 1	Graph points that represent a real-world situation	**5.G.A.2**
BACKWARD 2	Display a numerical data set in a dot plot	**6.SP.B.4**
FORWARD 1	Compare linear, quadratic, and exponential models for two-variabl…	**HSS-ID.B.6**
FORWARD 2	Fit an exponential or quadratic function to data and use it to make	**HSS-ID.B.6.a**

SET 53 • 8.SP.A.1

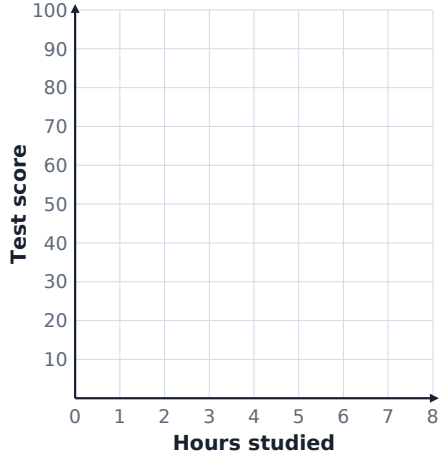
MAIN

Grade 8 main question • Grade 8 • Statistics & Probability • 8.SP.A.1

Seven students recorded (hours studied, test score):

(1, 50), (2, 60), (3, 60), (4, 70), (5, 80), (6, 80), (7, 90)

Make a scatter plot of the data on the grid.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • MAIN • Q1/5

1837

QUESTION ID

S53-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Points at (1, 50), (2, 60), (3, 60), (4, 70), (5, 80), (6, 80), and (7, 90)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

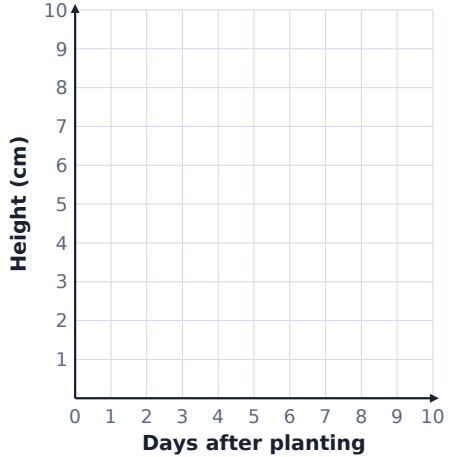
ID S53-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • MAIN • Q1/5

A bean plant was measured on different days (days after planting, height in cm):

(2, 1), (4, 3), (5, 4), (6, 4), (8, 6), (9, 7), (10, 8)

Make a scatter plot of the data on the grid.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • MAIN • Q2/5

1838

QUESTION ID

S53-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Points at (2, 1), (4, 3), (5, 4), (6, 4), (8, 6), (9, 7), and (10, 8)

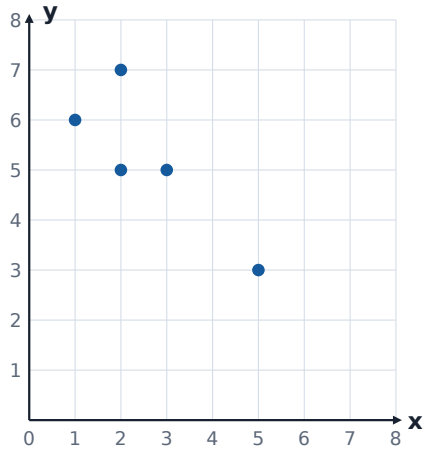
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • MAIN • Q2/5

The scatter plot should show the data (1, 6), (2, 5), (3, 5), (5, 3), and (7, 2), but one point is plotted wrong.

Which point is wrong, and where should it be?

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • MAIN • Q3/5

1839

QUESTION ID

S53-M-Q3

TYPE

Short answer

CORRECT ANSWER

The pair (7, 2) is plotted at (2, 7). It should be 7 units right and 2 units up, at (7, 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • MAIN • Q3/5

Ten students recorded their shoe sizes and heights to make a scatter plot.

What does each point on the scatter plot show?

A. How many students have each shoe size

B. One student's shoe size and height

C. The mean shoe size of the class

D. One student's shoe size only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • MAIN • Q4/5

1840

QUESTION ID

S53-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. One student's shoe size and height

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • MAIN • Q4/5

True or false?

A scatter plot of hours slept and test scores puts two points on the grid for each student, one for each variable.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • MAIN • Q5/5

1841

QUESTION ID

S53-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Each student is one point: (hours slept, test score).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • MAIN • Q5/5

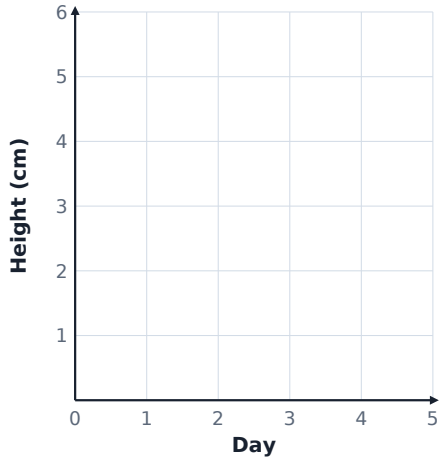
SET 53 • 8.SP.A.1

BACKWARD 1

Graph points that represent a real-world situation • Grade 5 • Geometry • 5.G.A.2

A plant was 2 cm tall on day 1, 4 cm tall on day 2, and 5 cm tall on day 3.

Plot the points (day, height) on the grid.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 53 • BACKWARD 1 • Q1/5

1843

QUESTION ID

S53-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Points at (1, 2), (2, 4), and (3, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 53 • BACKWARD 1 • Q1/5

A graph shows points (day, height in cm).

What does the point (4, 3) mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 53 • BACKWARD 1 • Q2/5

1844

QUESTION ID

S53-B1-Q2

TYPE

Short answer

CORRECT ANSWER

On day 4, the height was 3 cm.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 53 • BACKWARD 1 • Q2/5

Points on a graph are (hours, miles).

Which point shows 5 hours and 2 miles?

A. (5, 5)

B. (2, 5)

C. (5, 2)

D. (2, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 53 • BACKWARD 1 • Q3/5

1845

QUESTION ID

S53-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (5, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 53 • BACKWARD 1 • Q3/5

True or false?

To plot (3, 6), start at the origin, move 3 units right, then 6 units up.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 53 • BACKWARD 1 • Q4/5

1846

QUESTION ID

S53-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 53 • BACKWARD 1 • Q4/5

True or false?

The point (0, 4) is on the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 53 • BACKWARD 1 • Q5/5

1847

QUESTION ID

S53-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE (0, 4) is on the y-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 53 • BACKWARD 1 • Q5/5

SET 53 • 8.SP.A.1

BACKWARD 2

Display a numerical data set in a dot plot • Grade 6 • Statistics & Probability • 6.SP.B.4

The numbers of pets of 10 students are 0, 1, 1, 2, 2, 2, 3, 3, 4, and 1.

Make a dot plot of the data on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q1/5

1849

QUESTION ID

S53-B2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Dots: 1 above 0, 3 above 1, 3 above 2, 2 above 3, 1 above 4, and none above 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

How many students are shown in the dot plot?



Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q2/5

1850

QUESTION ID

S53-B2-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q2/5

Which display shows each value of a data set as a dot above a number line?

A. A box plot

B. A bar graph

C. A histogram

D. A dot plot

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q3/5

1851

QUESTION ID

S53-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A dot plot

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q3/5

True or false?

In a dot plot, a value that appears 3 times has 3 dots stacked above it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q4/5

1852

QUESTION ID

S53-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q4/5

True or false?

A dot plot shows two different measurements for each student.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q5/5

1853

QUESTION ID

S53-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A dot plot shows one variable.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q5/5

SET 53 • 8.SP.A.1

FORWARD 1

Compare linear, quadratic, and exponential models for two-variable data, using technology • High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6

Data: (0, 3), (1, 6), (2, 12), (3, 24).

Does a linear or an exponential model fit these data better? Explain using differences and ratios, and give the model.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 53 • FORWARD 1 • Q1/5

1855

QUESTION ID

S53-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Exponential: the differences (3, 6, 12) are not equal, but each y-value is 2 times the one before, so y = 3 · 2^x.

GRADING NOTE Both the reason and the model are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-F1-Q1**

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 53 • FORWARD 1 • Q1/5

Use a calculator or spreadsheet to find the least-squares line for the data (1, 2), (2, 3), (3, 5), (4, 6). Round the slope and the y-intercept to the nearest tenth.

Line: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 53 • FORWARD 1 • Q2/5

1856

QUESTION ID

S53-F1-Q2

TYPE

Short answer

CORRECT ANSWER

y = 1.4x + 0.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-F1-Q2**

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 53 • FORWARD 1 • Q2/5

Which model fits the data (1, 1), (2, 4), (3, 9), (4, 16) exactly?

A. $y = x^2$

B. y = 2^x

C. $y = 5x - 5$

D. $y = 4x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q3

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 53 • FORWARD 1 • Q3/5

1857

QUESTION ID

S53-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $y = x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-F1-Q3**

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 53 • FORWARD 1 • Q3/5

True or false?

A scatter plot that curves upward more and more steeply may be fit better by an exponential or quadratic function than by a line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 53 • FORWARD 1 • Q4/5

1858

QUESTION ID

S53-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 53 • FORWARD 1 • Q4/5

True or false?

A line with residuals 0, 3, −5, and 4 fits the same data better than a curve with residuals 0, 0.1, −0.2, and 0.1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 53 • FORWARD 1 • Q5/5

1859

QUESTION ID

S53-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The curve's residuals are much smaller.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 53 • FORWARD 1 • Q5/5

SET 53 • 8.SP.A.1

FORWARD 2

Fit an exponential or quadratic function to data and use it to make predictions • High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

A bacteria count is 100 at hour 0, 200 at hour 1, 400 at hour 2, and 800 at hour 3.

Write a function that fits the data.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q1

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 53 • FORWARD 2 • Q1/5

1861

QUESTION ID

S53-F2-Q1

TYPE

Short answer

CORRECT ANSWER

100 · 2^x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-F2-Q1**

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 53 • FORWARD 2 • Q1/5

A bacteria count fits y = 100 · 2^x after x hours.

Predict the count at hour 6.

Count: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q2

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 53 • FORWARD 2 • Q2/5

1862

QUESTION ID

S53-F2-Q2

TYPE

Short answer

CORRECT ANSWER

6,400

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-F2-Q2**

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 53 • FORWARD 2 • Q2/5

Which function fits the data exactly?

x	0	1	2	3	4
y	0	1	4	9	16

A. $y = x^2$

B. $y = 4x$

C. $y = x + 4$

D. $y = 2^x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q3

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 53 • FORWARD 2 • Q3/5

1863

QUESTION ID

S53-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $y = x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q3

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 53 • FORWARD 2 • Q3/5

True or false?

The data (0, 5), (1, 10), (2, 20), and (3, 40) fit $y = 5 \cdot 2^x$ exactly.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q4

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 53 • FORWARD 2 • Q4/5

1864

QUESTION ID

S53-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q4

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 53 • FORWARD 2 • Q4/5

True or false?

Data whose y-values go up by the same amount at each step are best fit by an exponential function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q5

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 53 • FORWARD 2 • Q5/5

1865

QUESTION ID

S53-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Equal differences mean a linear function fits.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q5

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 53 • FORWARD 2 • Q5/5

SET 54 • 8.SP.A.1 • STATISTICS & PROBABILITY

Describe clustering, outliers, and positive, negative, linear, or nonlinear association in a scatter plot

MAIN	Grade 8 main question	**8.SP.A.1**
BACKWARD 1	Describe a data distribution by its center, spread, and shape	**6.SP.A.2**
BACKWARD 2	Describe the overall pattern of data and any striking deviations	**6.SP.B.5.c**
FORWARD 1	Distinguish between correlation and causation	**HSS-ID.C.9**
FORWARD 2	Interpret the correlation coefficient of a linear fit	**HSS-ID.C.8**

SET 54 • 8.SP.A.1

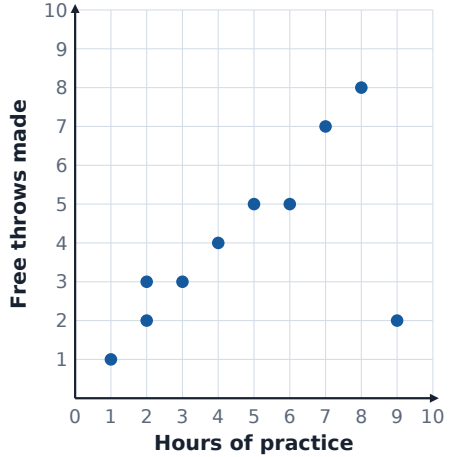
MAIN

Grade 8 main question • Grade 8 • Statistics & Probability • 8.SP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Describe the association in the scatter plot (positive, negative, or none; linear or nonlinear), and name any outlier.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 54 • MAIN • Q1/5

1868

QUESTION ID
S54-M-Q1

TYPE
Short answer

CORRECT ANSWER
A positive linear association; the point (9, 2) is an outlier.

GRADING NOTE Both the association and the outlier are required.

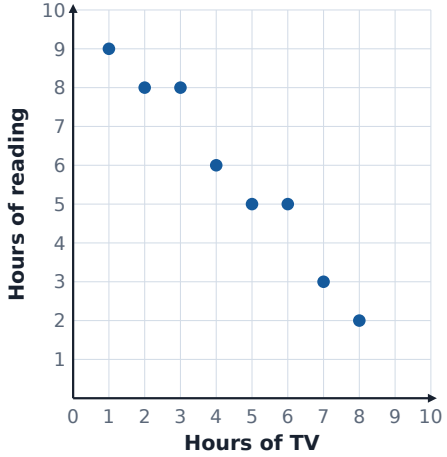
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 54 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Describe the association in the scatter plot.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 54 • MAIN • Q2/5

1869

QUESTION ID

S54-M-Q2

TYPE

Short answer

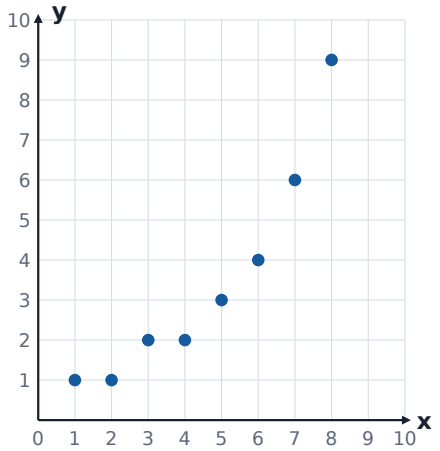
CORRECT ANSWER

A negative linear association: as hours of TV increase, hours of reading decrease.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 54 • MAIN • Q2/5

Which best describes the association in the scatter plot?



A. Positive and linear

B. No association

C. Negative and nonlinear

D. Positive and nonlinear

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 54 • MAIN • Q3/5

1870

QUESTION ID

S54-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Positive and nonlinear

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q3

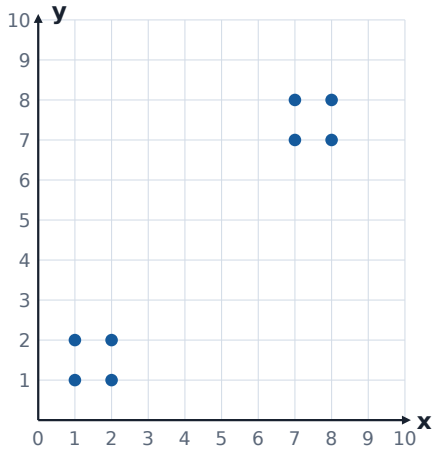
Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 54 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The points in the scatter plot form two groups.

Describe the two clusters.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 54 • MAIN • Q4/5

1871

QUESTION ID

S54-M-Q4

TYPE

Short answer

CORRECT ANSWER

One cluster has x and y from 1 to 2; the other has x and y from 7 to 8.

GRADING NOTE Any description that locates both clusters is correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 54 • MAIN • Q4/5

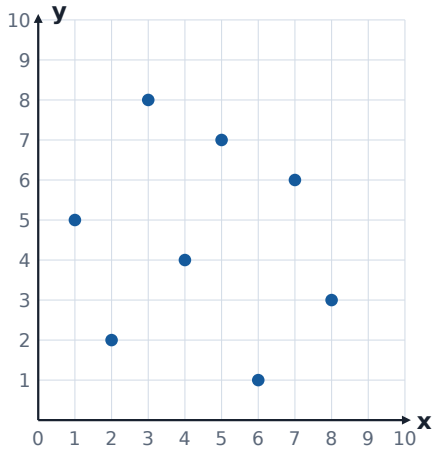
ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The scatter plot shows a negative association.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 54 • MAIN • Q5/5

1872

QUESTION ID

S54-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The points show no association.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 54 • MAIN • Q5/5

SET 54 • 8.SP.A.1

BACKWARD 1

Describe a data distribution by its center, spread, and shape • Grade 6 • Statistics & Probability • 6.SP.A.2

Is the data in the dot plot symmetric or skewed?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 54 • BACKWARD 1 • Q1/5

1874

QUESTION ID

S54-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Symmetric, centered at 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 54 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the range of the data in the dot plot?



Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 54 • BACKWARD 1 • Q2/5

1875

QUESTION ID

S54-B1-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 54 • BACKWARD 1 • Q2/5

Which value is far from the rest of the data?

3, 4, 4, 5, 5, 6, 18

A. 18

B. 6

C. 3

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 54 • BACKWARD 1 • Q3/5

1876

QUESTION ID

S54-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 54 • BACKWARD 1 • Q3/5

True or false?

The data set 2, 3, 3, 4, 4, 4, 5, 5, 6 is symmetric about 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 54 • BACKWARD 1 • Q4/5

1877

QUESTION ID

S54-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 54 • BACKWARD 1 • Q4/5

True or false?

The range of a data set measures its center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 54 • BACKWARD 1 • Q5/5

1878

QUESTION ID

S54-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The range measures spread.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 54 • BACKWARD 1 • Q5/5

SET 54 • 8.SP.A.1

BACKWARD 2

Describe the overall pattern of data and any striking deviations • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the median.

4, 6, 7, 7, 8, 9, 30

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 54 • BACKWARD 2 • Q1/5

1880

QUESTION ID

S54-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 54 • BACKWARD 2 • Q1/5

Which value is a striking deviation from the pattern?

4, 6, 7, 7, 8, 9, 30

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 54 • BACKWARD 2 • Q2/5

1881

QUESTION ID

S54-B2-Q2

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 54 • BACKWARD 2 • Q2/5

The data are 10, 12, 12, 13, 14, and 40.

Which measure of center is changed more by the value 40?

A. The mean

B. The median

C. Both are changed equally

D. Neither is changed

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 54 • BACKWARD 2 • Q3/5

1882

QUESTION ID

S54-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. The mean

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 54 • BACKWARD 2 • Q3/5

True or false?

Removing a value that is much larger than the others lowers the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 54 • BACKWARD 2 • Q4/5

1883

QUESTION ID

S54-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 54 • BACKWARD 2 • Q4/5

True or false?

An outlier always changes the median by a large amount.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 54 • BACKWARD 2 • Q5/5

1884

QUESTION ID

S54-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The median depends on the middle value, so it changes little.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 54 • BACKWARD 2 • Q5/5

SET 54 • 8.SP.A.1

FORWARD 1

Distinguish between correlation and causation • High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9

ANSWER KEY Reduced copy of the question page, with its answer

Data show that cities with more ice-cream sales also have more sunburns.

Do these data show that buying ice cream causes sunburns? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 54 • FORWARD 1 • Q1/5

1886

QUESTION ID

S54-F1-Q1

TYPE

Short answer

CORRECT ANSWER

No. The data show an association, not a cause. Something else, such as hot, sunny weather, could explain both.

GRADING NOTE Any reason that separates association from cause is accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F1-Q1**

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 54 • FORWARD 1 • Q1/5

A study finds that students who sleep more tend to have higher grades.

Can you conclude that more sleep causes higher grades? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 54 • FORWARD 1 • Q2/5

1887

QUESTION ID

S54-F1-Q2

TYPE

Short answer

CORRECT ANSWER

No. The data show an association, not a cause; a randomized experiment is needed to show cause.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 54 • FORWARD 1 • Q2/5

Which kind of study can show that one variable causes a change in another?

A. An observational study

B. A randomized experiment

C. A survey of volunteers

D. A scatter plot of survey data

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q3

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 54 • FORWARD 1 • Q3/5

1888

QUESTION ID

S54-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. A randomized experiment

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q3

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 54 • FORWARD 1 • Q3/5

True or false?

Two variables can be strongly associated without one causing the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 54 • FORWARD 1 • Q4/5

1889

QUESTION ID

S54-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 54 • FORWARD 1 • Q4/5

True or false?

A strong correlation shows that one variable causes the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 54 • FORWARD 1 • Q5/5

1890

QUESTION ID

S54-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 54 • FORWARD 1 • Q5/5

SET 54 • 8.SP.A.1

FORWARD 2

Interpret the correlation coefficient of a linear fit • High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

Use a calculator or spreadsheet to find the correlation coefficient r for the data (1, 2), (2, 4), (3, 5), (4, 4), (5, 6). Round to the nearest hundredth, and describe the association.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q1

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 54 • FORWARD 2 • Q1/5

1892

QUESTION ID

S54-F2-Q1

TYPE

Short answer

CORRECT ANSWER

r ≈ 0.85: a fairly strong positive linear association.

GRADING NOTE Both the value and the description are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F2-Q1**

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 54 • FORWARD 2 • Q1/5

Which shows the stronger linear association, r = 0.45 or r = −0.81? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q2

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 54 • FORWARD 2 • Q2/5

1893

QUESTION ID

S54-F2-Q2

TYPE

Short answer

CORRECT ANSWER

r = −0.81, because its absolute value is closer to 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F2-Q2**

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 54 • FORWARD 2 • Q2/5

Points lie almost exactly on a line that rises from left to right.

Which value of r fits?

A. −0.98

B. 0.98

C. 0

D. 0.10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q3

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 54 • FORWARD 2 • Q3/5

1894

QUESTION ID

S54-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 0.98

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q3

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 54 • FORWARD 2 • Q3/5

True or false?

The correlation coefficient r is always between −1 and 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q4

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 54 • FORWARD 2 • Q4/5

1895

QUESTION ID

S54-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q4

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 54 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

r = −0.9 shows a weaker linear association than r = 0.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q5

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 54 • FORWARD 2 • Q5/5

1896

QUESTION ID
S54-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE |−0.9| is greater than |0.5|, so r = −0.9 is stronger.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q5

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 54 • FORWARD 2 • Q5/5

SET 55 • 8.SP.A.2 • STATISTICS & PROBABILITY

Informally fit a straight line to a scatter plot and judge how closely it fits

MAIN	Grade 8 main question	**8.SP.A.2**
BACKWARD 1	Decide from a table or graph whether a relationship is proportion…	**7.RP.A.2.a**
BACKWARD 2	Write an equation for a relationship shown in a table or graph	**6.EE.C.9**
FORWARD 1	Fit a least-squares line with technology and compare candidate lin…	**HSS-ID.B.6.c**
FORWARD 2	Judge the fit of a line by finding and plotting residuals	**HSS-ID.B.6.b**

SET 55 • 8.SP.A.2

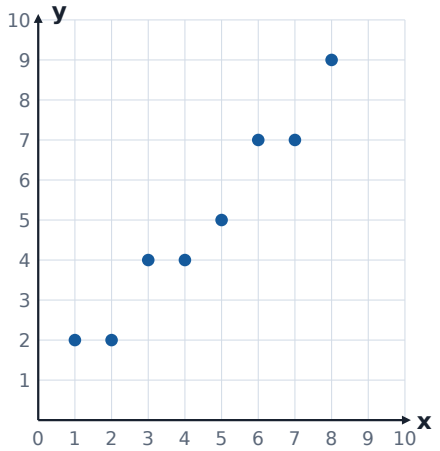
MAIN

Grade 8 main question • Grade 8 • Statistics & Probability • 8.SP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Draw a straight line that fits the data. Then write an equation of your line.

Equation: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 55 • MAIN • Q1/5

1899

QUESTION ID

S55-M-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A straight line close to the points, with about as many points above it as below it, for example through (0, 0.5) and (8, 8.5)

Written: The equation of the line drawn, about y = x + 0.5

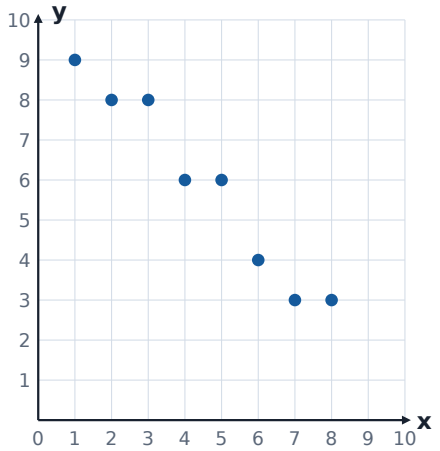
GRADING NOTE Slopes from about 0.8 to 1.2 and y-intercepts from about 0 to 1.5 are accepted when they match the line drawn.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 55 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Draw a straight line that fits the data. Use your line to estimate y when x = 9.



Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 55 • MAIN • Q2/5

1900

GRADING NOTE Accept an estimate that matches the student's own line.

QUESTION ID

S55-M-Q2

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A straight line close to the points, with about as many points above it as below it, for example through (0, 10) and (8, 2)

Written: About 1 to 2, read from the line drawn (y = −x + 10 gives 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 55 • MAIN • Q2/5

Two students drew lines for the same scatter plot. Line A has every point above it. Line B has about half the points above it and half below, all close to it.

Which line fits better? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q3

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 55 • MAIN • Q3/5

1901

QUESTION ID

S55-M-Q3

TYPE

Short answer

CORRECT ANSWER

Line B, because the points are close to it and balanced on both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q3

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 55 • MAIN • Q3/5

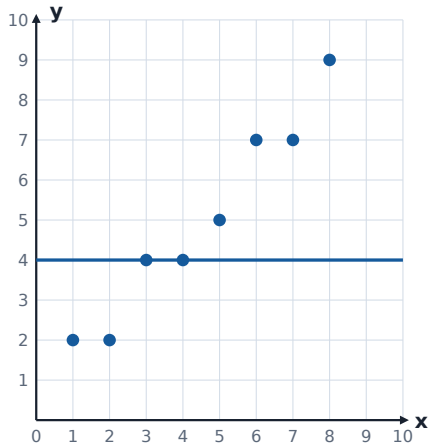
Why is the line on the scatter plot a poor fit?

A. It is not long enough.

B. It does not follow the upward trend of the points.

C. It starts at the y-axis.

D. It passes through too many points.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q4

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 55 • MAIN • Q4/5

1902

QUESTION ID

S55-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. It does not follow the upward trend of the points.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q4

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 55 • MAIN • Q4/5

True or false?

A line that fits a scatter plot well must pass through every point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q5

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 55 • MAIN • Q5/5

1903

QUESTION ID

S55-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A good line is close to the points; it may pass through none of them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q5

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 55 • MAIN • Q5/5

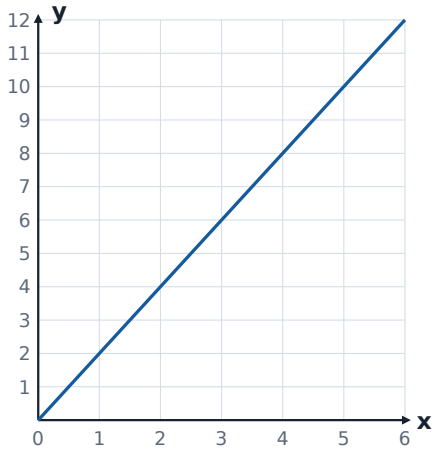
SET 55 • 8.SP.A.2

BACKWARD 1

Decide from a table or graph whether a relationship is proportional • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

ANSWER KEY Reduced copy of the question page, with its answer

Is the relationship shown in the graph proportional? Explain.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 55 • BACKWARD 1 • Q1/5

1905

QUESTION ID

S55-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. The graph is a straight line through the origin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 55 • BACKWARD 1 • Q1/5

Is the relationship in the table proportional? Explain.

x	1	2	4
y	3	6	13

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 55 • BACKWARD 1 • Q2/5

1906

QUESTION ID

S55-B1-Q2

TYPE

Short answer

CORRECT ANSWER

No. 3 ÷ 1 = 3 and 6 ÷ 2 = 3, but 13 ÷ 4 is not 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 55 • BACKWARD 1 • Q2/5

Which graph shows a proportional relationship?

A. A curve through (0, 0) and (2, 4)

B. A straight line through (0, 2) and (2, 6)

C. A horizontal line through (0, 3)

D. A straight line through (0, 0) and (2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 55 • BACKWARD 1 • Q3/5

1907

QUESTION ID

S55-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A straight line through (0, 0) and (2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 55 • BACKWARD 1 • Q3/5

True or false?

A straight line through (0, 0) and (3, 9) shows a proportional relationship.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 55 • BACKWARD 1 • Q4/5

1908

QUESTION ID

S55-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 55 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Every straight-line graph shows a proportional relationship.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 55 • BACKWARD 1 • Q5/5

1909

QUESTION ID

S55-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The line must also pass through the origin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 55 • BACKWARD 1 • Q5/5

SET 55 • 8.SP.A.2

BACKWARD 2

Write an equation for a relationship shown in a table or graph • Grade 6 • Expressions & Equations • 6.EE.C.9

ANSWER KEY Reduced copy of the question page, with its answer

Write an equation for y in terms of x.

x	1	2	3
y	4	8	12

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 55 • BACKWARD 2 • Q1/5

1911

QUESTION ID

S55-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 55 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write an equation for y in terms of x.

x	0	1	2	3
y	2	3	4	5

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 55 • BACKWARD 2 • Q2/5

1912

QUESTION ID

S55-B2-Q2

TYPE

Short answer

CORRECT ANSWER

x + 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 55 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A line passes through (0, 0), (1, 5), and (2, 10).

Which equation does it show?

A. y = 5x

B. y = 10x

C. x = 5y

D. y = x + 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 55 • BACKWARD 2 • Q3/5

1913

QUESTION ID

S55-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. y = 5x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 55 • BACKWARD 2 • Q3/5

True or false?

The points (1, 3), (2, 6), and (3, 9) fit y = 3x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 55 • BACKWARD 2 • Q4/5

1914

QUESTION ID

S55-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 55 • BACKWARD 2 • Q4/5

True or false?

The points (0, 1), (1, 2), and (2, 3) fit y = 2x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 55 • BACKWARD 2 • Q5/5

1915

QUESTION ID

S55-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They fit y = x + 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 55 • BACKWARD 2 • Q5/5

SET 55 • 8.SP.A.2

FORWARD 1

Fit a least-squares line with technology and compare candidate lines • High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c

Use a calculator or spreadsheet to find the least-squares line for the data (1, 3), (2, 5), (3, 6), (4, 8), (5, 10), (6, 11). Round the slope and the y-intercept to the nearest tenth.

Line: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 55 • FORWARD 1 • Q1/5

1917

QUESTION ID

S55-F1-Q1

TYPE

Short answer

CORRECT ANSWER

y = 1.6x + 1.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F1-Q1**

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 55 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Two lines are proposed for the data (1, 3), (2, 5), (3, 6), (4, 8): y = 2x + 1 and y = 1.6x + 1.5.

Find the sum of the squared residuals for each line. Which line fits better?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 55 • FORWARD 1 • Q2/5

1918

QUESTION ID

S55-F1-Q2

TYPE

Short answer

CORRECT ANSWER

y = 2x + 1:
0 + 0 + 1 + 1 = 2.
y = 1.6x + 1.5:
0.01 + 0.09 + 0.09 + 0.01 = 0.2. So y = 1.6x + 1.5 fits better.

GRADING NOTE Both sums and the conclusion are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 55 • FORWARD 1 • Q2/5

A least-squares line for hours studied x and test score y is y = 4.2x + 61. A student who studied 0 hours scored 70.

What is that student's residual?

A. 70

B. 9

C. 61

D. −9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q3

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 55 • FORWARD 1 • Q3/5

1919

QUESTION ID

S55-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q3

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 55 • FORWARD 1 • Q3/5

True or false?

The least-squares line makes the sum of the squared residuals as small as possible.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 55 • FORWARD 1 • Q4/5

1920

QUESTION ID

S55-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 55 • FORWARD 1 • Q4/5

True or false?

A line of fit should always be drawn through the first and last data points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 55 • FORWARD 1 • Q5/5

1921

QUESTION ID

S55-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It should follow the whole trend, not just two points.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 55 • FORWARD 1 • Q5/5

SET 55 • 8.SP.A.2

FORWARD 2

Judge the fit of a line by finding and plotting residuals • High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b

A line of fit is y = 2x + 1. A data point is (3, 8).

Find the residual (actual y minus predicted y).

Residual: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q1

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 55 • FORWARD 2 • Q1/5

1923

QUESTION ID

S55-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F2-Q1**

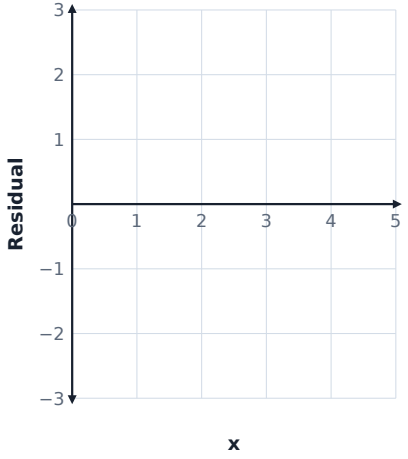
High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 55 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A line of fit is y = 2x + 1 for the data (1, 4), (2, 4), (3, 8), and (4, 9).

Find each residual (actual y minus predicted y) and plot the residuals against x on the grid.

Residuals: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q2

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 55 • FORWARD 2 • Q2/5

1924

QUESTION ID

S55-F2-Q2

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: Points at (1, 1), (2, −1), (3, 1), and (4, 0)

Written: 1, −1, 1, and 0

GRADING NOTE Grade both: the four residuals and the four plotted points.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q2

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 55 • FORWARD 2 • Q2/5

A residual plot shows small residuals scattered above and below 0 with no pattern.

What does this suggest?

A. A curve must be used.

B. Every prediction is exact.

C. A linear model is a poor choice.

D. A linear model is appropriate for the data.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q3

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 55 • FORWARD 2 • Q3/5

1925

QUESTION ID

S55-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A linear model is appropriate for the data.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q3

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 55 • FORWARD 2 • Q3/5

True or false?

A residual of 0 means the data point lies on the line of fit.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q4

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 55 • FORWARD 2 • Q4/5

1926

QUESTION ID

S55-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q4

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 55 • FORWARD 2 • Q4/5

True or false?

A residual plot with a clear U shape shows that a linear model fits well.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q5

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 55 • FORWARD 2 • Q5/5

1927

QUESTION ID

S55-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A U shape suggests a curve fits better.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q5

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 55 • FORWARD 2 • Q5/5

SET 56 • 8.SP.A.3 • STATISTICS & PROBABILITY

Use the equation of a linear model to interpret the slope and intercept and to make predictions

MAIN	Grade 8 main question	**8.SP.A.3**
BACKWARD 1	Explain what a point on the graph of a proportional relationship m…	**7.RP.A.2.d**
BACKWARD 2	Evaluate a formula for given values of its variable	**6.EE.A.2.c**
FORWARD 1	Interpret the slope and intercept of a linear model in context	**HSS-ID.C.7**
FORWARD 2	Evaluate reports and predictions based on data	**HSS-IC.B.6**

SET 56 • 8.SP.A.3

MAIN

Grade 8 main question • Grade 8 • Statistics & Probability • 8.SP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A line of fit for the height h (inches) of a plant after w weeks is h = 1.5w + 4.

What do the slope and the intercept mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 56 • MAIN • Q1/5

1930

QUESTION ID

S56-M-Q1

TYPE

Short answer

CORRECT ANSWER

Slope: the plant grows about 1.5 inches per week. Intercept: it was about 4 inches tall at week 0.

GRADING NOTE Both meanings are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 56 • MAIN • Q1/5

A line of fit for the gas g (gallons) left in a car's tank after m miles is g = 14 − 0.04m.

Predict the gas left after 200 miles.

Gas left: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 56 • MAIN • Q2/5

1931

QUESTION ID

S56-M-Q2

TYPE

Short answer

CORRECT ANSWER

6 gallons

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 56 • MAIN • Q2/5

A model for the water level w (feet) of a pond t days after the rain stops is $w = 12 - 0.3t$.

What does -0.3 mean?

A. The level rises about 0.3 foot each day.

B. The pond is empty after 0.3 day.

C. The level starts at 0.3 foot.

D. The level drops about 0.3 foot each day.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q3

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 56 • MAIN • Q3/5

1932

QUESTION ID

S56-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The level drops about 0.3 foot each day.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q3

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 56 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A model for the score s after h hours of practice is $s = 6h + 58$.

Predict the score after 5 hours of practice.

Score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q4

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 56 • MAIN • Q4/5

1933

QUESTION ID

S56-M-Q4

TYPE

Short answer

CORRECT ANSWER

88

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q4

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 56 • MAIN • Q4/5

True or false?

In the model y = 3.2x + 15 for the cost of a taxi ride of x miles, 15 is the cost per mile.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q5

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 56 • MAIN • Q5/5

1934

QUESTION ID

S56-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 15 is the starting fee; 3.2 is the cost per mile.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q5

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 56 • MAIN • Q5/5

SET 56 • 8.SP.A.3

BACKWARD 1

Explain what a point on the graph of a proportional relationship means • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d

The graph of cost (dollars) against pounds is proportional and passes through (1, 3).

What does (1, 3) mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 56 • BACKWARD 1 • Q1/5

1936

QUESTION ID

S56-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1 pound costs $3; this is the unit rate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 56 • BACKWARD 1 • Q1/5

A proportional graph of miles against hours passes through (0, 0).

What does (0, 0) mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 56 • BACKWARD 1 • Q2/5

1937

QUESTION ID

S56-B1-Q2

TYPE

Short answer

CORRECT ANSWER

In 0 hours, 0 miles are traveled.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 56 • BACKWARD 1 • Q2/5

A proportional graph of (gallons, cost) passes through (1, 4).

What is the unit rate?

A. $5 per gallon

B. $4 per gallon

C. $0.25 per gallon

D. $1 per gallon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 56 • BACKWARD 1 • Q3/5

1938

QUESTION ID

S56-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $4 per gallon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 56 • BACKWARD 1 • Q3/5

True or false?

On a proportional graph of (hours, pages), the point (1, 20) means 20 pages are read per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 56 • BACKWARD 1 • Q4/5

1939

QUESTION ID

S56-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 56 • BACKWARD 1 • Q4/5

True or false?

On a proportional graph of (tickets, cost), the point (2, 18) means that one ticket costs $18.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 56 • BACKWARD 1 • Q5/5

1940

QUESTION ID
S56-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE 2 tickets cost $18, so one ticket costs 18 ÷ 2 = $9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 56 • BACKWARD 1 • Q5/5

SET 56 • 8.SP.A.3

BACKWARD 2

Evaluate a formula for given values of its variable • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Evaluate 1.5w + 4 when w = 6.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 56 • BACKWARD 2 • Q1/5

1942

QUESTION ID

S56-B2-Q1

TYPE

Short answer

CORRECT ANSWER

13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 56 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate 300 − 40t when t = 3.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 56 • BACKWARD 2 • Q2/5

1943

QUESTION ID

S56-B2-Q2

TYPE

Short answer

CORRECT ANSWER

180

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 56 • BACKWARD 2 • Q2/5

Evaluate 6h + 58 when h = 2.

A. 62

B. 128

C. 66

D. 70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 56 • BACKWARD 2 • Q3/5

1944

QUESTION ID

S56-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 56 • BACKWARD 2 • Q3/5

True or false?

When x = 4, 3x + 5 = 35.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 56 • BACKWARD 2 • Q4/5

1945

QUESTION ID

S56-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 3 × 4 + 5 = 17.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 56 • BACKWARD 2 • Q4/5

True or false?

When t = 2, 10 − 3t = 14.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 56 • BACKWARD 2 • Q5/5

1946

QUESTION ID

S56-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 10 − 6 = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 56 • BACKWARD 2 • Q5/5

SET 56 • 8.SP.A.3

FORWARD 1

Interpret the slope and intercept of a linear model in context • High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7

A model for the number n of ice pops sold on a day with a high temperature of t °F is n = 4t − 200.

What does the slope mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 56 • FORWARD 1 • Q1/5

1948

QUESTION ID

S56-F1-Q1

TYPE

Short answer

CORRECT ANSWER

About 4 more ice pops are sold for each 1 °F rise in temperature.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 56 • FORWARD 1 • Q1/5

1948

A model for ice-pop sales is $n = 4t - 200$, where t is the temperature in °F.

Does the intercept -200 make sense in this situation? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 56 • FORWARD 1 • Q2/5

1949

QUESTION ID

S56-F1-Q2

TYPE

Short answer

CORRECT ANSWER

No. It would mean −200 ice pops sold at 0 °F, but sales cannot be negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 56 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A model for a phone's battery level b (percent) after h hours of use is $b = 100 - 12.5h$.

What does 100 mean?

A. The battery lasts 100 hours.

B. The battery charges 100% each hour.

C. The battery loses 100% each hour.

D. The battery starts at 100%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q3

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 56 • FORWARD 1 • Q3/5

1950

QUESTION ID

S56-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The battery starts at 100%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q3

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 56 • FORWARD 1 • Q3/5

True or false?

In the model y = 0.8x + 2, y increases by about 0.8 for each increase of 1 in x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 56 • FORWARD 1 • Q4/5

1951

QUESTION ID

S56-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 56 • FORWARD 1 • Q4/5

True or false?

In a linear model, the intercept always has a sensible meaning in the situation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 56 • FORWARD 1 • Q5/5

1952

QUESTION ID

S56-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 56 • FORWARD 1 • Q5/5

SET 56 • 8.SP.A.3

FORWARD 2

Evaluate reports and predictions based on data • High School (Grade 10 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

A model of children's heights for ages 2 to 10 is $h = 2.5a + 31$ inches at age a.

A report uses it to predict the height of a 40-year-old. Is that reasonable? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q1

High School (Grade 10 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 56 • FORWARD 2 • Q1/5

1954

QUESTION ID

S56-F2-Q1

TYPE

Short answer

CORRECT ANSWER

No. It predicts 131 inches (about 11 feet). The model fits only ages 2 to 10, so using it far outside the data is not reliable.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F2-Q1**

High School (Grade 10 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 56 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A line of fit was made from data collected between 60 °F and 90 °F, and the data points lie close to the line.

A report uses it to predict sales at 70 °F. Is that reasonable? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q2

High School (Grade 10 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 56 • FORWARD 2 • Q2/5

1955

QUESTION ID

S56-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. 70 °F is inside the range of the data, and the line fits the data closely.

GRADING NOTE A reason about the data range is required; noting how well the line fits is also expected.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F2-Q2**

High School (Grade 10 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 56 • FORWARD 2 • Q2/5

A report says: "People who own more books live longer, so buying books makes you live longer."

What is the main flaw?

A. It counts too few books.

B. It reports a positive association.

C. It treats an association as a cause.

D. It uses a line of fit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q3

High School (Grade 10 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 56 • FORWARD 2 • Q3/5

1956

QUESTION ID

S56-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. It treats an association as a cause.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q3

High School (Grade 10 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 56 • FORWARD 2 • Q3/5

True or false?

Predictions far outside the range of the data are less reliable than predictions inside it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q4

High School (Grade 10 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 56 • FORWARD 2 • Q4/5

1957

QUESTION ID

S56-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q4

High School (Grade 10 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 56 • FORWARD 2 • Q4/5

True or false?

A survey of volunteers always represents the whole population.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q5

High School (Grade 10 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 56 • FORWARD 2 • Q5/5

1958

QUESTION ID

S56-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q5

High School (Grade 10 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 56 • FORWARD 2 • Q5/5

SET 57 • 8.SP.A.4 • STATISTICS & PROBABILITY

Construct a two-way table summarizing data on two categorical variables

MAIN	Grade 8 main question	**8.SP.A.4**
BACKWARD 1	Report the number of observations in a data set	**6.SP.B.5.a**
BACKWARD 2	Solve "how many more" and "how many in all" problems with a sc…	**3.MD.B.3**
FORWARD 1	Find joint, marginal, and conditional relative frequencies in a two…	**HSS-ID.B.5**
FORWARD 2	Use a two-way table to find conditional probabilities and decide…	**HSS-CP.A.4**

SET 57 • 8.SP.A.4

MAIN

Grade 8 main question • Grade 8 • Statistics & Probability • 8.SP.A.4

60 students were asked whether they have a curfew and whether they have chores. 36 have a curfew. 24 of the students with a curfew have chores. 8 of the students without a curfew have chores.

Complete the two-way table.

	Chores	No chores	Total
Curfew			
No curfew			
Total			

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • MAIN • Q1/5

1961

QUESTION ID

S57-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Curfew: 24 chores, 12 no chores, 36 total. No curfew: 8 chores, 16 no chores, 24 total. Totals: 32 chores, 28 no chores, 60 in all.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • MAIN • Q1/5

50 students were asked whether they prefer summer or winter. 22 are in Grade 7 and 28 are in Grade 8. 15 of the Grade 7 students and 18 of the Grade 8 students prefer summer.

Complete the two-way table.

	Summer	Winter	Total
Grade 7			
Grade 8			
Total			

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • MAIN • Q2/5

1962

QUESTION ID

S57-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Grade 7: 15 summer, 7 winter, 22 total. Grade 8: 18 summer, 10 winter, 28 total. Totals: 33 summer, 17 winter, 50 in all.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

90 people were asked whether they exercise daily and whether they eat breakfast. 40 exercise daily. 30 of the people who exercise daily eat breakfast. 50 of the 90 people eat breakfast.

Complete the two-way table.

	Breakfast	No breakfast	Total
Exercises			
Does not			
Total			

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • MAIN • Q3/5

1963

QUESTION ID

S57-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

Exercises: 30 breakfast, 10 no breakfast, 40 total. Does not: 20 breakfast, 30 no breakfast, 50 total. Totals: 50 breakfast, 40 no breakfast, 90 in all.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Ten students answered two questions: "Do you have a pet?" and "Do you have a sibling?" Their answers (pet, sibling) are:

yes–yes, yes–no, no–no, yes–yes, no–yes, yes–yes, no–no, no–yes, yes–no, no–no

Complete the two-way table.

	Sibling	No sibling	Total
Pet			
No pet			
Total			

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • MAIN • Q4/5

1964

QUESTION ID

S57-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Pet: 3 sibling, 2 no sibling, 5 total. No pet: 2 sibling, 3 no sibling, 5 total. Totals: 5 sibling, 5 no sibling, 10 in all.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A completed two-way table for 60 students has exactly two rows, Curfew and No curfew, and every student is in one of them. Its row totals can be 36 and 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • MAIN • Q5/5

1965

QUESTION ID

S57-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Every student is in exactly one of the two rows, so the row totals must add to 60.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • MAIN • Q5/5

SET 57 • 8.SP.A.4

BACKWARD 1

Report the number of observations in a data set • Grade 6 • Statistics & Probability • 6.SP.B.5.a

In a survey, 12 students chose soccer, 9 chose basketball, and 7 chose tennis.

How many students were surveyed?

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 57 • BACKWARD 1 • Q1/5

1967

QUESTION ID

S57-B1-Q1

TYPE

Short answer

CORRECT ANSWER

28

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 57 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

How many observations are shown in the dot plot?



Observations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 57 • BACKWARD 1 • Q2/5

1968

QUESTION ID

S57-B1-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 57 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Each of 25 students named a favorite fruit. 5 different fruits were named.

How many observations are in the data set?

A. 5

B. 125

C. 25

D. 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 57 • BACKWARD 1 • Q3/5

1969

QUESTION ID

S57-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 57 • BACKWARD 1 • Q3/5

True or false?

If 14 students answer yes and 11 answer no, there are 25 observations.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 57 • BACKWARD 1 • Q4/5

1970

QUESTION ID

S57-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 57 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The data set 3, 3, 5, 7 has 3 observations.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 57 • BACKWARD 1 • Q5/5

1971

QUESTION ID

S57-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has 4 observations.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 57 • BACKWARD 1 • Q5/5

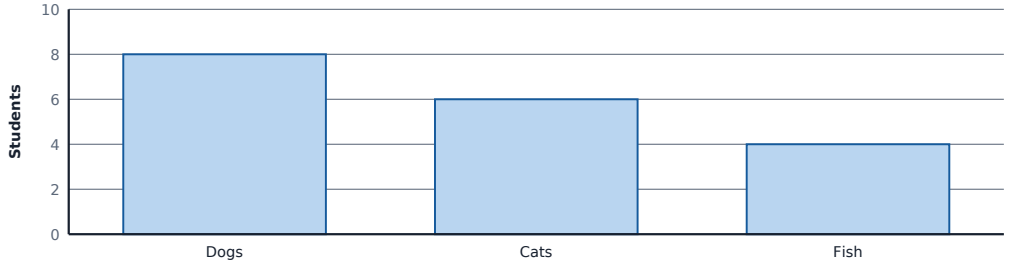
SET 57 • 8.SP.A.4

BACKWARD 2

Solve "how many more" and "how many in all" problems with a scaled bar graph • Grade 3 • Measurement & Data • 3.MD.B.3

ANSWER KEY Reduced copy of the question page, with its answer

How many more students chose dogs than fish?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 2 • Q1/5

1973

QUESTION ID

S57-B2-Q1

TYPE

Short answer

CORRECT ANSWER

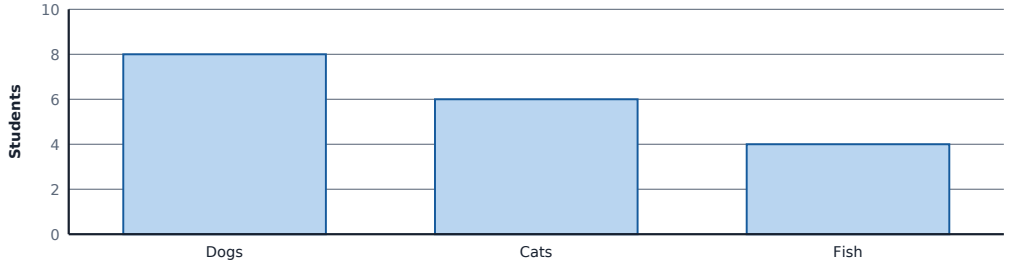
4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

How many students chose a pet in all?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 2 • Q2/5

1974

QUESTION ID

S57-B2-Q2

TYPE

Short answer

CORRECT ANSWER

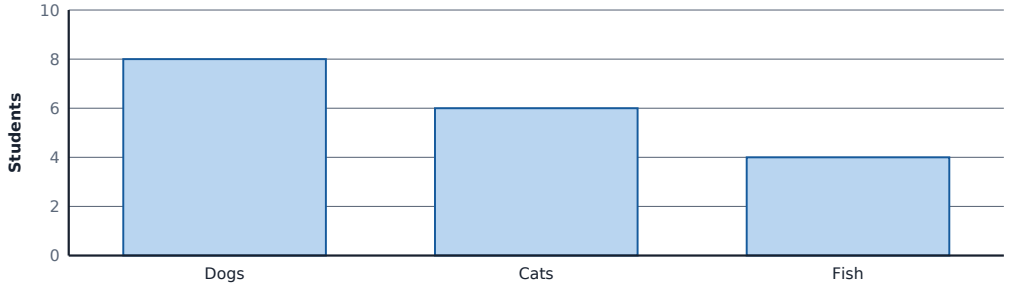
18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

How many fewer students chose fish than cats?



A. 4 **B.** 6 **C.** 10 **D.** 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 2 • Q3/5

1975

QUESTION ID

S57-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2

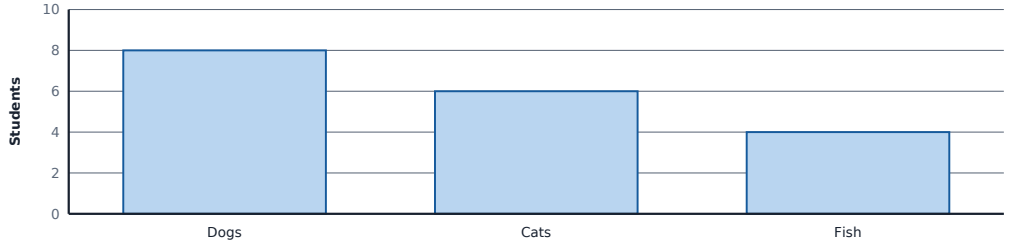
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Fewer students chose cats than fish.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 2 • Q4/5

1976

QUESTION ID

S57-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

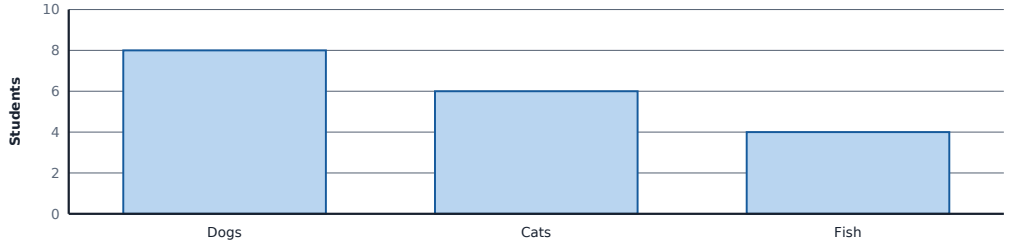
GRADING NOTE 6 chose cats and 4 chose fish.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 2 • Q4/5

True or false?

The same number of students chose dogs and cats.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 2 • Q5/5

1977

QUESTION ID

S57-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 2 • Q5/5

SET 57 • 8.SP.A.4

FORWARD 1

Find joint, marginal, and conditional relative frequencies in a two-way table • High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

What is the joint relative frequency of students who have a curfew and have chores?

	Chores	No chores	Total
Curfew	24	12	36
No curfew	8	16	24
Total	32	28	60

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 1 • Q1/5

1979

QUESTION ID

S57-F1-Q1

TYPE

Short answer

CORRECT ANSWER

0.4

GRADING NOTE $\frac{24}{60}$ or 40% is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S57-F1-Q1**

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 1 • Q1/5

What is the marginal relative frequency of students who have chores? Round to the nearest hundredth.

	Chores	No chores	Total
Curfew	24	12	36
No curfew	8	16	24
Total	32	28	60

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 1 • Q2/5

1980

QUESTION ID

S57-F1-Q2

TYPE

Short answer

CORRECT ANSWER

0.53

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 1 • Q2/5

What is the conditional relative frequency of having chores among students with no curfew?

	Chores	No chores	Total
Curfew	24	12	36
No curfew	8	16	24
Total	32	28	60

A. $\frac{2}{3}$ **B.** $\frac{1}{4}$ **C.** $\frac{1}{3}$ **D.** $\frac{2}{15}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q3

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 1 • Q3/5

1981

QUESTION ID

S57-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q3

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 1 • Q3/5

True or false?

The marginal relative frequency of students with a curfew is 0.36.

	Chores	No chores	Total
Curfew	24	12	36
No curfew	8	16	24
Total	32	28	60

A. True **B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 1 • Q4/5

1982

QUESTION ID

S57-F1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is $\frac{36}{60} = 0.6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 1 • Q4/5

True or false?

The joint relative frequencies in the four inner cells add to 2.

	Chores	No chores	Total
Curfew	24	12	36
No curfew	8	16	24
Total	32	28	60

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 1 • Q5/5

1983

QUESTION ID

S57-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They add to 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 1 • Q5/5

SET 57 • 8.SP.A.4

FORWARD 2

Use a two-way table to find conditional probabilities and decide whether events are independent • High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

ANSWER KEY Reduced copy of the question page, with its answer

A student from the table is chosen at random.

Find P(in club | Grade 8).

	In club	Not in club	Total
Grade 7	12	28	40
Grade 8	18	42	60
Total	30	70	100

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q1

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 57 • FORWARD 2 • Q1/5

1985

QUESTION ID

S57-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.3

GRADING NOTE $\frac{18}{60}$ is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q1

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 57 • FORWARD 2 • Q1/5

Are being in the club and being in Grade 7 independent? Explain with conditional probabilities.

	In club	Not in club	Total
Grade 7	12	28	40
Grade 8	18	42	60
Total	30	70	100

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q2

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 57 • FORWARD 2 • Q2/5

1986

QUESTION ID

S57-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. P(club | Grade 7) = $\frac{12}{40}$ = 0.3 and P(club | Grade 8) = $\frac{18}{60}$ = 0.3, both equal to P(club) = 0.3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q2

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 57 • FORWARD 2 • Q2/5

A student from the table is chosen at random.

Find P(Grade 7 | in club).

	In club	Not in club	Total
Grade 7	12	28	40
Grade 8	18	42	60
Total	30	70	100

A. 0.6 **B.** 0.12 **C.** 0.4 **D.** 0.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q3

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 57 • FORWARD 2 • Q3/5

1987

QUESTION ID

S57-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 0.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S57-F2-Q3**

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 57 • FORWARD 2 • Q3/5

True or false?

For a student chosen at random from the table, P(not in club) = 0.7.

	In club	Not in club	Total
Grade 7	12	28	40
Grade 8	18	42	60
Total	30	70	100

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q4

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 57 • FORWARD 2 • Q4/5

1988

QUESTION ID

S57-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q4

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 57 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

For the table, P(in club | Grade 7) equals P(Grade 7 | in club).

	In club	Not in club	Total
Grade 7	12	28	40
Grade 8	18	42	60
Total	30	70	100

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q5

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 57 • FORWARD 2 • Q5/5

1989

QUESTION ID

S57-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are 0.3 and 0.4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S57-F2-Q5**

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 57 • FORWARD 2 • Q5/5

SET 58 • 8.SP.A.4 • STATISTICS & PROBABILITY

Use relative frequencies in a two-way table to describe a possible association

MAIN	Grade 8 main question	**8.SP.A.4**
BACKWARD 1	Find a percent of a quantity and the percent one number is of ano…	**6.RP.A.3.c**
BACKWARD 2	Use ratio language to describe two quantities	**6.RP.A.1**
FORWARD 1	Recognize conditional probability and independence in everyday…	**HSS-CP.A.5**
FORWARD 2	Find a conditional probability as the fraction of B's outcomes that…	**HSS-CP.B.6**

SET 58 • 8.SP.A.4

MAIN

Grade 8 main question • Grade 8 • Statistics & Probability • 8.SP.A.4

ANSWER KEY Reduced copy of the question page, with its answer

For each row, what percent of the people use a phone at night?

	Phone at night	No phone at night	Total
Sleeps 8+ hours	12	28	40
Sleeps under 8 hours	36	24	60
Total	48	52	100

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 58 • MAIN • Q1/5

1992

QUESTION ID

S58-M-Q1

TYPE

Short answer

CORRECT ANSWER

Sleeps 8+ hours: 30%. Sleeps under 8 hours: 60%.

GRADING NOTE Both percents are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 58 • MAIN • Q1/5

Is there evidence of an association between sleep and phone use at night? Explain using the row relative frequencies.

	Phone at night	No phone at night	Total
Sleeps 8+ hours	12	28	40
Sleeps under 8 hours	36	24	60
Total	48	52	100

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 58 • MAIN • Q2/5

1993

QUESTION ID

S58-M-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. 60% of those who sleep under 8 hours use a phone at night, but only 30% of those who sleep 8+ hours do.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 58 • MAIN • Q2/5

Which conclusion do the row relative frequencies support?

	Breakfast	No breakfast	Total
Grade 7	30	20	50
Grade 8	45	30	75
Total	75	50	125

A. There is a strong association because 45 is greater than 30.

B. Grade 7 students are more likely to eat breakfast.

C. Grade 8 students are more likely to eat breakfast.

D. There is no association: 60% of each grade eat breakfast.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 58 • MAIN • Q3/5

1994

QUESTION ID

S58-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. There is no association: 60% of each grade eat breakfast.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 58 • MAIN • Q3/5

What percent of the people who use a phone at night sleep under 8 hours?

	Phone at night	No phone at night	Total
Sleeps 8+ hours	12	28	40
Sleeps under 8 hours	36	24	60
Total	48	52	100

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 58 • MAIN • Q4/5

1995

QUESTION ID

S58-M-Q4

TYPE

Short answer

CORRECT ANSWER

75%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 58 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Grade 8 students are more likely to eat breakfast than Grade 7 students, because 45 is greater than 30.

	Breakfast	No breakfast	Total
Grade 7	30	20	50
Grade 8	45	30	75
Total	75	50	125

A. True **B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 58 • MAIN • Q5/5

1996

QUESTION ID

S58-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Both grades have 60%; compare relative frequencies, not counts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 58 • MAIN • Q5/5

SET 58 • 8.SP.A.4

BACKWARD 1

Find a percent of a quantity and the percent one number is of another • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

ANSWER KEY Reduced copy of the question page, with its answer

What percent of 40 is 12?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 58 • BACKWARD 1 • Q1/5

1998

QUESTION ID

S58-B1-Q1

TYPE

Short answer

CORRECT ANSWER

30%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 58 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find 60% of 75.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 58 • BACKWARD 1 • Q2/5

1999

QUESTION ID

S58-B1-Q2

TYPE

Short answer

CORRECT ANSWER

45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 58 • BACKWARD 1 • Q2/5

1999

ANSWER KEY Reduced copy of the question page, with its answer

36 out of 48 is what percent?

A. 75%

B. 48%

C. 12%

D. 36%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 58 • BACKWARD 1 • Q3/5

2000

QUESTION ID

S58-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 75%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 58 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

15 out of 60 is 15%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 58 • BACKWARD 1 • Q4/5

2001

QUESTION ID

S58-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 15 out of 60 is 25%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 58 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

20% of 50 is 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 58 • BACKWARD 1 • Q5/5

2002

QUESTION ID

S58-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 20% of 50 is 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 58 • BACKWARD 1 • Q5/5

SET 58 • 8.SP.A.4

BACKWARD 2

Use ratio language to describe two quantities • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1

In a group, 12 people use a phone at night and 28 do not.

Write the ratio of users to non-users in simplest form.

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 58 • BACKWARD 2 • Q1/5

2004

QUESTION ID

S58-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3 : 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 58 • BACKWARD 2 • Q1/5

A class has 18 students who walk to school and 12 who ride.

For every 3 students who walk, how many ride?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 58 • BACKWARD 2 • Q2/5

2005

QUESTION ID

S58-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 58 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In a survey, 30 people eat breakfast and 20 do not.

Which statement describes the ratio?

A. For every 3 who eat breakfast, 2 do not.

B. For every 1 who eats breakfast, 30 do not.

C. For every 3 who eat breakfast, 5 do not.

D. For every 2 who eat breakfast, 3 do not.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 58 • BACKWARD 2 • Q3/5

2006

QUESTION ID

S58-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. For every 3 who eat breakfast, 2 do not.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 58 • BACKWARD 2 • Q3/5

2006

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The ratio 24 : 16 is equivalent to 3 : 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 58 • BACKWARD 2 • Q4/5

2007

QUESTION ID

S58-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 58 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The ratio 10 : 15 is equivalent to 3 : 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 58 • BACKWARD 2 • Q5/5

2008

QUESTION ID

S58-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 10 : 15 is equivalent to 2 : 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 58 • BACKWARD 2 • Q5/5

2008

SET 58 • 8.SP.A.4

FORWARD 1

Recognize conditional probability and independence in everyday situations • High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5

Explain the difference between P(sleeps under 8 hours | uses a phone at night) and P(uses a phone at night | sleeps under 8 hours).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 58 • FORWARD 1 • Q1/5

2010

QUESTION ID

S58-F1-Q1

TYPE

Short answer

CORRECT ANSWER

The first is the chance that a night phone user sleeps under 8 hours; the second is the chance that a person who sleeps under 8 hours uses a phone at night. The condition (the group you look in) is different.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S58-F1-Q1**

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 58 • FORWARD 1 • Q1/5

2010

ANSWER KEY Reduced copy of the question page, with its answer

A fair coin is flipped twice, and the flips are independent.

Is P(second flip is heads | first flip is heads) equal to P(second flip is heads)? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 58 • FORWARD 1 • Q2/5

2011

QUESTION ID

S58-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. Both are $\frac{1}{2}$; the first flip does not change the second.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 58 • FORWARD 1 • Q2/5

2011

Which pair of events is most likely independent?

A. Smoking and getting lung disease

B. Rolling a 6 on a fair number cube and flipping heads on a fair coin

C. Rain today and carrying an umbrella today

D. Studying more and getting a higher grade

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q3

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 58 • FORWARD 1 • Q3/5

2012

QUESTION ID

S58-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Rolling a 6 on a fair number cube and flipping heads on a fair coin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q3

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 58 • FORWARD 1 • Q3/5

True or false?

If two events are independent, knowing that one happened does not change the probability of the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 58 • FORWARD 1 • Q4/5

2013

QUESTION ID

S58-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 58 • FORWARD 1 • Q4/5

True or false?

P(A | B) is always equal to P(B | A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 58 • FORWARD 1 • Q5/5

2014

QUESTION ID

S58-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 58 • FORWARD 1 • Q5/5

SET 58 • 8.SP.A.4

FORWARD 2

Find a conditional probability as the fraction of B's outcomes that are also in A • High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6

ANSWER KEY Reduced copy of the question page, with its answer

A fair number cube numbered 1 to 6 is rolled.

Given that the roll is even, what is the probability that it is greater than 3?

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q1

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 58 • FORWARD 2 • Q1/5

2016

QUESTION ID

S58-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$\frac{2}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S58-F2-Q1**

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 58 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Two fair coins are flipped, and the flips are independent.

Given that at least one coin shows heads, what is the probability that both show heads?

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q2

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 58 • FORWARD 2 • Q2/5

2017

QUESTION ID

S58-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$\frac{1}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S58-F2-Q2**

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 58 • FORWARD 2 • Q2/5

Of 100 people, 48 use a phone at night, and 36 of those sleep under 8 hours.

A night phone user is chosen at random. What is P(sleeps under 8 hours | uses a phone at night)?

A. 0.75

B. 0.36

C. 0.48

D. 0.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q3

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 58 • FORWARD 2 • Q3/5

2018

QUESTION ID

S58-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 0.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S58-F2-Q3**

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 58 • FORWARD 2 • Q3/5

True or false?

A card is drawn at random from cards numbered 1 to 10.

Given that the number is odd, the probability that it is 5 or less is $\frac{3}{5}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q4

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 58 • FORWARD 2 • Q4/5

2019

QUESTION ID

S58-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q4

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 58 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A fair number cube numbered 1 to 6 is rolled.

Given that the roll is less than 3, the probability that it is 1 is $\frac{1}{6}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q5

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 58 • FORWARD 2 • Q5/5

2020

QUESTION ID

S58-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only 1 and 2 are possible, so it is $\frac{1}{2}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S58-F2-Q5**

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 58 • FORWARD 2 • Q5/5